basics - math

basics

Apr 20, 2026

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This material is part of the **basics-books project**. It is also available as a .pdf document.

Contents.

- **Linear Algebra.** …, matrix factorization,…; basics of lots of numerical methods; **Vector and Tensor Algebra** provides the mathematical framework for manipulating vectors and tensors, the mathematical objects meant to represent **absolute quantities** - invariant under coordinate transformations - fundamental in geometry and physics.
- **Multivariable Calculus** provides the tools for working with continuous functions of many variables; **Differential Geometry** studies smooth curves, surfaces, volumes (and manifolds in general), within the framework of a Riemann structure, providing a metric to define distances, angles, curvatures and other geometric properties; **Vector and Tensor Calculus** studies vector and tensor fields defined on a manifold, along with their differentiation and integration, and thus being essential for a mature approach to geometry and physics;
- **Functional analysis** and **Complex Calculus** (complex analysis, transforms,…)
- ODEs
- PDEs
- Calculus of Variations: theoretical background; Lagrange multiplier method for constraints; sensitivty; gradient-based methods,…
- Optimization

Part I

Linear Algebra

CHAPTER

ONE

INTRODUCTION TO LINEAR ALGEBRA

- Matrices:
  - properties
  - representation of linear transformations
  - factorizations
  - linear systems

todo:

- existence of solution of equilibria of LTI and their stability, in *LTI:Equilibria* and in *LTI:Stability*

CHAPTER

TWO

MATRICES

$\mathbf{A} \in \mathbb{K}^{m,n}$ with usually $\mathbb{K}^{m,n} = \mathbb{R}^{m,n}$ or $\mathbb{C}^{m,n}$

Hermitian matrix. The Hermitian matrix $\mathbf{A}^*$ of a matrix $\mathbf{A}$ is the transpose and complex conjugate matrix (if $\mathbb{K} = \mathbb{C}$),

$$[\mathbf{A}^*]_{ij} = A^*_{ji} \ ,$$

with the notation of a^* for the complex conjugate of a numerical quantity.

2.1 Subspaces

2.1.1 Range, Image

$$R(\mathbf{A}) = \{\mathbf{y} \in \mathbb{K}^m \mid \exists \mathbf{x} \in \mathbb{K}^m \ , \text{ s.t. } \mathbf{A}\mathbf{x} = \mathbf{y}\}$$

The range of a matrix $\mathbf{A}$ is the linear space built on the columns of $\mathbf{A}$, since the operation $\mathbf{A}\mathbf{x}$ represents nothing but a linear combination of the columns of matrix.

2.1.2 Null, Kernel

$$K(\mathbf{A}) = \{\mathbf{x} \in \mathbb{K}^n \mid \mathbf{A}\mathbf{x} = \mathbf{0}\}$$

2.2 Theorem

2.2.1 Orthogonality of $R(\mathbf{A})$ and $K(\mathbf{A}^*)$

The following holds,

$$R(\mathbf{A}) \perp K(\mathbf{A}^*) \ ,$$

meaning that $\forall \mathbf{u} \in R(\mathbf{A})$ and $\forall \mathbf{v} \in K(\mathbf{A}^*)$, $\mathbf{u}^*\mathbf{v} = 0$.

Proof.

$$\mathbf{u} = \mathbf{A}\mathbf{x}$$
$$\mathbf{0} = \mathbf{A}^*\mathbf{v}$$

and thus, premultiplication by $\mathbf{x}^*$ of the second relation gives

$$0 = \mathbf{x}^*\mathbf{0} = \underbrace{\mathbf{x}^*\mathbf{A}^*}_{=(\mathbf{A}\mathbf{x})^*=\mathbf{u}^*} \mathbf{v} = \mathbf{u}^*\mathbf{v} \,.$$

This theorem becomes quite useful, e.g. for constrained linear systems and projections… (e.g. N-S, or other constrained linear systems…)

todo add links

CHAPTER

THREE

MATRIX FACTORIZATIONS

- **Singular Value Decomposition (SVD)**
- **Spectral decomposition** Eigenvalues, eigenvectors; Jordan canonical formula…

3.1 Gershgorin circle theorem

Let $\mathbf{A}$ a complex n, n matrix. Let $R_i = \sum_{j \neq i} |a_{ij}|$, and $D(a_{ii}, R) \subseteq \mathbb{C}$ the closed disc centered at a_{ii} with radius R_i in the complex plane.

Every eigenvalue of $\mathbf{A}$ lies within at least one of the Gershgorin discs $D(a_{ii}, R_i)$.

Proof

Let λ an eigenvalue of $\mathbf{A}$. Thus, it exsits a vector $\mathbf{v}$ s.t. $\mathbf{A}\mathbf{v} = \lambda \mathbf{v}$, or

$$\sum_j a_{ij} v_j = \lambda v_i \ .$$

Moving a_{ii} from the LHS to the RHS, it follows

$$\sum_{j \neq i} a_{ij} v_j = (\lambda - a_{ii})\, v_i \ .$$

Selecting i s.t. $|v_i| \geq |v_j|$, $\forall j$ it follows that

$$|\lambda - a_{ii}| = \left| \sum_{j \neq i} a_{ij} \frac{v_j}{v_i} \right| \leq \sum_{j \neq i} \left| a_{ij} \frac{v_j}{v_i} \right| \leq \sum_{j \neq i} |a_{ij}| \leq R_i \ .$$

3.2 Spectral radius

Spectral radius of a matrix $\mathbf{A}$ is defined as the maximum of the absolute value of its eigenvalues

$$\rho(\mathbf{A}) = \max |\lambda(\mathbf{A})| \ .$$

- **QR**
- **LU**
- **Schur**

- **Cholesky** Symmetric positive definite matrices have Choleski decomposition,

$$\mathbf{M} = \mathbf{L}\mathbf{L}^* \ ,$$

with $\mathbf{L}$ lower triangular matrix. And thus quite easy to "invert", for solving linear systems.

3.3 Singular Value Decomposition

Singular value decomposition of a matrix $\mathbf{A} \in \mathbb{C}^{m,n}$

$$\mathbf{A}_{(m,n)} = (\text{SVD}) = \mathbf{U}_{(m,m)}\mathbf{S}_{(m,n)}\mathbf{V}^*_{(n,n)}$$

with $\mathbf{U}^*\mathbf{U} = \mathbf{I}_{(m,m)}$, and $\mathbf{V}^*\mathbf{V} = \mathbf{I}_{(n,n)}$, $\mathbf{S} = \text{diag}\{\sigma_i\}$, $\sigma_i \geq 0$.

Exploiting the definition of matrix product, the SVD of matrix A can be written as

$$\mathbf{A} = \sum_{i=\min(m,n)} \sigma_i \mathbf{u}_i \mathbf{v}_i^* \ ,$$

see also economic decomposition below.

3.3.1 Properties

Relation with *range* and *kernel* the matrix.

$$\begin{aligned} \text{R}(\mathbf{A}) &= \{\mathbf{u}_i | \sigma_i > 0\} \\ \text{K}(\mathbf{A}) &= \{\mathbf{v}_i | \sigma_i = 0\} \\ \text{R}(\mathbf{A}^*) &= \{\mathbf{v}_i | \sigma_i > 0\} \\ \text{K}(\mathbf{A}^*) &= \{\mathbf{u}_i | \sigma_i = 0\} \end{aligned}$$

It's immediate to prove $R(\mathbf{A}) \perp K(\mathbf{A}^*)$.

Full or economic decomposition. In general the $\mathbf{S}$ of the full decompsition in not square.

$$\mathbf{A} = (\text{SVD}) = \mathbf{U}_{(m,m)}\mathbf{S}_{(m,n)}\mathbf{V}^*_{(n,n)} = \mathbf{U}^e_{(m,k)}\mathbf{S}^e_{(k,k)}\mathbf{V}^{e*}_{(k,n)} \ ,$$

with $k = \min(m, n)$.

3.3.2 Applications

Solution of under-determined linear systems

Norms and optimization

If L^2-norm is used for vector norms, see *Example 3.3.1*

$$\text{Find} \max_{||\mathbf{x}||=1} ||\mathbf{A}\mathbf{x}||^2$$

$$\mathbf{x}^*\mathbf{A}^*\mathbf{A}\mathbf{x} = \mathbf{x}^*\mathbf{V}\mathbf{S}^* \underbrace{\mathbf{U}^*\mathbf{U}}_{\mathbf{I}} \mathbf{S} \underbrace{\mathbf{V}^*\mathbf{x}}_{\mathbf{z}}$$

and $\mathbf{z} = \mathbf{V}^*\mathbf{x}$ is unitary as well (since $1 = \mathbf{x}^*\mathbf{x} = \mathbf{z}^*\mathbf{V}^*\mathbf{V}\mathbf{z} = \mathbf{z}^*\mathbf{z}$).

After defining $\mathbf{S}_2 := \mathbf{S}^*\mathbf{S}$, the problem thus becomes

$$\text{Find} \max_{||\mathbf{z}||=1} \mathbf{z}^*\mathbf{S}_2\mathbf{z}$$

Manipulating the objective function as $\sum_i \sigma_i^2 |z_i|^2$, the constraint optimization problem has global maximum σ_1^2 (sorted singular values from the largest to the smalles) when $\mathbf{z}_1 = (1, 0, 0, \dots, 0)^T$. Going back to the original variable, optimal condition

- is achieved for $\mathbf{x}_1 = \mathbf{v}_1$;
- has value $\max_{||\mathbf{x}||=1} ||\mathbf{A}\mathbf{x}|| = \sigma_1^2$
- and the response of the system is $\mathbf{y}_1 = \sigma_1 \mathbf{u}_1$ as

$$\mathbf{y}_1 := \mathbf{A}\mathbf{x}_1 = \mathbf{U}\mathbf{S}\mathbf{V}^*\mathbf{v}_1 = \sum_k (\sigma_k \mathbf{u}_k \mathbf{v}_k^*)\, \mathbf{v}_1 = \sigma_1 \mathbf{u}_1 \ .$$

Example 3.3.1 (Other norms - variations of the L^2-norm)

This kind of problem may results as the discrete counterpart of a continuous problem, as an example from *finite element methods*, where $\mathbf{x}$, $\mathbf{y}$ contain the coefficients of the basis functions. In this case, the discrete counterpart of the continous norm-measure of the continuous fields may involve a "mass matrix" (symmetric, definite positive - and thus with Choloeski factorization…),

$$\int_{\Omega_x} |x(\vec{r})|^2 d\vec{r} \simeq \mathbf{x}^*\mathbf{M}_x\mathbf{x}$$
$$\int_{\Omega_y} |y(\vec{r})|^2 d\vec{r} \simeq \mathbf{y}^*\mathbf{M}_y\mathbf{y}$$

Continuous and discrete optimization problems are

$$\text{Find} \max_{|x(\vec{r})|_{L^2(\Omega_x)}=1} |y|^2_{L^2(\Omega_y)}$$

$$\text{Find} \max_{\mathbf{x}^*\mathbf{M}_x\mathbf{x}=1} \mathbf{x}^*\mathbf{A}^*\mathbf{M}_y\mathbf{A}\mathbf{x}$$

with the relation $\mathbf{y} = \mathbf{A}\mathbf{x}$ between the discrete input and output.

This problem can be easily (and efficiently?) recast to the standard form of the problem, using *Cholesky decomposition* of matrix $\mathbf{M}_x = \mathbf{L}_x\mathbf{L}_x^*$, with the definition of the variable $\mathbf{z} = \mathbf{L}_x^*\mathbf{x}$

$$\text{Find} \max_{||\mathbf{z}||=1} \mathbf{z}^*\mathbf{L}_x^{-1}\mathbf{A}^*\mathbf{L}_y\mathbf{L}_y^*\mathbf{A}\mathbf{L}_x^{-*}\mathbf{z}$$

This problem can be efficiently solved with iterative algorithms to compute the SVD of the matrix $\widetilde{\mathbf{A}} := \mathbf{L}_y^*\mathbf{A}\mathbf{L}_x^{-*}$, that doesn't need the expensive full inversion of a matrix but only its action on a vector (instead of evaluating the inverse, a linear system - here triangular! Easier to solve - can be efficiently solved). Algorithms like **Arnoldi algorithm** evaluates the largest eigenvalues or singular values(if no options to set other goals) and the corresponding eigenvectors and singular vectors, alternating **power iterations** and **orthogonalization**. Here power iteration to evaluate the action of the matrix $\widetilde{\mathbf{A}} = \mathbf{L}_y^*\mathbf{A}\mathbf{L}_x^{-*}$ on a generic vector $\mathbf{z}$ is made of the following steps:

1. solution of the linear system $\mathbf{L}_x^*\mathbf{a} = \mathbf{z} \rightarrow \ \mathbf{a} = \dots$
2. matrix-vector multiplication $\mathbf{b} = \mathbf{A}\mathbf{a}$
3. matrix-vector multiplication $\mathbf{c} = \mathbf{L}_y^*\mathbf{b}$

Once the SVD is solved, with $\mathbf{z}_1 = \mathbf{v}_1$

$$\mathbf{x}_1 = \mathbf{L}_x^{-*}\mathbf{v}_1$$
$$\mathbf{y}_1 = \mathbf{A}\mathbf{x}_1 = \sigma_1\mathbf{L}_y^{-*}\mathbf{u}_1$$

CHAPTER

FOUR

LINEAR SYSTEMS

The linear system

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

with $\mathbf{A} \in \mathbb{R}^{m,n}$, $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^m$ has solution if there exists (at least) a vector $\tilde{\mathbf{x}}$ whose product with $\mathbf{A}$ gives $\mathbf{b}$.

Condition for the existence of a solution. A solution exists if $\mathbf{b}$ belongs to the *range of* $\mathbf{A}$.

Uniqueness of a solution. If a solution $\tilde{\mathbf{x}}$ exists, it's unique if the *kernel of* $\mathbf{A}$ is empty, $K(\mathbf{A}) = \emptyset$. If the kernel is not empty,

$$\mathbf{b} = \mathbf{A}(\tilde{\mathbf{x}} + \mathbf{u}) = \mathbf{A}\tilde{\mathbf{x}} + \underbrace{\mathbf{A}\mathbf{u}}_{=\mathbf{0}} ,$$

for $\forall \mathbf{u} \in K(\mathbf{A})$, and thus an infinite number of solutions exists. Given a vector basis of the kernel $\mathbf{K}(\mathbf{A})$, where $\dim(K(\mathbf{A})) = n_K$, $\{\mathbf{u}_1, \dots, \mathbf{u}_{n_K}\}$, the general solution has n_K "degrees of arbitrarieness", since the general solution of the problem is

$$\tilde{\mathbf{x}} + \sum_{i=1}^{n_K} a_i \mathbf{u}_i = \tilde{\mathbf{x}} + \mathbf{U}\mathbf{a} .$$

todo treat under-, det-, over-determined lin sys

CHAPTER

FIVE

SPECTRAL DECOMPOSITION

- **Introduction.** What's spectral (or eigenvalue) decomposition? Arbitrarieness. When is it possible? Special matrices. Properties and theorems (*Cayley-Hamilton*)

$$\mathbf{A}\mathbf{u}_i = s_i \mathbf{u}_i$$

- **Generalized spectral decompositions.** Generalized spectral decomposition of first order systems; generalized spectral deocmposition of second order systems[1]

$$\mathbf{A}\mathbf{u}_i = s_i \mathbf{B}\mathbf{u}_i$$

$$\left[s_i^2 \mathbf{M} + s_i \mathbf{C} + \mathbf{K}\right] \mathbf{u}_i = \mathbf{0} \; .$$

- **Sensitivity of spectral decomposition.** …

5.1 Spectral decomposition of symmetric matrices

In this section the spectral decomposition of symmetric matrices (or Hermitian, if defined on the complex field), $\mathbf{A} = \mathbf{A}^*$, is discussed.

Theorem 5.1.1 (Spectral theorem of a symmetric matrix)

A symmetric matrix has a spectral decomposition, with a complete set of unit orthogonal vectors and real eigenvalues,

$$\mathbf{A} = \mathbf{R}\mathbf{S}\mathbf{R}^* \; ,$$

with $\mathbf{R} = [\, \mathbf{r}_1 \mid \ldots \mid \mathbf{r}_n \,]$, and $\mathbf{r}_i^* \mathbf{r}_k = \delta_{ik}$, and $\mathbf{S} = \mathrm{diag}\{s_i\}$, $s_i \in \mathbb{R}$.

[1] It's likely that mainly underdamped systems with simultaneously diagonalizable matrices will be treated, as this kind of systems often arises in structural dynamics of elastic structures with small structural damping.

Proof

todo *By construction…*

$$
\begin{aligned}
\mathbf{A} &= \mathbf{R}\mathbf{S}\mathbf{R}^* = \\
&= \begin{bmatrix}\mathbf{r}_1 & \dots & \mathbf{r}_n\end{bmatrix} \operatorname{diag}\{s_i\} \begin{bmatrix}\mathbf{r}_1^* \\ \dots \\ \mathbf{r}_n^*\end{bmatrix} = \\
&= \begin{bmatrix}s_1\mathbf{r}_1 & \dots & s_n\mathbf{r}_n\end{bmatrix} \begin{bmatrix}\mathbf{r}_1^* \\ \dots \\ \mathbf{r}_n^*\end{bmatrix} = \\
&= s_1\mathbf{r}_1\mathbf{r}_1^* + \dots + s_n\mathbf{r}_n\mathbf{r}_n^* \ .
\end{aligned}
$$

Properties.

- Left and right eigenvectors are the same. This immediately follows from the symmetry of the matrix and the definition of the left and right eigenvectors

$$
\begin{aligned}
\mathbf{A}\mathbf{r}_k &= s_k\mathbf{r}_k \\
\mathbf{l}_j^*\mathbf{A} &= s_j^*\mathbf{l}_i^* \quad \rightarrow \quad \mathbf{A}\mathbf{l}_j = s_j\mathbf{l}_j \ ,
\end{aligned}
$$

- Eigenvalues are real, as

$$
0 = \mathbf{l}_k^* \left(\mathbf{A}\mathbf{r}_k\right) - \left(\mathbf{l}_k^*\mathbf{A}\right)\mathbf{r}_k = \left(s_k - s_k^*\right)\mathbf{l}_k^*\mathbf{r}_k = 2\,\mathrm{im}\{s_k\}\ \underbrace{\mathbf{l}_k^*\mathbf{r}_k}_{=|\mathbf{r}_k|^2} \ ,
$$

  and thus $\mathrm{im}\{s_k\} = 0$.

- Eigenvectors with different eigenvalues are orthogonal, as

$$
0 = \mathbf{l}_j^* \left(\mathbf{A}\mathbf{r}_k\right) - \left(\mathbf{l}_j^*\mathbf{A}\right)\mathbf{r}_k = \left(s_k - s_j^*\right)\mathbf{l}_j^*\mathbf{r}_k \ ,
$$

  and thus $\mathbf{l}_j^*\mathbf{r}_k = 0$ for $j \neq k$, if $s_j \neq s_k$. Left and right eigenvectors are orthogonal w.r.t. matrix $\mathbf{A}$ as well.

5.2 Sensitivity of spectral decomposition

Matrices involved in a eigenvalue problem can be function of some parameters p. Sensitivity analysis evaluates first-order changes[1] of eigenvalues and eigenvectors following a increment of the parameter $p = \overline{p} + \Delta p$,

$$
\begin{aligned}
s_i(p) &= s_i(\overline{p}) + \Delta p\, s_{i/p}(\overline{p}) + o(\Delta p) \\
\mathbf{u}_i(p) &= \mathbf{u}_i(\overline{p}) + \Delta p\, \mathbf{u}_{i/p}(\overline{p}) + o(\Delta p) \ .
\end{aligned}
$$

Here the terms $s_{i/p}(\overline{p})$ and $\mathbf{u}_{i/p}(\overline{p})$ are the **sensitivity** of the ith eigenvalue and eigenvector respectively w.r.t. to an increment Δp of the variable p, starting from the reference configuration determined by the value $\overline{p}$ of the parameter.

Rank of sensitivity. Sensitivity of eigenvalue to a scalar parameter is a scalar quantity. Sensitivity of eivenvector to a scalar parameter is a vector quantity. If the parameter $\mathbf{p}$ is a vector quantity, sensitivity of eigenvalue w.r.t. $\mathbf{p}$ is a vector quantity, and sensitivity of eigenvector w.r.t. $\mathbf{p}$ is a matrix (or tensor, sometimes…) quantity.

[1] First order changes should be meant as derivatives w.r.t. the parameter p, evaluated for the reference value of $\overline{p}$ that appear in a Taylor series of the quantity of interest, $s(p_i) = s(\overline{p}_i) + \Delta p_k\, \partial_{p_k} s(\overline{p}_i) + o(\Delta p_i)$.

5.2.1 Generalized eigenvalue problem (first order)

Sensitivity of eigenvalue and right eigenvector to a parameter p of matrices $\mathbf{A}(p)$, $\mathbf{B}(p)$ read

$$s_{i/p} = \mathbf{w}_i^* \left(\mathbf{A}_{/p} - s_i \mathbf{B}_{/p}\right) \mathbf{u}_i \tag{5.1}$$

$$\mathbf{u}_{i/p} = \sum_{a \notin \{i\}} \mathbf{w}_a^* \left(-\mathbf{A}_{/p} + s_i \mathbf{B}\right) \mathbf{u}_i \frac{1}{s_a - s_i} \mathbf{u}_a \ ,$$

having exploited normalization condition $\mathbf{w}_i^* \mathbf{B} \mathbf{u}_i = 1$, being $\mathbf{u}_i$ and $\mathbf{w}_i$ the[2] right and left eigenvectors associated with eigenvalue s_i.

Right and left eigenvalue problem

Right and left eigenvalue problems are defined as

$$\begin{aligned} \mathbf{A}\mathbf{u}_i &= s_i \mathbf{B}\mathbf{u}_i \\ \mathbf{v}_j^* \mathbf{A} &= s_j \mathbf{v}_j^* \mathbf{B} \end{aligned} \tag{5.2}$$

Property 5.2.1 (Eigenvalues of the right and left spectral problems)

Left and right eigenvalue problems have the same set of eigenvalues.

Proof

todo

Property 5.2.2 (Orthogonality conditions of right and left spectral problems)

Left and right eigenvectors associated with different eigenvalues are orthogonal w.r.t. the matrices of the system, i.e.

$$\begin{aligned} \mathbf{v}_j^* \mathbf{B} \mathbf{u}_i &= b^{(i)} \delta_{ij} \\ \mathbf{v}_j^* \mathbf{A} \mathbf{u}_i &= a^{(i)} \delta_{ij} \ , \end{aligned}$$

with $a^{(i)} = s_i b^{(i)}$. Parameters $a^{(i)}$, $b^{(i)}$ are not uniquely determined, and one of them (or their combination) can be used in a **normalization condition** removing arbitrarieness of eigenvectors to a multiplicative factor.

Proof

Starting from the left and right eigenvalue problems (5.2) for two different eigenvalues $s_i \neq s_j$, and multiplying (on the left) the right eigenvalue problem by $\mathbf{v}_j^*$ and multiplying (on the right) the left eigenvalue problem by $\mathbf{u}_i$

$$\begin{aligned} \mathbf{v}_j^* \mathbf{A} \mathbf{u}_i &= s_i \mathbf{v}_j^* \mathbf{B} \mathbf{u}_i \\ \mathbf{v}_j^* \mathbf{A} \mathbf{u}_i &= s_j \mathbf{v}_j^* \mathbf{B} \mathbf{u}_i \end{aligned} \tag{5.3}$$

[2] Eigenvalues with algebraic and geometric multiplicity larger than one, $m_i^{(a)} = m_i^{(g)} > 1$, have the associated eigenvectors that spans a subspace of dimension m_i. In this subspace, it's always possible to define a set of orthogonal vectors. **todo** write it better; refer to introduction to spectral decomposition… **todo** treat sensitivity to parameters of eigenvalues and eigenvectors with multiplicity larger than one…

and subracting the two equations

$$0 = (s_i - s_j)\mathbf{v}_j^*\mathbf{B}\mathbf{u}_i$$

$$\rightarrow \qquad 0 = \mathbf{v}_j^*\mathbf{B}\mathbf{u}_i \quad \text{if } s_i \neq s_j \ , \tag{5.4}$$

as the left-hand-side terms are the same, and $s_i \neq s_j$. As (5.4) holds, from (5.3) it also follows that

$$\rightarrow \qquad 0 = \mathbf{v}_j^*\mathbf{A}\mathbf{u}_i \quad \text{if } s_i \neq s_j \ . \tag{5.5}$$

These relations don't hold for left and right eigenvectors associated with the same eigenvalues, and thus these conditions can be used as normalization conditions.

Sensitivity of eigenvalue

The derivative w.r.t. parameter p of the right eigenvalue problem

$$\mathbf{A}(p)\mathbf{u}_i(p) = s_i(p)\mathbf{B}(p)\mathbf{u}_i(p) \ ,$$

reads

$$0 = (\mathbf{A}_{/p} - s_{i/p}\mathbf{B} - s_i\mathbf{B}_{/p})\mathbf{u}_i + (\mathbf{A} - s_i\mathbf{B})\mathbf{u}_{i/p} \ .$$

Multiplying on the left by the left eigenvector $\mathbf{w}_i$, and recalling that the left eigenvalue problem reads $0 = \mathbf{w}_i^* \left(\mathbf{A} - s_i\mathbf{B}\right)$, the last term is identically zero and

$$0 = \mathbf{w}_i^*(\mathbf{A}_{/p} - s_{i/p}\mathbf{B} - s_i\mathbf{B}_{/p})\mathbf{u}_i \ ,$$

so that it's possible to find the expression of the eigenvalue sensitivity as

$$s_{i/p} = \frac{\mathbf{w}_i \left(\mathbf{A}_{/p} - s_i\mathbf{B}_{/p}\right) \mathbf{u}_i}{\mathbf{w}_i^*\mathbf{B}\mathbf{u}_i} = \mathbf{w}_i \left(\mathbf{A}_{/p} - s_i\mathbf{B}_{/p}\right) \mathbf{u}_i \ , \tag{5.6}$$

where the last step exploits the normalization condition $\mathbf{w}_i^*\mathbf{B}\mathbf{u}_i = b^{(i)} = 1$.

Sensitivity of eigenvector

The sensitivity of the eigenvector $\mathbf{u}_{i/p}$ is the solution of the linear system that can be obtained from the derivative of the eigenvalue problem,

$$(\mathbf{A} - s_i\mathbf{B})\mathbf{u}_{i/p} = \underbrace{-\left(\mathbf{A}_{/p} - s_{i/p}\mathbf{B} - s_i\mathbf{B}_{/p}\right)\mathbf{u}_i}_{=:\mathbf{b}_i} \ ,$$

once the sensitivity of the eigenvector $s_{i/p}$ is known from expression (5.6).

Property 5.2.3 (Linear system is singular, $\mathbf{K}(A) = \{\mathbf{u}_i\}$)

Linear system is singular as s_i is an eigenvalue of the problem, $|\mathbf{A} - s_i\mathbf{B}| = 0$, and the kernel of the matrix is the space spanned by the right eigenvectors associated with s_i, as

$$\left(\mathbf{A} - s_i\mathbf{B}\right)\mathbf{u}_i = \mathbf{0} \ .$$

As the matrix of the linear system is singular, the linear system has **no solution if** the right-hand side is not in the range of the matrix of the system, i.e. $\mathbf{b}_i \notin \mathrm{R}(\mathbf{A} - s_i\mathbf{B})$.

Fortunately, the RHS belongs to the range of the matrix, $\mathbf{b}_i \in \mathrm{R}(\mathbf{A} - s_i\mathbf{B})$, as it's proved in the box below.

Property 5.2.4 ($\mathbf{b}_i \in \mathbf{R}(\mathbf{A} - s_i\mathbf{B})$)

If $\mathbf{b}_i$ belongs to $\mathrm{R}(\mathbf{A} - s_i\mathbf{B})$, it's orthogonal to $\mathrm{K}(\mathbf{A}^* - s_i^*\mathbf{B}^*)$ and viceversa, by theorem … **todo** add link

Kernel of $\mathbf{A}^* - s_i^*\mathbf{B}^*$ is spanned by the left eigenvectors $\mathbf{v}_i$. Thus, using expression of the eigenvalue sensitivity in evaluating the product

$$\begin{aligned}
\mathbf{v}_i^*\mathbf{b}_i &= -\mathbf{v}_i^* \left(\mathbf{A}_{/p} - s_{i/p}\mathbf{B} - s_i\mathbf{B}_{/p})\mathbf{u}_i\right) = \\
&= -\mathbf{v}_i^* \left(\mathbf{A}_{/p} - s_i\mathbf{B}_{/p})\mathbf{u}_i\right) + \mathbf{v}_i^* \left(\mathbf{A}_{/p} - s_i\mathbf{B}_{/p}\right)\mathbf{u}_i \underbrace{\mathbf{v}_i^*\mathbf{B}\mathbf{u}_i}_{=1} = \\
&= 0 \ ,
\end{aligned}$$

it's readily proved that $\mathbf{v}_i^*\mathbf{b}_i = 0$. **todo** treat the case where eigenvalue multiplicity is larger than 1.

As the singular linear system has at least a solution, it has infinite solution as the kernel of the matrix is not empty. If $\widetilde{\mathbf{u}_{/p}}$ is a solution of the system, adding a linear combination of the vectors of $\mathrm{K}(\mathbf{A} - s_i\mathbf{B}) = \{\mathbf{u}_i\}$ produces a solution as well,

$$\widetilde{\mathbf{u}_{i/p}} + \mathbf{U}_i\beta \ ,$$

as $(\mathbf{A} - s_i\mathbf{B})\mathbf{U}_i = \mathbf{0}$, being $\mathbf{U}_i$ the matrix whose columns are the right eigenvectors $\mathbf{u}_i$ associated with eigenvalue s_i, spanning the kernel of the matrix.

In order to remove this arbitrariness, it's possible to introduce the orthogonality condition $\mathbf{V}_i^*\mathbf{B}\widetilde{\mathbf{u}_{i/p}} = \mathbf{0}$, setting the coefficients of the linear combination of the vectors of the kernel $\beta = \mathbf{0}$.

Property 5.2.5 (Augmented linear system)

$$\begin{bmatrix} \mathbf{A} - s_i\mathbf{B} & \mathbf{U}_i \\ \mathbf{V}_i^*\mathbf{B} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \widetilde{\mathbf{u}_{i/p}} \\ \beta \end{bmatrix} = \begin{bmatrix} \mathbf{b}_i \\ \mathbf{0} \end{bmatrix}$$

If everything goes right, the coefficients β must be zero.

Property 5.2.6 (Decomposition of the soluton using the modal basis)

Writing the solution as a linear combination of the right eigenvectors,

$$\widetilde{\mathbf{u}_{i/p}} = \mathbf{U}_{\notin i}\alpha + \mathbf{U}_i\beta \ ,$$

it's possible to solve for α and for β and then retrieve the solution of the underdetermined linear system. Projecting the linear system on $\mathbf{V}_{\notin i}$, and prescibing the orthogonality condition $\mathbf{V}_i^*\mathbf{B}\widetilde{\mathbf{u}_{i/p}} = \mathbf{0}$ on the solution, two decoupled sets of equations for α and β appears,

$$\begin{cases} \mathbf{V}_{\notin i}^* \left(\mathbf{A} - s_i\mathbf{B}\right) \mathbf{U}_{\notin i}\alpha = \mathbf{V}_{\notin i}^*\mathbf{b}_i \\ \mathbf{V}_i^*\mathbf{B}\mathbf{U}_i\beta = \mathbf{0} \ , \end{cases}$$

as $\left(\mathbf{A} - s_i\mathbf{B}\right) \mathbf{U}_i = \mathbf{0}$ and $\mathbf{V}_i^*\mathbf{B}\mathbf{U}_{\notin i} = \mathbf{0}$. Exploiting diagonalization properties, the system can be written as two uncoupled diagonal systems

$$\begin{cases} \mathrm{diag}\left\{a^{(j)\notin i} - s_i b^{(j)\notin i}\right\} \alpha = \mathbf{V}_{\notin i}^*\mathbf{b}_i \\ \mathrm{diag}\left\{b^{(i)}\right\} \beta = \mathbf{0} \ , \end{cases}$$

or, exploiting normalization condition $\mathbf{v}_i^* \mathbf{B} \mathbf{u}_i = 1$, and thus $b^{(j)} = 1$ and $a^{(j)} = s_j$,

$$\begin{cases} \text{diag}\left\{ s_{j \notin i} - s_i \right\} \alpha = \mathbf{V}_{\notin i}^* \mathbf{b}_i \\ \mathbf{I}\, \beta = \mathbf{0} \,, \end{cases}$$

The solution reads

$$\begin{cases} \alpha = \text{diag}\left\{ \dfrac{1}{s_{j \notin i} - s_i} \right\} \mathbf{V}_{\notin i}^* \mathbf{b}_i \\ \beta = \mathbf{0} \,, \end{cases}$$

and thus

$$\begin{aligned} \tilde{\mathbf{u}}_{i/p} &= \mathbf{U}_{\notin i} \text{diag}\left\{ \frac{1}{s_{j \notin i} - s_i} \right\} \mathbf{V}_{\notin i}^* \mathbf{b}_i = \\ &= \mathbf{U}_{\notin i} \text{diag}\left\{ \frac{1}{s_{j \notin i} - s_i} \right\} \mathbf{V}_{\notin i}^* (-(\mathbf{A}_{/p} - s_{i/p} \mathbf{B} - s_i \mathbf{B}_{/p}) \mathbf{u}_i) = \\ &= \mathbf{U}_{\notin i} \text{diag}\left\{ \frac{1}{s_{j \notin i} - s_i} \right\} \mathbf{V}_{\notin i}^* (-(\mathbf{A}_{/p} - s_i \mathbf{B}_{/p}) \mathbf{u}_i) = \\ &= \sum_{j \notin i} \frac{\mathbf{v}_j^* (-\mathbf{A}_{/p} + s_i \mathbf{B}_{/p}) \mathbf{u}_i}{s_j - s_i} \mathbf{u}_j \,. \end{aligned}$$

Some algebra with components

$$\begin{aligned} u_{a/p}^i &= \sum_{b \neq i} u_a^{(b)} \delta_{bc} D^{(c)} v_d^{(c)\,*} b_d^{(i)} = \\ &= \sum_{b \neq i} u_a^{(b)} \delta_{bc} D^{(c)} v_d^{(c)\,*} M_{de} u_e^{(i)} = \\ &= \sum_{b \neq i} u_a^{(b)} D^{(b)} v_d^{(b)\,*} M_{de} u_e^{(i)} = \\ &= \sum_{b \neq i} C^{b,i} u_a^{(b)} \,, \end{aligned}$$

with $C^{b,i} = D^{(b)} v_d^{(b)\,*} M_{de/p} u_e^{(i)} = \frac{\mathbf{v}_b^* \mathbf{M} \mathbf{u}_i}{s_b - s_i}$, and $\mathbf{M} = -\mathbf{A}_{/p} + s_i \mathbf{B}_{/p}$.

5.2.2 Generalized eigenvalue problem (second order)

Right and left eigenvalue problem

CHAPTER

SIX

CAYLEY-HAMILTON THEOREM

Every square matrix $\mathbf{A}$ satisfies its own characteristic equation,

$$\mathbf{A}^n + a_{n-1}\mathbf{A}^{n-1} + \cdots + a_1\mathbf{A} + a_0\mathbf{I} = \mathbf{0} \ .$$

todo Proof using the adjugate matrix and the identity (**prove it!**) $\det(\mathbf{M})\,\mathbf{I} = \mathbf{M}\,\mathrm{adj}(\mathbf{M})$, and the telescopic sum. This identity comes from the properties of the determinant. Need to write a section about it.

Part II

Multivariable Calculus

CHAPTER

SEVEN

INTRODUCTION TO MULTI-VARIABLE CALCULUS

7.1 Function

7.2 Limit

7.3 Derivatives

7.4 Integrals

7.5 Theorems

7.5.1 Green's lemma

$$\begin{aligned}\int_S \frac{\partial F}{\partial y} dxdy &= -\oint_{\partial S} F dx \\ \int_S \frac{\partial G}{\partial x} dxdy &= \oint_{\partial S} G dy\end{aligned}$$

Proof for simple domains.

In a simple domain in x, so that the closed contour ∂S is delimited by the curves $y = Y_1(x)$, $y = Y_2(x) > Y_1(x)$, for $x \in [x_1, x_2]$,

$$\begin{aligned}\int_S \frac{\partial F}{\partial y} dxdy &= \int_{x=x_1}^{x_2} \int_{y=Y_1(x)}^{Y_2(x)} \frac{\partial F}{\partial y} dy\, dx = \\ &= \int_{x=x_1}^{x_2} \left[F(x, Y_2(x)) - F(x, Y_1(x))\right] dx = \\ &= -\int_{x=x_1}^{x_2} F(x, Y_1(x)) - \int_{x=x_2}^{x_1} F(x, Y_2(x)) dx = \\ &= -\oint_{\partial S} F(x, y) dx\end{aligned}$$

Proof for generic domain

todo

In a simple domain in y, so that the closed contour ∂S is delimited by the curves $x = X_1(y)$, $x = X_2(y) > X_1(y)$ for $y \in [y_1, y_2]$,

$$\begin{aligned}
\int_S \frac{\partial G}{\partial x} dxdy &= \int_{y=y_1}^{y_2} \int_{x=X_1(y)}^{X_2(y)} \frac{\partial G}{\partial x} dx\, dy = \\
&= \int_{y=y_1}^{y_2} \left[G(X_2(y), y) - G(X_1(y), y)\right] dy = \\
&= \int_{y=y_1}^{y_2} G(X_1(y), y) dy + \int_{y=y_2}^{y_1} G(X_2(y), y) dy = \\
&= \oint_{\partial S} G(x, y) dy
\end{aligned}$$

Part III

Differential Geometry

CHAPTER

EIGHT

INTRODUCTION TO DIFFERENTIAL GEOMETRY

8.1 Differential geometry in E^3

8.1.1 Curves

Parametric representation of curve in 3-dimensional (Euclidean) space E^3

$$\vec{r}(q)$$

Differential, $d\vec{r}$.

$$d\vec{r}(q) = \vec{r}\,'(q)\, dq\ .$$

Arc-length parameter, s. So that $ds = |d\vec{r}(s)|$ and thus

$$|d\vec{r}(s)| = |\vec{r}\,'(s)|\,|ds| \qquad \rightarrow \qquad |\vec{r}\,'(s)| = 1 \qquad \rightarrow \qquad \vec{r}\,'(s) = \hat{t}(s)\ .$$

Frenet basis. Using arc-length parameter, Frenet basis is naturally defined as the set $\{\hat{t}, \hat{n}, \hat{b}\}$:

- tangent unit vector, $\hat{t}(s) = \vec{r}\,'(s)$,
- normal unit vector, $\hat{r}''(s) = \hat{t}\,'(s) =: \kappa(s)\,\hat{n}(s)$, with $\kappa(s)$ local curvature
- binormal unit vector, $\hat{b}(s) = \hat{t}(s) \times \hat{n}(s)$

Using a general parameter, t, with some abuse of notation $\vec{r}(t) = \vec{r}(s(t))$ and indicating $\dot{()} = \frac{d}{dt}$,

- $\dot{\vec{r}} = \frac{ds}{dt}\frac{d\vec{r}}{ds} = \dot{s}\hat{t}$
- $\ddot{\vec{r}} = \dfrac{d}{dt}\dot{\vec{r}} = \dfrac{d}{dt}\left(\dot{s}\hat{t}\right) = \ddot{s}\hat{t} + \dfrac{ds}{dt}\dfrac{d}{ds}\hat{t} = \ddot{s}\hat{t} + \dot{s}^2\kappa\,\hat{n}$

Osculator circle. Circle with $R(s) = \frac{1}{\kappa(s)}$, in plane orthogonal to $\hat{b}(s)$, passing through $\vec{r}(s)$, and thus center in $\vec{r}_C(s) = \vec{r}(s) + \hat{n}R(s)$. Its parametric representation using its arc-length parameter p, with $\vec{r}(p=0) = \vec{r}(s)$ reads

$$\vec{r}(p) = \vec{r}_C(s) + R(s)\left[-\cos\left(\frac{p}{R(s)}\right)\hat{n}(s) + \sin\left(\frac{p}{R(s)}\right)\hat{t}(s)\right]\ .$$

Its first and second order derivatives w.r.t. the arc-length p evaluated in $p = 0$, i.e. $\vec{r} = \vec{r}(s)$ read:

- first derivative in $p = 0$,

$$\hat{t}(p)\big|_{p=0} = \vec{r}\,'(p)\big|_{p=0} = \left[\sin\left(\frac{p}{R(s)}\right)\hat{n}(s) + \cos\left(\frac{p}{R(s)}\right)\hat{t}(s)\right]\Bigg|_{p=0} = \hat{t}(s)\ ,$$

  i.e. the osculator circle has the same tangent as the curve in the point.

- second derivative in $p = 0$,

$$\kappa(p)\hat{n}(p)|_{p=0} = \vec{r}''(p)|_{p=0} = \frac{1}{R(s)}\left[\cos\left(\frac{p}{R(s)}\right)\hat{n}(s) - \sin\left(\frac{p}{R(s)}\right)\hat{t}(s)\right]\Bigg|_{p=0} = \frac{1}{R(s)}\hat{n}(s) = \kappa(s)\hat{n}(s)\ ,$$

  i.e. the osculator circle has the same normal vector and curvature as the curve in the point.

8.1.2 Surfaces

$$\vec{r}(q^1, q^2)$$

$$d\vec{r} = \frac{\partial \vec{r}}{\partial q^1}\, dq^1 + \frac{\partial \vec{r}}{\partial q^2}\, dq^2 = \vec{b}_1\, dq^1 + \vec{b}_2\, dq^2$$

A third vector $\vec{b}_3 := \hat{n}$ can be defined so that $|\hat{n}| = 1$ and $\hat{n} \cdot \vec{b}_i = 0$, $i = 1 : 2$. For $i = 1 : 2$, $k = 1 : 2$

$$\frac{\partial \vec{b}_i}{\partial q^j} = \Gamma^k_{ij}\vec{b}_k = \Gamma^1_{ij}\vec{b}_1 + \Gamma^2_{ij}\vec{b}_2 + \Gamma^3_{ij}\vec{b}_3$$

so that

$$\Gamma^k_{ij} = \vec{b}^k \cdot \frac{\partial \vec{b}_i}{\partial q^j}$$

Normal vector.

$$\vec{n}(q^1, q^2) = \frac{\partial \vec{r}}{\partial q^1}(q^1, q^2) \times \frac{\partial \vec{r}}{\partial q^2}(q^1, q^2) = \vec{b}_1(q^1, q^2) \times \vec{b}_2(q^1, q^2)$$

Tangent plane.

$$(\vec{r} - \vec{r}(q^1, q^2)) \cdot \vec{n}(q^1, q^2) = 0$$

Length of elementary segment.

$$\begin{aligned}
|d\vec{r}|^2 &= d\vec{r} \cdot d\vec{r} = \\
&= \left(\vec{b}_1\, dq^1 + \vec{b}_2\, dq^2\right) \cdot \left(\vec{b}_1\, dq^1 + \vec{b}_2\, dq^2\right) = \\
&= g_{11}\, dq^1\, dq^1 + g_{12}\, dq^1\, dq^2 + g_{21}\, dq^2\, dq^1 + g_{22}\, dq^2\, dq^2 = g_{ij}\, dq^i\, dq^j
\end{aligned}$$

Second order approximation.

$$\begin{aligned}
\vec{r}(q^1 + dq^1, q^2 + dq^2) &= \vec{r}(q_1, q_2) + \frac{\partial \vec{r}}{\partial q^i}\, dq^i + \frac{\partial^2 \vec{r}}{\partial q^i \partial q^j}\, dq^i\, dq^j = \\
&= \vec{r}(q_1, q_2) + \vec{b}_i\, dq^i + \vec{b}_k \Gamma^k_{ij}\, dq^i\, dq^j + \hat{n}\, \Gamma^3_{ij}\, dq^i\, dq^j
\end{aligned}$$

so that

$$\begin{aligned}
\left[\vec{r}(q^1 + dq^1, q^2 + dq^2) - \vec{r}(q^1, q^2)\right] \cdot \hat{n} &= \Gamma^3_{ij}\, dq^i\, dq^j = \\
&= \hat{n} \cdot \frac{\partial^2 \vec{r}}{\partial q^i \partial q^j}\, dq^i\, dq^j = \\
&= \hat{n} \cdot \frac{\partial^2 \vec{r}}{\partial q^i \partial q^j}\, \vec{b}^i \cdot \vec{b}_k dq^k\, \vec{b}^j \cdot \vec{b}_l dq^l = \\
&= \underbrace{dq^k \vec{b}_k}_{d\vec{r}} \cdot \left[\hat{n} \cdot \frac{\partial^2 \vec{r}}{\partial q^i \partial q^j}\vec{b}^i \otimes \vec{b}^j\right] \cdot \underbrace{dq^l \vec{b}_l}_{d\vec{r}}
\end{aligned}$$

Curvature tensor.

Part IV

Vector and Tensor Algebra and Calculus

CHAPTER

NINE

TENSOR ALGEBRA

This section introduces tensor algebra

Warning: This introduction is meant for **spaces with inner product** that allows to introduce quite naturally the useful concept of the *reciprocal basis* of a given basis of $\mathcal{V}$, belonging to the same space $\mathcal{V}$, somehow avoiding the complications coming from the introduction of dual space $\mathcal{V}^*$ and basis, and everything is required for that.

9.1 Vector space $\mathcal{V}$

A vector space is an algebraic structure with:

- a **set** $\mathcal{V}$, whose elements are called **vectors**, here indicated with bold symbols $\mathbf{v} \in \mathcal{V}$
- a **field** K (usually $\mathbb{R}$ or $\mathbb{C}$), whose elements are calles **scalars**,
- 2 **operations**, closed w.r.t. the set $\mathcal{V}$:
  - **vector sum**, $\mathbf{u} + \mathbf{v} \in \mathcal{V}$ for $\forall \mathbf{u},\ \mathbf{v} \in \mathcal{V}$
  - **multiplication of a scalar and a vector**, $a\mathbf{v} \in \mathcal{V}$ for $\forall a \in K,\ \mathbf{u} \in \mathcal{V}$

  with properties discussed below **todo** …

9.1.1 Operations (I)

Sum

…

Multiplication by a scalar

…

Definition 9.1.1 (Linear combination)

A linear combination of n vectors $\{\mathbf{v}_i\}_{i=1:n}$, $\mathbf{v}_i \in \mathcal{V}$, is the weighted sum

$$a^1\mathbf{v}_1 + a^2\mathbf{v}_2 + \cdots + a^n\mathbf{v}_n = a^i\mathbf{v}_i \in \mathcal{V}\ ,$$

having used Einstein's summation convention over repeated indices. Here the position of the indices has no particular meaning, but it'll have soon in the following sections.

Inner product

The existence of an inner product is not a requirement of a vector space

…

An inner product in a vectors space $\mathcal{V}$ over field K is an operation $\cdot : \mathcal{V} \times \mathcal{V} \to K$,

$$\mathbf{u} \cdot \mathbf{v} \,,$$

with the following properties: **todo**

9.1.2 Basis of a vector space $\mathcal{V}$

Definition 9.1.2 (Basis)

A basis is a minimal set of vectors of $\mathcal{V}$ that can represent all the elements of $\mathcal{V}$.

Definition 9.1.3 (Dimension of a vector space)

The dimension of a vector space $\mathcal{V}$ is the number of the elements of a basis of the space.

Definition 9.1.4 (Reciprocal basis)

In a inner product space, the reciprocal basis of a given basis $\{\mathbf{b}_a\}_{a=1:d}$ is the set of vectors $\{\mathbf{b}^b\}_{b=1:d}$, s.t.

$$\mathbf{b}^b \cdot \mathbf{b}_a = \delta^b_a \,.$$

Definition 9.1.5 ("Metric tensor")

$$\begin{aligned} g_{ab} &:= \mathbf{b}_a \cdot \mathbf{b}_b \\ g^{ab} &:= \mathbf{b}^a \cdot \mathbf{b}^b \end{aligned}$$

The following holds

$$\begin{aligned} \mathbf{b}_a &= g_{ab}\mathbf{b}^b \quad (1) \\ \mathbf{b}^a &= g^{ab}\mathbf{b}_b \quad (2) \end{aligned} \tag{9.1}$$

"Proof"

Taking the dot product with $\mathbf{b}_c$ of the relation (9.1)(1),

$$\mathbf{b}_c \cdot \mathbf{b}_a = g_{ab} \underbrace{\mathbf{b}_c \cdot \mathbf{b}^b}_{=\delta^b_c} = g_{ac} \ .$$

9.1.3 Change of basis

Let T^b_a the elements of the matrix of change of basis, representing the vectors of the basis $\{\tilde{b}_a\}$ as linear combinations of the vectors of the basis $\{\mathbf{b}_b\}$,

$$\tilde{\mathbf{b}}_b = T^a_b \mathbf{b}_a \ .$$

Inverse transformation. Let $\widetilde{T}$ be the elements of the inverse transformation,

$$\mathbf{b}_c = \widetilde{T}^b_c \tilde{\mathbf{b}}_b = \widetilde{T}^b_c T^a_b \mathbf{b}_a \ ,$$

and thus

$$\widetilde{T}^b_c T^a_b = \delta^a_c \ .$$

Transformation of the reciprocal basis. todo

Transformation of components

A vector $\mathbf{v}$ can be represented in different basis, as different linear combinations of the elements of those bases,

$$\mathbf{v} = v^a \mathbf{b}_a = \tilde{v}^b \tilde{\mathbf{b}}_b \ .$$

Given the rules of change of basis, the rule of transformation of components immediately follwos

$$\begin{aligned} \tilde{\mathbf{b}}_b &= T^a_b \mathbf{b}_a \quad & \tilde{v}^b &= \widetilde{T}^b_a v^a \\ \mathbf{b}_b &= \widetilde{T}^a_b \tilde{\mathbf{b}}_a \quad & v^b &= T^b_a \tilde{v}^a \end{aligned}$$

Proof

$$\mathbf{v} = v^a \mathbf{b}_a = \underbrace{v^a \widetilde{T}^b_a}_{\tilde{v}^b} \tilde{\mathbf{b}}_b = \tilde{v}^b \tilde{\mathbf{b}}_b$$

or

$$\mathbf{v} = \tilde{v}^b \tilde{\mathbf{b}}_b = \underbrace{\tilde{v}^b \widetilde{T}^b_c}_{v^c} \mathbf{b}_c = v^c \mathbf{b}_c$$

It's clear that the vectors of the bases and the components follow inverse transformations to preserve the **invariance of vector w.r.t. a change of basis**: the vector $\mathbf{v}$ doesn't change if we change our description of it, by changing a basis.

Matrix of change of basis as a tensor (todo maybe later? Tensor not introduced yet here)

The rule of transformation between different basis can be interpreted using dot product between tensors and vectors

$$\tilde{\mathbf{b}}_b = T^a_b \mathbf{b}_a = T^a_c \mathbf{b}_a \delta^c_b = T^a_c \mathbf{b}_a \otimes \mathbf{b}^c \cdot \mathbf{b}_b = (T^a_c \mathbf{b}_a \otimes \mathbf{b}^c) \cdot \mathbf{b}_b = \mathbb{T} \cdot \mathbf{b}_b$$

Inverse and transpose

Interpreting the indices of the transformation matrix as the indices of rows and columns of a 2x2 matrix, transformation of components involves the transpose of the inverse matrix, as the indices a, b are swapped.

Example 9.1.1

Example 9.1.2 (Rotation)

As the inverse of a rotation is its transpose, $\widetilde{T}^b_a = T^a_b$. So the rule of transformation of components follows the same rule of change of basis. As an example, let the transformation between two Cartesian bases be the rotation

$$\begin{aligned} \tilde{\mathbf{x}} &= \cos\theta\,\mathbf{x} + \sin\theta\,\mathbf{y} \\ \tilde{\mathbf{y}} &= -\sin\theta\,\mathbf{x} + \cos\theta\,\mathbf{y} \end{aligned} \quad , \quad \begin{aligned} \mathbf{x} &= \cos\theta\,\tilde{\mathbf{x}} - \sin\theta\,\tilde{\mathbf{y}} \\ \mathbf{y} &= \sin\theta\,\tilde{\mathbf{x}} + \cos\theta\,\tilde{\mathbf{y}} \end{aligned}$$

Let a vector

$$\begin{aligned} \mathbf{v} &= v_x\mathbf{x} + v_y\mathbf{y} = \\ &= v_x\left(\cos\theta\,\tilde{\mathbf{x}} - \sin\theta\,\tilde{\mathbf{y}}\right) + v_y\left(\sin\theta\,\tilde{\mathbf{x}} + \cos\theta\,\tilde{\mathbf{y}}\right) = \\ &= \tilde{v}_x\tilde{\mathbf{x}} + \tilde{v}_y\tilde{\mathbf{y}} = \\ &= \tilde{v}_x\left(\cos\theta\,\mathbf{x} + \sin\theta\,\mathbf{y}\right) + \tilde{v}_y\left(-\sin\theta\,\mathbf{x} + \cos\theta\,\mathbf{y}\right) \end{aligned}$$

and thus

$$\begin{aligned} \tilde{v}_x &= \cos\theta\,v_x + \sin\theta\,v_y \\ \tilde{v}_y &= -\sin\theta\,v_x + \sin\theta\,v_y \end{aligned} \quad , \quad \begin{aligned} v_x &= \cos\theta\,\tilde{v}_x - \sin\theta\,\tilde{v}_y \\ v_y &= \sin\theta\,\tilde{v}_x + \sin\theta\,\tilde{v}_y \end{aligned}$$

9.1.4 Operations

Tensor product of vectors

$$\mathbf{v}_{(1)} \otimes \mathbf{v}_{(2)} \otimes \cdots \otimes \mathbf{v}_{(r)}$$

or writing each vector as a linear combination of the elements of a basis $\mathbf{b}_a$,

$$\mathbf{v}_{(i)} = v^{a_i}_{(i)}\mathbf{b}_{a_i}\ , \ldots$$

$$\begin{aligned} \mathbf{v}_{(1)} \otimes \mathbf{v}_{(2)} \otimes \cdots \otimes \mathbf{v}_{(3)} &= \left(v^{a_1}_{(1)}\mathbf{b}_{a_1}\right) \otimes \left(v^{a_2}_{(2)}\mathbf{b}_{a_2}\right) \otimes \cdots \otimes \left(v^{a_r}_{(r)}\mathbf{b}_{a_r}\right) = \\ &= v^{a_1}_{(1)}\,v^{a_2}_{(2)}\,\ldots\,v^{a_r}_{(r)}\mathbf{b}_{a_1} \otimes \mathbf{b}_{a_2} \otimes \cdots \otimes \mathbf{b}_{a_r} \end{aligned}$$

The result of the tensor product of r vectors is a rank-r tensor, $\mathbb{T}$, as it will be clear below, with components

$$T^{a_1 a_2 \ldots a_r} = v^{a_1}_{(1)}\,v^{a_2}_{(2)}\,\ldots\,v^{a_r}_{(r)}\ .$$

9.2 Space of tensors

Definition 9.2.1 (Tensor as a multilinear map)

A rank-r tensor $\mathbf{A}$ can be defined as a multilinear map, acting on r vectors[1] $\mathbf{v}^{(i)} \in \mathcal{V}$, $i = 1 : r$

$$\mathbf{A}\left(\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(r)}\right) : \underbrace{\mathcal{V} \times \cdots \times \mathcal{V}}_{r \text{ times}} \to K.$$

todo

- Tensor definition by components, and action on the vectors
- Reference to a similar definition of a vector: a vector $\mathbf{v} \in \mathcal{V}$, vector space with inner product, is uniquely determined by the value of its scalar product $\mathbf{u} \cdot \mathbf{v}$ with every vector $\mathbf{u} \in \mathcal{V}$ or, analogously, with every vector in a set of n linearly independent vectors $\left\{\mathbf{u}^{(i)}\right\}$, $i = 1 : n$, with $n = \dim(\mathcal{V})$.
- Definition of components w.r.t. a basis

9.2.1 Operations (I)

Sum

Sum of tensors with the same rank, $\mathbf{A}$, $\mathbf{B} \in \mathcal{V}$,

$$\mathbf{A} + \mathbf{B} \in \mathcal{V}^r$$

todo Components

Multiplication by a scalar

Multiplication by a scalar $a \in K$, of a r-rank tensor $\mathbf{A}$,

$$a\mathbf{A} \in \mathcal{V}^r$$

todo Components

Vector space of tensors

The set of tensors of a given rank with the operations of sum and multiplication by a scalar defined above forms a vector space.

[1] This introduction has no ambition iof being the most general — and precise? — introduction to tensors: as already stated somewhere else, existence of inner product is assumed even it's not necessary for the most general treatment of tensors; moreover, no distinction is made between dual space $\mathcal{V}^*$ of covectors — space of linear forms on V — and V itself;…

9.2.2 Basis

Change of basis and rule of transformation of components - classical definition of a tensor

$$
\begin{aligned}
\tilde{\mathbf{b}}_{i_a} &= T^{i_a}_{i_b} \mathbf{b}_{i_b} \\
\mathbf{b}_{i_a} &= \widetilde{T}^{i_b}_{i_a} \tilde{\mathbf{b}}_{i_b} \\
\mathbf{A} = A^{i_1 \dots i_p} \mathbf{b}_{i_1} \dots \mathbf{b}_{i_p} &= \\
= A^{i_1 \dots i_p} \left(T^{j_1}_{i_1} \tilde{\mathbf{b}}_{j_1} \right) \dots \left(T^{j_p}_{i_p} \tilde{\mathbf{b}}_{j_p} \right) &= \\
= A^{i_1 \dots i_p} T^{j_1}_{i_1} \dots T^{j_p}_{i_p} \tilde{\mathbf{b}}_{j_1} \dots \tilde{\mathbf{b}}_{j_p} &= \\
= \widetilde{A}^{j_1 \dots j_p} \tilde{\mathbf{b}}_{j_1} \dots \tilde{\mathbf{b}}_{j_p} &,
\end{aligned}
$$

with

$$
\widetilde{A}^{j_1 \dots j_p} = A^{i_1 \dots i_p} T^{j_1}_{i_1} \dots T^{j_p}_{i_p} \ . \tag{9.2}
$$

Definition 9.2.2 (Classical definition of a tensor)

Relation (9.2) is the "historical" definition of a tensor, through the law of transformation of its components following a change of basis.

9.2.3 Operations (II)

Tensor product

The tensor product of a p-rank tensor $\mathbf{A}$ and a q-rank tensor $\mathbf{B}$ is the $(p+q)$-rank tensor $\mathbf{A} \otimes \mathbf{B} = \mathbf{AB}$, that can be defined using component representation in a given basis,

$$
\begin{aligned}
\mathbf{A} \otimes \mathbf{B} &= \left(A^{a_1 \dots a_p} \mathbf{b}_{a_1} \dots \mathbf{b}_{a_p} \right) \otimes \left(B^{b_1 \dots b_q} \mathbf{b}_{b_1} \dots \mathbf{b}_{b_q} \right) = \\
&= A^{a_1 \dots a_{p-1} a_p} \, B^{b_1 \, b_2 \dots b_q} \mathbf{b}_{a_1} \dots \mathbf{b}_{a_p} \mathbf{b}_{b_1} \dots \mathbf{b}_{b_q}
\end{aligned}
$$

Dot product

The dot product of a p-rank tensor $\mathbf{A}$ and a q-rank tensor $\mathbf{B}$ is the $(p+q-2)$-rank tensor $\mathbf{A} \cdot \mathbf{B}$, that can be defined using component representation in a given basis,

$$
\begin{aligned}
\mathbf{A} \cdot \mathbf{B} &= \left(A^{a_1 \dots a_p} \mathbf{b}_{a_1} \dots \mathbf{b}_{a_p} \right) \cdot \left(B_{b_1 \dots b_q} \mathbf{b}^{b_1} \dots \mathbf{b}^{b_q} \right) = \\
&= A^{a_1 \dots a_p} \, B_{b_1 \dots b_q} \mathbf{b}_{a_1} \dots \underbrace{\mathbf{b}_{a_p} \cdot \mathbf{b}^{b_1}}_{=\delta^{b_1}_{a_p}} \dots \mathbf{b}^{b_q} = \\
&= A^{a_1 \dots a_{p-1} k} \, B_{k \, b_2 \dots b_q} \mathbf{b}_{a_1} \dots \mathbf{b}_{a_{p-1}} \mathbf{b}^{b_2} \dots \mathbf{b}^{b_q}
\end{aligned}
$$

Contraction

Contraction of a pair of index of a p-rank tensor $\mathbf{A}$ returns a $p-2$-rank tensor defined as

$$C_a^b(\mathbf{A}) = ...$$

Exterior product

todo *see exterior algebra*

9.2.4 Invariants

9.3 Exterior algebra

$$\wedge$$

9.3.1 Exterior product

Generalization of the vector product

CHAPTER

TEN

TENSOR CALCULUS IN EUCLIDEAN SPACES

This section deals with tensor calculus in Euclidean space or on manifolds embedded in Euclidean spaces, focusing on d-dimensional spaces with $d \leq 3$, with *inner product*.

This section may rely on results of *differential geometry*.

10.1 Coordinates

A set of parameters $\{q^a\}_{a=1:d}$ to represent vector (or point) in space,

$$\vec{r}(q^a)$$

if $\vec{r} \in E^d$, $a = 1 : d$.

In E^3,

- **Coordinate lines**, 2-parameter family of lines, keeping 2 coordinates constant. As an example, coordinate lines with constant q^2, q^3

$$\vec{r}_1(q^1) = \vec{r}(q^1, \bar{q}^2, \bar{q}^3) \ .$$

- **Coordinate surfaces,** 1-parameter family of surfaces, keeping 1 coordinate constant. As an example, coordinate surfaces with constant q^1,

$$\vec{r}_{23}(q^2, q^3) = \vec{r}(\bar{q}^1, q^2, q^3) \ .$$

Definition 10.1.1 (Regular parametrization)

If $\frac{\partial \vec{r}}{\partial q^a} \neq 0$.

10.1.1 Natural basis

Definition 10.1.2 (Natural basis)

Vectors of natural basis

$$\vec{b}_a := \frac{\partial \vec{r}}{\partial q^a}$$

Definition 10.1.3 (Reciprocal basis (todo move to Tensor Algebra))

Given a basis $\{\vec{b}_a\}_a$, its reciprocal basis the set of vector $\{\vec{b}^b\}_b$ defined as

$$\vec{b}^b \cdot \vec{b}_a = \delta_a^b \ ,$$

being δ_a^b Kronecker delta.

Definition 10.1.4 (Christoffel symbols)

Christoffel symbols (of the 2nd kind) are defined as the components of the partial derivatives of the vectors of a natural basis w.r.t. the coordinates referred to the natural basis itself,

$$\frac{\partial \vec{b}_a}{\partial q^b} = \Gamma^c_{ab} \vec{b}_c \tag{10.1}$$

Properties of Christoffel symbols

Exploiting the definition of reciprocal basis, Christoffel symbols can be written as

$$\Gamma^c_{ab} = \vec{b}^c \cdot \frac{\partial \vec{b}_a}{\partial q^b} \ .$$

Symmetry. Symmetry os the lower indices

$$\Gamma^c_{ab} = \Gamma^c_{ba} \ ,$$

readily follows Schwartz theorem about partial derivatives

$$\frac{\partial \vec{b}_a}{\partial q^b} = \frac{\partial}{\partial q^c}\frac{\partial \vec{r}}{\partial q^a} = \frac{\partial}{\partial q^a}\frac{\partial \vec{r}}{\partial q^b} = \frac{\partial \vec{b}_b}{\partial q^a}$$

10.2 Fields

Function of the points in space $F : E^d \to V^r$, being V^r a space of tensors of order r.

10.3 Differential operators

10.3.1 Directional derivative

$$F(\vec{r}) = F\left(\vec{r}\left(q^a\right)\right) = f(q^a)$$

$$f(q^a + \beta \Delta q^a) = F(\vec{r}(q^a + \beta \Delta q^a))$$

$$\vec{r}(q^a) + \alpha \vec{v} = \vec{r}(q^a + \beta \Delta q^a) \sim \vec{r}(q^a) + \frac{\partial \vec{r}}{\partial q^b} \beta \Delta q^b$$

$$\alpha \vec{v} \sim \beta \frac{\partial \vec{r}}{\partial q^b}(q^a)\, \Delta q^b = \beta \vec{b}_b(q^a) \Delta q^b \qquad \to \qquad \Delta q^b = \frac{\alpha}{\beta} \vec{b}^b(q^a) \cdot \vec{v}$$

The directional derivative for an arbitrary vector $\vec{v} \in V$

$$\frac{d}{d\alpha} F(\vec{r} + \alpha\vec{v})\Big|_{\alpha=0}$$

is evaluated as the limit for $\alpha \to 0$ of the incremental ratio

$$\begin{aligned}
\frac{F(\vec{r} + \alpha\vec{v}) - F(\vec{r})}{\alpha} &\sim \frac{f(q^a + \beta\Delta q^a) - f(q^a)}{\alpha} = \\
&\sim \frac{1}{\alpha}\frac{\partial f}{\partial q^b}(q^a)\beta\Delta q^b = \\
&\sim \vec{v}\cdot\vec{b}^b(q^a)\frac{\partial f}{\partial q^b}(q^a) = \\
&= \vec{v}\cdot\nabla F(\vec{r})
\end{aligned}$$

10.3.2 Gradient

The gradient is the differential operator is the first-order differential operator appearing in the definition of the directional derivative, $\nabla F(\vec{r})$. It takes a tensor field $F(\vec{r})$ of order r and gives a tensor field $\nabla F(\vec{r})$ of order $r+1$. Given a set of coordinates $\{q^a\}_{a=1:d}$, the gradient can be written using the reciprocal basis of the natural basis as

$$\nabla F(\vec{r}) = \vec{b}^b(\vec{r})\frac{\partial F}{\partial q^b}(\vec{r}) \tag{10.2}$$

Examples. …

Example 10.3.1 (Gradient of a scalar field - with general coordinates q^a)

Applying the definition (10.2) of gradient operator, it readily follows

$$\nabla F = \vec{b}^a\frac{\partial F}{\partial q^a}$$

Example 10.3.2 (Gradient of a vector field - with general coordinates q^a)

Applying the definition (10.2) of gradient operator, rule for the derivative of a product and the definition (10.1) of Christoffel symbols to write derivatives of base vectors,

$$\begin{aligned}
\nabla F &= \vec{b}^a\frac{\partial}{\partial q^a}\left(F^b\vec{b}_b\right) = \\
&= \vec{b}^a\left[\frac{\partial F^b}{\partial q^a}\vec{b}_b + F^b\frac{\partial\vec{b}_b}{\partial q^a}\right] = \\
&= \vec{b}^a\left[\frac{\partial F^b}{\partial q^a}\vec{b}_b + F^b\,\Gamma^c_{ab}\,\vec{b}_c\right] = \\
&= \vec{b}^a\otimes\vec{b}_b\left[\frac{\partial F^b}{\partial q^a} + \Gamma^b_{ac}\,F^c\right].
\end{aligned}$$

Example 10.3.3 (Gradient of a 2nd-order tensor field - with general coordinates q^a)

Applying the definition (10.2) of gradient operator, rule for the derivative of a product and the definition (10.1) of Christoffel symbols to write derivatives of base vectors,

$$
\begin{aligned}
\nabla F &= \vec{b}^a \frac{\partial}{\partial q^a}\left(F^{bc}\vec{b}_b \otimes \vec{b}_c\right) = \\
&= \vec{b}^a \left[\frac{\partial F^{bc}}{\partial q^a}\vec{b}_b\,\vec{b}_c + F^{bc}\frac{\partial \vec{b}_b}{\partial q^a}\,\vec{b}_c + F^{bc}\vec{b}_b\frac{\partial \vec{b}_c}{\partial q^a}\right] = \\
&= \vec{b}^a \left[\frac{\partial F^{bc}}{\partial q^a}\vec{b}_b\,\vec{b}_c + F^{bc}\Gamma^d_{ab}\,\vec{b}_d\,\vec{b}_c + F^{bc}\,\Gamma^{ac}_d\,\vec{b}_b\,\vec{b}_d\right] = \\
&= \vec{b}^a \otimes \vec{b}_b \otimes \vec{b}_c \left[\frac{\partial F^{bc}}{\partial q^a} + \Gamma^b_{ad}\,F^{dc} + \Gamma^c_{ad}\,F^{bd}\right] .
\end{aligned}
$$

10.3.3 Divergence

Divergence opearator is a first-order differential operator that can be defined as the contraction of the first two indices of the gradient,

$$
\nabla \cdot F = C^2_1\left(\nabla F\right) .
$$

It takes a tensor field $F(\vec{r})$ of order $r \geq 1$ and gives a tensor field $\nabla \cdot F(\vec{r})$ of order $r - 1 \geq 0$.

Example 10.3.4 (Divergence of a vector field - with general coordiantes q^a)

Applying contraction to the gradient of a vector field, it readily follows,

$$
\begin{aligned}
\nabla \cdot \left(F^b\vec{b}_b\right) &= C^2_1\left(\nabla F\right) = \\
&= C^2_1\left(\vec{b}^a \otimes \vec{b}_b\left[\frac{\partial F^b}{\partial q^a} + \Gamma^b_{ac}\,F^c\right]\right) = \\
&= \frac{\partial F^a}{\partial q^a} + \Gamma^a_{ac}\,F^c
\end{aligned}
$$

Example 10.3.5 (Divergence of a 2nd-order tensor field - with general coordiantes q^a)

Applying contraction to the gradient of a vector field, it readily follows,

$$
\begin{aligned}
\nabla \cdot \left(F^{bc}\vec{b}_b \otimes \vec{b}_c\right) &= C^2_1\left(\nabla F\right) = \\
&= C^2_1\left(\vec{b}^a \otimes \vec{b}_b \otimes \vec{b}_c\left[\frac{\partial F^{bc}}{\partial q^a} + \Gamma^b_{ad}\,F^{dc} + \Gamma^c_{ad}\,F^{bd}\right]\right) = \\
&= \vec{b}_c\left[\frac{\partial F^{ac}}{\partial q^a} + \Gamma^a_{ad}\,F^{dc} + \Gamma^c_{ad}\,F^{ad}\right]
\end{aligned}
$$

10.3.4 Laplacian

Laplacian operator is second-order differential operator that can be defined as the divergence of the gradient,

$$\Delta F = \nabla^2 F = \nabla \cdot \nabla F \; .$$

Example 10.3.6 (Laplacian of a scalar field - with general coordinates q^a)

$$\begin{aligned}
\nabla \cdot \nabla F &= C_1^2 \left[\nabla \left(\nabla F \right) \right] = \\
&= C_1^2 \left[\nabla \left(\vec{b}^a \frac{\partial F}{\partial q^a} \right) \right] = \\
&= C_1^2 \left[\nabla \left(\vec{b}_b \, g^{ab} \frac{\partial F}{\partial q^a} \right) \right] = \\
&= C_1^2 \left[\vec{b}^c \frac{\partial}{\partial q^c} \left(\vec{b}_b \, g^{ab} \frac{\partial F}{\partial q^a} \right) \right] = \\
&= C_1^2 \left\{ \vec{b}^c \left[\vec{b}_b \frac{\partial}{\partial q^c} \left(g^{ab} \frac{\partial F}{\partial q^a} \right) + g^{ab} \frac{\partial F}{\partial q^a} \frac{\partial \vec{b}_b}{\partial q^c} \right] \right\} = \\
&= C_1^2 \left\{ \vec{b}^c \left[\vec{b}_b \frac{\partial}{\partial q^c} \left(g^{ab} \frac{\partial F}{\partial q^a} \right) + g^{ab} \frac{\partial F}{\partial q^a} \Gamma^d_{bc} \, \vec{b}_d \right] \right\} = \\
&= C_1^2 \left\{ \vec{b}^c \, \vec{b}_b \left[\frac{\partial}{\partial q^c} \left(g^{ab} \frac{\partial F}{\partial q^a} \right) + g^{ad} \, \Gamma^b_{cd} \frac{\partial F}{\partial q^a} \right] \right\} = \\
&= \frac{\partial}{\partial q^b} \left(g^{ab} \frac{\partial F}{\partial q^a} \right) + g^{ad} \, \Gamma^b_{bd} \frac{\partial F}{\partial q^a} \; .
\end{aligned}$$

Example 10.3.7 (Laplacian of a vector field - with general coordinates q^a)

$$\begin{aligned}
\nabla \cdot \nabla F &= C_1^2 \left[\nabla \left(\nabla F \right) \right] = \\
&= C_1^2 \left\{ \nabla \left[\vec{b}^a \, \vec{b}_b \left(\frac{\partial F^b}{\partial q^a} + \Gamma^b_{ac} F^c \right) \right] \right\} = \\
&= C_1^2 \left\{ \nabla \left[\vec{b}_c \, \vec{b}_b \, g^{ac} \left(\frac{\partial F^b}{\partial q^a} + \Gamma^b_{ac} F^c \right) \right] \right\} = \\
&= C_1^2 \left\{ \nabla \cdot \left((\nabla F)^{cb} \, \vec{b}_c \, \vec{b}_b \right) \right\} = \\
&= C_1^2 \left\{ \vec{b}^a \, \vec{b}_c \, \vec{b}_b \left[\frac{\partial (\nabla F)^{cb}}{\partial q^a} + \Gamma^c_{ad} (\nabla F)^{db} + \Gamma^b_{ad} (\nabla F)^{cd} \right] \right\} = \\
&= \vec{b}_b \left[\frac{\partial (\nabla F)^{ab}}{\partial q^a} + \Gamma^a_{ad} (\nabla F)^{db} + \Gamma^b_{ad} (\nabla F)^{ad} \right] = \; .
\end{aligned}$$

10.3.5 Curl

10.4 Integrals in E^d, $d \leq 3$

10.4.1 Line integrals

Density

Integrals

$$\int_{\vec{r}\in\gamma} F(\vec{r})$$

represent the summation of contributions $F(\vec{r})$ over elementary segments of path γ, whose dimension is $|d\vec{r}|$, i.e. implicitly means

$$\int_{\vec{r}\in\gamma} F(\vec{r}) = \int_{\vec{r}\in\gamma} F(\vec{r})\,|d\vec{r}|\ .$$

Given a regular parametrization of the curve $\vec{r}(q^1)$ (with increasing q^1 so that $|dq^1| = dq^1$), and the differential $d\vec{r} = \vec{r}\,'(q^1)\,dq^1$, the integral can be written as an integral in the parameter q^1

$$\int_{q=q^1_a}^{q^1_b} F(\vec{r}(q^1))\,|\vec{r}\,'(q^1)|\,dq^1\ ,$$

with $\vec{r}(q^1_a)$, $\vec{r}(q^1_b)$ the extreme points of path γ.

Work

Integrals

$$\int_{\vec{r}\in\gamma} F(\vec{r})\cdot\hat{t}(\vec{r})$$

implicitly mean

$$\int_{\vec{r}\in\gamma} F(\vec{r})\cdot\hat{t}(\vec{r}) = \int_{\vec{r}\in\gamma} F(\vec{r})\cdot\hat{t}(\vec{r})|d\vec{r}| = \int_{\vec{r}\in\gamma} F(\vec{r})\cdot d\vec{r}\ ,$$

as $\hat{t} = \frac{d\vec{r}}{|d\vec{r}|}$. Given a regular parametrization of the curve $\vec{r}(q^1)$ (with increasing q^1 so that $|dq^1| = dq^1$), and the differential $d\vec{r} = \vec{r}\,'(q^1)\,dq^1$, the integral can be written as an integral in the parameter q^1

$$\int_{q^1=q^1_a}^{q^1_b} F(\vec{r}(q^1))\cdot\vec{r}\,'(q^1)\,dq^1$$

10.4.2 Surface integrals

Given two coordinates q^1, q^2 describing a surface, $\vec{r}(q^1,q^2)$ the elementary surface with unit normal reads

$$\hat{n}\,dS = d\vec{r}_1\times d\vec{r}_2 = \frac{\partial\vec{r}}{\partial q^1}\times\frac{\partial\vec{r}}{\partial q^2}\,dq^1\,dq^2\ ,$$

and the elementary surface thus reads

$$|dS| = |\hat{n}dS| = \left|\frac{\partial\vec{r}}{\partial q^1}\times\frac{\partial\vec{r}}{\partial q^2}\,dq^1\,dq^2\right|$$

Density

Integrals

$$\int_{\vec{r}\in S} F(\vec{r})$$

implicitly mean

$$\int_{\vec{r}\in S} F(\vec{r}) = \int_{\vec{r}\in S} F(\vec{r})|dS| \ .$$

Given regular parametrization of the surface, $\vec{r}(q^1,\, q^2),\ (q^1, q^2) \in Q^{12}$, the integral can be written as the multi-dimensional integral in coordinates $q^1,\ q^2$,

$$\int_{\vec{r}\in S} F(\vec{r}) = \int_{(q^1,q^2)\in Q^{12}} F(\vec{r}(q^1,q^2)) \left| \frac{\partial \vec{r}}{\partial q^1} \times \frac{\partial \vec{r}}{\partial q^2}\, dq^1\, dq^2 \right|$$

Flux

Integrals

$$\int_{\vec{r}\in S} \hat{n}(\vec{r}) \cdot F(\vec{r})$$

implicitly mean

$$\int_{\vec{r}\in S} \hat{n}(\vec{r}) \cdot F(\vec{r}) = \int_{\vec{r}\in S} \hat{n}(\vec{r}) \cdot F(\vec{r})|dS|$$

Given regular parametrization of the surface, $\vec{r}(q^1,\, q^2),\ (q^1, q^2) \in Q^{12}$, the integral can be written as the multi-dimensional integral in coordinates $q^1,\ q^2$,

$$\int_{\vec{r}\in S} \hat{n}(\vec{r}) \cdot F(\vec{r}) = \int_{(q^1,q^2)\in Q^{12}} \frac{\partial \vec{r}}{\partial q^1} \times \frac{\partial \vec{r}}{\partial q^2} \cdot F(\vec{r}(q^1,q^2))\, dq^1\, dq^2$$

10.4.3 Volume

$$dV = \frac{\partial \vec{r}}{\partial q^1} \cdot \frac{\partial \vec{r}}{\partial q^2} \times \frac{\partial \vec{r}}{\partial q^3}\, dq^1\, dq^2\, dq^3 \ .$$

Density

Integrals

$$\int_{\vec{r}\in V} F(\vec{r})$$

implicitly mean

$$\int_{\vec{r}\in V} F(\vec{r}) = \int_{\vec{r}\in V} F(\vec{r})\, |dV| \ .$$

Given regular parametrization of the volume, $\vec{r}(q^1,\, q^2,\, q^3),\ (q^1, q^2, q^3) \in Q$, the integral can be written as the multi-dimensional integral in coordinates $q^1,\ q^2,\ q^3$,

$$\int_{\vec{r}\in V} F(\vec{r})|dV| = \int_{(q^1,q^2,q^3)\in Q} F(\vec{r}(q^1,q^2,q^3)) \left| \frac{\partial \vec{r}}{\partial q^1} \cdot \frac{\partial \vec{r}}{\partial q^2} \times \frac{\partial \vec{r}}{\partial q^3}\, dq^1\, dq^2\, dq^3 \right| \ .$$

10.4.4 Theorems

Two useful lemmas

The next lemma forms the foundation of the well-known *divergence theorem* and *gradient theorem*:
the proof of these two theorems is based on a straightforward repeated use of this lemma.
Given how simple this lemma is and how frequently it is applied in writing balances and, more generally, in integration by parts, it is very convenient to remember this simple result.

Theorem 10.4.1 (Lemma 1.)

Under the assumptions of *Green's lemma in the plane*,

$$\int_V \frac{\partial A}{\partial x_i} = \oint_S A n_i$$

Proof

The reasoning closely follows the one used for the proof of *Green's lemma in the plane*.
For $\partial A/\partial z$:

$$\begin{aligned}\int_V \frac{\partial A}{\partial z} &= \int_R \int_{z=f_1(x,y)}^{z=f_2(x,y)} \frac{\partial A}{\partial z} dz dx dy = \\ &= \int_R [A(x,y,f_2(x,y)) - A(x,y,f_1(x,y))] dx dy\end{aligned}$$

The most complex step is transitioning from the integral over $(x,y) \in R$ to the surface integral over S, the boundary of volume V: the infinitesimal area element dR in the xy-plane is equal to $dR = dxdy$; the drawing and the proof refer to a *simple* volume (as in the case of Green's lemma in the plane, the results can be generalized to domains of arbitrary shape).
It is possible to divide the surface S into two "halves" $S^+ : z = f_2(x,y)$ and $S^- : z = f_1(x,y)$ such that $S^+ \cup S^- = S$, and the outward normal has positive and negative z-components respectively ($S^+ : \hat{\mathbf{n}} \cdot \hat{\mathbf{z}} > 0$, $S^- : \hat{\mathbf{n}} \cdot \hat{\mathbf{z}} < 0$).
The surface element dR is also the projection of the surface element dS onto the xy-plane: in general, dS will not be parallel to the xy-plane and thus will be larger than dR. It's not difficult to show that:

$$dxdy = dR = \begin{cases} dS \hat{\mathbf{z}} \cdot \hat{\mathbf{n}} & \text{on } S^+ \\ -dS \hat{\mathbf{z}} \cdot \hat{\mathbf{n}} & \text{on } S^- \end{cases}$$

We can now continue the proof:

$$\begin{aligned}&\int_R [A(x,y,f_2(x,y)) - A(x,y,f_1(x,y))] dx dy = \\ &= \int_{S^+} A \hat{\mathbf{n}} \cdot \hat{\mathbf{z}} dS + \int_{S^-} A \hat{\mathbf{n}} \cdot \hat{\mathbf{z}} dS = \\ &= \oint_S A \hat{\mathbf{z}} \cdot \hat{\mathbf{n}} dS = \\ &= \oint_S A n_z dS\end{aligned}$$

Just as the previous lemma forms the basis for the proof of the *gradient theorem* and the *divergence theorem*,
the following lemma forms the basis for the proof of the *curl theorem*.

Theorem 10.4.2 (Lemma 2.)

Under the assumptions of *Green's lemma in the plane*,

$$\int_S [\nabla \times (A\hat{\mathbf{e}}_\mathbf{i})] \cdot \hat{\mathbf{n}} = \oint_\gamma A dx_i$$

Proof

For $A\hat{\mathbf{e}}_\mathbf{x}$, we have $\nabla \times (A\hat{\mathbf{e}}_\mathbf{x}) = \partial A/\partial z \hat{\mathbf{e}}_\mathbf{y} - \partial A/\partial y \hat{\mathbf{e}}_\mathbf{z}$. The surface S is written in parametric form as: $\mathbf{r} = x\hat{\mathbf{e}}_\mathbf{x} + y\hat{\mathbf{e}}_\mathbf{y} + z(x,y)\hat{\mathbf{e}}_\mathbf{z}$. The vector $\partial \mathbf{r}/\partial y = \hat{\mathbf{e}}_\mathbf{y} + \partial z/\partial y \hat{\mathbf{e}}_\mathbf{z}$ is tangent to the surface S and hence perpendicular to the normal $\hat{\mathbf{n}}$:

$$0 = \hat{\mathbf{n}} \cdot \left(\hat{\mathbf{e}}_\mathbf{y} + \frac{\partial z}{\partial y}\hat{\mathbf{e}}_\mathbf{z}\right)$$

Now writing $[\nabla \times (A\hat{\mathbf{e}}_\mathbf{x})] \cdot \hat{\mathbf{n}}$:

$$[\nabla \times (A\hat{\mathbf{e}}_\mathbf{x})] \cdot \hat{\mathbf{n}} = \frac{\partial A}{\partial z}\hat{\mathbf{e}}_\mathbf{y} \cdot \hat{\mathbf{n}} - \frac{\partial A}{\partial y}\hat{\mathbf{e}}_\mathbf{z} \cdot \hat{\mathbf{n}} = -\left[\frac{\partial A}{\partial z}\frac{\partial z}{\partial y} + \frac{\partial A}{\partial y}\right]\hat{\mathbf{e}}_\mathbf{z} \cdot \hat{\mathbf{n}}$$

Recognizing that $\frac{\partial A(x,y,z(x,y))}{\partial y} = \frac{\partial A}{\partial z}\frac{\partial z}{\partial y} + \frac{\partial A}{\partial y}$, we can write:

$$\int_S [\nabla \times (A\hat{\mathbf{e}}_\mathbf{x})] \cdot \hat{\mathbf{n}} = -\int_S \frac{\partial A}{\partial y} \underbrace{\hat{\mathbf{e}}_\mathbf{z} \cdot \hat{\mathbf{n}} dS}_{dR = dxdy} = -\int_R \frac{\partial A}{\partial y} dxdy = \int_\gamma A dx$$

Gradient theorem

…**todo** *assumptions*…

$$\int_V \nabla f = \oint_{\partial V} f\hat{n}$$

Proof for simple domains V

This result immediately follows from Lemma 1 *Theorem 10.4.1*

$$\oint_{\partial V} f\hat{n} = \hat{x}_i \oint_{\partial V} f n_i = \hat{x}_i \int_V \partial_i f = \int_V \hat{x}_i \partial_i f = \int_V \nabla \vec{f}\,,$$

having (1) exploited the freedom to put unit vectors of the Cartesian basis inside the integrals, as they're unifrom - constant in space -, and (2) recognized the expression of the *gradient of a scalar field expressed using Cartesian coordinates*, as shown in *Example 10.5.1*.

Divergence theorem

…**todo** *assumptions*…

$$\int_V \nabla \cdot \vec{f} = \oint_{\partial V} \vec{f} \cdot \hat{n}$$

Proof

This result immediately follows from Lemma 1 *Theorem 10.4.1*

$$\oint_{\partial V} \vec{f} \cdot \hat{n} = \oint_{\partial V} f_i n_i = \int_V \partial_i f_i = \int_V \nabla \cdot \vec{f} \,,$$

having (1) exploited the freedom to put unit vectors of the Cartesian basis inside the integrals, as they're unifrom - constant in space -, and (2) recognized the expression of the *divergence of a vector field expressed using Cartesian coordinates*, as shown in *Example 10.5.4*.

Curl theorem

…**todo** *assumptions*…

$$\int_S \left[\nabla \times \vec{f}\right] \cdot \hat{n} = \oint_{\partial S} \vec{f} \cdot \hat{t}$$

Proof

This proof seamlessly follows from Lemma *Theorem 10.4.2*, applied to all the Cartesian contributions of the vector field

$$\vec{f} = f_x \hat{x} + f_y \hat{y} + f_z \hat{z} \,,$$

as

$$\int_S \left(\nabla \times \vec{f}\right) \cdot \hat{n} = \int_S (\nabla \times f_i \hat{x}_i) \cdot \hat{n} = \oint_{\partial S} f_i t_i = \oint_{\partial S} \vec{f} \cdot \hat{t} \,.$$

10.5 Tensor Calculus in Euclidean Spaces - Cartesian coordinates in E^3

Using Cartesian coordinates $(q^1, q^2, q^3) = (r, \theta, z)$ and Cartesian base vectors (uniform in space, so that their derivatives are zero), a point in Euclidean vector space E^3 can be represented as

$$\vec{r} = x\,\hat{x} + y\,\hat{y} + z\,\hat{z} \,.$$

10.5.1 Natural basis, reciprocal basis, metric tensor, and Christoffel symbols

Cartesian coordinates in Euclidean spaces are a very special coordinate system, with reciprocal basis everywhere coinciding with natural basis, with uniform basis in space (zero second-order derivative of space w.r.t. coordinates, and thus zero first order derivative of base vectors, and thus identically zero Christoffel symbols), and components of the metric tensor equal to the identity matrix

$$\begin{cases} \vec{b}_1 = \vec{b}^1 = \hat{x} \\ \vec{b}_2 = \vec{b}^2 = \hat{y} \\ \vec{b}_3 = \vec{b}^3 = \hat{z} \end{cases}$$

$$[g_{ab}] = [g^{ab}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Gamma^c_{ab} = 0 \qquad , \qquad \forall a, b, c = 1 : 3 \,.$$

10.5.2 Differential operators

Gradient

Example 10.5.1 (Gradient of a scalar field)

$$\nabla F = \hat{x}\,\partial_x F + \hat{y}\,\partial_y F + \hat{z}\,\partial_z F_z$$

Example 10.5.2 (Gradient of a vector field)

$$\begin{aligned}\nabla F &= \nabla(F_x\hat{x} + F_y\hat{y} + F_z\hat{z}) = \\ &= \hat{x}\otimes\hat{x}\,\partial_x F_x + \hat{x}\otimes\hat{y}\,\partial_x F_y + \hat{x}\otimes\hat{z}\,\partial_x F_z + \\ &+ \hat{y}\otimes\hat{x}\,\partial_y F_x + \hat{y}\otimes\hat{y}\,\partial_y F_y + \hat{y}\otimes\hat{z}\,\partial_y F_z + \\ &+ \hat{z}\otimes\hat{x}\,\partial_z F_x + \hat{z}\otimes\hat{y}\,\partial_z F_y + \hat{z}\otimes\hat{z}\,\partial_z F_z +\end{aligned}$$

Example 10.5.3 (Gradient of a 2nd-order tensor field)

Directional derivative

Divergence

Example 10.5.4 (Divergence of a vector field)

$$\begin{aligned}\nabla\cdot F &= \nabla\cdot\left(F_x\,\hat{x} + F_y\,\hat{y} + F_z\,\hat{z}\right) = \\ &= \partial_x F_x + \partial_y F_y + \partial_z F_z\,.\end{aligned}$$

Example 10.5.5 (Divergence of a 2nd-order tensor field)

$$\begin{aligned}\nabla\cdot F &= \nabla\cdot(F_{ab}\vec{e}_a\otimes\vec{e}_b) = \\ &= \vec{e}_c\frac{\partial F_{ab}}{\partial q^a} = \\ &= \hat{x}\left[\partial_x F_{xx} + \partial_y F_{yx} + \partial_z F_{zx}\right] + \\ &+ \hat{y}\left[\partial_x F_{xy} + \partial_y F_{yy} + \partial_z F_{zy}\right] + \\ &+ \hat{z}\left[\partial_x F_{xz} + \partial_y F_{yz} + \partial_z F_{zz}\right]\;.\end{aligned}$$

Laplacian

Example 10.5.6 (Laplacian of a scalar field)

$$\nabla^2 F = \partial_{xx} F + \partial_{yy} F + \partial_{zz} F$$

Example 10.5.7 (Laplacian of a vector field)

10.6 Tensor Calculus in Euclidean Spaces - cylindrical coordinates in E^3

10.6.1 Cylindrical coordiantes and cylindrical coordinates

Using cylindrical coordinates $(q^1, q^2, q^3) = (r, \theta, z)$ and cylindrical base vectors (uniform in space, so that their derivatives are zero), a point in Euclidean vector space E^3 can be represented as

$$\vec{r} = r\cos\theta\,\hat{x} + r\sin\theta\,\hat{y} + z\,\hat{z}\,.$$

10.6.2 Natural basis, reciprocal basis, metric tensor, and Christoffel symbols

Natural basis

Natural basis reads

$$\begin{cases} \vec{b}_1 = \dfrac{\partial\vec{r}}{\partial q^1} = \dfrac{\partial\vec{r}}{\partial r} = \cos\theta\,\hat{x} + \sin\theta\,\hat{y} \\ \vec{b}_2 = \dfrac{\partial\vec{r}}{\partial q^2} = \dfrac{\partial\vec{r}}{\partial \theta} = -r\sin\theta\,\hat{x} + r\cos\theta\,\hat{y} \\ \vec{b}_3 = \dfrac{\partial\vec{r}}{\partial q^3} = \dfrac{\partial\vec{r}}{\partial z} = \hat{z} \end{cases}$$

Metric tensor

Covariant components of metric tensors,

$$g_{ab} = \vec{b}_a \cdot \vec{b}_b\,,$$

can be collected in the diagonal matrix

$$[g_{ab}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}\,,$$

while its contra-variant components can be collected in the inverse matrix (easy to compute, since $[g_{ab}]$ is diagonal),

$$[g^{ab}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & 1 \end{bmatrix}\,.$$

Reciprocal basis

Reciprocal basis is readily evaluated using $\vec{b}^a = g^{ab}\,\vec{b}_b$,

$$\begin{cases} \vec{b}^1 = \cos\theta\,\hat{x} + \sin\theta\,\hat{y} \\ \vec{b}^2 = -\dfrac{1}{r}\sin\theta\,\hat{x} + \dfrac{1}{r}\cos\theta\,\hat{y} \\ \vec{b}^3 = \hat{z} \end{cases}$$

Physical basis

Since metric tensor is diagonal, the cylindrical coordinate system is orthogonal, and its natural and reciprocal basis are orthogonal. A unit orthogonal basis, usually named **physical basis** with unit vector with no physical dimension, is evalated by normalization process,

$$\begin{cases} \hat{r} = \hat{b}_1 = \dfrac{\vec{b}_1}{g_{11}} = \dfrac{\vec{b}^1}{g^{11}} = \cos\theta\,\hat{x} + \sin\theta\,\hat{y} \\ \hat{\theta} = \hat{b}_2 = \dfrac{\vec{b}_2}{g_{22}} = \dfrac{\vec{b}^2}{g^{22}} = -\sin\theta\,\hat{x} + \cos\theta\,\hat{y} \\ \hat{z} = \hat{b}_3 = \dfrac{\vec{b}_3}{g_{33}} = \dfrac{\vec{b}^3}{g^{33}} = \hat{z}\,. \end{cases}$$

Derivatives of natural basis and Christoffel symbols

Derivatives of the natural basis read

$$\begin{aligned} \frac{\partial\vec{b}_1}{\partial q^1} &= \vec{0} \\ \frac{\partial\vec{b}_2}{\partial q^2} &= -r\cos\theta\,\hat{x} - r\sin\theta\,\hat{y} = -q^1\,\vec{b}_1 \\ \frac{\partial\vec{b}_3}{\partial q^3} &= \vec{0} \\ \frac{\partial\vec{b}_2}{\partial q^1} = \frac{\partial\vec{b}_1}{\partial q^2} &= -\sin\theta\,\hat{x} + \cos\theta\hat{y} = \frac{1}{q^1}\,\vec{b}_2 \\ \frac{\partial\vec{b}_3}{\partial q^1} = \frac{\partial\vec{b}_1}{\partial q^3} &= \vec{0} \\ \frac{\partial\vec{b}_3}{\partial q^2} = \frac{\partial\vec{b}_2}{\partial q^3} &= \vec{0} \end{aligned}$$

so that non-zero Christoffel symbols of a cylindrical coordinate system are

$$\begin{aligned} \Gamma^2_{12} = \Gamma^2_{21} &= \frac{1}{q^1} \\ \Gamma^1_{22} &= -q^1\,. \end{aligned}$$

10.6.3 Differential operators

Gradient

Example 10.6.1 (Gradient of a scalar field)

$$
\begin{aligned}
\nabla F &= \vec{b}^a \frac{\partial F}{\partial q^a} = \\
&= \hat{b}_a \, g^{aa} \, \frac{\partial F}{\partial q^a} = \\
&= \hat{r} \, \frac{\partial F}{\partial r} + \hat{\theta} \, \frac{1}{r} \, \frac{\partial F}{\partial \theta} + \hat{z} \frac{\partial F}{\partial z} \, .
\end{aligned}
$$

Example 10.6.2 (Gradient of a vector field)

$$
\begin{aligned}
\nabla F &= \vec{b}^a \otimes \vec{b}_b \left[\frac{\partial F^b}{\partial q^a} + \Gamma^b_{ac} \, F^c \right] = \\
&= \cdots = \\
&= \vec{b}^1 \otimes \vec{b}_1 \, \partial_1 F^1 && + \vec{b}^1 \otimes \vec{b}_2 \, \left[\partial_1 F^2 + \Gamma^2_{12} F^2 \right] && + \vec{b}^1 \otimes \vec{b}_3 \, \partial_1 F^3 \\
&+ \vec{b}^2 \otimes \vec{b}_1 \, \left[\partial_2 F^1 + \Gamma^1_{22} F^2 \right] && + \vec{b}^2 \otimes \vec{b}_2 \, \left[\partial_2 F^2 + \Gamma^2_{21} F^1 \right] && + \vec{b}^2 \otimes \vec{b}_3 \, \partial_2 F^3 \\
&+ \vec{b}^3 \otimes \vec{b}_1 \, \partial_3 F^1 && + \vec{b}^3 \otimes \vec{b}_2 \, \partial_3 F^2 && + \vec{b}^3 \otimes \vec{b}_3 \, \partial_3 F^3 \\
&= \hat{r} \otimes \hat{r} \, \partial_r F_r && + \hat{r} \otimes \hat{\theta} \, \frac{1}{r} \left[\partial_r (r F_\theta) + F_\theta \right] && + \hat{r} \otimes \hat{z} \, \partial_r F_z \\
&+ \hat{\theta} \otimes \hat{r} \, \frac{1}{r} \left[\partial_\theta F_r - r F_\theta \right] && + \hat{\theta} \otimes \hat{\theta} \, \left[\partial_\theta \left(\frac{F_\theta}{r} \right) + \frac{F_r}{r} \right] && + \hat{\theta} \otimes \hat{z} \, \frac{1}{r} \partial_\theta F_z \\
&+ \hat{z} \otimes \hat{r} \, \partial_z F_x && + \hat{z} \otimes \hat{\theta} \, \frac{1}{r} \partial_\theta F_y && + \hat{z} \otimes \hat{z} \, \partial_z F_z \, .
\end{aligned}
$$

Example 10.6.3 (Gradient of a 2nd-order tensor field)

Directional derivative

Divergence

Example 10.6.4 (Divergence of a vector field)

$$
\begin{aligned}
\nabla \cdot \vec{F} &= \frac{\partial F^a}{\partial q^a} + \Gamma^a_{ac} \, F^c = \\
&= \frac{\partial F_r}{\partial r} + \frac{\partial}{\partial \theta} \left(\frac{F_\theta}{r} \right) + \frac{F_\theta}{r} + \frac{\partial F_z}{\partial z} \, .
\end{aligned}
$$

Divergence of a 2nd-order tensor field

Using the general formula of the divergence of a 2nd-order tensor field (see *Divergence*)

$$\nabla \cdot \left(F^{bc} \vec{b}_b \otimes \vec{b}_c \right) = C_1^2 \left(\nabla F \right) = \\ = \vec{b}_c \left[\frac{\partial F^{ac}}{\partial q^a} + \Gamma^a_{ad} F^{dc} + \Gamma^c_{ad} F^{ad} \right]$$

the contravariant components in the natural basis induced by cylindrical coordinates of the divergence of a second order tensor reads

$$\begin{aligned} \nabla \cdot \mathbb{F} = \\ &= \vec{b}_1 \left[\frac{\partial F^{11}}{\partial q^1} + \frac{\partial F^{21}}{\partial q^2} + \frac{\partial F^{31}}{\partial q^3} + \Gamma^2_{21} F^{11} + \Gamma^1_{22} F^{22} \right] + \\ &+ \vec{b}_2 \left[\frac{\partial F^{12}}{\partial q^1} + \frac{\partial F^{22}}{\partial q^2} + \frac{\partial F^{32}}{\partial q^3} + \Gamma^2_{21} F^{12} + \Gamma^2_{21} F^{12} + \Gamma^2_{21} F^{21} \right] + \\ &+ \vec{b}_3 \left[\frac{\partial F^{13}}{\partial q^1} + \frac{\partial F^{23}}{\partial q^2} + \frac{\partial F^{33}}{\partial q^3} + \Gamma^2_{21} F^{13} \right] = \\ &= \vec{b}_1 \left[\frac{\partial F^{11}}{\partial q^1} + \frac{\partial F^{21}}{\partial q^2} + \frac{\partial F^{31}}{\partial q^3} + \frac{1}{q^1} F^{11} - q^1 F^{22} \right] + \\ &+ \vec{b}_2 \left[\frac{\partial F^{12}}{\partial q^1} + \frac{\partial F^{22}}{\partial q^2} + \frac{\partial F^{32}}{\partial q^3} + \frac{2}{q^1} F^{12} + \frac{1}{q^1} F^{21} \right] + \\ &+ \vec{b}_3 \left[\frac{\partial F^{13}}{\partial q^1} + \frac{\partial F^{23}}{\partial q^2} + \frac{\partial F^{33}}{\partial q^3} + \frac{1}{q^1} F^{13} \right] . \end{aligned}$$

Next, it's easy to exploit the definition of the coordinates $\left(q^1, q^2, q^3\right) = (r, \theta, z)$ and the relation between natural and physical basis and components to get

$$\begin{aligned} \nabla \cdot \mathbb{F} = \\ &= \hat{r} \left[\frac{\partial \left(F^{rr}\right)}{\partial r} + \frac{\partial \left(\frac{1}{r} F^{\theta r}\right)}{\partial \theta} + \frac{\partial F^{zr}}{\partial z} + \frac{1}{r} F^{rr} - r \frac{1}{r^2} F^{\theta\theta} \right] + \\ &+ r\hat{\theta} \left[\frac{\partial \left(\frac{1}{r} F^{r\theta}\right)}{\partial r} + \frac{1}{r} \frac{\partial \left(\frac{1}{r} F^{\theta\theta}\right)}{\partial \theta} + \frac{\partial F^{z\theta}}{\partial z} + \frac{2}{r} \frac{1}{r} F^{r\theta} + \frac{1}{r} \frac{1}{r} F^{\theta r} \right] + \\ &+ \hat{z} \left[\frac{\partial F^{rz}}{\partial r} + \frac{\partial \left(\frac{1}{r} F^{\theta z}\right)}{\partial \theta} + \frac{\partial F^{zz}}{\partial z} + \frac{1}{r} F^{rz} \right] . \end{aligned}$$

or, after few algebraic manipulations,

$$\begin{aligned} \nabla \cdot \mathbb{F} = \\ &= \hat{r} \left[\frac{\partial F^{rr}}{\partial r} + \frac{1}{r} \frac{\partial F^{\theta r}}{\partial \theta} + \frac{\partial F^{zr}}{\partial z} + \frac{1}{r} F^{rr} - \frac{1}{r} F^{\theta\theta} \right] + \\ &+ \hat{\theta} \left[\frac{\partial F^{r\theta}}{\partial r} + \frac{1}{r} \frac{\partial F^{\theta\theta}}{\partial \theta} + \frac{\partial F^{z\theta}}{\partial z} + \frac{1}{r} F^{r\theta} + \frac{1}{r} F^{\theta r} \right] + \\ &+ \hat{z} \left[\frac{\partial F^{rz}}{\partial r} + \frac{1}{r} \frac{\partial F^{\theta z}}{\partial \theta} + \frac{\partial F^{zz}}{\partial z} + \frac{1}{r} F^{rz} \right] . \end{aligned}$$

Laplacian

Example 10.6.5 (Laplacian of a scalar field)

Example 10.6.6 (Laplacian of a vector field)

10.7 Tensor Calculus in Euclidean Spaces - Spehrical coordinates in E^3

Using spherical coordinates $(q^1, q^2, q^3) = (r, \phi, \theta)$ and spherical base vectors (uniform in space, so that their derivatives are zero), a point in Euclidean vector space E^3 can be represented as

$$\vec{r} = r\cos\theta\sin\phi\,\hat{x} + r\sin\theta\sin\phi\,\hat{y} + r\cos\phi\,\hat{z}\,.$$

10.7.1 Natural basis, reciprocal basis, metric tensor, and Christoffel symbols

10.7.2 Differential operators

Gradient

Example 10.7.1 (Gradient of a scalar field)

Example 10.7.2 (Gradient of a vector field)

Example 10.7.3 (Gradient of a 2nd-order tensor field)

Directional derivative

Divergence

Example 10.7.4 (Divergence of a vector field)

Example 10.7.5 (Divergence of a 2nd-order tensor field)

Laplacian

Example 10.7.6 (Laplacian of a scalar field)

Example 10.7.7 (Laplacian of a vector field)

CHAPTER

ELEVEN

TENSOR INVARIANTS

11.1 Rank-2 tensors

- Characteristic polynomial (*this definition needs that determinant exists*)

$$0 = \det\left(\mathbf{A} - s\mathbf{I}\right) \ .$$

Using a unit normal basis (**todo** generalization required?)

$$0 = \begin{vmatrix} A_{11} - s & A_{12} & A_{13} \\ A_{21} & A_{22} - s & A_{23} \\ A_{31} & A_{32} & A_{33} - s \end{vmatrix} = -s^3 + s^2 I_1 - s I_2 + I_3 \ ,$$

with

$$\begin{aligned} I_1 &= \mathrm{tr}(\mathbf{A}) = A_{11} + A_{22} + A_{33} \\ I_2 &= \frac{1}{2}\left(\mathrm{tr}(\mathbf{A})^2 - \mathrm{tr}(\mathbf{A}^2)\right) = \cdots = A_{11}A_{22} + A_{11}A_{33} + A_{22}A_{33} - A_{31}A_{13} - A_{12}A_{21} - A_{32}A_{23} \\ I_3 &= \det(\mathbf{A}) = ... \end{aligned}$$

The coefficients of the characteristic polynomial are invariant under transformation of coordinates (**todo** here only referring to Cartesian basis, and thus orthogonal transformations?)

- **Trace**…
- Determinant…

CHAPTER

TWELVE

UNITARY AND ROTATION TENSORS

Some notes in the introduction to classical mechanics: Rotations: Tensor formalism for rotations.

todo Move the mathematical treatment here?

todo Discuss vector operations that are invariant under 3-dimensional rotation and other unitary tensor applications (i.e. reflections). These properties are useful to prove the general expression of *isotropic tensors*.

12.1 Invariant operations under rotations and unitary transformations

Proof that/if:

- the only independent invariant operations producing a scalar with **two** vectors **u**, **v** are

$$\begin{aligned}
&\mathbf{u}\cdot\mathbf{u} = |\mathbf{u}|^2 \\
&\mathbf{v}\cdot\mathbf{v} = |\mathbf{v}|^2 \\
&\mathbf{u}\cdot\mathbf{v}
\end{aligned}$$

- the only independent invariant operations producing a scalar with **three** vectors **u**, **v**, **w** are

$$\mathbf{u}\times\mathbf{v}\cdot\mathbf{w}$$

- the only independent invariant operations producing a scalar with **four** vectors $\{\mathbf{u}^{(i)}\}_{i=1:4}$ are

$$\mathbf{u}^{(k_1)}\cdot\mathbf{u}^{(k_2)}\ \mathbf{u}^{(k_3)}\cdot\mathbf{u}^{(k_4)}\ ,$$

  with every index k_j independently ranging from 1 to 4.

CHAPTER

THIRTEEN

ISOTROPIC TENSORS

This section provides the definition of isotropic tensors, the general forms of low-rank tensors (from rank-2 to rank-4 tensors) that may be useful in some fields of science (e.g. in continuum mechanics, in the constitutive law relating stress tensor with strain tensor in isotropic linear elastic media or stress tensor with strain velocity tensor in Newtonian fluids).

Definition 13.1 (Isotropic tensor)

An isotropic tensor is …

The general form of the low-rank isotropic tensors in three-dimensional spaces, dim $(\mathcal{V}) = 3$ is proved using the definition of a rank-r tensor as a multilinear map acting on r vectors, and exploiting the invariance of some vector operations under rotations.

13.1 Rank-2 isotropic tensors

The most general rank-2 isotropic tensor is a scalar multiple of the identity tensor, $\mathbf{A} = a\mathbf{I}$.

Proof

The action of a rank-2 isotropic tensor in $\mathcal{V}$ on every pair of vectors $\mathbf{u}$, $\mathbf{v}$ can be a function of $\mathbf{u} \cdot \mathbf{v}$ only, i.e. the only vector operation:

- producing a scalar
- invariant under rotations
- involving the two vectors $\mathbf{u}$, $\mathbf{v}$ (i.e. the tensor is a multilinear map, $\mathbf{A}(\mathbf{u}, \mathbf{v})$)

Now, using coordinates, the general expression of the isotropic tensor is retrieved comparing the expression built under the isotropic assumption,

$$\begin{aligned}\mathbf{A}(\mathbf{u}, \mathbf{v}) &= a\mathbf{u} \cdot \mathbf{v} = \\ &= a\left(u^i \mathbf{b}_i\right) \cdot \left(v^j \mathbf{b}_j\right) = \\ &= a u^i v^j g_{ij}\end{aligned}$$

with the general expression of the action of the stress tensor on the vectors $\mathbf{u}$, $\mathbf{v}$

$$\mathbf{A}(\mathbf{u}, \mathbf{v}) = A_{ij} \mathbf{b}^i \mathbf{b}^j \left(u^k \mathbf{b}_k, v^l \mathbf{b}_l\right) = A_{ij} \delta^i_k \delta^j_l u^k v^l = A_{ij} u^i v^j \ ,$$

so that the *covariant components* read

$$A_{ij} = a g_{ij}$$

and the tensor can be written as

$$\begin{aligned}\mathbf{A} = A_{ij}\mathbf{b}^i\mathbf{b}^j &= a g_{ij}\mathbf{b}^i\mathbf{b}^j = \\ &= a\mathbf{b}^i\mathbf{b}_i = a\delta_i^j\mathbf{b}^i\mathbf{b}_j = \\ &= a g^{ij}\mathbf{b}_j\mathbf{b}_i = a g^{ij}\mathbf{b}_i\mathbf{b}_j \ .\end{aligned}$$

Components of the identity tensor

The identity tensor can be defined as a rank-2 tensor whose dot product with any arbitrary $\mathbf{v} \in \mathbf{V}$ gives

$$\mathbf{I} \cdot \mathbf{v} = \mathbf{v} \ .$$

The identity tensor can be written using any vector basis $\{\mathbf{b}_i\}_i$, and its reciprocal $\{\mathbf{b}^j\}$ as

$$\mathbf{I} = \mathbf{b}_i\mathbf{b}^i = \mathbf{b}^i\mathbf{b}_i = g_{ij}\mathbf{b}^i\mathbf{b}^j = g^{ij}\mathbf{b}_i\mathbf{b}_j \ ,$$

as it's immediate to prove by direct computation

$$\begin{aligned}\mathbf{b}_i\mathbf{b}^i \cdot \mathbf{v} &= \mathbf{b}_i\mathbf{b}^i \cdot (v^k\mathbf{b}_k) = \mathbf{b}_i\delta_k^i v^k = v^i\mathbf{b}_i = \mathbf{v} \\ \mathbf{b}^i\mathbf{b}_i \cdot \mathbf{v} &= \mathbf{b}^i\mathbf{b}_i \cdot (v_k\mathbf{b}^k) = \mathbf{b}^i\delta_i^k v_k = v_i\mathbf{b}^i = \mathbf{v} \\ g_{ij}\mathbf{b}^i\mathbf{b}^j \cdot \mathbf{v} &= g_{ij}\mathbf{b}^i\mathbf{b}^j \cdot (v^k\mathbf{b}_k) = \mathbf{b}^i\delta_k^j v^k g_{ij} = \mathbf{b}^i \underbrace{v^k g_{ik}}_{=v_i} = \mathbf{v} \\ g^{ij}\mathbf{b}_i\mathbf{b}_j \cdot \mathbf{v} &= g^{ij}\mathbf{b}_i\mathbf{b}_j \cdot (v_k\mathbf{b}^k) = \mathbf{b}_i\delta_j^k v_k g^{ij} = \mathbf{b}_i \underbrace{v_k g^{ik}}_{=v^i} = \mathbf{v}\end{aligned}$$

13.2 Rank-3 isotropic tensors

todo

- Invariant object: $\mathbf{u} \times \mathbf{v} \cdot \mathbf{w}$
- Retrieve components, to find Levi-Civita symbols
- Discuss invariance under rotation and symmetry (pseudo-tensor)

13.3 Rank-4 isotropic tensors

Proof

The action of a rank-4 isotropic tensor in $\mathcal{V}$ on every set of 4 vectors $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$, $\mathbf{x}$ can be a function of the products of scalar products

$$\begin{aligned}&\mathbf{u} \cdot \mathbf{v}\,\mathbf{w} \cdot \mathbf{x} \\ &\mathbf{u} \cdot \mathbf{w}\,\mathbf{v} \cdot \mathbf{x} \\ &\mathbf{u} \cdot \mathbf{x}\,\mathbf{v} \cdot \mathbf{w} \ ,\end{aligned}$$

i.e. any possible combinations through scalar products of different vectors

- producing a scalar
- invariant under rotations

- involving all the 4 vectors **u**, **v**, **w**, **x** (i.e. the tensor is a multilinear map, $\mathbf{A}(\mathbf{u},\mathbf{v},\mathbf{w},\mathbf{x})$)

Now, using coordinates, the general expression of the isotropic tensor is retrieved comparing the expression built under the isotropic assumption,

$$\begin{aligned}\mathbf{A}(\mathbf{u},\mathbf{v}) &= a\mathbf{u}\cdot\mathbf{v}\,\mathbf{w}\cdot\mathbf{x} + b\mathbf{u}\cdot\mathbf{w}\,\mathbf{v}\cdot\mathbf{x} + c\mathbf{u}\cdot\mathbf{x}\,\mathbf{v}\cdot\mathbf{w} = \\ &= a\,g_{ij}g_{kl}u^i v^j w^k x^l + b\,g_{ij}g_{kl}u^i w^j v^k x^l + c\,g_{ij}g_{kl}x^i v^j v^k w^l = \\ &= \left(a\,g_{ij}g_{kl} + b\,g_{ik}g_{jl} + c\,g_{il}g_{jk}\right)u^i v^j w^k x^l\ ,\end{aligned}$$

with the general expression of the action of the stress tensor on the vectors **u**, **v**

$$\begin{aligned}\mathbf{A}(\mathbf{u},\mathbf{v}) &= A_{ijkl}\mathbf{b}^i\mathbf{b}^j\mathbf{b}^k\mathbf{b}^l\left(u^m\mathbf{b}_m, v^n\mathbf{b}_n, u^p\mathbf{b}_p, v^q\mathbf{b}_q\right) = \\ &= A_{ijkl}\delta^i_m\delta^j_n\delta^k_p\delta^l_q u^m v^n w^p x^q = A_{ijkl}u^i v^j w^k x^l\ ,\end{aligned}$$

so that the *covariant components* read

$$A_{ijkl} = a\,g_{ij}g_{kl} + b\,g_{ik}g_{jl} + c\,g_{il}g_{jk}\ ,$$

and the tensor can be written as

$$\mathbf{A} = A_{ijkl}\mathbf{b}^i\mathbf{b}^j\mathbf{b}^k\mathbf{b}^l = \left(a\,g_{ij}g_{kl} + b\,g_{ik}g_{jl} + c\,g_{il}g_{jk}\right)\mathbf{b}^i\mathbf{b}^j\mathbf{b}^k\mathbf{b}^l =$$

The most general liner isotropic relation between two rank-2 tensors reads **A**, **B**

$$\begin{aligned}\mathbf{A} &= a\,\mathrm{tr}\,(\mathbf{B})\,\mathbf{I} + b\,\mathbf{B} + c\,\mathbf{B}^T \\ &= a\,\mathrm{tr}\,(\mathbf{B})\,\mathbf{I} + \tilde{b}\,\mathbf{B}^s + \tilde{c}\,\mathbf{B}^a\ ,\end{aligned}$$

being $\mathbf{B}^s = \frac{1}{2}\left(\mathbf{B}+\mathbf{B}^T\right)$, and $\mathbf{B}^a = \frac{1}{2}\left(\mathbf{B}-\mathbf{B}^T\right)$ the symmetric and the anti-symmetric parts of the tensor **B** respectively.

Linear isotropic relation between two rank-2 tensors

$$\begin{aligned}\mathbf{A} &= \mathbf{Q}:\mathbf{B} \\ \mathbf{Q}:\mathbf{B} &= \left(Q_{ijkl}\mathbf{b}^i\mathbf{b}^j\mathbf{b}^k\mathbf{b}^l\right):\left(B^{mn}\mathbf{b}_m\mathbf{b}_n\right) = \\ &= Q_{ijkl}B^{kl}\mathbf{b}^i\mathbf{b}^j = \\ &= \left(a\,g_{ij}g_{kl} + b\,g_{ik}g_{jl} + c\,g_{il}g_{jk}\right)B^{kl}\mathbf{b}^i\mathbf{b}^j = \\ &= \left(a\,g_{ij}B_l{}^l + b\,B_{ij} + c\,B_{ji}\right)\mathbf{b}^i\mathbf{b}^j = \\ &= \left(a\,g_{ij}B_l{}^l + b\,B_{ij} + c\,B_{ji}\right)\mathbf{b}^i\mathbf{b}^j =\end{aligned}$$

depends on the three constant a, b, c of the isotropic tensor.

The most general liner isotropic relation between two symmetric rank-2 tensors reads **A**, **B**

$$\mathbf{A} = a\,\mathrm{tr}\,(\mathbf{B})\,\mathbf{I} + \tilde{b}\,\mathbf{B}\ ,$$

as the anti-symmetric part of a symmetric tensor is identically zero.

Linear isotropic relation between two symmetric rank-2 tensors

If $B_{ij} = B_{ji}$, the double-dot product becomes

$$\mathbf{A} = \mathbf{Q}:\mathbf{B} = \left(a\,g_{ij}B_l{}^l + d\,B_{ij}\right)\mathbf{b}^i\mathbf{b}^j\ ,$$

depends only on two constants a, d, having defined $d = b + c$. This is a common condition in continuum mechanics, where some constitutive law links two symmetric rank-2 tensors, like the stress and strain tensors for linear elastic media, or the stress and strain velocity tensors for Newtonian fluids.

CHAPTER

FOURTEEN

TIME DERIVATIVE OF INTEGRALS OVER MOVING DOMAINS

Some results about time derivatives of integrals over moving domains are collected here. These results are useful for writing balance equations of physical quantities in integral form over arbitrary domains, like:

- integral form of balance equations in continuum mechanics, see Continuum Mechanics:Governing Equations:Integral balance equations for arbitrary domains
- integral form of Maxwell's equations governing classical electromagnetism, see Electromagnetism:Principles of Classical Electromagnetism:derivation of balance equations for arbitary volume, starting from equations for a control volume

Link to hand-written notes.

14.1 Volume density

Reynolds transport theorem. Given a volume $V(t)$ with boundary $\partial V(t)$, whose points $\vec{r} \in \partial V(t)$ have velocity $\vec{v}_b$,

$$\frac{d}{dt}\int_{V(t)} f = \int_{V(t)} \frac{\partial f}{\partial t} + \oint_{\partial V(t)} f \vec{v}_b \cdot \hat{n} \, .$$

"Proof"

$$\begin{aligned}
\frac{d}{dt}\int_{v(t)} f(\vec{r},t)\,dt &= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\left[\int_{v(t+\Delta t)} f(\vec{r},t+\Delta t) - \int_{v(t)} f(\vec{r},t)\right] = \\
&= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\left[\int_{v(t)} \{f(\vec{r},t+\Delta t) - f(\vec{r},t)\} + \int_{\Delta v(t;\Delta t)} f(\vec{r},t)\right] = \\
&= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\left[\int_{v(t)} \left\{\Delta t\,\frac{\partial f}{\partial t}(\vec{r},t) + o(\Delta t)\right\} + \oint_{\partial v(t)} \{\Delta t\,\vec{v}_b\cdot\hat{n}\,f(\vec{r},t) + o(\Delta t)\}\right] = \\
&= \int_{v(t)} \frac{\partial f}{\partial t} + \oint_{\partial v(t)} f\vec{v}_b\cdot\hat{n}\,.
\end{aligned}$$

14.2 Flux across a surface

$$\frac{d}{dt}\int_{S(t)} \vec{f}\cdot\hat{n} = \int_{S(t)} \frac{\partial \vec{f}}{\partial t}\cdot\hat{n} + \int_{S(t)} \nabla\cdot\vec{f}\,\vec{v}_b\cdot\hat{n} - \int_{\partial S(t)} \vec{v}_b\times\vec{f}\cdot\hat{t}$$

"Proof"

$$\begin{aligned}
\frac{d}{dt}\int_{s(t)} \vec{f}(\vec{r},t)\cdot\hat{n}\,dt &= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\left[\int_{s(t+\Delta t)} \vec{f}(\vec{r},t+\Delta t)\cdot\hat{n}(\vec{r},t+\Delta t) - \int_{s(t)} f(\vec{r},t)\cdot\hat{n}(\vec{r},t)\right] = \\
&= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\left[\int_{s(t+\Delta t)} \vec{f}(\vec{r},t+\Delta t)\cdot\hat{n}(\vec{r},t+\Delta t) - \int_{s(t)} f(\vec{r},t)\cdot\hat{n}(\vec{r},t) + \right. \\
&\qquad \left. - \int_{s(t)} \vec{f}(\vec{r},t+\Delta t)\cdot\hat{n}(\vec{r},t) + \int_{s(t)} \vec{f}(\vec{r},t+\Delta t)\cdot\hat{n}(\vec{r},t)\right] = \\
&= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\left[\int_{s(t)} \Delta t\frac{\partial \vec{f}}{\partial t}\cdot\hat{n} + o(\Delta t) + \oint_{\partial\Delta v(t;\Delta t)} \vec{f}\cdot\hat{n} - \oint_{\partial s(t)} \Delta t\vec{f}\cdot\hat{t}\times\vec{v}_b\right] = \\
&= \int_{s(t)} \frac{\partial \vec{f}}{\partial t}\cdot\hat{n} + \int_{s} \nabla\cdot\vec{f}\,\vec{v}_b\cdot\hat{n} - \oint_{\partial s(t)} \vec{v}_b\times\vec{f}\cdot\hat{t}\,.
\end{aligned}$$

having used

$$\oint_{\partial\Delta v(t;\Delta t)} \vec{f}\cdot\hat{n} = \int_{\Delta v} \nabla\cdot\vec{f} = \Delta t\int_{s} \nabla\cdot\vec{f}\,\hat{n}\cdot\vec{v}_b + o(\Delta t)$$

14.3 Work line integral along a line

$$\frac{d}{dt}\int_{\ell(t)} \vec{f}\cdot\hat{t} = \int_{\ell(t)} \frac{\partial \vec{f}}{\partial t}\cdot\hat{t} + \int_{\ell(t)} \nabla\times\vec{f}\cdot\vec{v}_b\times\hat{t} + \vec{f}_B\cdot\vec{v}_B - \vec{f}_A\cdot\vec{v}_A$$

"Proof"

$$\begin{aligned}
\frac{d}{dt}\int_{\ell(t)} \vec{f}(\vec{r},t)\cdot\hat{t}(\vec{r},t)\,dt &= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\left[\int_{\ell(t+\Delta t)} \vec{f}(\vec{r},t+\Delta t)\cdot\hat{t}(\vec{r},t+\Delta t) - \int_{\ell(t)} \vec{f}(\vec{r},t)\cdot\hat{t}(\vec{r},t)\right] = \\
&= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\left[\int_{\ell(t+\Delta t)} \vec{f}(\vec{r},t+\Delta t)\cdot\hat{t}(\vec{r},t+\Delta t) - \int_{\ell(t)} f(\vec{r},t)\cdot\hat{t}(\vec{r},t) + \right. \\
&\qquad \left. - \int_{\ell(t)} \vec{f}(\vec{r},t+\Delta t)\cdot\hat{t}(\vec{r},t) + \int_{\ell(t)} \vec{f}(\vec{r},t+\Delta t)\cdot\hat{t}(\vec{r},t)\right] = \\
&= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\left[\int_{\ell(t+\Delta t)} \Delta t\,\frac{\partial \vec{f}}{\partial t}(\vec{r},t+\Delta t)\cdot\hat{t}(\vec{r},t+\Delta t) + \right. \\
&\qquad \left. + \oint_{\partial\Delta s(t)} f(\vec{r},t)\cdot\hat{t}(\vec{r},t) - \Delta t\vec{f}_A\cdot\vec{v}_A + \Delta t\vec{f}_B\cdot\vec{v}_B + o(\Delta t)\right] = \\
&= \int_{\ell(t)} \frac{\partial \vec{f}}{\partial t}\cdot\hat{t} + \int_{\ell(t)} \nabla\times\vec{f}\cdot\vec{v}_b\times\hat{t} + \vec{f}_B\cdot\vec{v}_B - \vec{f}_A\cdot\vec{v}_A\,.
\end{aligned}$$

having used

$$\oint_{\partial\Delta s(t;\Delta t)} \vec{f}\cdot\hat{t} = \int_{\Delta s} \hat{n}\cdot\nabla\times\vec{f} = \Delta t\int_{\ell}\nabla\times\vec{f}\cdot\hat{v}_b\times\vec{t} + o(\Delta t)$$

CHAPTER

FIFTEEN

CALCULUS IDENTITIES

Here some calculus identities are listed and proved, often using Cartesian coordinates and index notation. While these identities are independent on the choice of the coordinates - as calculus is - a generic set of coordinates can be used to prove them, and Cartesian coordinates are the most convenint choice.

$$\Delta\vec{v} = \nabla(\nabla\cdot\vec{v}) - \nabla\times\nabla\times\vec{v}\,. \tag{15.1}$$

Proof.

Uusing the identity

$$\varepsilon_{ijk}\varepsilon_{ilm} = \delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}$$

and index notation for Cartesian coordinates,

$$\begin{aligned}
\left(\nabla\times\nabla\times\vec{v}\right)_i &= \hat{e}_i\cdot\left(\nabla\times\nabla\times\vec{v}\right) = \\
&= \varepsilon_{ijk}\,\partial_j(\varepsilon_{klm}\,\partial_l v_m) = \\
&= \varepsilon_{kij}\,\varepsilon_{klm}\partial_{jl}v_m = \\
&= \left(\delta_{il}\,\delta_{jm} - \delta_{im}\,\delta_{jl}\right)\partial_{jl}v_m = \\
&= \partial_{ij}\,v_j - \partial_{jj}\,v_i = \\
&= \hat{e}_i\cdot\left(\nabla(\nabla\cdot\vec{v}) - \Delta\vec{v}\right) = \qquad = \left(\nabla(\nabla\cdot\vec{v}) - \Delta\vec{v}\right)_i\,.
\end{aligned}$$

Part V

Functional Analysis

CHAPTER

SIXTEEN

INTRODUCTION TO FUNCTIONAL ANALYSIS

- Lebesgue integral
- L^p, H^p function spaces
- Banach and Hilbert spaces

CHAPTER

SEVENTEEN

DIRAC'S DELTA

Dirac's delta $\delta(x)$ is a distribution, or generalized function, with the following properties

1. $$\int_D \delta(x - x_0)\, dx = 1 \quad \text{if } x_0 \in D \tag{17.1}$$

2. $$\int_D f(x)\delta(x - x_0)\, dx \quad \text{if } x_0 \in D \tag{17.2}$$

for $\forall f(x)$ "regular" **todo** *what does regular mean?*

17.1 Dirac's delta in terms of regular functions

17.1.1 Piece-wise constant

$$\delta(x) \sim r_\varepsilon(x) = \begin{cases} \frac{1}{\varepsilon} & x \in [-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}] \\ 0 & \text{otherwise} \end{cases}$$

Properties - proof.

1. Unitariety

$$\int_{x=-\infty}^{\infty} r_\varepsilon(x - x_0)\, dx = \int_{x=x_0-\frac{\varepsilon}{2}}^{x_0+\frac{\varepsilon}{2}} \frac{1}{\varepsilon}\, dx = 1 \ ,$$

   for $\forall \varepsilon$;

2. Shift property, using mean-value theorem of continuous functions

$$\int_{x=-\infty}^{\infty} r_\varepsilon(x - x_0) f(x)\, dx = \int_{x=x_0-\frac{\varepsilon}{2}}^{x_0+\frac{\varepsilon}{2}} \frac{1}{\varepsilon} f(x)\, dx = \frac{1}{\varepsilon}\varepsilon f(\xi) \ ,$$

   with $\xi \in [x_0 - \frac{\varepsilon}{2}, x_0 + \frac{\varepsilon}{2}]$, for the mean value theorem. As $\varepsilon \to 0$, $\xi \to x_0$, and thus

$$\int_{x=-\infty}^{\infty} r_\varepsilon(x - x_0) f(x)\, dx \to f(x_0)$$

17.1.2 Piecewise-linear

$$\delta(x) \sim t_\varepsilon(x) = \begin{cases} \frac{2}{\varepsilon}\left(1 - \frac{2|x|}{\varepsilon}\right) & x \in [-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}] \\ 0 & \text{otherwise} \end{cases}$$

Properties - proof

1. Unitariety

$$\int_{x=-\infty}^{\infty} t_\varepsilon(x - x_0)\, dx = \int_{x=x_0-\frac{\varepsilon}{2}}^{x_0+\frac{\varepsilon}{2}} \frac{2}{\varepsilon}\left(1 - \frac{2|x|}{\varepsilon}\right) dx = \frac{1}{2}\varepsilon\frac{2}{\varepsilon} = 1\ ,$$

   for $\forall\varepsilon$;

2. Shift property, using mean-value integration scheme in $x \in [x_0 - \frac{\varepsilon}{2}, x_0]$, $x \in [x_0, x_0 + \frac{\varepsilon}{2}]$ (**todo** *why?*)

$$\begin{aligned}
\int_{x=-\infty}^{\infty} t_\varepsilon(x - x_0) f(x)\, dx &= \int_{x=x_0-\frac{\varepsilon}{2}}^{x_0+\frac{\varepsilon}{2}} \frac{2}{\varepsilon}\left(1 - \frac{2|x - x_0|}{\varepsilon}\right) f(x)\, dx = \\
&= \int_{x=x_0-\frac{\varepsilon}{2}}^{x_0} \frac{2}{\varepsilon}\left(1 - \frac{2|x - x_0|}{\varepsilon}\right) f(x)\, dx + \int_{x=x_0}^{x_0+\frac{\varepsilon}{2}} \frac{2}{\varepsilon}\left(1 - \frac{2|x - x_0|}{\varepsilon}\right) f(x)\, dx = \\
&= \frac{\varepsilon}{2}\frac{2}{\varepsilon}\left(1 - \frac{2}{\varepsilon}\frac{\varepsilon}{4}\right) f\left(x_0 - \frac{\varepsilon}{4}\right) dx + \frac{\varepsilon}{2}\frac{2}{\varepsilon}\left(1 - \frac{2}{\varepsilon}\frac{\varepsilon}{4}\right) f\left(x_0 + \frac{\varepsilon}{4}\right) dx = \\
&= \frac{1}{2} f\left(x_0 - \frac{\varepsilon}{4}\right) + \frac{1}{2} f\left(x_0 + \frac{\varepsilon}{4}\right)
\end{aligned}$$

   As $\varepsilon \to 0$

$$\int_{x=-\infty}^{\infty} t_\varepsilon(x - x_0) f(x)\, dx \to f(x_0)$$

17.1.3 Gaussian approximation

For $\alpha \to +\infty$,

$$\varphi_\alpha(x) = \sqrt{\frac{\alpha}{\pi}} e^{-\alpha x^2} \sim \delta(x)$$

Properties - proof

Fourier transform of $\varphi_\alpha(x)$ reads

$$\begin{aligned}
\mathcal{F}\{\varphi_\alpha(x)\}(k) &= \int_{x=-\infty}^{+\infty} \varphi_\alpha(x) e^{-ikx}\, dx = \\
&= \int_{x=-\infty}^{+\infty} \sqrt{\frac{\alpha}{\pi}} e^{-\alpha x^2} e^{-ikx}\, dx = \\
&= \sqrt{\frac{\alpha}{\pi}} \int_{x=-\infty}^{+\infty} e^{-\alpha\left(x + i\frac{k}{2\alpha}\right)^2} dx\, e^{-\frac{k^2}{4\alpha}} = \\
&= \sqrt{\frac{\alpha}{\pi}} \sqrt{\frac{\pi}{\alpha}}\, e^{-\frac{k^2}{4\alpha}} = e^{-\frac{k^2}{4\alpha}}\ ,
\end{aligned}$$

for $\alpha \to +\infty$,

$$\mathcal{F}\{\varphi_\alpha(x)\}(k) \to 1$$

Fourier transform of Dirac's delta is 1, as shown in (20.3), thus $\varphi_\alpha(x) \to \delta(x)$ for $\alpha \to +\infty$.

17.1.4 Fourier anti-transform

For $a \to +\infty$,

$$\delta(x) \sim \int_{y=-a}^{+a} e^{i2\pi yx}\, dy = \frac{1}{2\pi} \int_{k=-2\pi a}^{2\pi a} e^{ikx}\, dk\ , \tag{17.3}$$

or

$$\delta \sim 2 \int_{y=0}^{a} \cos(2\pi yx)\, dy\ .$$

Proof of the equilvanece

$$\delta(x) \sim \frac{1}{2\pi} \int_{k=-2\pi a}^{2\pi a} e^{ikx}\, dk = \frac{1}{2\pi} \left(\int_{k=-2\pi a}^{0} e^{ikx}\, dk + \int_{0}^{k=2\pi a} e^{ikx}\, dk \right) = \frac{1}{2\pi} \int_{k=0}^{2\pi a} \left(e^{ikx} + e^{ikx}\right)\, dx = \frac{1}{\pi} \int_{x=0}^{2\pi a} \cos(kx)\, dk$$
$$= \int_{y=-a}^{+a} e^{i2\pi yx}\, dy = \cdots = \int_{y=0}^{a} \left(e^{i2\pi yx} + e^{i2\pi yx}\right) dy = 2 \int_{y=0}^{a} \cos(2\pi yx)\, dy\ .$$

17.1.5 sinc(x) approximation

For $a \to +\infty$

$$\delta(x) \sim \frac{\sin(2\pi xa)}{\pi x}$$

Proof

Directly follows from integral of the approximation (17.3)

$$\int_{y=-a}^{+a} e^{i2\pi yx}\, dy = \frac{1}{i2\pi x}\ e^{i2\pi yx}\Big|_{y=-a}^{+a} = \frac{1}{\pi x} \frac{e^{i2\pi ax} - e^{-i2\pi ax}}{2i} = \frac{\sin(2\pi xa)}{\pi x}$$

17.1.6 Fourier series

For $x \in [-\pi, \pi]$, and $N \to +\infty$, *Fourier series* of Dirac's delta (train with period 2π) reads

$$\delta(x) \sim \frac{1}{2\pi} \sum_{n=-N}^{N} e^{inx} = \frac{1}{2\pi} \frac{\sin\left(\left(N + \frac{1}{2}\right) x\right)}{\sin\left(\frac{x}{2}\right)}$$

or the T-periodic Dirac's delta train,

$$\delta(x) \sim \frac{1}{T} \sum_{n=-N}^{N} e^{in\frac{2\pi}{T}x}\ .$$

todo *Write the proof of the last expression, using the relation between complex exponentials and cosine and sine*

Proof

Coefficients of the Fourier series of Dirac's delta (train with period $T = 2\pi$) are evaluated using the expression (20.2)

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} \delta(t) e^{-in\frac{2\pi}{2\pi}t} = \frac{1}{2\pi} \ ,$$

and thus the complex Fourier series (20.1) of Dirac's delta reads

$$\delta(x) \sim \sum_{n=-\infty}^{+\infty} c_n e^{in\frac{2\pi}{T}x} = \frac{1}{2\pi} \sum_{n=-\infty}^{+\infty} e^{inx}$$

Obs. here, integration interval $[-\pi, \pi]$ to "avoid troubles" with Dirac's delta on the extreme points of the interval (it would give $1/2$ and $1/2$ contributions on both extremes…)

It's possible to write the T-periodic Dirac's delta train as

$$\delta(x) \sim \sum_{n=-\infty}^{+\infty} c_n e^{in\frac{2\pi}{T}x} = \frac{1}{T} \sum_{n=-\infty}^{+\infty} e^{in\frac{2\pi}{T}x}$$

Integral $I = \int_{-\infty}^{+\infty} e^{-\alpha x^2}\, dx$

$$\begin{aligned}
I^2 &= \int_{x=-\infty}^{+\infty} e^{-\alpha x^2}\, dx \int_{y=-\infty}^{+\infty} e^{-\alpha y^2}\, dy = \\
&= \int_{x=-\infty}^{+\infty} \int_{y=-\infty}^{+\infty} e^{-\alpha(x^2+y^2)}\, dx\, dy = \\
&= \int_{\theta=0}^{2\pi} \int_{r=0}^{+\infty} e^{-\alpha r^2}\, r\, dr\, d\theta = \\
&= 2\pi \frac{1}{2\alpha} \int_{r=0}^{+\infty} e^{-\alpha r^2} d\left(\alpha r^2\right) = \\
&= \frac{\pi}{\alpha} \left[-e^{\alpha r^2}\right] \Big|_{r=0}^{+\infty} = \frac{\pi}{\alpha} \ .
\end{aligned}$$

Part VI

Complex Calculus

CHAPTER

EIGHTEEN

COMPLEX ANALYSIS

18.1 Complex functions, $f : \mathbb{C} \to \mathbb{C}$

A complex function f of complex variable $z = x + iy$, $f : \mathbb{C} \to \mathbb{C}$, can be written as

$$f(z) = \tilde{u}(z) + i\tilde{v}(z) = u(x, y) + iv(x, y) \ ,$$

as the sum of its real part $u(z)$ and i times its imaginary part $v(x, y)$. Here $x, y \in \mathbb{R}$, while $\tilde{u}(z), \tilde{v}(z) : \mathbb{C} \to \mathbb{R}$ and $u(x, y), v(x, y) : \mathbb{R}^2 \to \mathbb{R}$. With some abuse of notation, tilde won't be always explicitly written when arguments of real and imaginary parts of f functions won't be written.

18.1.1 Limit

$$\lim_{z \to z_0} f(z) = f(z_0) \qquad , \qquad \forall \varepsilon > 0 \ \exists \delta > 0 \ \text{ s.t. } \ |f(z) - f(z_0)| < \delta \ \forall z \text{ s.t. } |z - z_0| < \varepsilon, \ z \neq z_0 \ .$$

18.1.2 Derivative

Using the definition of *limit of complex functions*, the derivative of a function $f : \mathbb{C} \to \mathbb{C}$, if it exists, is the limit of incremental ratio,

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \ .$$

18.1.3 Line Integrals

Given a line $\gamma \in \mathbb{C}$, whose parametric form is $z(s)$, with regular parametrization with parameter $s \in [s_0, s_1]$,

$$\int_\gamma f(z) \, dz = \int_{s=s_0}^{s_1} f(z(s)) \, z'(s) \, ds \ .$$

18.2 Holomorphic Functions - Analytic Functions

Definition 18.2.1

A holomorphic function is a function whose *derivative* exists.

Examples of analytic functions. todo…

18.2.1 Cauchy-Riemann conditions

For a holomorphic function $f(z) = u(x,y) + iv(x,y)$, Cauchy-Riemann conditions

$$\begin{cases} u_{/x} = v_{/y} \\ u_{/y} = -v_{/x} \end{cases}$$

hold. The evaluation of the derivative once with $\Delta z = \Delta x$ and once with $\Delta z = i\Delta y$

$$\begin{aligned} f'(z) &= \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} = \\ &= \begin{cases} \lim\limits_{\Delta x \to 0} \dfrac{f(x+\Delta x, y) - f(x,y)}{\Delta x} = \lim\limits_{\Delta x \to 0} \dfrac{u(x+\Delta x, y) + iv(x+\Delta x, y) - u(x,y) - iv(x,y)}{\Delta x} = u_{/x} + iv_{/x} \\ \lim\limits_{\Delta y \to 0} \dfrac{f(x, y+\Delta y) - f(x,y)}{i\Delta y} = \lim\limits_{\Delta y \to 0} \dfrac{u(x, y+\Delta y) + iv(x, y+\Delta y) - u(x,y) - iv(x,y)}{i\Delta y} = -iu_{/y} + v_{/y} \end{cases} \end{aligned}$$

provides the proof.

18.2.2 Cauchy Theorem

For a holomorphic function f, $f : \Omega \subseteq \mathbb{C} \to \mathbb{C}$

$$\oint_\gamma f(z)\, dz = 0 \,,$$

for $\forall \gamma \subset \Omega$. Proof follows from *Green's lemma*, and *Cauchy-Riemann conditions*

$$\begin{aligned} \oint_\gamma f(z) dz &= \oint_\gamma (u(x,y) + iv(x,y))\,(dx + idy) = \\ &= \oint_\gamma (udx - vdy) + i \oint_\gamma (udy + vdx) = \\ &= -\int_S \left(\underbrace{u_{/y} + v_{/x}}_{=0} \right) dx\, dy + i \int_S \left(\underbrace{u_{/x} - v_{/y}}_{=0} \right) dx\, dy = 0 \,. \end{aligned}$$

18.3 Useful integrals

18.3.1 Independence of line integral for holomorphic functions

For a function $f(z)$ analytic in D, the line integral on paths $\ell_{ab,i}$ with the same extreme points a, b contained in D is independent on the path, but only depends on the extreme points a, b,

$$\int_{\ell_{ab,1}} f(z)\,dz = \int_{\ell_{ab,2}} f(z)\,dz$$

The proof readily follows, using *Cauchy theorem* applied to a function $f(z) : D \subseteq \mathbb{C} \to \mathbb{C}$, analytic in D, and splitting the closed path γ into two paths ℓ_1, ℓ_2 with the same extreme points, $\gamma = \ell_1 \cup (-\ell_2)$

$$0 = \oint_\gamma f(z)\,dz = \int_{\ell_1} f(z)\,dz + \int_{-\ell_2} f(z)\,dz = \int_{\ell_1} f(z)\,dz - \int_{\ell_2} f(z)\,dz\,.$$

18.3.2 Sum and difference of line integrals

18.3.3 Integral of z^n

Given a path γ embracing $z = 0$ only once in counter-clockwise direction, and $n \in \mathbb{Z}$

$$\oint_\gamma z^n\,dz = \begin{cases} 2\pi i & \text{if } n = -1 \\ 0 & \text{otherwise} \end{cases}$$

Since z^n is analytic everywhere (**todo** *prove it! Add a section with proofs for common functions*) except for $z = 0$, it's possible to evaluate the integral on a circle with center $z = 0$ and radius R. Using polar expression of the complex numbers on the circle, $z = Re^{i\theta}$, $\theta \in [0, 2\pi]$, R const, the differential becomes $dz = iRe^{i\theta}d\theta$ and the integral

$$\begin{aligned}
\oint_\gamma z^n\,dz &= \int_{\theta=0}^{2\pi} \left(Re^{i\theta}\right)^n iRe^{i\theta}d\theta = \\
&= i\int_{\theta=0}^{2\pi} R^{n+1}e^{i(n+1)\theta}d\theta = \\
&= \begin{cases} \text{if } n = -1 & : \quad i2\pi \\ \text{otherwise} & : \quad iR^{n+1}\dfrac{1}{i(n+1)}\,e^{i(n+1)\theta}\Big|_{\theta=0}^{2\pi} = \dfrac{R^{n+1}}{n+1}(1-1) = 0 \end{cases}
\end{aligned}$$

18.4 Meromorphic functions

Definition 18.4.1

A meromorphic function in a domain is a function holomorphic everywhere except for a (finite?) number of poles. **check**

18.4.1 Singularities

Definition 18.4.2 (Pole)

A pole of order n of a function $f(z)$ is a complex number a so that

$$f(z) = \frac{\phi(z)}{(z-a)^n} \ ,$$

with $\phi(z)$ holomorphic in $\phi(a) \neq 0$

Examples. …

Definition 18.4.3 (Branch)

Examples. $f(z) = z^{\frac{1}{2}}$

Definition 18.4.4 (Removable singularities)

Example. $f(z) = \frac{\sin z}{z}$

Other irregularities.

18.4.2 Laurent Series

Given a function $f(z)$, in a disk $D_{a,\varepsilon} : 0 < |z-a| < \varepsilon$, its Laurent series centered in a is the convergent (to $f(z)$, **todo** *which type of convergnence?*) series

$$f(z) \sim \sum_{n=-\infty}^{+\infty} a_n (z-a)^n \ , \tag{18.1}$$

with

$$a_n = \frac{1}{2\pi i} \int_\gamma f(z)\,(z-a)^{-(n+1)}\,dz \tag{18.2}$$

and γ embracing $z = a$ once counter-clockwise. Proof follows immediately inserting the expressions of the coefficients a_n and using the *integral of z^n*. Evaluating the integral (18.2) of the coefficients of the Laurent series, using (18.1) to replace $f(z)$ with its series

$$\begin{aligned}
a_n &= \frac{1}{2\pi i} \oint_\gamma \sum_{m=-\infty}^{+\infty} a_m (z-a)^m (z-a)^{-(n+1)} = \\
&= \frac{1}{2\pi i} \oint_\gamma \sum_{m=-\infty}^{+\infty} a_m (z-a)^{m-n-1}\,dz = \\
&= \frac{1}{2\pi i} \oint_\gamma a_n\, z^{-1}\,dz = \\
&= a_n \ .
\end{aligned}$$

todo *Some freestyle with function and its convergent series…add some detail, and the meaning of convergence*

18.4.3 Cauchy formula

For an analytic function $f(z)$,

$$f(a) = \frac{1}{2\pi i} \oint_\gamma \frac{f(z)}{z-a}\, dz$$

Proof readily follows using the *integral of z^n* on the Taylor series of $\frac{f(z)}{z-a}$ whose 0th order term reads $f(a)$,

$$\frac{1}{2\pi i} \oint_\gamma \frac{f(a) + \sum_{m=1}^{+\infty} f'(a)(z-a)^m}{z-a}\, dz = \frac{1}{2\pi i} \oint_\gamma \frac{f(a)}{z-a}\, dz = f(a)\frac{2\pi i}{2\pi i} = f(a)\ .$$

18.4.4 Residues

Definition 18.4.5 (Residue)

The residue of function f in a, $\text{Res}(f,a)$ is a complex number R so that $f(z) - \frac{R}{(z-a)}$ has analytic antiderivative in a disk $D_{a,\varepsilon}:\ 0 < |z-a| < \varepsilon$.

todo Explain this definition. Couldn't be possible to use $\text{Res}(f,a) = \frac{1}{2\pi i}\oint_\gamma f(z)\, dz = a_{-1}$ instead?

Properties.

- If $f(z)$ is analytic in $D_{a,\varepsilon}$ and has a pole of order n in $z = a$, its Laurent series has $a_m = 0$ for $m < n$ and reads

$$f(z) = \sum_{m=-n}^{+\infty} a_m (z-a)^m\ , \tag{18.3}$$

  with $a_{-n} \neq 0$. Since $f(z)$ has a pole of order n in $z = a$, it can be written as

$$f(z) = \frac{\phi(z)}{(z-a)^n}\ ,$$

  with $\phi(z)$ analytic in $D_{a,\varepsilon}$ and $\phi(a) \neq 0$. Since $\phi(z)$ is analytic, it has a Taylor series (or a Laurent series with non-negative powers),

$$\phi(z) \sim \sum_{m=0}^{+\infty} b_m (z-a)^m\ ,$$

  (**todo** *prove it! Extension of the real case. Add a link to the proof*) and thus

$$f(z) \sim \sum_{m=0}^{+\infty} b_m (z-a)^{m-n} = \sum_{m=-n}^{+\infty} b_{m+n}(z-a)^m = \sum_{m=-n}^{+\infty} a_m (z-a)^m\ ,$$

  with $a_m = b_{m+n}$.

- For simple closed path γ (embracing a only once counter-clokwise) in $D_{a,\varepsilon}$,

$$\oint_\gamma f(z)\, dz = 2\pi i a_{-1} = 2\pi i \text{Res}(f,a) \tag{18.4}$$

  The proof readily follows, using the *integral of z^n* and Laurent series (18.1) of $f(z)$,

$$\oint_\gamma f(z)\, dz = \oint_\gamma \sum_{m=-\infty}^{+\infty} a_m (z-a)^m\, dz = 2\pi i a_{-1}\ .$$

- For a pole a of order n, the following holds

$$a_{-1} = \frac{1}{(n+1)!} \lim_{z \to a} \frac{d^{n-1}}{dz^{n-1}} \left[(z-a)^n f(z) \right]$$

The proof follows using Laurent series {eq}`eq:laurent:pole-n} for a function with pole of order n, and evaluating the $(n-1)^{th}$ order derivative

$$\begin{aligned}
\frac{d^{n-1}}{dz^{n-1}} \left[(z-a)^n f(z) \right] &= \frac{d^{n-1}}{dz^{n-1}} \left[(z-a)^n \sum_{m=-n}^{+\infty} a_n (z-a)^m \right] = \\
&= \frac{d^{n-1}}{dz^{n-1}} \left[\sum_{m=-n}^{+\infty} a_n (z-a)^{m+n} \right] = \\
&= \frac{d^{n-1}}{dz^{n-1}} \left[\sum_{m=0}^{+\infty} a_{m-n} (z-a)^m \right] = \\
&= \frac{d^{n-2}}{dz^{n-2}} \left[\sum_{m=0}^{+\infty} m a_{m-n} (z-a)^{m-1} \right] = \\
&= \frac{d^{n-3}}{dz^{n-3}} \left[\sum_{m=0}^{+\infty} m(m-1) a_{m-n} (z-a)^{m-2} \right] = \\
&= \cdots = \\
&= \left[\sum_{m=0}^{+\infty} m!\, a_{m-n} (z-a)^{m-n+1} \right]
\end{aligned}$$

and then letting $z \to a$, so that only the term with $m - n + 1 = 0$ survives

$$\lim_{z \to a} \frac{d^{n-1}}{dz^{n-1}} \left[(z-a)^n \sum_{m=-n}^{+\infty} a_n (z-a)^m \right] = (n-1)!\, a_{-1} \, .$$

18.4.5 Residue Theorem

Theorem 18.4.1 (Residue Theorem)

Given $f(z)$ with a finite number of poles $p_n \in D$, then

$$\int_\gamma f(z)\, dz = 2\pi i \sum_n I(\gamma, p_n) \mathrm{Res}(f, p_n) \, ,$$

being γ a path in D, and $I(\gamma, p_n)$ the winding index of the path γ around pole p_n (+1 for each counter-clockwise loop, -1 for each clockwise loop).

The proof readily follows extending the result for a single pole (18.4) to general number of poles and general paths γ embracing (with sign) each pole p_n $I(\gamma, p_n)$ times, with the same techinques shown in section *Sum and difference of line integrals*.

18.4.6 Cauchy argument principle

18.4.7 Evaluation of integrals

18.4.8 Inverse Laplace Transform

Given Laplace transform

$$F(s) := \mathcal{L}\{f(t)\}(s) := \int_{t=0^-}^{+\infty} f(t)e^{-st}\,dt\ ,$$

the inverse transform can be evaluated as

$$f(t) = \mathcal{L}^{-1}\{F(s)\}(t) := \lim_{T\to+\infty} \frac{1}{2\pi i}\int_{s=a-iT}^{a+iT} e^{st}F(s)\,ds\ ,$$

with $a > \mathrm{Re}\{p_n\}$ (**todo** *why?*) for each pole of the function $F(s)$, evaluated on the vertical line $s = a+iy$, $y \in [-T, T]$, $ds = idy$,

$$\begin{aligned}
\lim_{T\to+\infty} \frac{1}{2\pi i}\int_{s=a-iT}^{a+iT} e^{st}F(s)\,ds &= \lim_{T\to+\infty} \frac{1}{2\pi i}\int_{s=a-iT}^{a+iT} e^{st}\int_{\tau=0^-}^{+\infty} f(\tau)e^{-s\tau}\,d\tau\,ds = \\
&= \lim_{T\to+\infty} \frac{1}{2\pi i}\int_{y=-T}^{T} e^{(a+iy)t}\int_{\tau=0^-}^{+\infty} f(\tau)e^{-(a+iy)\tau}\,d\tau\,idy = \\
&= \lim_{T\to+\infty} \frac{1}{2\pi}\int_{y=-T}^{T}\int_{\tau=0^-}^{+\infty} e^{iy(t-\tau)}e^{a(t-\tau)}f(\tau)\,d\tau\,dy = \\
&= \ldots \\
&= \int_{\tau=0^-}^{+\infty} \delta(t-\tau)e^{a(t-\tau)}f(\tau)d\tau = f(t)\ .
\end{aligned}$$

having used the transform of *Dirac's delta* $\delta(t) = \frac{1}{2\pi}\int_{\omega=-\infty}^{+\infty} e^{-j\omega t}\,d\omega$.

todo *Ohter approach: if $a >$ Re$\{p_n\}$, the contour built with the vertical line with real part a and the arc of circumference on its…*

CHAPTER

NINETEEN

LAPLACE TRANSFORM

Definition of Laplace transform. Definition of Laplace transform, inverse transform, properties and theorems.

Applications of Laplace transform. Solution of ODEs,…

19.1 Definition and Properties

19.1.1 Definition of Lapalce transform

$$\mathcal{L}\left\{f(t)\right\}(s) := \int_{t=0^-}^{+\infty} e^{-st} f(t)\, dt = F(s)\ .$$

19.1.2 Inverse transform

$$f(t) = \mathcal{L}^{-1}\left\{F(s)\right\} = ...$$

19.1.3 Properties

Linearity.

$$\mathcal{L}\{af(t) + bg(t)\}(s) = aF(s) + bG(s)$$

***Dirac delta*.**

$$\mathcal{L}\left\{\delta(t)\right\} = \int_{t=0^-}^{+\infty} \delta(t)\, e^{st}\, dt = 1$$

Time delay. If $f(t) = 0$ for $t < 0$ ("causality"), for $\tau > 0$,

$$\mathcal{L}\{f(t-\tau)\}(s) = e^{-s\tau} F(s)$$

Proof readily follows direct computation with change of variable $z = t - \tau$, $dt = dz$

$$\mathcal{L}\{f(t-\tau)\}(s) = \int_{t=0^-}^{+\infty} f(t-\tau) e^{-st}\, dt = \int_{z=-\tau}^{+\infty} f(z) e^{-sz}\, dz\, e^{-s\tau} = \int_{z=0}^{+\infty} f(z) e^{-sz}\, dz\, e^{-s\tau} = e^{-s\tau} F(s)\ .$$

"Frequency shift"

$$\mathcal{L}\{f(t)e^{at}\}(s) = F(s-a)$$

Direct computation gives

$$\mathcal{L}\{f(t)e^{at}\}(s) = \int_{t=0^-}^{+\infty} f(t)e^{at}e^{-st}\,dt = \int_{t=0^-}^{+\infty} f(t)e^{-(s-a)t}\,dt = F(s-a)$$

Derivative.

$$\mathcal{L}\{f'(t)\}(s) = sF(s) - f(0^-)\ .$$

Proof readily follows direct computation, with integration by parts

$$\mathcal{L}\{f'(t)\}(s) = \int_{t=0^-}^{+\infty} f'(t)e^{-st}\,dt = [f(t)e^{-st}]\,|_{t=0^-}^{+\infty} + s\int_{t=0^-}^{+\infty} f(t)e^{-st}\,dt = sF(s) - f(0^-)\ ,$$

provided that $\lim_{s\to+\infty} f(t)e^{-st} = 0$.

Integral.

$$\mathcal{L}\left\{\int_{\tau=0}^{t} f(\tau)\,d\tau\right\}(s) = \frac{1}{s}F(s)\ .$$

Proof readily follows direct computation, with integration by parts

$$\mathcal{L}\left\{\int_{\tau=0^-}^{t} f(\tau)\,d\tau\right\}(s) = \int_{t=0^-}^{+\infty}\int_{\tau=0^-}^{t} f(\tau)\,d\tau e^{-st}\,dt = \left[-\frac{e^{-st}}{s}\int_{\tau=0^-}^{t} f(\tau)\,d\tau\right]_{t=0}^{+\infty} + \frac{1}{s}\int_{t=0}^{+\infty} f(t)e^{-st}\,dt = \frac{1}{s}F(s)\ ,$$

provided that $\int_{\tau=0^-}^{0} f(\tau)d\tau = 0$ and $\lim_{t\to+\infty}\frac{e^{-st}}{s}\int_{\tau=0^-}^{+\infty} f(\tau)\,d\tau = 0$.

Convolution.

$$\begin{aligned}
\mathcal{L}\left\{f(t) * g(t)\right\} &= \int_{t=0^-}^{+\infty}\int_{\tau=-\infty}^{+\infty} f(t-\tau)g(\tau)\,d\tau\,e^{-st}\,dt = \qquad (1)\\
&= \int_{\tau=-\infty}^{+\infty}\int_{z=-\tau^-}^{+\infty} f(z)g(\tau)\,e^{-s(z+\tau)}\,d\tau\,dz = (2)\\
&= \int_{z=0^-}^{+\infty} f(z)\,e^{-sz}\,dz\int_{\tau=0^-}^{+\infty} g(\tau)e^{-s\tau} =\\
&= \mathcal{L}\{f(t)\}(s)\,\mathcal{L}\{g(t)\}(s)\ .
\end{aligned} \tag{19.1}$$

having performed the change of coordinates $z = t-\tau$, $\tau = \tau$, with unitary Jacobian,

$$\frac{\partial(t,\tau)}{\partial(z,\tau)} = \partial_z t\partial_\tau\tau - \partial_z\tau\partial_z t = 1\cdot 1 - 1\cdot 0 = 1,$$

given the proper description of the domain of integration summarised in the extremes of integration in (1), and causality - i.e. all the functions $f(t)$ are identically zero for $t < 0$ - in (2).

Initial value. If …

$$f(0^+) = \lim_{s\to+\infty} sF(s)$$

From direct computation,

$$\begin{aligned}
\lim_{s\to+\infty} sF(s) &= \lim_{s\to+\infty} s\int_{t=0^-}^{+\infty} f(t)\,e^{-st}\,dt =\\
&= \lim_{s\to+\infty}\left\{\left[s\left(-\frac{e^{-st}}{s}\right)f(t)\right]\Bigg|_{t=0}^{+\infty} + \int_{t=0}^{+\infty} e^{-st}f'(t)\,dt\right\} =\\
&= \lim_{s\to+\infty}\left\{\left[-e^{-st}f(t)\right]\Bigg|_{t=0}^{+\infty} + \int_{t=0}^{+\infty} e^{-st}f'(t)\,dt\right\} =\\
&= f(0)\ ,
\end{aligned}$$

provided that $\lim_{s\to+\infty}\lim_{t\to+\infty} e^{-st}f(t)=0$ and $\lim_{s\to+\infty}\int_{t=0}^{+\infty} e^{-st}f'(t)\,dt=0$.

Final value. If …

$$f(+\infty)=\lim_{s\to 0} sF(s)$$

From direct computation (**todo** *check and/or explain proof*),

$$\begin{aligned}
\lim_{s\to 0} sF(s) &= \lim_{s\to 0} s\int_{t=0^-}^{+\infty} f(t)\,e^{-st}\,dt = \\
&= \lim_{s\to 0}\left\{\left[s\left(-\frac{e^{-st}}{s}\right)f(t)\right]\Bigg|_{t=0}^{+\infty}+\int_{t=0}^{+\infty} e^{-st}f'(t)\,dt\right\} = \\
&= \lim_{s\to 0}\left\{\left[-e^{-st}f(t)\right]\Bigg|_{t=0}^{+\infty}+\int_{t=0}^{+\infty} e^{-st}f'(t)\,dt\right\} = \\
&= f(0)+f(+\infty)-f(0)=f(+\infty)\ ,
\end{aligned}$$

provided that $\lim_{s\to 0}\lim_{t\to+\infty} e^{-st}f(t)=0$.

19.2 Applications of Laplace Transform

19.2.1 Ordinary differential equations

Exploiting the properties of transforming derivation into product by the complex variable s, Laplace transform can be a useful tool in solving *Ordinary differential equations*.

Linear ODEs are treated in details in the section about *Linear Time-Invariant Systems*.

First-order linear differential equation with constant coefficients

A Cauchy problem governed by a first-order linear differential equation with constant coefficients reads

$$\begin{cases}\dot{u}+au=f(t) & , \quad \text{for } t>0\\ u(0^-)=u_0\ . & \end{cases}$$

Comments. 1) For linear differential equations with constant coefficients, a time shift is always possible to set initial conditions in $t_0=0$; 2) dealing with **impulsive forces** (more on this in *LTI systems: impulsive forces*), some attention needs to be paid to the time of initial conditions: if impulsive forces at $t=0$ may exist, initial conditions need to be set at $t=0^-$; the effect of the impulsive force is equivalent to a jump in initial conditions from $t=0^-$ and $t=0^+$.

General solution in time domain

Multiplying the ODE by e^{at},

$$e^{at}f(t)=e^{at}\left(\dot{u}+au\right)=\frac{d}{dt}\left(e^{at}u(t)\right)$$

and integrating from 0^- to a generic time t — after changing the dummy integration variable from τ to t —

$$e^{at}u(t)-u(0^-)=\int_{\tau=0^-}^{t} e^{a\tau}f(\tau)d\tau\ ,$$

and thus

$$u(t) = e^{-at}u(0^-) + \int_{\tau=0^-}^{t} e^{-a(t-\tau)} f(\tau) d\tau \ .$$

The solution of the problem can be also computed using Laplace transform:

1. transforming the problem from time to Laplace domain: the differential problem with unknown $u(t)$ in time is transformed into an algebraic problem $\hat{u}(s)$. Using Laplace **transform of the time derivative**

$$s\hat{u}(s) + u_0 + a\hat{u}(s) = \hat{f}(s) \ ,$$

2. solving the algebraic problem in Laplace domain for $\hat{u}(s)$

$$\hat{u}(s) = \frac{1}{s+a}u_0 + \frac{1}{s+a}f(s) \ .$$

3. transforming back the solution in time domain, $u(t) = \mathcal{L}^{-1}\{\hat{u}(s)\}$,

$$\begin{aligned} u(t) &= \mathcal{L}^{-1}\{\hat{u}(s)\}(t) = \\ &= \mathcal{L}^{-1}\left\{\frac{u_0}{s+a}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s+a}\hat{f}(s)\right\} = \\ &= e^{-at}u_0 + \int_{0^-}^{t} e^{-a(t-\tau)} f(\tau) d\tau \ , \end{aligned}$$

having used the inverse transform of the exponential $\mathcal{L}\{e^{ct}\} = \int_{t=0^-}^{+\infty} e^{ct}e^{-st}\, dt = \frac{1}{s-c}$, and the inverse transform of the convolution

$$\hat{f}(s)\hat{g}(s) = \mathcal{L}\{f(t) * g(t)\}(s) = \int_{t=0^-}^{+\infty} \left\{ \int_{\tau=0^-}^{t} f(\tau)g(t-\tau)d\tau \right\} e^{-st} dt \ .$$

CHAPTER

TWENTY

FOURIER TRANSFORMS

Fourier transforms are linear transformations of functions usually relating a physical domain of time and/or space, with a domain of frequency and/or wave-vectors.

Fourier transforms can be useful in:

- highlighting the frequency content of functions
- solving problems: sometimes, it can be easier to transform a problem in frequency domain, solve it in frequency domain, and transform the solution back to the physical domain

Contents.

Fourier series. Fourier series is defined for finite-domain or periodic, time-continuous functions, or - more generally - continuous functions in the physical domain.

Fourier transform. Fourier transform is defined for infinite-domain non-periodic, time-continuous functions, or - more generally - continuous functions in the physical domain.

Relations between Fourier transforms and sampling. Fourier series, Fourier transform, discrete time Fourier transform and discrete Fourier transforms are presented, and their relations discussed. Fundamental results about **evenly-spaced sampling** seamlessly follows, as **Shannon-Nyquist theorem**, *Theorem 20.3.1*, shows.

Different Fourier transforms exist, depending if the original function is:

- time discrete/time continuous
- periodic/non-periodic

20.1 Fourier Series

For a T-periodic function,

$$g(t) \sim \frac{a_0}{2} + \sum_{n=1}^{+\infty} \left[a_n \cos\left(n \frac{2\pi}{T} t\right) + b_n \sin\left(n \frac{2\pi}{T} t\right) \right] ,$$

todo Prove it with properties of integrals of sin and cos over $t \in [0, T]$; prove convergence to average value at jumps

The exponential form reads

$$g(t) \sim \sum_{n=-\infty}^{+\infty} c_n e^{in \frac{2\pi}{T} t} , \tag{20.1}$$

where

$$c_n = \frac{1}{T} \int_{t=0}^{T} f(t)\, e^{-in\frac{2\pi}{T}t} \,. \tag{20.2}$$

Proof

Exploiting the properties of integrals of complex exponentials with $k \in \mathbb{Z}$

$$\int_{t=0}^{T} e^{ik\frac{2\pi}{T}t}\, dt = \begin{cases} \frac{1}{ik\frac{2\pi}{T}} \left[e^{ik\frac{2\pi}{T}t}\right]\Big|_{t=0}^{T} = 0 & \text{if } k \neq 0 \\ T & \text{if } k = 0 \end{cases}$$

$$\int_{t=0}^{T} f(t) e^{-im\frac{2\pi}{T}t}\, dt \sim \int_{t=0}^{T} \sum_{n=-\infty}^{+\infty} c_n e^{in\frac{2\pi}{T}t} e^{-im\frac{2\pi}{T}t} \sim \sum_{n=-\infty}^{+\infty} c_n \sim \int_{t=0}^{T} e^{i(n-m)\frac{2\pi}{T}t} \sim T\, c_m \,.$$

20.2 Fourier Transform

Contents: *definition*; *properties*; *inverse transform*; *Plancherel's theorem*; *uncertainty relation*

20.2.1 Definition

…

$$\mathcal{F}\left\{g(t)\right\}(f) := \int_{t=-\infty}^{+\infty} g(t)\, e^{-i2\pi ft}\, dt.$$

20.2.2 Properties

Linearity

***Dirac delta*.**

$$\mathcal{L}\left\{\delta(t)\right\} = \int_{t=-\infty}^{+\infty} \delta(t)\, e^{-i2\pi ft}\, dt = 1 \tag{20.3}$$

Time delay.

Derivative.

Integral.

Initial value.

Final value.

Property 20.2.1 (Transform of convolution)

$$\mathcal{F}\left\{a * b(t)\right\}(f) = \mathcal{F}\left\{a(t)\right\}(f)\, \mathcal{F}\left\{b(t)\right\}(f)$$

Proof.

$$\begin{aligned}\mathcal{F}\left\{a * b(t)\right\}(f) &= \int_{t=-\infty}^{+\infty}\int_{\tau=-\infty}^{+\infty} a(\tau)b(t-\tau)\,d\tau\, e^{-i2\pi ft}\,dt = \qquad (1)\\ &= \int_{\tau=-\infty}^{+\infty} a(\tau)\, e^{-i2\pi f\tau}\,d\tau \int_{z=-\infty}^{+\infty} b(z)\, e^{-i2\pi fz} = \\ &= \mathcal{F}\{a(t)\}(f)\,\mathcal{F}\{b(t)\}(f)\ ,\end{aligned}$$

having used (1) transformation of coordinates $(z,\tau) = (t-\tau,\tau)$ with unit Jacobian

$$\frac{\partial(z,\tau)}{\partial(t,\tau)} = \begin{vmatrix}1 & -1\\ 0 & 1\end{vmatrix} = 1\ .$$

20.2.3 Inverse Fourier Transform

Under the assumptions …**todo**, the inverse Fourier transform reads

$$\mathcal{F}^{-1}\left\{G(f)\right\}(t) := \int_{f=-\infty}^{+\infty} G(f)\, e^{i2\pi ft}\,df.$$

Proof using Dirac's delta expression.

$$\begin{aligned}\mathcal{F}^{-1}\left\{G(f)\right\}(t) := \int_{f=-\infty}^{+\infty} G(f)\, e^{i2\pi ft}\,df &= \int_{f=-\infty}^{+\infty}\int_{\tau=-\infty}^{+\infty} g(\tau)e^{-i2\pi f\tau}\, e^{i2\pi ft}\,df = \\ &= \int_{f=-\infty}^{+\infty}\int_{\tau=-\infty}^{+\infty} g(\tau)e^{-i2\pi f\tau}\, e^{i2\pi ft}\,df = \\ &= \int_{f=-\infty}^{+\infty}\int_{\tau=-\infty}^{+\infty} g(\tau)e^{i2\pi f(t-\tau)}\,df = \\ &= \int_{\tau=-\infty}^{+\infty} g(\tau)\delta(t-\tau)\,d\tau = g(t)\ .\end{aligned}$$

Proof using dominated convergence theorem and Fubini's lemma.

Proof. By the *dominated convergence theorem*, it follows that

$$\begin{aligned}\int_{\mathbb{R}} e^{i2\pi x\xi}F(\xi)\,d\xi &= \lim_{\varepsilon\to 0}\int_{\mathbb{R}} \underbrace{e^{-\pi\varepsilon^2\xi^2+i2\pi x\xi}}_{G(\xi;x,\varepsilon)} F(\xi)\,d\xi = \\ &= \lim_{\varepsilon\to 0}\int_{\mathbb{R}} g(y;x,\varepsilon)f(y)\,dy = \\ &= \lim_{\varepsilon\to 0}\int_{\mathbb{R}} \varphi_\varepsilon(x-y)\,f(y)\,dy = \\ &= \int_{\mathbb{R}} \delta(x-y)\,f(y)\,dy = f(x)\end{aligned}$$

Lemma 1. The Fourier transform of function $\varphi(t) := e^{-\pi|t|^2}$ reads

$$\begin{aligned}
\mathcal{F}\{\varphi(t)\}(\omega) &= \int_{t=-\infty}^{+\infty} \varphi(t) e^{-i\omega t}\, dt = \\
&= \int_{t=-\infty}^{+\infty} e^{-\pi|t|^2} e^{-i\omega t}\, dt = \\
&= \int_{t=-\infty}^{+\infty} e^{-\pi\left(t^2 + i\frac{\omega}{\pi}t - \frac{\omega^2}{4\pi^2}\right)}\, dt\, e^{-\frac{\omega^2}{4\pi^2}} = \\
&= \int_{t=-\infty}^{+\infty} e^{-\pi\left(t + i\frac{\omega}{2\pi}\right)^2}\, dt\, e^{-\frac{\omega^2}{4\pi}} = \\
&= e^{-\frac{\omega^2}{4\pi}} ,
\end{aligned}$$

having evaluated *the integral* $\int_{-\infty}^{+\infty} e^{-\alpha x^2}$ with $\alpha = \pi$. **todo** *justify the result for complex exponential. Use Bromwich contour integrals*

Lemma 2. Fourier transform of $f(\alpha t)$, $\alpha > 0$

$$\mathcal{F}\{f(\alpha t)\}(\omega) = \int_{\mathbb{R}} f(\alpha t) e^{-j\omega t}\, dt = \int_{\tau\in\mathbb{R}} f(\tau) e^{-j\frac{\omega}{\alpha}\tau}\, d\tau \frac{1}{\alpha} = \frac{1}{\alpha} F\left(\frac{\omega}{\alpha}\right)$$

Lemma 3. $\frac{1}{\varepsilon}\varphi\left(\frac{t}{\varepsilon}\right) \to \delta(x)$ for $\varepsilon \to 0$

$$\mathcal{F}\left\{\frac{1}{\varepsilon}\varphi\left(\frac{t}{\varepsilon}\right)\right\}(\omega) = \frac{1}{\varepsilon}\varepsilon e^{-\frac{\omega^2}{4\pi\varepsilon^2}} = e^{-\frac{\omega^2}{4\pi\varepsilon^2}}$$

0. Fourier transform

$$G(f) = \int_{t=-\infty}^{\infty} e^{-i\omega t} g(t)\, dt$$

1.

$$g(t) = e^{i\alpha t}\psi(t)$$

$$\mathcal{F}\{g(t)\}(\omega) = \int_{t=-\infty}^{+\infty} g(t) e^{-i\omega t}\, dt = \int_{t=-\infty}^{+\infty} \psi(t) e^{i\alpha t} e^{-i\omega t}\, dt = \int_{t=-\infty}^{+\infty} \psi(t) e^{-i(\omega-\alpha)t}\, dt = \mathcal{F}\{\psi(t)\}(\omega-\alpha) .$$

2.

$$\psi(t) = \phi(\alpha t)$$

$$\mathcal{F}\{\psi(t)\} = \int_{t=-\infty}^{+\infty} \psi(t) e^{-i\omega t}\, dt = \int_{t=-\infty}^{+\infty} \phi(\alpha t) e^{-i\omega t}\, dt = \int_{\tau=-\infty}^{+\infty} \phi(\tau) e^{-i\frac{\omega}{\alpha}\tau}\, \frac{d\tau}{\alpha} = \frac{1}{\alpha}\mathcal{F}\{\phi(t)\}\left(\frac{\omega}{\alpha}\right) .$$

3. Fubini's theorem

4.

$$\varphi(t) := e^{-\pi t^2}$$

$$\mathcal{F}\{\varphi(t)\} = \int_{t=-\infty}^{+\infty} \varphi(t) e^{-i\omega t}\, dt = \int_{t=-\infty}^{+\infty} e^{-\pi t^2} e^{-i\omega t}\, dt$$

$$0 = \oint_{\gamma} e^{-\alpha|z|^2}\, dz = \int_{\dots} \dots$$

$$z = Re^{i\theta}, \quad dz = iRe^{i\theta} d\theta$$

$$\int_{C/4} e^{-\alpha|z|^2}\, dz = \int_{\theta=0}^{\frac{\pi}{2}} e^{-\alpha R^2} iRe^{i\theta} d\theta = iRe^{-\alpha R^2} \frac{e^{-i\theta}}{i} \Big|_{\theta=0}^{\frac{\pi}{2}}$$

$$\int_{t=0}^{+\infty} e^{-\pi t^2} e^{-i\omega t}\, dt = \int_{t=0}^{+\infty} e^{-\left(\pi t^2 + i\omega t - \frac{\omega^2}{4\pi}\right)}\, dt\, e^{-\frac{\omega^2}{4\pi}} =$$
$$= \int_{t=0}^{+\infty} e^{-\pi\left(t + i\frac{\omega}{2\pi}\right)^2}\, dt\, e^{-\frac{\omega^2}{4\pi}}$$

5. $\varphi_\varepsilon(t) = \frac{1}{\varepsilon^n}\varphi\left(\frac{t}{\varepsilon}\right)$, $t \in \mathbb{R}^n$, is an approximation of Dirac's delta for $\varepsilon \to 0$, so that

$$\lim_{\varepsilon\to 0} \int_{t=-\infty}^{+\infty} \varphi_\varepsilon(t-\tau) f(t)\, dt = f(\tau)$$
$$\lim_{\varepsilon\to 0} \int_{t=-\infty}^{+\infty} \varphi_\varepsilon(t)\, dt = 1$$

As the Fourier transform $\mathcal{F}\left\{\varphi_\varepsilon(t)\right\}(\omega) \to 1$ for $\varepsilon \to 0$, then $\varphi_\varepsilon(t) \to \delta(t)$.

20.2.4 Plancherel's theorem

…assumptions…**todo**

$$\int_{f=-\infty}^{+\infty} |G(f)|^2\, df = \int_{t=-\infty}^{+\infty} |g(t)|^2\, dt \tag{20.4}$$

and

$$\int_{f=-\infty}^{+\infty} A^*(f)\, G(f)\, df = \int_{t=-\infty}^{+\infty} a^*(t)\, g(t)\, dt\ . \tag{20.5}$$

Proof of Plancherel's thm for the magnitude

$$\begin{aligned}
\int_{f=-\infty}^{+\infty} |G(f)|^2\, df &= \int_{f=-\infty}^{+\infty} G(f)^* G(f)\, df = \\
&= \int_{f=-\infty}^{+\infty} \left(\int_{t_1=-\infty}^{+\infty} g(t_1) e^{-i2\pi f t_1} dt_1\right)^* \left(\int_{t_2=-\infty}^{+\infty} g(t_2) e^{-i2\pi f t_2} dt_2\right) df = \\
&= \int_{t_1,t_2=-\infty}^{+\infty} g^*(t_1)\, g(t_2) \int_{f=-\infty}^{+\infty} e^{i2\pi f(t_1-t_2)}\, df\, dt_1\, dt_2 = && (1) \\
&= \int_{t_1,t_2=-\infty}^{+\infty} g^*(t_1)\, g(t_2)\, \delta(t_1-t_2)\, dt_1\, dt_2 = && (2) \\
&= \int_{t_1=-\infty}^{+\infty} g^*(t_1)\, g(t_1)\, dt_1 = && (3) \\
&= \int_{t_1=-\infty}^{+\infty} |g(t_1)|^2\, dt_1\ .
\end{aligned}$$

having used (1) the approximation (17.3) of Dirac's delta, and (2) property (17.2) of Dirac's delta, and (3) the expression of the absolute value of complex functions $g^*(t_1)g(t_1) = |g(t_1)|^2$.

Proof of Plancherel's thm for the product of functions

$$
\begin{aligned}
\int_{f=-\infty}^{+\infty} A^*(f)\, G(f)\, df &= \int_{f=-\infty}^{+\infty} G(f)^* G(f)\, df = \\
&= \int_{f=-\infty}^{+\infty} \left(\int_{t_1=-\infty}^{+\infty} a(t_1) e^{-i2\pi f t_1} dt_1 \right)^* \left(\int_{t_2=-\infty}^{+\infty} g(t_2) e^{-i2\pi f t_2} dt_2 \right) df = \\
&= \int_{t_1,t_2=-\infty}^{+\infty} a^*(t_1)\, g(t_2) \int_{f=-\infty}^{+\infty} e^{i2\pi f(t_1-t_2)}\, df\, dt_1\, dt_2 = \qquad (1) \\
&= \int_{t_1,t_2=-\infty}^{+\infty} a^*(t_1)\, g(t_2)\, \delta(t_1-t_2)\, dt_1\, dt_2 = \qquad (2) \\
&= \int_{t_1=-\infty}^{+\infty} a^*(t_1)\, g(t_1)\, dt_1 \ .
\end{aligned}
$$

having used (1) the approximation (17.3) of Dirac's delta, and (2) property (17.2) of Dirac's delta.

20.2.5 Uncertainty relation

An uncertainty relation holds linking standard deviations of a probability density function in time domain and a probability density function built with its Fourier transform. From this very same relation, Heisenberg uncertainty relation between position and momentum in Quantum Mechanics seamlessly follows.

Given a function $g(t)$ whose square of the absolute value is normalized to one, and thus it can be used as a probability density function in time domain,

$$\int_{t=-\infty}^{+\infty} |g(t)|^2\, dt = \int_{t=-\infty}^{+\infty} g^*(t)\, g(t)\, dt = 1 \ .$$

for *Plancherel's theorem*, the square of the magnitude of Fourier transform $G(f)$ is unitary as well,

$$\int_{f=-\infty}^{+\infty} |G(f)|^2\, df = \int_{f=-\infty}^{+\infty} G^*(f)\, G(f)\, df = 1 \ ,$$

and thus it can be interpreted as a probability density function in frequency domain. The following uncertainty relation holds

$$\sigma_{t,g}^2 \sigma_{f,G}^2 \geq \left(\frac{1}{2\pi} \frac{1}{2} \right)^2 \ ,$$

or in terms of pulsation $\omega = 2\pi\, f$,

$$\sigma_{t,g}^2 \sigma_{\omega,G}^2 \geq \left(\frac{1}{2} \right)^2 \ ,$$

Heisenberg uncertainty relation in quantum mechanics

Space and momentum representation of the state function Ψ are related by the transformation,

$$\langle x|\Psi\rangle := \psi(x,t) = \int_{p=-\infty}^{+\infty} \psi_p(p,t)\, e^{i\frac{p}{\hbar}x}\, dp \ ,$$

as it's shown in the section Quantum Mechanics:From position to momentum representation. The wave number reads $k = \frac{p}{\hbar}$. Starting from the uncertainty relation between the space coordinate x and the wave number k,

$$\sigma_x \sigma_k \geq \frac{1}{2} \ ,$$

Heisenberg uncertainty principle for position and momentum (for the same Cartesian coordinates) reads

$$\sigma_x \sigma_p \geq \frac{\hbar}{2} \, .$$

Proof of the uncertainty relation

Assuming zero average $\overline{t} = 0$, $\overline{f} = 0$ (see below for proof without this assumption)

$$\begin{aligned}
\sigma_{t,g}^2 \sigma_{f,G}^2 &= \int_{t=-\infty}^{+\infty} |t|^2 \; g^*(t)\, g(t)\, dt \int_{f=-\infty}^{+\infty} |f|^2 \; G^*(f)\, G(f)\, df = \\
&= \int_{t=-\infty}^{+\infty} |t\, g(t)|^2 \; dt \int_{f=-\infty}^{+\infty} |f\, G(f)|^2 \; df = && (1) \\
&= \int_{t=-\infty}^{+\infty} |t\, g(t)|^2 \; dt \int_{t=-\infty}^{+\infty} \left| -\frac{i}{2\pi}\, \dot{g}(t) \right|^2 \; dt \geq && (2) \\
&= \left| \int_{t=-\infty}^{+\infty} -t\, g^*(t) \frac{i}{2\pi}\, \dot{g}(t)\, dt \right|^2 \geq && (3) \\
&= \left(\frac{1}{2\pi}\right)^2 \left(\frac{1}{2}\right)^2
\end{aligned}$$

having used in

(1)

$$\mathcal{F}\{\dot{g}(t)\}(f) = \int_{t=-\infty}^{+\infty} \dot{g}(t) e^{-i2\pi f t}\, dt = \cdots = i2\pi\, f\, G(f) \; ,$$

$$f\, G(f) = -i\, \frac{\mathcal{F}\{\dot{g}(t)\}}{2\pi}$$

and thus *Plancherel's theorem*

$$\begin{aligned}
\int_{f=-\infty}^{+\infty} |f\, G(f)|^2\, df &= \int_{f=-\infty}^{+\infty} \left| -\frac{i}{2\pi}\, \mathcal{F}\{\dot{g}(t)\} \right|^2 \; df = \\
&= \int_{t=-\infty}^{+\infty} \left| -\frac{i}{2\pi}\, \dot{g}(t) \right|^2 \; dt
\end{aligned}$$

in (2) Cauchy-Schwartz inequality,

(3)

$$\begin{aligned}
a &:= \int_{t=-\infty}^{+\infty} t g^*(t) \dot{g}(t)\, dt = \\
&= \underbrace{[t\, g^*(t)\, g(t)]\, |_{-\infty}^{+\infty}}_{=0} - \int_{t=-\infty}^{+\infty} \frac{d}{dt}\, (t g^*(t))\, g(t)\, dt = \\
&= -\int_{t=-\infty}^{+\infty} g^*(t)\, g(t)\, dt - \int_{t=-\infty}^{+\infty} t \dot{g}^*(t) g(t)\, dt = \\
&= -1 - a^* \, .
\end{aligned}$$

$$-1 = a + a^* = 2\,\mathrm{re}\{a\}\ ,$$

and thus

$$|a|^2 \geq \mathrm{re}\{a\}^2 = \frac{1}{4}\ .$$

If $\overline{t} \neq 0$, or $\overline{f} \neq 0$,

$$\begin{aligned}
\sigma_{t,g}^2 \sigma_{f,G}^2 &= \int_{t=-\infty}^{+\infty} \left|t - \overline{t}\right|^2 g^*(t)\, g(t)\, dt \int_{f=-\infty}^{+\infty} \left|f - \overline{f}\right|^2 G^*(f)\, G(f)\, df = \\
&= \int_{t=-\infty}^{+\infty} \left|(t - \overline{t})\, g(t)\right|^2 dt \int_{f=-\infty}^{+\infty} \left|(f - \overline{f})\, G(f)\right|^2 df = \qquad (1) \\
&= \ldots
\end{aligned}$$

20.3 Relations between Fourier transforms

Some freestyle in changing order of summations and integrals, and use of generalized functions here…check it!

Different Fourier transforms exist, depending if the original function is:

- time discrete/time continuous
- periodic/non-periodic

namely,

- FS, Fourier series: time continuous, periodic function (or finite domain, with a periodic extension)
- FT, Fourier transform: time continuous, non-periodic function
- DTFT, discrete-time Fourier transform: time didscrete, infinite-length sequence
- DFT, discrete Fourier transform: time discrete, finite-length sequence (and then with a periodic extension)

20.3.1 Fourier transform of integrable functions

$$F(\nu) := \mathcal{F}\left\{f(t)\right\}(\nu) := \int_{t=-\infty}^{+\infty} f(t) e^{-i2\pi\nu t}\, dt\ ,$$

20.3.2 Fourier transform of the sum of shifted integrable functions

The infinite sum of a shifted integrable function is defined as

$$\tilde{f}_T(t) = \sum_{n=-\infty}^{+\infty} f(t - nT)\ .$$

Its Fourier transform reads

$$
\begin{aligned}
\mathcal{F}\left\{\tilde{f}_T(t)\right\}(\nu) &= \int_{t=-\infty}^{+\infty} \tilde{f}_T(t) e^{-i2\pi\nu t}\, dt = \\
&= \sum_{n=-\infty}^{+\infty} \int_{t=-\infty}^{+\infty} f(t-nT) e^{-i2\pi\nu t}\, dt = \qquad (1) \\
&= \sum_{n=-\infty}^{+\infty} F(\nu) e^{-i2\pi\nu nT} = \qquad (2) \\
&= F(\nu) \sum_{n=-\infty}^{+\infty} e^{-i2\pi\nu nT} = \qquad (3) \\
&= \Delta\nu\, F(\nu)\, \text{III}_{\Delta\nu}(\nu)\ ,
\end{aligned}
$$

having used properties of Fourier transform of shifted function in (1), and the properties of Dirac's comb in (3), having defined the frequency resolution

$$\Delta\nu := \frac{1}{T}\ .$$

This Fourier transform is proportional to the Fourier transform of the original function, sampled in frequency with elementary frequency $\Delta\nu$.

20.3.3 Fourier transform of the a function sampled with a Dirac comb - DTFT

Fourier transform of the original function sampled with $\Delta t\, \text{III}_{\Delta t}(t)$ reads

$$
\begin{aligned}
\mathcal{F}\left\{\Delta t\, f(t)\, \text{III}_{\Delta t}(t)\right\} &= \Delta t \int_{t=-\infty}^{+\infty} f(t)\, \text{III}_{\Delta t}(t) e^{-i2\pi\nu t}\, dt = \\
&\sim \Delta t \frac{1}{\Delta t} \int_{t=-\infty}^{+\infty} f(t) \sum_{n=-\infty}^{+\infty} e^{in\frac{2\pi}{\Delta t}t} e^{-i2\pi\nu t}\, dt = \\
&= \sum_{n=-\infty}^{+\infty} \int_{t=-\infty}^{+\infty} f(t)\, e^{-i2\pi(\nu - n\overline{\nu})t}\, dt = \qquad (20.6) \\
&= \sum_{n=-\infty}^{+\infty} F\left(\nu - n\overline{\nu}\right) = \text{DTFT}(f(t); \Delta t)\ ,
\end{aligned}
$$

i.e. equals the periodic sum of the Fourier of the original function, with period

$$\overline{\nu} := \frac{1}{\Delta t}\ .$$

From this last sentence and from the *symmetry properties of Fourier transform*, **Nyquist-Shannon sampling theorem** follows seamlessly.

Theorem 20.3.1 (Nyquist-Shannon sampling theorem)

In order to **avoid aliasing** the sampling frequency must be twice the maxiumum[1] frequency in the signal,

$$\nu_s \geq 2\nu_{max}\ .$$

---

[1] Usually there's no such a frequency above which the signal is exactly zero, but usually there's a frequency above which the spectrum of the signal is approximately zero, i.e. below a threshold where it can be treated as zero, and introduce no aliasing.

todo check alternative expressions if using the definition of train of impulses instead of the Fourier series of Dirac's comb.

$$
\begin{aligned}
&= \Delta t \int_{t=-\infty}^{+\infty} f(t) \sum_{k=-\infty}^{+\infty} \delta(t - k\Delta t)\, e^{-i2\pi\nu t}\, dt = \\
&= \Delta t \sum_{k=-\infty}^{+\infty} f(k\Delta t) e^{-i2\pi\nu k\Delta t} = \mathrm{DTFT}\left(f(t); \Delta t\right)
\end{aligned}
\tag{20.7}
$$

20.3.4 Fourier transform of the sum of shifted integral functions sampled with a Dirac comb

Fourier transform of the periodic sum

$$
\Delta t\, \tilde{f}(t)\, \mathrm{III}_{\Delta t}(t) = \Delta t \sum_{n=-\infty}^{+\infty} f(t - nT)\, \mathrm{III}_{\Delta t}(t)
$$

reads

$$
\begin{aligned}
\mathcal{F}\left\{\Delta t\, \tilde{f}(t)\, \mathrm{III}_{\Delta t}(t)\right\}(\nu) &= \Delta t \int_{t=-\infty}^{+\infty} \sum_{n=-\infty}^{+\infty} f(t - nT) \sum_{k=-\infty}^{+\infty} \delta(t - k\Delta t)\, e^{-i2\pi\nu t}\, dt = \\
&= \Delta t \sum_{n=-\infty}^{+\infty} \sum_{k=-\infty}^{+\infty} f(k\Delta t - nT)\, e^{-i2\pi\nu k\Delta t} =
\end{aligned}
$$

and defining $k\Delta\tau_n := k\Delta t - nT$,

$$
\begin{aligned}
&= \Delta t \sum_{n=-\infty}^{+\infty} \sum_{k=-\infty}^{+\infty} f(k\Delta\tau_n) e^{-i2\pi\nu k\Delta\tau_n} e^{-i2\pi\nu nT} = \\
&= \underbrace{\Delta t \sum_{k=-\infty}^{+\infty} f(k\Delta\tau_n) e^{-i2\pi\nu k\Delta\tau_n}}_{=\mathrm{DTFT}(f(t),\Delta t)} \underbrace{\sum_{n=-\infty}^{+\infty} e^{-i2\pi\nu nT}}_{=\Delta\nu\, \mathrm{III}_{\Delta\nu}(\nu)} = \\
&= \mathrm{DTFT}(f(t), \Delta t)\, \Delta\nu\, \mathrm{III}_{\Delta\nu}(\nu)\ .
\end{aligned}
$$

todo check! check the change of coordinates that makes DTFT appear

todo check! what follows or, using the relation between Δt and $T = N\Delta t$, $\Delta\nu = \frac{1}{T}$, and thus

$$
\Delta t\, \Delta\nu = \Delta t\, \frac{1}{T} = \frac{1}{N}\ ,
$$

it follows

$$
= \frac{1}{N} \sum_{k=-\infty}^{+\infty} f(k\Delta\tau_n) e^{-i2\pi\nu k\Delta\tau_n}\, \mathrm{III}_{\Delta\nu}(\nu)\ .
$$

20.3.5 Useful properties

Dirac's comb $\mathrm{III}_T(t)$

Dirac comb $\mathrm{III}_T(t)$ is defined as a train of Dirac's delta

$$
\mathrm{III}_T(t) = \sum_{m=-\infty}^{+\infty} \delta(t - mT)\ .
$$

Coefficients (20.2) of the Fourier series (20.1) of a T-periodic train of Dirac delta for $t \in \left[-\frac{T}{2}, \frac{T}{2}\right]$, read

$$c_n = \frac{1}{T} \int_{t=0}^{T} \delta(t)\, e^{-in\frac{2\pi}{T}t} = \frac{1}{T} \ ,$$

and thus the Fourier series of Dirac comb $\mathrm{III}_T(t)$ reads

$$\mathrm{III}_T(t) = \sum_{m=-\infty}^{+\infty} \delta(t - mT) \sim \frac{1}{T} \sum_{n=-\infty}^{+\infty} e^{in\frac{2\pi}{T}t} \ .$$

Symmetry of Fourier transform

Part VII

Calculus of Variations

CHAPTER

TWENTYONE

INTRODUCTION TO CALCULUS OF VARIATIONS

Calculus of variation deals with variations - i.e. “small changes” - of functions and functionals.

The meaning of the term functional may vary on the subfield of interest. In the field of calculus of variation, a **functional** can be defined as a function of function, i.e. a function whose argument is another function.

Fields and applications

Fields and applications related to calculus of variations (give some examples below):

- gradient-based techniques like some methods in:
  - optimization, either free or constrained (via Lagrange multiplier methods)
  - sensitivity
- classical mechanics and physics in general:
  - analytical mechanics: Lagrangian formulation and Hamiltonian formulation of classical mechanics
- …

Examples

- Lagrange equations for general problem
- examples:
  - brachistochrone for minimum time,…
  - catenary, i.e. static solution of wire and cables with neglibile bending stiffness
  - isoperimetric inequality, i.e. circle is the plane closed curve with given perimeter enclosing the largest area
- sensitivity of results to parameters. Some interesting sensitivity, both in time and trasnformed domains
  - characteristics of a system:
    - equilibria
    - eigenvalues
    - …
- optimal control methods

21.1 Lagrange equations

Given the functional S, with arguments a function $q(t)$ and the independent variable t,

$$S[q(t), t] = \int_{t=t_0}^{t_1} L(\dot{q}(t),\, q(t),\, t)\, dt$$

its variation w.r.t. the function $q(t)$ reads

$$\delta S[q(t), t] = \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \left(S[q(t) + \varepsilon w(t),\, t] - S[q(t),\, t] \right)$$

where the function $w(t)$ is arbitrary, among those satisfying the constraint of the problems: as an example here, if the function $q(t)$ has prescribed values q^* for some values of the independent variable, t^*, the variation $w(t)$ of the function $q(t)$ is zero there, $w(t^*)$ so that the variated function $q(t) + \varepsilon w(t)$ satisfies the constraint as well, i.e. $q(t^*) + \varepsilon w(t^*) = q^*$.

Variation involves only small changes of function arguments, since these ones are the elements that can be effectively changed, while the independent variable is not.

Direct computation of the variation gives

$$\begin{aligned}
\delta S[q(t), t] &= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \left(S[q(t) + \varepsilon w(t),\, t] - S[q(t),\, t] \right) = \\
&= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \left(\int_{t=t_0}^{t_1} L(\dot{q}(t) + \varepsilon w(t),\, q(t) + \varepsilon w(t),\, t) - \int_{t=t_0}^{t_1} L(\dot{q}(t),\, q(t),\, t)\, dt \right) = \\
&= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \int_{t=t_0}^{t_1} \left(L(\dot{q}(t) + \varepsilon w(t),\, q(t) + \varepsilon w(t),\, t) - L(\dot{q}(t),\, q(t),\, t) \right)\, dt = \\
&= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \int_{t=t_0}^{t_1} \left\{ L(\dot{q}(t),\, q(t),\, t) + \varepsilon \left[\frac{\partial L}{\partial \dot{q}} \dot{w}(t) + \frac{\partial L}{\partial q} w(t) \right] + o(\varepsilon) - L(\dot{q}(t),\, q(t),\, t) \right\}\, dt = \\
&= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \int_{t=t_0}^{t_1} \left\{ \varepsilon \left[\frac{\partial L}{\partial \dot{q}} \dot{w}(t) + \frac{\partial L}{\partial q} w(t) \right] + o(\varepsilon) \right\}\, dt = \\
&= \int_{t=t_0}^{t_1} \left\{ \frac{\partial L}{\partial \dot{q}} \dot{w}(t) + \frac{\partial L}{\partial q} w(t) \right\}\, dt = \\
&= \left[w(t) \frac{\partial L}{\partial \dot{q}} \right]\Bigg|_{t=t_0}^{t_1} + \int_{t=t_0}^{t_1} \left\{ -\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) + \frac{\partial L}{\partial q} \right\} w(t)\, dt\ .
\end{aligned}$$

The solution depends on the boundary conditions at the extreme points t_0, t_1. **If** the value of the function $q(t)$ is prescribed in t_0 and t_1, $q(t_0) = q_0$, $q(t_1) = q_1$, then its variation is zero, $w(t_0) = w(t_1) = 0$, for the reason that has been discussed above. The variation of the functional with prescribed boundary values of the argument function thus reads

$$\delta S[q(t), t] = \int_{t=t_0}^{t_1} \left\{ -\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) + \frac{\partial L}{\partial q} \right\} \delta q(t)\, dt\ ,$$

having called $w(t) =: \delta q(t)$ to stress that is the variation of function $q(t)$. This notation - it's just notation, it has no special properties - could be useful if the functional depends on several arguments.

Stationary conditions, $\delta S = 0$. Stationary condition of the functional S implies that $\delta S = 0$ for all the possible variations of the argument function, $\forall \delta q(t)$. This condition implies that the integrand is identically zero, i.e. **Lagrange equations**,

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0\ ,$$

Higher-order derivatives

Method 1. If the Lagrangian function L depends on higher order derivatives,

$$L\left(q^{(n)}(t),\, q^{(n-1)}(t),\, \dots,\, q'(t),\, q(t),\, t\right)$$

it's possible to recast the problem defining the n-dimensional function, $\mathbf{q}(t)$,

$$\mathbf{q}(t) = \left(q^0(t), q^1(t), \dots, q^{n-1}(t)\right) := \left(q(t), q'(t), \dots, q^{(n-1)}(t)\right) \ .$$

With some abuse of notation in L, the functional S can be recasted as

$$\begin{aligned} S[q(t), t] &= \int_{t=t_0}^{t_1} L(q^{(n)}(t),\, \dots,\, q(t),\, t)\, dt = \\ &= \int_{t=t_0}^{t_1} L(\dot{\mathbf{q}}(t),\, \mathbf{q}(t),\, t)\, dt \ . \end{aligned}$$

todo *Add constraints on components of* $\mathbf{q}(t)$*?*

Repeating the computation, the variation of the functional reads

$$\delta S[\mathbf{q}(t), t] = \left[\delta\mathbf{q}^T(t)\frac{\partial L}{\partial \dot{\mathbf{q}}}\right]\Bigg|_{t=t_0}^{t_1} + \int_{t=t_0}^{t_1} \delta\mathbf{q}^T(t)\, \left\{-\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\mathbf{q}}}\right) + \frac{\partial L}{\partial \mathbf{q}}\right\} dt \ .$$

Method 2. …

21.1.1 Euler-Beltrami equation

If the Lagrangian function L is not an explicit function of the independent variable, $L(q'(x), q(x))$, Euler-Berltrami equation follows from the derivative of the Lagrangian,

$$\begin{aligned} \frac{dL}{dx} &= \frac{\partial L}{\partial q'} q''(x) + \frac{\partial L}{\partial q} q' = \\ &= \frac{\partial L}{\partial q'} q'' + \frac{d}{dx}\left(\frac{\partial L}{\partial q'}\right) q' = \\ &= \frac{d}{dx}\left(\frac{\partial L}{\partial q'} q'\right) \ , \end{aligned}$$

and thus

$$\frac{d}{dx}\left[L - q'\frac{\partial L}{\partial q'}\right] = 0 \qquad \rightarrow \qquad L - q'\frac{\partial L}{\partial q'} = C \quad \text{const.}$$

Note 1. While Lagrange equations are a set of N equations if the functional depends on N argument functions $q_k(t)$, $k = 1 : N$, Euler-Beltrami equation is an equation only. Indeed for multiple argument functions

$$\begin{aligned} \frac{dL}{dx} &= \frac{\partial L}{\partial q_k'} q_k''(x) + \frac{\partial L}{\partial q_k} q_k' = \\ &= \frac{\partial L}{\partial q_k'} q_k'' + \frac{d}{dx}\left(\frac{\partial L}{\partial q_k'}\right) q_k' = \frac{d}{dx}\left(\frac{\partial L}{\partial q_k'} q_k'\right) \ , \end{aligned}$$

where Einstein's summation notation of repeated index is used. Euler-Beltrami thus reads

$$L(q_l'(x), q_l(x)) - q_k'(x)\frac{\partial L}{\partial q_k'}(q_l'(x), q_l(x)) = C \ .$$

Note 2. If the Lagrangian function is an explicit function of the independent variable x, $L(q'(x), q(x), x)$, it's not hard to realize that the derivative of the Lagrangian function, along with the use of the Lagrange equation, gives

$$\frac{d}{dx}\left[L - q'\frac{\partial L}{\partial q'}\right] = \frac{\partial L}{\partial x} \ .$$

Example 21.1.1 (Euler-Beltrami with $L(q'(x), q(x), x)$, Hamiltonian, energy and E.Noether)

Euler-Beltrami equation shows that if $L(q'(x), q(x))$, thus $L - q'\partial_{q'}L$ is constant, (or an integral of motion in dynamics). In analytical mechanics (Lagrange mechanics, Hamiltonian mechanics), Lagrangian and Hamiltonian functions of a system read

$$\begin{aligned} L(\dot{q}_k(t), q_k(t), t) &= T(\dot{q}, q, t) + U(q, t) \\ H(p, q, t) &:= p_k\dot{q}_k - L = \dot{q}_k\frac{\partial L}{\partial \dot{q}^k} - L \ , \end{aligned}$$

having used the common definition of the generalized momenta $p_k := \frac{\partial L}{\partial \dot{q}^k}$. It should be immediate to realize that the Hamiltonian is just the quantity appearing in Euler-Beltrami equation (or in its "modified version" if $\partial_t L \neq 0$), and thus

$$\frac{dH}{dt} = \frac{\partial L}{\partial t} \ .$$

In mechanics, if $\partial_t L = 0$, the Hamiltonian is a constant of motion. In this case, it can be prove that the Hamiltonian is equal to the eneergy of the system.

Classical examples.

Example 21.1.2 (Brachistochrone)

Find the trajectory…

- Elementary length: $ds = v\,dt$
- Energy: $E(y) = \frac{1}{2}mv^2 - mgy + C$. Setting $E = 0$ at starting point, from rest, at $y_0 = 0$, it implies $C = 0$; thus $v = \sqrt{2gy}$
- $x(s)$, $y(s)$,
- $ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + y'^2(x)}\,dx$

$$T = \int_{t_0}^{t_1} dt = \int_{s_0}^{s_1} \frac{ds}{v} = \int_{x_0}^{x_1} \frac{\sqrt{1 + y'^2(x)}}{\sqrt{2gy(x)}}\,dx$$

The Lagrangian doesn't explicitly depend on x, thus Euler-Beltrami equation can be used. Partial derivative of the Lagrangian function w.r.t. q' reads

$$\frac{\partial L}{\partial y'} = ... = \frac{1}{\sqrt{2gy}\sqrt{1 + y'^2}}y' \ ,$$

and thus Euler-Beltrami equation reads

$$C = L - q'\frac{\partial L}{\partial y'} = \frac{1 + y'^2 - y'^2}{\sqrt{2gy}\sqrt{1 + y'^2}} = \frac{1}{\sqrt{2gy}\sqrt{1 + y'^2}}$$

Squaring $2gC^2 = \frac{1}{y(1+y'^2)}$, it's possible to write

$$y(x) = \frac{1}{2gC^2(1 + y'^2(x))} \ ,$$

Making the substitution $y(x)' = ...$

Example 21.1.3 (Catenary)

Example 21.1.4 (Isoperimetric problem)

Part VIII

Ordinary Differential Equations

CHAPTER

TWENTYTWO

INTRODUCTION TO ORDINARY DIFFERENTIAL EQUATIONS

- Theorems
- General solution of linear ODEs

Links

- *System theory*
- *Control theory*

CHAPTER

TWENTYTHREE

LINEAR SYSTEMS

23.1 General solution

The general solution of the initial value problem

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{x}(t_0) &= \mathbf{x}_0\end{aligned} \tag{23.1}$$

with no impulsive forcing in t_0 reads

$$\mathbf{x}(t) = \Phi(t, t_0)\mathbf{x_0} + \int_{\tau=t_0}^{t} \Phi(t, \tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau \tag{23.2}$$

with

$$\begin{cases} \frac{\partial}{\partial t}\Phi(t, t_0) = \mathbf{A}(t)\Phi(t, t_0) \\ \Phi(t, t_0) = \mathbf{I}\,, \end{cases}$$

or alternatively, $\partial_{t_0}\Phi(t, t_0) = \Phi(t, t_0)\mathbf{A}(t_0)$.

General solution - details

Pre-multiplication of equation (23.1) by $\phi(t)$, with $\dot{\phi}(t) = -\phi(t)\mathbf{A}(t)$ gives

$$\begin{aligned}\phi(t)\dot{\mathbf{x}} - \phi(t)\mathbf{A}\mathbf{x} &= \phi(t)\mathbf{B}\mathbf{u} \\ \phi(t)\dot{\mathbf{x}} + \dot{\phi}(t)\mathbf{x} &= \phi(t)\mathbf{B}\mathbf{u} \\ \frac{d}{dt}\left(\phi(t)\mathbf{x}\right) &= \phi(t)\mathbf{B}\mathbf{u}\end{aligned}$$

and integration - without impulsive forces in $t = t_0$, that would cause jump in initial conditions, see *section about impulsive force* -

$$\phi(t)\mathbf{x} - \phi(t_0)\mathbf{x_0} = \int_{\tau=t_0}^{t} \phi(\tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau$$

and thus

$$\mathbf{x}(t) = \phi^{-1}(t)\phi(t_0)\mathbf{x_0} + \int_{\tau=t_0}^{t} \phi^{-1}(t)\phi(\tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau$$

or, defining $\Phi(a,b) := \phi^{-1}(a)\phi(b)$,

$$\mathbf{x}(t) = \Phi(t,t_0)\mathbf{x_0} + \int_{\tau=t_0}^{t} \Phi(t,\tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau$$

with $\Phi(t,t_0)$ satisfying the initial value problem

$$\begin{cases} \frac{\partial}{\partial t}\Phi(t,t_0) = \mathbf{A}(t)\Phi(t,t_0) \\ \Phi(t,t_0) = \mathbf{I}\,. \end{cases}$$

The initial conditions follows from the solution, as the integral identically vanishes as $t \to t_0$ if there's no impulsive force in t_0. The matrix differential equation follows from the definition of the function $\Phi(t,\tau)$ in terms of $\phi(t)$ and from the time derivative of the inverse of a matrix

$$\frac{d}{dt}\phi^{-1} = -\phi^{-1}\dot{\phi}\phi^{-1}\,,$$

as $\mathbf{0} = \frac{d}{dt}\mathbf{I} = \frac{d}{dt}\left(\phi^{-1}\phi\right)$.

Time derivative of $\Phi(t,\tau)$ w.r.t. the first argument reads

$$\begin{aligned} \partial_t\Phi(t,\tau) &= \partial_t\left(\phi^{-1}(t)\phi(\tau)\right) = \\ &= \frac{d}{dt}\phi^{-1}(t)\,\phi(\tau) = \\ &= -\phi^{-1}(t)\dot{\phi}(t)\phi^{-1}(t)\,\phi(\tau) = \\ &= \underbrace{\phi^{-1}(t)\phi(t)}_{=\mathbf{I}}\,\mathbf{A}(t)\,\underbrace{\phi^{-1}(t)\,\phi(\tau)}_{\Phi(t,\tau)} = \\ &= \mathbf{A}(t)\Phi(t,\tau)\,. \end{aligned}$$

Derivative w.r.t. the second argument simply reads

$$\begin{aligned} \partial_\tau\Phi(t,\tau) &= \phi^{-1}(t)\frac{d}{d\tau}\phi(\tau) = \\ &= \phi^{-1}(t)\phi(\tau)\mathbf{A}(\tau) = \\ &= \Phi(t,\tau)\mathbf{A}(\tau)\,. \end{aligned}$$

23.2 Stochastic response

23.2.1 Correlation matrix

Using the general expression of the solution (23.2) of the linear system (23.1), the following relation holds for the correlation $\mathbf{R_{xx}}(t,\tau) := \mathbb{E}[\mathbf{x}(t)\mathbf{x}^*(\tau)]$,

$$\mathbf{R_{xx}}(t,\tau) = \Phi(t,t_0)\mathbf{P}_0\Phi^*(t,t_0) + \int_{\lambda=t_0}^{t}\int_{\chi=t_0}^{\tau} \Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{R_{uu}}(\lambda,\chi)\mathbf{B}^*(\chi)\Phi^*(\tau,\chi)d\lambda\,d\chi\,,$$

or for $\mathbf{P_{xx}}(t) := \mathbf{R_{xx}}(t,t)$

$$\mathbf{P_{xx}}(t) = \Phi(t,t_0)\mathbf{P}_0\Phi^*(t,t_0) + \int_{\lambda=t_0}^{t}\int_{\chi=t_0}^{t} \Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{R_{uu}}(\lambda,\chi)\mathbf{B}^*(\chi)\Phi^*(t,\chi)d\lambda\,d\chi\,.$$

Details

$$
\begin{aligned}
\mathbf{R}_{\mathbf{xx}}(t,\tau) &= \mathbb{E}\left[\mathbf{x}(t)\mathbf{x}^*(\tau)\right] = \\
&= \mathbb{E}\left[\Phi(t,t_0)\mathbf{x_0}\mathbf{x_0}^*\Phi^*(t,t_0)\right] + \\
&+ \mathbb{E}\left[\int_{\lambda=t_0}^{t} \Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{u}(\lambda)d\lambda\, \mathbf{x_0}^*\Phi^*(t,t_0)\right] + \\
&+ \mathbb{E}\left[\Phi(t,t_0)\mathbf{x_0}\int_{\lambda=t_0}^{t} \mathbf{u}^*(\lambda)\mathbf{B}^*(\lambda)\Phi^*(t,\lambda)d\lambda\right] + \\
&+ \mathbb{E}\left[\int_{\lambda=t_0}^{t} \Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{u}(\lambda)d\lambda \left(\int_{\chi=t_0}^{\tau} \Phi(\tau,\chi)\mathbf{B}(\chi)\mathbf{u}(\chi)d\chi\right)^*\right] ,
\end{aligned}
$$

and applying the expectation opearator only on stochastic functions,

$$
\begin{aligned}
\mathbf{R}_{\mathbf{xx}}(t,\tau) &= \Phi(t,t_0)\mathbb{E}\left[\mathbf{x_0}\mathbf{x_0}^*\right]\Phi^*(t,t_0) + \\
&+ \int_{\lambda=t_0}^{t} \Phi(t,\lambda)\mathbf{B}(\lambda)\mathbb{E}\left[\mathbf{u}(\lambda)\mathbf{x}_0^*\right]d\lambda\Phi^*(t,t_0) + \\
&+ \Phi(t,t_0)\int_{\lambda=t_0}^{t} \mathbb{E}\left[\mathbf{x_0}\mathbf{u}^*(\lambda)\right]\mathbf{B}^*(\lambda)\Phi^*(t,\lambda)d\lambda + \\
&+ \int_{\lambda=t_0}^{t}\int_{\chi=t_0}^{\tau} \Phi(t,\lambda)\mathbf{B}(\lambda)\mathbb{E}\left[\mathbf{u}(\lambda)\mathbf{u}^*(\chi)\right]\mathbf{B}^*(\chi)\Phi^*(\tau,\chi)d\lambda\, d\chi .
\end{aligned}
$$

Assuming uncorrelated input noise and uncertainty on the initial conditions, $\mathbb{E}[\mathbf{u}(t)\mathbf{x}_0^*] = \mathbf{0}$, and thus

$$
\mathbf{R}_{\mathbf{xx}}(t,\tau) = \Phi(t,t_0)\mathbf{P}_0\Phi^*(t,t_0) + \int_{\lambda=t_0}^{t}\int_{\chi=t_0}^{\tau} \Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{R}_{\mathbf{uu}}(\lambda,\chi)\mathbf{B}^*(\chi)\Phi^*(\tau,\chi)d\lambda\, d\chi ,
$$

with the definition

$$
\mathbf{P}_{\mathbf{xx}}(t) := \mathbf{R}_{\mathbf{xx}}(t,t) .
$$

If the input is a white noise, $\mathbf{R}_{\mathbf{uu}}(t,\tau) = \mathbf{U}(t)\delta(t-\tau)$,

$$
\mathbf{P}_{\mathbf{xx}}(t) = \Phi(t,t_0)\mathbf{P}_0\Phi^*(t,t_0) + \int_{\lambda=t_0}^{t} \Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{U}(\lambda)\mathbf{B}^*(\lambda)\Phi^*(\tau,\lambda)d\lambda
$$

Details

$$
\begin{aligned}
&\int_{\lambda=t_0}^{t}\int_{\chi=t_0}^{\tau} \Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{R}_{\mathbf{uu}}(\lambda,\chi)\mathbf{B}^*(\chi)\Phi^*(\tau,\chi)d\lambda\, d\chi = \\
&= \int_{\lambda=t_0}^{t}\int_{\chi=t_0}^{\tau} \Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{U}(\lambda)\delta(\lambda-\chi)\mathbf{B}^*(\chi)\Phi^*(\tau,\chi)d\lambda\, d\chi = \\
&= \int_{\lambda=t_0}^{\min\{t,\tau\}} \Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{U}(\lambda)\mathbf{B}^*(\lambda)\Phi^*(\tau,\lambda)d\lambda ,
\end{aligned}
$$

as

$$
\int_{\chi=t_0}^{\tau} \delta(\lambda-\chi)\mathbf{B}^*(\chi)\Phi^*(\tau,\chi)d\chi = \begin{cases} \mathbf{B}^*(\lambda)\Phi^*(\tau,\lambda) & , \quad \text{if } \lambda \in [t_0,\tau] \\ 0 & , \quad \text{otherwise} \end{cases}
$$

Lyapunov equation of the correlation matrix

Matrix $\mathbf{P}(t)$ satisfies the following Lyapunov equation,

$$\dot{\mathbf{P}} = \mathbf{A}\mathbf{P} + \mathbf{P}\mathbf{A}^* + \mathbf{B}\mathbf{U}\mathbf{B}^* \ . \tag{23.3}$$

Details

$$\begin{aligned}
\frac{d\mathbf{P}}{dt} &= \partial_t \Phi \mathbf{P}_0 \Phi + \Phi \mathbf{P}_0 \partial_t \Phi + \underbrace{\Phi(t,t)}_{\mathbf{I}} \mathbf{B}(t)\mathbf{U}(t)\mathbf{B}^*(t) \underbrace{\Phi(t,t)}_{=\mathbf{I}} + \\
&+ \int_{\lambda=t_0}^{t} \left\{ \partial_t \Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{U}(\lambda)\mathbf{B}^*(\lambda)\Phi^*(\tau,\lambda) + \Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{U}(\lambda)\mathbf{B}^*(\lambda)\partial_t \Phi^*(\tau,\lambda) \right\} d\lambda = \\
&= \mathbf{B}(t)\mathbf{U}(t)\mathbf{B}^*(t) + \mathbf{A}(t) \underbrace{\left[\Phi(t,t_0)\mathbf{P}_0\Phi^*(t,t_0) + \int_{\lambda=t_0}^{t} \Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{U}(\lambda)\mathbf{B}^*(\lambda)\Phi(t,\lambda) \right]}_{=\mathbf{P}(t)} + [\ldots]^* \mathbf{A}^*(t) = \\
&= \mathbf{A}\mathbf{P} + \mathbf{P}\mathbf{A}^* + \mathbf{B}\mathbf{U}\mathbf{B}^* \ .
\end{aligned}$$

Part IX

System Theory

CHAPTER

TWENTYFOUR

INTRODUCTION TO SYSTEM THEORY

System theory shifts the focus from solving individual equations to analyzing the interaction between inputs, states, and outputs. …

Links

- *Ordinary differential equations* …
- *Control theory* aims at getting the desired behavior from a given model

- **State-space representation.**
- **Linear systems.**
- **Stability**
- **Controllability and Observability.**

CHAPTER

TWENTYFIVE

LINEAR TIME-INVARIANT SYSTEMS

A linear time invariant system is governed by a linear ODE with constant coefficients. These equations can be recast as a first order system of ODEs,

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \\ \mathbf{x}(0^-) = \mathbf{x}_0 \end{cases}$$

Exploiting the *properties of matrix exponential* the general expression of the state can be written as the sum of the free response to initial condition and the forced response.

$$\begin{aligned} \mathbf{x}(t) &= e^{\mathbf{A}t}\mathbf{x}_0 + \int_{\tau=0^-}^{t} e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)\,d\tau \\ \mathbf{y}(t) &= \mathbf{C}e^{\mathbf{A}t}\mathbf{x}_0 + \mathbf{C}\int_{\tau=0^-}^{t} e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)\,d\tau + \mathbf{D}\mathbf{u}(t) \end{aligned}$$

Proof in time domain

Multipying by $e^{-\mathbf{A}t}$,

$$\begin{aligned} e^{-\mathbf{A}t}(\dot{x}(t) - \mathbf{A}\mathbf{x}(t)) &= e^{-\mathbf{A}t}\mathbf{B}\mathbf{u}(t) \\ \frac{d}{dt}\left(\mathbf{x}e^{-\mathbf{A}t}\right) &= e^{-\mathbf{A}t}\mathbf{B}\mathbf{u}(t) \end{aligned}$$

and integrating from 0^- to a generic time value t,

$$e^{-\mathbf{A}t}\mathbf{x}(t) - \mathbf{x}_0 = \int_{\tau=0^-}^{t} e^{-\mathbf{A}\tau}\mathbf{B}\mathbf{u}(\tau)\,d\tau$$

The state $\mathbf{x}(t)$ can be written as the sum of the free response and a force response. The general expression of the state and the output as a function reads

$$\begin{aligned} \mathbf{x}(t) &= e^{\mathbf{A}t}\mathbf{x}_0 + \int_{\tau=0^-}^{t} e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)\,d\tau \\ \mathbf{y}(t) &= \mathbf{C}e^{\mathbf{A}t}\mathbf{x}_0 + \mathbf{C}\int_{\tau=0^-}^{t} e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)\,d\tau + \mathbf{D}\mathbf{u}(t) \end{aligned}$$

Laplace domain. The *Laplace transform* of the problem reads

$$\begin{cases} s\hat{\mathbf{x}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}\hat{\mathbf{u}} + \mathbf{x}_0 \\ \hat{\mathbf{y}} = \mathbf{C}\hat{\mathbf{x}} + \mathbf{D}\hat{\mathbf{u}} \end{cases}$$

$$
\begin{aligned}
(s\mathbf{I}-\mathbf{A})\hat{\mathbf{x}} &= \mathbf{B}\hat{\mathbf{u}} + \mathbf{x}_0 \\
\hat{\mathbf{x}}(s) &= (s\mathbf{I}-\mathbf{A})^{-1}\mathbf{x}_0 + (s\mathbf{I}-\mathbf{A})^{-1}\mathbf{B}\hat{\mathbf{u}}(s) \\
\hat{\mathbf{y}}(s) &= \mathbf{C}(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{x}_0 + \left[\mathbf{C}(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{B}+\mathbf{D}\right]\hat{\mathbf{u}}(s)
\end{aligned}
$$

Performing inverse Laplace transform allows to go back to time domain (just use Laplace inverse transform of a matrix exponential, and the formula (19.1) for Laplace transform of convolution).

25.1 Impulsive force

The effect of an impulsive force $\mathbf{u}_\delta\delta(t)$ at time $t=0$ is equivalent to an instantaneous change in the initial state, $\Delta\mathbf{x}_0 = \mathbf{B}\mathbf{u}_\delta$, from time 0^- before the impulse to time 0^+ after the impulse. Splitting the input $\mathbf{u}(t)$ as the sum of impulsive input and regular input,

$$
\begin{aligned}
\mathbf{u}(t) &= \mathbf{u}_r(t) + \mathbf{u}_\delta\delta(t) \\
\hat{\mathbf{u}}(s) &= \hat{\mathbf{u}}_r(s) + \mathbf{u}_\delta
\end{aligned}
$$

the solution in **time domain** reads

$$
\begin{aligned}
\mathbf{x}(t) &= e^{\mathbf{A}t}\mathbf{x}_0 + \int_{\tau=0^-}^{t} e^{\mathbf{A}(t-\tau)}\mathbf{B}\left(\mathbf{u}_r(\tau)+\mathbf{u}_\delta\delta(\tau)\right)\,d\tau = \\
&= e^{\mathbf{A}t}\left(\mathbf{x}_0+\mathbf{B}\mathbf{u}_\delta\right) + \int_{\tau=0^-}^{t} e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}_r(\tau)\,d\tau \\
\mathbf{y}(t) &= \mathbf{C}e^{\mathbf{A}t}\left(\mathbf{x}_0+\mathbf{B}\mathbf{u}_\delta\right) + \int_{\tau=0^-}^{t} \mathbf{C}e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}_r(\tau)\,d\tau + \mathbf{D}\mathbf{u}_r(t) + \mathbf{D}\mathbf{u}_\delta(t)
\end{aligned}
$$

while in **Laplace domain** reads

$$
\begin{aligned}
\hat{\mathbf{x}}(s) &= (s\mathbf{I}-\mathbf{A})^{-1}\mathbf{B}\hat{\mathbf{u}}_r(s) + (s\mathbf{I}-\mathbf{A})^{-1}\left(\mathbf{x}_0+\mathbf{B}\mathbf{u}_\delta\right) \\
\hat{\mathbf{y}}(s) &= \left[\mathbf{C}(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{B}+\mathbf{D}\right]\hat{\mathbf{u}}_r(s) + \mathbf{C}(s\mathbf{I}-\mathbf{A})^{-1}\left(\mathbf{x}_0+\mathbf{B}\mathbf{u}_\delta\right) + \mathbf{D}\mathbf{u}_\delta
\end{aligned}
$$

25.2 Properties

Matrix exponential.

$$
e^{\mathbf{A}t} = \sum_{k=0}^{+\infty}\frac{\mathbf{A}^k t^k}{k!}\ .
$$

Assuming it's possible swap derivative operator and summation (when?), it's possible to write

$$
\frac{d}{dt}e^{\mathbf{A}t} = \frac{d}{dt}\sum_{k=0}^{+\infty}\frac{\mathbf{A}^k t^k}{k!} = \sum_{k=1}^{+\infty} k t^{k-1}\frac{\mathbf{A}^k t^{k-1}}{k!} = \mathbf{A}e^{\mathbf{A}t}\ .
$$

Laplace transform of exponential matrix.

$$
\begin{aligned}
\mathcal{L}\left\{e^{\mathbf{A}t}\right\}(s) &:= \int_{t=0^-}^{+\infty} e^{\mathbf{A}t}e^{-st}\,dt = \\
&= \int_{t=0^-}^{+\infty} e^{(-s\mathbf{I}+\mathbf{A})t}\,dt = \\
&= (-s\mathbf{I}+\mathbf{A})^{-1}\ e^{(-s\mathbf{I}+\mathbf{A})t}\Big|_{t=0^-}^{+\infty} = \\
&= (s\mathbf{I}-\mathbf{A})^{-1}\ ,
\end{aligned}
$$

for all the values of s for which $-s\mathbf{I} + \mathbf{A}$ is asymptotically stable, i.e. has all the eignevalues (thus, assuming that the matrix $\mathbf{A}$ can be diagonalizable. What happens if not? Exploit other matrix decompositions to draw conclusions) with negative real parts, and thus for all the values of $s > \max \mathrm{re}\{s_k(\mathbf{A})\}$, as it's shown in *Example 25.2.1*

Example 25.2.1 (Asymptotic stability of a matrix A)

An $N \times N$ diagnonalizable matrix $\mathbf{A}$,

$$\mathbf{A}\mathbf{v}_k = \mathbf{v}_k\, s_k \tag{25.1}$$

has all the eigenvalues with negative real part, $\mathrm{re}\,\{s_k\} < 0$, $\forall k = 1 : N$.

The eigenvalues of a matrix $a\mathbf{I} + \mathbf{A}$ are $a + s_k$, while the eigenvectors are the same as those of the matrix $\mathbf{A}$. This can be easily proved adding $a\mathbf{I}\mathbf{v}_k$ to both sides of equation (25.1),

$$(a\mathbf{I} + \mathbf{A})\,\mathbf{v}_k = \mathbf{v}_k(a + s_k)\ .$$

Transform of the convolution.

CHAPTER

TWENTYSIX

LTI SYSTEM RESPONSE

Usually the response of LTI to 3 different input are studied

- integrable input, being response to non-zero initial condition equivalent to an impulsive force at time $t = 0$, see *equivalence impulsive force - istantanteous change of state*
- periodic input
- stochastic input

26.1 LTI

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \end{cases}$$

with initial conditions, $\mathbb{x}(0) = \mathbf{x}_0$.

26.2 Response of LTI systems

The response of a LTI can be written as the sum of a free and forced response.

$$\begin{aligned} \mathbf{x}(t) &= e^{\mathbf{A}t}\mathbf{x}_0 + \int_{\tau=0}^{t} e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)\, d\tau \\ \mathbf{y}(t) &= \mathbf{C}\, e^{\mathbf{A}t}\mathbf{x}_0 + \int_{\tau=0}^{t} \mathbf{C}e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)\, d\tau + \mathbf{D}\mathbf{u}(t) \end{aligned} \tag{26.1}$$

Proof

todo

Impulse response, $\mathbf{H}(t)$

Functions

$$\begin{aligned} \mathbf{H}_x(t) &:= e^{\mathbf{A}t}\mathbf{B} \\ \mathbf{H}(t) &:= \mathbf{C}e^{\mathbf{A}t}\mathbf{B} + \mathbf{D}\delta(t) \end{aligned}$$

are defined **impulse response functions** for the state $\mathbf{x}$ and the output $\mathbf{y}$, as they determine with the evolution of the state and the output of a system with zero initial condition $\mathbf{x}_0 = \mathbf{0}$ as a consequence of a impulsive force $\mathbf{u}_\delta(t) = \mathbf{u}\delta(t)$. Using Lagrange formula (26.1)

$$\begin{aligned} \mathbf{x}(t) &= e^{\mathbf{A}t}\mathbf{B}\mathbf{u} &&= \mathbf{H}_x(t)\mathbf{u} \\ \mathbf{y}(t) &= \left[\mathbf{C}e^{\mathbf{A}t}\mathbf{B} + \mathbf{D}\delta(t)\right]\mathbf{u} &&= \mathbf{H}(t)\mathbf{u} \end{aligned} \tag{26.2}$$

Convolution

Integral in Lagrange formula (26.1) is the convolution of the impulse response function and the input. Changing the integration variable $z = t - \tau$, $d\tau = -dz$, $\tau = 0 : \ z = t$, $\tau = t, \ z = 0$ (and going back from z to τ as the symbol for the dummy integration variable), it's possible to swap the arguments of the impulsive force and the forcing,

$$\begin{aligned} \mathbf{x}(t) &= e^{\mathbf{A}t}\mathbf{x}_0 + \int_{\tau=0}^{t} \mathbf{H}_x(\tau)\mathbf{u}(t-\tau)\, d\tau \\ \mathbf{y}(t) &= \mathbf{C}\, e^{\mathbf{A}t}\mathbf{x}_0 + \int_{\tau=0}^{t} \mathbf{H}(\tau)\mathbf{u}(t-\tau)\, d\tau \end{aligned} \tag{26.3}$$

Linear but not time-invariant

todo *The response of linear system, but not time-invariant can be written with the same expression used by Lagrange fomrula (26.1), with $\mathbf{H}(t)$ not equal to …, but solution of the ODE…*

$$\begin{aligned} \Phi\left(\dot{\mathbf{x}} - \mathbf{A}\mathbf{x}\right) &= \Phi\mathbf{B}\mathbf{u} \\ \frac{d}{dt}\left(\Phi\mathbf{x}\right) &= \Phi\mathbf{B}\mathbf{u}\ , \end{aligned}$$

with $\dot{\Phi} = -\Phi\mathbf{A}$, and proper initial conditions $\Phi(t, 0) = \mathbf{I}$.

…

$$\mathbf{x}(t) = \Phi(t, 0)\mathbf{x}_0 + \int_{\tau=0}^{t} \Phi(t, \tau)\mathbf{B}(\tau)\mathbf{u}(\tau)\, d\tau\ .$$

26.3 Equilibria

Equilibrium solution of a ODE is a stationary solution, independent from time, $\dot{\bar{\mathbf{x}}} = \mathbf{0}$. With a steady forcing $\bar{\mathbf{u}}$,

$$\begin{cases} \mathbf{0} = \mathbf{A}\bar{\mathbf{x}} + \mathbf{B}\bar{\mathbf{u}} \\ \mathbf{y} = \mathbf{C}\bar{\mathbf{x}} + \mathbf{D}\bar{\mathbf{u}}\ . \end{cases}$$

If A is not singular, for every steady forcing $\bar{\mathbf{u}}$ only one equilibrium exists for the system, $\bar{\mathbf{x}} = \mathbf{A}^{-1}\mathbf{B}\bar{\mathbf{u}}$.

If A is singular…**todo** *rely on* Linear algebra

26.4 Stability

Asymptotic stability to initial perturbations. A system is stable under perturbation of initial conditions if the free response goes to zero as $t \to +\infty$,

$$\lim_{t \to +\infty} \mathbf{x}_{free}(t) = \mathbf{0} \ .$$

todo *for all perturbations*(?).

If matrix $\mathbf{A}$ is diagonalizable, a system is asymptotically stable if all the eigenvalues of the matrix $\mathbf{A}$ have negative real part, $\mathrm{re}\{\lambda_i(\mathbf{A})\} < 0$.

todo *rely on* Linear algebra

26.5 Response to integrable input - and Laplace transform

$$\begin{aligned} \mathbf{x}(s) &= (s\mathbf{I} - \mathbf{A})^{-1} \mathbf{x}_0 + \mathbf{H}_x(s)\mathbf{u}(s) \\ \mathbf{y}(s) &= \mathbf{C}\ (s\mathbf{I} - \mathbf{A})^{-1} \mathbf{x}_0 + \mathbf{H}(s)\mathbf{u}(s) \end{aligned} \tag{26.4}$$

26.6 Response to periodic input of stable systems - and Fourier transform

Forced response to periodic input of an stable system can be represented in Fourier domain replacing s with $j\omega$ (and dropping j once it's clear we're dealing with Fourier analysis)

$$\begin{aligned} \mathbf{x}(\omega) &= \mathbf{H}_x(\omega)\ \mathbf{u}(\omega) \\ \mathbf{y}(\omega) &= \mathbf{H}(\omega)\ \mathbf{u}(\omega) \end{aligned} \tag{26.5}$$

26.7 Response to stochastic input of stable systems

26.7.1 Assumptions

…

26.7.2 Expected value

$$\mu_{\mathbf{x}}(t) := \mathbb{E}[\mathbf{x}(t)]$$

Taking expected value of LTI equations

$$\begin{cases} \mathbb{E}[\dot{\mathbf{x}}] = \mathbf{A}\mathbb{E}[\mathbf{x}] + \mathbf{B}\mathbb{E}[\mathbf{u}] \\ \mathbb{E}[\mathbf{y}] = \mathbf{C}\mathbb{E}[\mathbf{x}] + \mathbf{D}\mathbb{E}[\mathbf{u}] \end{cases}$$

todo

- *is it possible to swap time derivative and expected value operators?*

- *does time derivative of a stochastic process always exist? What's the right way to interpret time derivative in stochastic equations? As an increment?*

$$d\mathbf{x} = \mathbf{A}\mathbf{x}\,dt + \mathbf{B}\mathbf{u}\,dt$$

so that if $\mathbf{u}(t) \sim \xi(t)$, then $\mathbf{u}\,dt \sim d\mathbf{W}$

26.8 Analysis of response of LTI in Fourier domain

Energy or power. For

- time integrable signals: don't average in time; energy
- periodic or stochastic non-integrable signals: average in time; power

Stationary stochastic process. Strong-sense (joint probability independent from time) or wide-sense (moments up to order m are independent on time)

Ergodic process. Average over random process can be replaced with average over time.

Example of stationary non-ergodic process

As an example of stationary, but not ergodic process is a random process whose realizations are

$$X_t = \theta\,, \qquad \forall t \in [0, T]$$

with θ drawn from a random variable Θ, with average μ. Average in time over a realization gives θ, while average over realizations gives μ for all t,

$$\begin{aligned} \text{Avg. in time over a realization:} \quad & \frac{1}{T}\int_{t=0}^{T} X_t\,dt = \frac{1}{T}\int_{t=0}^{T} \theta\,dt = \theta \\ \text{Avg. in probability over realizations:} \quad & \mathbb{E}[X_t] = \mathbb{E}[\Theta] = \mu\,. \end{aligned}$$

26.8.1 Stationary Ergodic processes

Auto-correlation

$$\mathbf{R}_{XX}(t_1, t_2) = \mathbb{E}\left[\mathbf{X}_{t_1}\,\mathbf{X}_{t_2}^*\right]$$

Auto-covariance

$$\mathbf{K}_{XX}(t_1, t_2) = \mathbb{E}\left[(\mathbf{X}_{t_1} - \mu_{t_1})(\mathbf{X}_{t_2} - \mu_{t_2})^*\right]$$

For stationary processes, both auto-correlation and auto-covariance don't depend on two time instant t_1, t_2 but only on the time-shift between them, $t_2 = t_1 + \tau$,

$$\mathbf{R}_{XX}(\tau) = \mathbb{E}\left[\mathbf{X}_t\,\mathbf{X}_{t+\tau}^*\right]\,, \qquad \forall t\,.$$

Autocorrelation and ESD of time integrable processes

With causal, $\mathbf{x}(t \leq 0) = 0$, and integrable processes, autocorrelation is defined as

$$\tilde{\mathbf{R}}_{xx}(\tau) = \int_{t=0}^{+\infty} \mathbf{x}(t)\mathbf{x}^*(t+\tau)\, dt \ ,$$

Energy Spectral Density (ESD) is defined as its Fourier transform

$$\tilde{\Phi}_{xx}(f) = \mathcal{F}\left\{\mathbf{R}_{xx}(\tau)\right\}(f) \ ,$$

Autocorrelation and PSD of infinite-time processes

Using average over time as an approximation of the probability average

$$\mathbb{E}[f(t)] \sim \lim_{T\to+\infty} \frac{1}{T} \int_{t=0}^{T} f(\tau)\, d\tau \ ,$$

Power Spectral Density (PSD), defined as the Fourier transform of the auto-correlation, reads

$$\begin{aligned}\Phi_{YY}(f) &= \mathcal{F}[\mathbf{R}_{YY}(\tau)](f) = \\ &= \int_{\tau=-\infty}^{+\infty} \mathbb{E}\left[\mathbf{y}(t)\mathbf{y}^*(t+\tau)\right] e^{-i2\pi f\tau}\, d\tau \ .\end{aligned}$$

For asymptotically stable systems subject to periodic forcing, response to non-homogenoeus initial conditions vanishes while the forced response can be written either in Fourier or in time domain, exploiting the property of the transform of convolution *Property 20.2.1*, as

$$\mathbf{y}(f) = \mathbf{H}(f)\,\mathbf{u}(f) \qquad , \qquad \mathbf{y}(t) = \mathbf{H} * \mathbf{u}(t) = \int_{\tau=-\infty}^{+\infty} \mathbf{H}(\tau)\mathbf{u}(t-\tau)\, d\tau \ .$$

Inserting harmonic response into the expression of PSD, it becomes

$$\begin{aligned}\Phi_{YY}(f) &= \int_{\tau=-\infty}^{+\infty} \mathbb{E}\left[\left(\int_{\xi=-\infty}^{+\infty} \mathbf{H}(\xi)\,\mathbf{u}(t-\xi)\, d\xi\right)\left(\int_{\chi=-\infty}^{+\infty} \mathbf{H}(\chi)\,\mathbf{u}(t+\tau-\chi)\, d\chi\right)^*\right] e^{-i2\pi f\tau}\, d\tau = && (1)\\ &= \int_{\tau=-\infty}^{+\infty}\int_{\xi=-\infty}^{+\infty}\int_{\chi=-\infty}^{+\infty} \mathbf{H}(\xi)\mathbb{E}\left[\mathbf{u}(t-\xi)\mathbf{u}^*(t+\tau-\chi)\right]\mathbf{H}^*(\chi)\, e^{-i2\pi f\tau}\, d\tau\, d\chi\, d\xi = && (2)\\ &= \int_{\tau=-\infty}^{+\infty}\int_{\xi=-\infty}^{+\infty}\int_{\chi=-\infty}^{+\infty} \mathbf{H}(\xi)\mathbb{E}\left[\mathbf{u}(t)\mathbf{u}^*(t+\tau-\chi+\xi)\right]\mathbf{H}^*(\chi)\, e^{-i2\pi f\tau}\, d\tau\, d\chi\, d\xi = && (3)\\ &= \int_{z=-\infty}^{+\infty}\int_{\xi=-\infty}^{+\infty}\int_{\chi=-\infty}^{+\infty} \mathbf{H}(\xi)\mathbb{E}\left[\mathbf{u}(t)\mathbf{u}^*(t+z)\right]\mathbf{H}^*(\chi)\, e^{-i2\pi f(z-\xi+\chi)}\, dz\, d\chi\, d\xi = \\ &= \int_{\xi=-\infty}^{+\infty} \mathbf{H}(\xi)\, e^{i2\pi f\xi}\, d\xi \int_{z=-\infty}^{+\infty} \mathbf{R}_{uu}(z)\, e^{-i2\pi f z}\, dz \int_{\chi=-\infty}^{+\infty} \mathbf{H}^*(\chi)\, e^{-i2\pi f\chi}\, d\chi = && (4)\\ &= \overline{\mathbf{H}}(f)\,\Phi_{uu}(f)\,\mathbf{H}^T(f) \ .\end{aligned}$$

having used (1) the fact that the impulse response function is deterministic, (2) time shift in the autocorrelation for a **stationary** random process $\mathbf{u}(t)$ as an input, and (3) the change of coordinates $(z, \xi, \chi) = (\tau - \chi + \xi, \xi, \chi)$, and (4) a slight notation abuse ndicating Fourier transform with the same symbols of the functions in time domain.

CHAPTER

TWENTYSEVEN

LTI: STABILITY AND FEEDBACK

Contents

- Stability of a SISO LTI, w.r.t. non-zero i.c. or impulsive input
-
-

27.1 Stability of a LTI - SISO

27.1.1 Transfer function

Let the SISO input-output relation of a LTI system be represented in Laplace domain by the transfer function $G(s)$,

$$y(s) = G(s)u(s)$$

For rational transfer functions

$$G(s) = k\frac{\prod_{n=1}^{N}(z_n - s)}{\prod_{d=1}^{D}(p_d - s)} = \frac{\sum_{n=1}^{N} a_n s^n}{\sum_{d=0}^{D} b_d s^d} \ ,$$

with z_n the zeros, p_d the poles, and $k = \frac{a_0}{b_0} = \lim_{s\to 0} G(s)$ the static gain. If the system is strictly proper, $N < D$, and the TF can be written as a **sum of partial functions**. As an example, for a transfer function with simple poles

$$G(s) = \sum_{d=1}^{N} \frac{A_d}{(p_d - s)} \ ,$$

while for a pole p_d with multiplicity $m_d > 1$ all the terms $\propto \frac{1}{(p_d - s)^{e_d}}$, with $e_d = 1 : m_d$ must be included, see example below.

$$G(s) = \sum_{d=1}^{N} \frac{A_d}{(p_d - s)^{e_d}} \ ,$$

Example 27.1.1 (Sum of partial fractions of a TF with poles with multiplicity > 1)

Let's write the rational function $G(s) = \frac{s+1}{(s+2)^3}$ as a sum of partial functions,

$$\begin{aligned} G(s) &= \frac{s+1}{(s+2)^3} = \\ &= \frac{A}{s+2} + \frac{B}{(s+2)^2} + \frac{C}{(s+2)^3} = \\ &= \frac{A(s+2)^2 + B(s+2) + C}{(s+2)^3} = \\ &= \frac{s^2(A) + s(4A+B) + 4A + 2B + C}{(s+2)^3} \,, \end{aligned}$$

The value of the coefficients A, B, C is computed comparing the first and the last expression of the numerator of $G(s)$

$$\begin{cases} s^2 : & A = 0 \\ s \ : & 4A + B = 1 \\ 1 \ : & 4A + 2B + C = 1 \end{cases} \qquad \rightarrow \qquad \begin{cases} A = 0 \\ B = 1 \\ C = -1 \end{cases}$$

and thus

$$G(s) = \frac{s+1}{(s+2)^3} = \frac{1}{(s+2)^2} - \frac{1}{(s+2)^3} \,.$$

27.1.2 Stability w.r.t. non-zero initial conditions - or w.r.t. implusive input - in time domain

Transfer function $G(s)$ represents the free the function w.r.t. impulsive input. Response in time domain can be evaluated as the inverse Laplace transform of the transfer function,

$$y(t) = \mathcal{L}^{-1}\left\{y(s)\right\} = \mathcal{L}^{-1}\left\{\sum_{n_d=1}^{D} \frac{R_d}{(s-p_d)}\right\} = \sum_{d=1}^{D} R_d \, e^{p_d t} \,.$$

If re$\{p_d\} < 0$ for $\forall d$, then the response is asymptotically stble for $t \rightarrow +\infty$, as $|e^{p_d t}| = e^{\text{re}\{p_d\}t} \rightarrow 0$, for $t \rightarrow +\infty$.

CHAPTER

TWENTYEIGHT

OBSERVABILITY AND DETECTABILITY

28.1 Time-continuous linear systems

The solution of the linear system

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$
$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

with initial conditions $\mathbf{x}(0) = \mathbf{x}_0$ reads

$$\mathbf{x}(t) = \Phi(t, t_0)\mathbf{x}_0 + \int_{\tau=0}^{t} \Phi(t, \tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau \ ,$$

and the output of the system reads

$$\begin{aligned} \mathbf{y}(t) &= \mathbf{C}(t)\mathbf{x}(t) + \mathbf{D}(t)\mathbf{u}(t) = \\ &= \mathbf{C}(t)\Phi(t, t_0)\mathbf{x}_0 + \mathbf{C}(t)\int_{\tau=0}^{t} \Phi(t, \tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau + \mathbf{D}(t)\mathbf{u}(t) \ . \end{aligned} \tag{28.1}$$

28.1.1 Observability

A system is observable if the initial state $\mathbf{x}_0$ can be uniquely determined, knowing the output $\mathbf{y}(t)$ and the input $\mathbf{u}(t)$ over a finite time interval.

A relation between initial conditions as a function of the output and the input of the system can be retrieved from the expression (28.1),

$$\mathbf{C}(t)\Phi(t, t_0)\mathbf{x}_0 = \mathbf{y}(t) - \left[\mathbf{C}(t)\int_{\tau=0}^{t} \Phi(t, \tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau + \mathbf{D}(t)\mathbf{u}(t)\right] =: \tilde{\mathbf{y}}(t) \ ,$$

with the definition of the known combination of input and output on the RHS as $\tilde{\mathbf{y}}(t)$. This system of equations is usually not square, and only involves two time instants t_0, t. If the latter equation is multiplied by $\Phi^T(\tau, t_0)\mathbf{C}^T(\tau)$ (least square solution; L_2 norm;...) and integrated from t_0 to t in order to use all the information from t_0 to t

$$\int_{\tau=t_0}^{t} \Phi^T(\tau, t_0)\mathbf{C}^T(\tau)\mathbf{C}(\tau)\Phi(\tau, t_0)d\tau\, \mathbf{x}_0 = \int_{\tau=t_0}^{t} \Phi^T(\tau, t_0)\mathbf{C}^T(\tau)\tilde{\mathbf{y}}(\tau)d\tau \ .$$

$$\mathbf{W}_o(t)\, \mathbf{x}_0 = \int_{\tau=t_0}^{t} \Phi^T(\tau, t_0)\mathbf{C}^T(\tau)\tilde{\mathbf{y}}(\tau)d\tau \ .$$

todo

- Discuss the **pullback** of the output to the state space (before $\sim$ least-square inversion)

28.1.2 Detectability

A system is observable if all the unobservable modes are naturally stable, i.e. they decay to zero.

todo *Check definition, and requirement for finite or infinite time range*

Optimal input and controllability Gramian

Let's solve the optimal output defined as the one with the maximum objective function $\frac{1}{2}\int_{\tau=0}^{t}|\mathbf{y}(\tau)|^2$, subject to the equations of motion as constraints. **todo** Here the final time t is assumed to be known, but in general it's not: include t in the optimization, or run several optimizations to find optimal t^*. The solution of the free system - or the forced system, with the $\tilde{\mathbf{y}}(t)$ defined as the combination of output and input above - reads

$$\mathbf{x}(t) = \Phi(t, t_0)\mathbf{x}_0 \ ,$$

so that the output is $\mathbf{y}(t) = \mathbf{C}\Phi(t, t_0)\mathbf{x}_0$. The objective function read

$$\begin{aligned} J(\mathbf{x}_0) &= \frac{1}{2}\int_{\tau=0}^{t}\mathbf{y}^T(\tau)\mathbf{y}(\tau)d\tau = \\ &= \frac{1}{2}\mathbf{x}_0^T\int_{\tau=0}^{t}\Phi^T(t, t_0)\mathbf{C}^T\mathbf{C}\Phi(t, t_0)\mathbf{x}_0 = \\ &= \frac{1}{2}\mathbf{x}_0^T\mathbf{W}_o(t)\mathbf{x}_0 \ . \end{aligned}$$

- If $\mathbf{x}_0 \in K(\mathbf{W}_o(t))$, i.e. belongs to the kernel of the **observatility Gramian**, thus $J(\mathbf{x}_0) = 0$, i.e. the initial conditions has no influence on the output.
- If $\mathbf{x}_0 \notin K(\mathbf{W}_o(t))$, i.e. does not belong to the kernel of the observatility Gramian, thus $J(\mathbf{x}_0) > 0$.
- using singular value decomposition, it's possible to find the initial condition producing the largest output. Let $\mathbf{W}_o = \mathbf{U}\Sigma\mathbf{U}^*$, the optimal initial condition with given L_2 norm, e.g. $\|\mathbf{x}_0\|_2 = 1$, is the singular vector $\mathbf{u}_1$ associated with the largest singular value σ_1, $\mathbf{x}_0 = \mathbf{u}_1$, and the optimal output is

$$\max_{\|\mathbf{x}_0\|_2=1} J(\mathbf{x}_0) = J(\mathbf{x}_0 = \mathbf{u}_1) = \frac{1}{2}\sigma_1 \ .$$

Observability Gramian satisfies the *Lyapunov equation*

$$\dot{\mathbf{W}}_o(t) = \mathbf{C}^T(t)\mathbf{C}(t) + \mathbf{A}^T(t)\mathbf{W}_o(t) + \mathbf{W}_o(t)\mathbf{A}(t) \ ,$$

with initial conditions $\mathbf{W}_o(0) = \mathbf{0}$, as the integral vanishes with the same values of the lower and upper extremes of integration.

Lyapunov equation for the controllability Graminan

$$\begin{aligned} \frac{d}{dt}\mathbf{W}_o(t) &= \frac{d}{dt}\int_{\tau=0}^{t}\Phi^T(t,\tau)\mathbf{C}^T(\tau)\mathbf{C}(\tau)\Phi(t,\tau)d\tau = \\ &= \mathbf{C}(t)\mathbf{C}^T(t) + \int_{\tau=0}^{t}\partial_t\Phi^T(t,\tau)\mathbf{C}^T(\tau)\mathbf{C}(\tau)\Phi(t,\tau)d\tau + \int_{\tau=0}^{t}\Phi^T(t,\tau)\mathbf{C}^T(\tau)\mathbf{C}(\tau)\partial_t\Phi(t,\tau)d\tau = \\ &= \mathbf{C}(t)\mathbf{C}^T(t) + \mathbf{A}(t)\mathbf{W}_c(t) + \mathbf{W}_c(t)\mathbf{A}^T(t) \ , \end{aligned}$$

as $\Phi(t,t) = \mathbf{I}$, and $\partial_t\Phi(t,\tau) = \mathbf{A}(t)\Phi(t,\tau)$.

Observability for LTI systems

For LTI systems $\Phi(t, t_0) = e^{\mathbf{A}(t-t_0)}$. The controllability Gramian becomes

$$\mathbf{W}_o(t) = \int_{\tau=0}^{t} e^{\mathbf{A}^T(t-\tau)}\mathbf{C}^T\mathbf{C}e^{\mathbf{A}(t-\tau)}d\tau$$

A vector $\mathbf{v}$ belongs to the null space of $\left(e^{\mathbf{A}^T t}\mathbf{C}^T\right)^T$ if

$$0 = \mathbf{C}e^{\mathbf{A}t}\mathbf{v}\ .$$

Using the definition of the exponential matrix $e^{\mathbf{A}t} := \sum_{n=0}^{+\infty} \frac{\mathbf{A}^n t^n}{n!}$, it follows

$$0 = \mathbf{C}\sum_{k=0}^{+\infty} \frac{\mathbf{A}^k t^k}{k!}\mathbf{v}$$

Thus…, a vector $\mathbf{v}$ is in the null-space of $\mathbf{W}_o$ (symmetric! Be precise and use *orthogonality between range and kernel of the adjoint* if it's orthogonal to the rows of the matrix

$$\mathcal{O} = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \dots \\ \mathbf{CA}^{n-1} \end{bmatrix}\ ,$$

i.e. $\mathcal{O}\mathbf{v} = \mathbf{0}$. If the row space of matrix $\mathcal{O}$ spans $\mathbb{R}^n$, then the system is fully observable.

28.2 Time-discrete linear systems

CHAPTER

TWENTYNINE

REACHABILITY AND CONTROLLABILITY

29.1 Time-continuous linear systems

The solution of the linear system

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$
$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

with initial conditions $\mathbf{x}(0) = \mathbf{x}_0$ reads

$$\mathbf{x}(t) = \Phi(t, t_0)\mathbf{x}_0 + \int_{\tau=0}^{t} \Phi(t, \tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau \ ,$$

29.1.1 Reachability

A state $\mathbf{x}_1$ is reachable if there's a control input $\mathbf{u}(t)$ that can move the system from the initial state $\mathbf{x}_0 = \mathbf{0}$ to the final state $\mathbf{x}_1$ in finite time.

29.1.2 Controllability

A state $\mathbf{x}_0$ is reachable if there's a control input $\mathbf{u}(t)$ that can move the system from the initial state $\mathbf{x}_0$ to the final state $\mathbf{x}_1 = \mathbf{0}$ in finite time.

Optimal input and controllability Gramian

Let's solve the optimal input defined as the one with the minimal cost $\frac{1}{2}\int_{\tau=0}^{t_1} |\mathbf{u}(\tau)|^2$, subject to the equations of motion as constraints. **todo** Here the final time t_1 is assumed to be known, but in general it's not: include t_1 in the optimization, or run several optimizations to find optimal t_1^*.

The cost function of the constrained problem reads

$$\widetilde{\mathcal{J}}(\mathbf{u}; \lambda) = \frac{1}{2}\int_{\tau=0}^{t_1} \mathbf{u}^T(\tau)\mathbf{u}(\tau)d\tau + \lambda^T \left(\mathbf{x}_1 - \int_{\tau=0}^{t_1} \Phi(t_1, \tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau \right) \ .$$

Variation w.r.t. λ gives the solution of the equation, i.e. the constraints. Variation w.r.t. the control $\mathbf{u}(\tau)$ gives the optimal control as a function of λ

$$\mathbf{u}(\tau) = \mathbf{B}^T(\tau)\Phi^T(t_1, \tau)\lambda \ .$$

Replacing this expression in the solution of the dynamical system, an expression of λ in terms of the final state $\mathbf{x}_1$ follows,

$$\mathbf{x}_1 = \int_{\tau=0}^{t_1} \Phi(t_1,\tau)\mathbf{B}(\tau)\mathbf{B}^T(\tau)\Phi^T(t_1,\tau)d\tau\lambda = \mathbf{W}_c(t_1)\lambda \ ,$$

having introduced the definition of the **controllability Gramian**, $\mathbf{W}_c(t_1)$.

- If $\mathbf{x}_1 \in R(\mathbf{W}_c(t_1))$ is in the range of the controllability Gramian, at least a value of the Lagrange multiplier λ that satisfies the relation - and thus a control driving the system from $\mathbf{0}$ to $\mathbf{x}_1$ - exists.
- If the controllability Gramian is full-rank, it's invertible, the solution of the problem is unique and reads

  $$\lambda = \mathbf{W}_c^{-1}(t_1)\mathbf{x}_1 \ ,$$

  and the cost of the control is

  $$J = \frac{1}{2}\mathbf{x}_1^T\mathbf{W}_c^{-1}\int_{\tau=0}^{t_1} \Phi(t_1,\tau)\mathbf{B}(\tau)\mathbf{B}^T(\tau)\Phi^T(t_1,\tau)d\tau\mathbf{W}_c^{-1}\mathbf{x}_1 = \frac{1}{2}\mathbf{x}_1^T\mathbf{W}_c^{-1}\mathbf{x}_1 \ .$$

- In general, using singular value decomposition of the Gramian $\mathbf{W}_c = \mathbf{U}\Sigma\mathbf{U}^*$ (as the Gramian is symmetric, $\mathbf{V} = \mathbf{U}$)

  $$\mathbf{u}_{\mathbf{x}_1} = \Sigma\mathbf{u}_\lambda \ ,$$

  with $\mathbf{u}_\mathbf{v} := \mathbf{U}^*\mathbf{v}$, the projection of the vector $\mathbf{v}$ onto the unit orthogonal base of $\mathbb{R}^n$. A solution $\mathbf{u}_\lambda$ exists if zero singular values $\sigma_i = 0$ correspond to null projection of the vector $\mathbf{x}_1$ onto the ith vector of the base $\mathbf{u}_i^*\mathbf{x}_1 = \left\{\mathbf{u}_{\mathbf{x}_1}\right\}_i = 0$. Otherwise a solution of the equation does not exist: in this situation, there's no control law driving the system into $\mathbf{x}_1$, an error in the final state exists, and the problem can be formulated as a optimization problem weighting both the control and the error (resulting in a least square solution of a linear system with no solution?).

Controllability Gramian satisfies the *Lyapunov equation*

$$\dot{\mathbf{W}}_c(t) = \mathbf{B}(t)\mathbf{B}^T(t) + \mathbf{A}(t)\mathbf{W}_c(t) + \mathbf{W}_c(t)\mathbf{A}^T(t) \ ,$$

with initial conditions $\mathbf{W}_c(0) = \mathbf{0}$, as the integral vanishes with the same values of the lower and upper extremes of integration.

Lyapunov equation for the controllability Graminan

$$\begin{aligned}\frac{d}{dt}\mathbf{W}_c(t) &= \frac{d}{dt}\int_{\tau=0}^{t} \Phi(t,\tau)\mathbf{B}(\tau)\mathbf{B}^T(\tau)\Phi^T(t,\tau)d\tau = \\ &= \mathbf{B}(t)\mathbf{B}^T(t) + \int_{\tau=0}^{t} \partial_t\Phi(t,\tau)\mathbf{B}(\tau)\mathbf{B}^T(\tau)\Phi^T(t,\tau)d\tau + \int_{\tau=0}^{t} \Phi(t,\tau)\mathbf{B}(\tau)\mathbf{B}^T(\tau)\partial_t\Phi^T(t,\tau)d\tau = \\ &= \mathbf{B}(t)\mathbf{B}^T(t) + \mathbf{A}(t)\mathbf{W}_c(t) + \mathbf{W}_c(t)\mathbf{A}^T(t) \ ,\end{aligned}$$

as $\Phi(t,t) = \mathbf{I}$, and $\partial_t\Phi(t,\tau) = \mathbf{A}(t)\Phi(t,\tau)$.

Controllability for LTI systems

For LTI systems $\Phi(t,t_0) = e^{\mathbf{A}(t-t_0)}$. The controllability Gramian becomes

$$\mathbf{W}_c(t) = \int_{\tau=0}^{t} e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{B}^T e^{\mathbf{A}^T(t-\tau)}d\tau$$

A vector $\mathbf{v}$ belongs to the null space of $\left(e^{\mathbf{A}t}\mathbf{B}\right)^T$ if

$$0 = \mathbf{v}^T e^{\mathbf{A}t}\mathbf{B} \; .$$

Using the definition of the exponential matrix $e^{\mathbf{A}t} := \sum_{n=0}^{+\infty} \frac{\mathbf{A}^n t^n}{n!}$, it follows

$$0 = \mathbf{v}^T \sum_{n=0}^{+\infty} \frac{\mathbf{A}^n t^n}{n!} \mathbf{B}$$

Thus…, a vector $\mathbf{v}$ is in the null-space of $\mathbf{W}_c$ (symmetric! Be precise and use *orthogonality between range and kernel of the adjoint* if it's orthogonal to the columns of the matrix

$$\mathcal{C} = \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \dots & \mathbf{A}^{n-1}\mathbf{B} \end{bmatrix} \; ,$$

i.e. $\mathbf{v}^T \mathcal{C} = \mathbf{0}$. If the column space of matrix $\mathcal{C}$ spans $\mathbb{R}^n$, then the system is fully controllable.

Obs. If $\mathbf{v}^T \mathbf{A}^n \mathbf{B}$ for $n = 0 : N-1$, thus it holds for all the integer exponent $n \geq N$, as a consequence of the *Cayley-Hamilton theorem*, i.e. $\mathbf{A}^n$ can be written as a linear combination of the powers $\mathbf{A}^k$, $k = 0 : n-1$.

29.2 Time-discrete linear systems

CHAPTER

THIRTY

LYAPUNOV EQUATION

Examples.

- *Controllability* Gramian
- *Observability* Gramian
- In optimal control
- Variance of the state in linear system forced by white Gaussian noise
- …

Part X

Control Theory

CHAPTER

THIRTYONE

INTRODUCTION TO CONTROL METHODS

System behavior is treated in detailed in *System Theory* section. Control theory targets a desired behavior of the system, ensuring stability and performance (in terms of desired output, error, response velocity, input effort, robustness to noise and model uncertainty,…)

Links

- *Ordinary differential equations*
- *System theory*

Table of contents

- Frequency domain control of linear systems
- State estimation and observer
- Optimal control
- System identification

Features

Domain.

- Frequency domain for linear systems: root locus, Nyquist criterion and Bode criteria,…
- Time domain: *optimal control*, robust optimal control,…

Full-state, partial-state feedback.

- Full-state feedback: the complete state of the system can be observed and used for the feedback
- Partial-state feedback

Observers, e.g. Kalman filter, tries to estimate the state of a system from its output. Later, this estimatio can be used for feedback.

Model-based/model-free control.

31.1 Frequency domain control

31.1.1 Stability of closed-loop SISO systems

Cauchy argument principle

For the *Cauchy argument principle*, the difference of argument of function $F(s)$ when s performs a counter-clockwise loop over th contour Γ enclosing poles p_n, and zeros z_d of the function $F(s)$ reads

$$\Delta\text{arg}\{F(s)\} = \sum_n \text{arg}(s - z_n) - \sum_d \text{arg}(s - p_d) = Z2\pi - P2\pi = (Z - P)2\pi \ ,$$

and thus the the diagram of function $F(s)$ performs

$$N = Z - P \tag{31.1}$$

loops around the origin $0 + i0$ of the complex plane.

Nyquist stability criterion

Nyquist stability criterion provides some conditions for the stability of a closed loop transfer function

$$G_c(s) := \frac{G(s)}{1 + G(s)} \ ,$$

being $G(s)$ the open loop transfer function, and the feedback with the opposite of the output ($y = G(s)\,(u - y)$).

The poles of the closed loop systems are the zeros of the function $1 + G(s)$. If the closed-loop system is asymptotically stable, it must have no pole with positive real part. As the complex variable s performs a clockwise (and thus, the signs of the relation change w.r.t. Cauchy argument criterion) loop over **Nyquist path** (semicircle in the RHS half-plane; then **todo** discuss the case where poles and zeros are on the imaginary axis…deform the path…). No zero of $1 + G(s)$ with positive real part means $Z = 0$. Using the relation between poles, zeros and loops around the origin given by Cauchy argument principle (31.1), and recalling the opposite direction, it follows that

$$N = P \ ,$$

in order to have $Z = 0$, i.e. the diagram of $1 + G(s)$ must perform N counter-clockwise loops around $0 + i0$ equal to the number of its poles with positive real parts. As the poles of $1 + G(s)$ are the same as the poles of $G(s)$, it's possible to formulate Nyquist criterion looking at $G(s)$, as

---

Theorem 31.1.1 (Nyquist theorem)

In order for the closed-loop system to be asymptotically stable, the diagram of the open-loop transfer function $G(s)$ must perform a number N of counter-clocwise loops around the *critical point* $-1 + i\,0$ equal to the number of its zeros with positve real part.

Bode stability criterion for minimal phase systems

Minimal phase systems

Definition 31.1.1 (Minimal phase systems)

…

…

Theorem 31.1.2 (Bode stability criterion)

…

Phase margin

Gain margin

31.2 Optimal control

- Deterministic/stochastic signals
- Full-state feedback, observer and full-state on observed state, output feedback

Deterministic signals.

- *Full-state feedback*, $\mathbf{u} = -\mathbf{K}\mathbf{x}$
- *Observer*, for state estimation
- *Full-state feed back on the observed state*, $\mathbf{u} = -\mathbf{K}\mathbf{o}$
- *Output feedback*, $\mathbf{u} = -\mathbf{K}\mathbf{y}$

Stochastic signals.

- *Kalman filter*, i.e. optimal observer w.r.t. white noise system exogenous input and measurement noise

31.2.1 Full-state feedback

Optimal control can be recast as a **constrained optimization problem**, J, where an extreme - optimum - of an objective function must be found, subject to constraints that include the equations of motion. Some constraints may be included into an augmented objective function $\widetilde{J}$ with the methods of *Lagrange multipliers*.

Different models - governing equations

- Generic ODE

$$\mathbf{M}\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$$

- Linear ODE for deterministic signals

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}_u\mathbf{u} + \mathbf{B}_d\mathbf{d} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}_u\mathbf{u} + \mathbf{D}_d\mathbf{d} + \mathbf{D}_r\mathbf{r} \end{cases}$$

- Linear SDE for stochastic signals. Exogenous inputs like disturbances **d** and measuerement noise **r** can be treated as stochastic processes. Thus, the equations are stochastic equations. As an example, if these noise are **white noise** they can be interpreted as the time derivative of Wiener processes, and the SDE can be written as

$$d\mathbf{x} = \mathbf{A}\mathbf{x}\,dt + \mathbf{B}_u\mathbf{u}\,dt + \mathbf{B}_d d\mathbf{w}_d \ ,$$

  begin "$d\mathbf{w}_d = \mathbf{d}\,dt$". While this may look like a useless trick, it helps us recallling that $|d\mathbf{w}_d|^2 \sim dt$, when evaluating covariances.

Different approaches to the solution

Different approaches to the solution can be used, and help for a detailed comprehension of the topics.

- Variational apporach to constrained optimization,

$$J(\mathbf{x}, \mathbf{u}) = \int_{\tau=t}^{T} C(\mathbf{x}(\tau), \mathbf{u}(\tau))d\tau + D(\mathbf{x}(T)) - \int_{\tau=0}^{T} \lambda^T \left(\dot{\mathbf{x}} - \mathbf{f}(\mathbf{x}, \mathbf{u})\right) \ ,$$

  where the equations of motion are constraints inserted in the objective function with the method of Lagrange multipliers.

- Hamilton-Bellman-Jacobi equation

$$V(\mathbf{x}_t, t; \mathbf{u}) = \int_{\tau=t}^{T} C(\mathbf{x}(\tau), \mathbf{u}(\tau))d\tau + D(\mathbf{x}(T)) \ ,$$

  with the dynamics of the system subject to the governing equation $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$, and the initial condition for the value function - the tail cost function - $x(t) = x_t$. These constraints can be explicitly applied if an expression of the solution of the dynamical equation exists, or they can be added with the method of Lagrange multipliers.

- …

A common choice of the running cost $C(\mathbf{x}, \mathbf{u})$ and the final cost $D(\mathbf{x}(T))$ are

$$C(\mathbf{x}(\tau), \mathbf{u}(\tau)) = \frac{1}{2}\left\{\mathbf{x}^T(\tau)\mathbf{Q}\mathbf{x}(\tau) + \mathbf{u}^T(\tau)\mathbf{R}\mathbf{u}(\tau)\right\}$$
$$D(\mathbf{x}(T)) = \frac{1}{2}\mathbf{x}^T(T)\mathbf{Q}_T\mathbf{x}(T) \ .$$

Finite time vs. Infinite time horizon.

…

Generic ODE without exogenous inputs

$$\mathbf{M}\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}) \ ,$$

with initial condition $\mathbf{x}(0) = \mathbf{x}_0$.

The objective function combines (weights) the error on a desired performance and the control input, in order to get the desired behavior with feasible control (that can be provided by actuators, without saturation, avoiding unnecessary high power input and too sharp behavior,…)

As an example, if the goal of the control $\mathbf{u}$ is to keep the system around $\mathbf{x} = \mathbf{0}$, the cost function to be minimized can be designed as

$$J = \int_{t=0}^{T} \frac{1}{2} \begin{bmatrix} \mathbf{x}^T & \mathbf{u}^T \end{bmatrix} \begin{bmatrix} \mathbf{Q} & \mathbf{S} \\ \mathbf{S}^T & \mathbf{R} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{u} \end{bmatrix} dt + \frac{1}{2}\mathbf{x}^T(T)\mathbf{Q}_T\mathbf{x}(T) \ ,$$

with $\mathbf{Q} \geq 0$, and $\mathbf{R} > 0$ and symmetric.

Constrained optimization

$$\widetilde{J}(\mathbf{x}, \mathbf{u}; \lambda) = \int_{\tau=0}^{T} C\left(\mathbf{x}(\tau), \mathbf{u}(\tau)\right) \, d\tau + D\left(\mathbf{x}(T)\right) - \int_{\tau=0}^{T} \lambda^T \left(\mathbf{M}\dot{\mathbf{x}} - \mathbf{f}(\mathbf{x}(\tau), \mathbf{u}(\tau))\right) \, d\tau \ .$$

$$\begin{aligned} 0 = \delta\widetilde{J} = & \int_{t=0}^{T} \delta\mathbf{x}^T \left(\partial_{\mathbf{x}} C + \partial_{\mathbf{x}}\mathbf{f}^T\lambda + \mathbf{M}^T\dot{\lambda}\right) d\tau + \\ & + \int_{t=0}^{T} \delta\mathbf{u}^T \left(\partial_{\mathbf{u}} C + \partial_{\mathbf{u}}\mathbf{f}^T\lambda\right) d\tau + \\ & + \delta\mathbf{x}^T(T)\partial_{\mathbf{x}_T} D + \\ & + \int_{t=0}^{T} \delta\lambda^T(\mathbf{M}\dot{\mathbf{x}} - \mathbf{f}(\mathbf{x}, \mathbf{u}))d\tau + \\ & - \delta\mathbf{x}\mathbf{M}^T\lambda\Big|_{t=0}^{T} \ . \end{aligned}$$

So that

$$\begin{aligned} \delta\lambda(t): \quad & \mathbf{M}\dot{\mathbf{x}} = \mathbf{f} \\ & \mathbf{x}(0) = \mathbf{x}_0 \\ \delta\mathbf{x}(t): \quad & \mathbf{M}^T\dot{\lambda} = -\partial_{\mathbf{x}}\mathbf{f}^T\lambda - \partial_{\mathbf{x}} C \\ \delta\mathbf{x}(T): \quad & \mathbf{M}^T\lambda(T) = \partial_{\mathbf{x}_T} D \\ \delta\mathbf{u}(t): \quad & \mathbf{0} = \partial_{\mathbf{u}} C + \partial_{\mathbf{u}}\mathbf{f}^T\lambda \ . \end{aligned} \tag{31.2}$$

The equations can be recast after the definition of the Hamiltonian of the system $H(\mathbf{x}, \mathbf{u}, \lambda) = C(\mathbf{x}, \mathbf{u}) + \lambda^T\mathbf{f}(\mathbf{x}, \mathbf{u})$ as

$$\begin{aligned} \delta\lambda(t): \quad & \mathbf{0} = -\mathbf{M}\dot{\mathbf{x}} + \partial_\lambda H = -\mathbf{M}\dot{\mathbf{x}} + \mathbf{f} \\ & \mathbf{x}(0) = \mathbf{x}_0 \\ \delta\mathbf{x}(t): \quad & \mathbf{M}^T\dot{\lambda} = -\partial_{\mathbf{x}} H \\ \delta\mathbf{x}(T): \quad & \mathbf{M}^T\lambda(T) = \partial_{\mathbf{x}_T} D \\ \delta\mathbf{u}(t): \quad & \mathbf{0} = \partial_{\mathbf{u}} H \ . \end{aligned} \tag{31.3}$$

Gradient descent

- Start from a control law $\mathbf{u}^{(0)}(t)$,
- the state equation is integrated forward in time to get $\mathbf{x}^{(0)}(t)$,
- Lagrange multiplier equation is integrated backward in time to get $\lambda^{(0)}(t)$,
- The control law is updated with an increment $\delta\mathbf{u}$ that's proportional to the gradient of the cost function (with "opposite direction" to get $\delta\widetilde{J} < 0$), i.e. $\mathbf{u}^{(1)}(t) = \mathbf{u}^{(0)}(t) + \delta\mathbf{u}^{(0)}(t)$ with $\delta\mathbf{u}^{(0)}(t) = -c\nabla_{\mathbf{u}(t)}J^{(0)}$, with a positive step $c > 0$ and $\nabla_{\mathbf{u}}J = \mathbf{S}^T\mathbf{x} + \mathbf{R}\mathbf{u} + \partial_{\mathbf{u}}\mathbf{f}^T\lambda$.
- Repeat previous steps with the updated control law, until convergence.

HJB equation - Evaluation equation

Without dynamical equations as a constraint introduced with Lagrange multipliers

$$V(\mathbf{x}_t, t) = \int_{\tau=t}^{T} C(\mathbf{x}(\tau), \mathbf{u}(\tau))d\tau + D(\mathbf{x}(T)) \ ,$$

subject to the dynamical equations of motion. The value function is a function of two arguments, $\mathbf{x}_t$, and t, and the first argument is a function of t. Thus the ordinary derivative w.r.t. time t reads

$$\frac{d}{dt}V(\mathbf{x}_t, t) = \partial_t V(\mathbf{x}_t, t) + \partial_{\mathbf{x}} V\,\dot{\mathbf{x}} = \partial_t V(\mathbf{x}_t, t) + \partial_{\mathbf{x}} V\,\mathbf{f} \ ,$$

or, using the definition and the rule of derivative for integrals

$$\frac{d}{dt}V(\mathbf{x}_t, t) = -C(\mathbf{x}(t), \mathbf{u}(t)) \ .$$

Comparing the two expressions of the time derivative, Hamilton-Bellman-Jacobi equation for evaluating a given control $\mathbf{u}$ follows

$$0 = \partial_t V(\mathbf{x}_t, t) + \partial_{\mathbf{x}} V(\mathbf{x}_t, t)\,\mathbf{f}(\mathbf{x}_t, \mathbf{u}_t) + C(\mathbf{x}_t, \mathbf{u}_t) \ .$$

With dynamical equations as a constraint introduced with Lagrange multipliers

$$V(\mathbf{x}_t, t; \lambda) = \int_{\tau=t}^{T} C(\mathbf{x}(\tau), \mathbf{u}(\tau))d\tau + D(\mathbf{x}(T)) - \int_{\tau=t}^{T} \lambda^T(\mathbf{M}\dot{\mathbf{x}} - \mathbf{f})d\tau \ ,$$

so that the variation (where's the variation w.r.t. t? Integration extreme is not prescribed) reads

$$\begin{aligned}
\delta V(\mathbf{x}_t, t; \mathbf{u}, \lambda) = &\int_{\tau=t}^{T} \left\{\delta\mathbf{x}^T\partial_{\mathbf{x}}C + \delta\mathbf{u}^T\partial_{\mathbf{u}}C\right\} d\tau + \delta\mathbf{x}_T\partial_{\mathbf{x}_T}D(\mathbf{x}(T)) + \\
&- \int_{\tau=t}^{T} \delta\lambda^T(\mathbf{M}\dot{\mathbf{x}} - \mathbf{f})d\tau + \int_{\tau=t}^{T} \delta\mathbf{x}^T\mathbf{M}^T\dot{\lambda}\,d\tau + \\
&- \delta\mathbf{x}^T(T)\mathbf{M}^T\lambda(T) + \delta\mathbf{x}_t^T\mathbf{M}^T\lambda(t) + \\
&+ \int_{\tau=t}^{T} \left\{\delta\mathbf{x}^T\partial_{\mathbf{x}}\mathbf{f}^T\lambda + \delta\mathbf{u}^T\partial_{\mathbf{u}}\mathbf{f}^T\lambda\right\} d\tau
\end{aligned}$$

All the variations but $\delta\mathbf{x}_t$ are identically zero if the equations (31.6) are satisfied, the variation w.r.t. $\delta\mathbf{x}_t$ shows that the sensitivity to the initial state in the value function $\mathbf{x}_t$ equals the value of the Lagrange multiplier $\lambda(t)$,

$$\nabla_{\mathbf{x}_t}V(\mathbf{x}_t, t; ...) = \lambda(t) \ ,$$

with $\lambda(t)$ part of the solution of the equations (31.6). Thus the Lagrange multiplier $\lambda(t)$ provides an information (first order, sensitivity) about the change in the value function $V(\mathbf{x}_t, t)$ for small changes of the intial state $\mathbf{x}_t$, i.e.

$$\delta_{\mathbf{x}_t} V(\mathbf{x}_t, t) \sim \delta \mathbf{x}_t^T \nabla_{\mathbf{x}_t} V + o(|\delta \mathbf{x}_t|) = \delta \mathbf{x}_t^T \lambda(t) + o(|\delta \mathbf{x}_t|) \ .$$

todo

- useful to exploit adjoint info for sensitivity

HJB equation - Optimality equation

$$\begin{aligned} V^*(\mathbf{x}_t, t) = \min_{\mathbf{u}} V(\mathbf{x}_t, t) = &= \min_{\mathbf{u}} \left\{ \int_{\tau=t}^{T} C(\mathbf{x}(\tau), \mathbf{u}(\tau)) d\tau + D(\mathbf{x}(T)) \right\} = \\ &= \int_{\tau=t}^{T} C(\mathbf{x}(\tau), \mathbf{u}^*(\tau)) d\tau + D(\mathbf{x}(T)) \ , \end{aligned}$$

subject to the dynamical equations of motion. The value function is a function of two arguments, $\mathbf{x}_t$, and t, and the first argument is a function of t. Thus the ordinary derivative w.r.t. time t reads

$$\frac{d}{dt} V^*(\mathbf{x}_t, t) = \partial_t V^*(\mathbf{x}_t, t) + \partial_{\mathbf{x}} V^* \, \dot{\mathbf{x}} = \partial_t V^*(\mathbf{x}_t, t) + \partial_{\mathbf{x}} V^* \, \mathbf{f}(\mathbf{x}_t, \mathbf{u}_t^*) \ ,$$

or, using the definition and the rule of derivative for integrals

$$\frac{d}{dt} V^*(\mathbf{x}_t, t) = -C(\mathbf{x}(t), \mathbf{u}^*(t)) \ .$$

Comparing the two expressions of the time derivative, Hamilton-Bellman-Jacobi equation for evaluating a given control $\mathbf{u}$ follows

$$0 = \partial_t V^*(\mathbf{x}_t, t) + \partial_{\mathbf{x}} V^*(\mathbf{x}_t, t) \, \mathbf{f}(\mathbf{x}_t, \mathbf{u}_t^*) + C(\mathbf{x}_t, \mathbf{u}_t^*) \ ,$$

or

$$0 = \partial_t V^*(\mathbf{x}_t, t) + \min_{\mathbf{u}} \left\{ \partial_{\mathbf{x}} V^*(\mathbf{x}_t, t) \, \mathbf{f}(\mathbf{x}_t, \mathbf{u}_t) + C(\mathbf{x}_t, \mathbf{u}_t) \right\} \ .$$

Linear system

Let the dynamical equation of a system be

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}_y \mathbf{x} + \mathbf{D}_y \mathbf{u} \\ \mathbf{z} = \mathbf{C}_z \mathbf{x} + \mathbf{D}_z \mathbf{u} \ , \end{cases}$$

with measurment output $\mathbf{y}$ and performance output $\mathbf{z}$, and with running and final cost

$$\begin{aligned} C(\mathbf{x}, \mathbf{u}) = &\frac{1}{2} \left(\mathbf{z}^T \mathbf{Q} \mathbf{z} + \mathbf{u}^T \mathbf{R} \mathbf{u} \right) = \\ &= \frac{1}{2} \begin{bmatrix} \mathbf{x} \\ \mathbf{u} \end{bmatrix}^T \begin{bmatrix} \mathbf{C}_z^T \mathbf{Q} \mathbf{C}_z & \mathbf{C}_z^T \mathbf{Q} \mathbf{D}_z \\ \mathbf{D}_z^T \mathbf{Q} \mathbf{C}_z & \mathbf{D}_z^T \mathbf{Q} \mathbf{D}_z + \mathbf{R} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{u} \end{bmatrix} = \\ &= \frac{1}{2} \begin{bmatrix} \mathbf{x} \\ \mathbf{u} \end{bmatrix}^T \begin{bmatrix} \widetilde{\mathbf{Q}} & \mathbf{S} \\ \mathbf{S}^T & \widetilde{\mathbf{R}} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{u} \end{bmatrix} \ , \end{aligned}$$

$$D(\mathbf{x}_T) = \frac{1}{2} \mathbf{x}^T(T) \mathbf{Q}_T \mathbf{x}(T) \ .$$

The equations (31.6) become

$$
\begin{aligned}
\delta\lambda(t): \quad & \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\
& \mathbf{x}(0) = \mathbf{x}_0 \\
\delta\mathbf{x}(t): \quad & \dot{\lambda} = -\mathbf{A}^T\lambda - \widetilde{\mathbf{Q}}\mathbf{x} - \mathbf{S}\mathbf{u} \\
\delta\mathbf{x}(T): \quad & \lambda(T) = \mathbf{Q}_T\mathbf{x}(T) \\
\delta\mathbf{u}(t): \quad & \mathbf{0} = \widetilde{\mathbf{R}}\mathbf{u} + \mathbf{S}^T\mathbf{x} + \mathbf{B}^T\lambda \ .
\end{aligned} \tag{31.4}
$$

The weights onf the control are definite positive, $\widetilde{\mathbf{R}} > 0$, and thus invertible. The control can be written as a function of the state and the co-state as

$$
\mathbf{u} = -\widetilde{\mathbf{R}}^{-1}\left(\mathbf{S}^T\mathbf{x} + \mathbf{B}^T\lambda\right) \ .
$$

Now the closed loop system reads

$$
\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\lambda} \end{bmatrix}
\begin{bmatrix} \mathbf{A} - \mathbf{B}\widetilde{\mathbf{R}}^{-1}\mathbf{S}^T & -\mathbf{B}\widetilde{\mathbf{R}}^{-1}\mathbf{B}^T \\ -\widetilde{\mathbf{Q}} + \mathbf{S}\widetilde{\mathbf{R}}^{-1}\mathbf{S}^T & -\mathbf{A}^T + \mathbf{S}\widetilde{\mathbf{R}}^{-1}\mathbf{B}^T \end{bmatrix}
\begin{bmatrix} \mathbf{x} \\ \lambda \end{bmatrix} \ ,
$$

with initial condition $\mathbf{x}(0) = \mathbf{x}_0$ and final condition $\lambda(T) = \mathbf{Q}_T\mathbf{x}(T)$. The solution can be written as

$$
\begin{bmatrix} \mathbf{x}(t) \\ \lambda(t) \end{bmatrix} = \Psi(t, t_0)\begin{bmatrix} \mathbf{x}(t_0) \\ \lambda(t_0) \end{bmatrix} = \begin{bmatrix} \Psi_{xx}(t, t_0) & \Psi_{x\lambda}(t, t_0) \\ \Psi_{\lambda x}(t, t_0) & \Psi_{\lambda\lambda}(t, t_0) \end{bmatrix}\begin{bmatrix} \mathbf{x}(t_0) \\ \lambda(t_0) \end{bmatrix}
$$

so that the state in T reads

$$
\begin{bmatrix} \mathbf{x}(T) \\ \lambda(T) \end{bmatrix} = \begin{bmatrix} \Psi_{xx}(T, t_0) & \Psi_{x\lambda}(T, t_0) \\ \Psi_{\lambda x}(T, t_0) & \Psi_{\lambda\lambda}(T, t_0) \end{bmatrix}\begin{bmatrix} \mathbf{x}(t_0) \\ \lambda(t_0) \end{bmatrix}
$$

so that

$$
\begin{aligned}
\mathbf{0} &= -\lambda_T + \Psi_{\lambda\lambda}(T, t_0)\lambda_0 + \Psi_{\lambda x}(T, t_0)\mathbf{x}_0 = \\
&= -\mathbf{Q}_T\mathbf{x}_T + \Psi_{\lambda\lambda}(T, t_0)\lambda_0 + \Psi_{\lambda x}(T, t_0)\mathbf{x}_0 = \\
&= -\mathbf{Q}_T\left[\Psi_{xx}(T, t_0)\mathbf{x}_0 + \Psi_{x\lambda}(T, t_0)\lambda_0\right] + \Psi_{\lambda\lambda}(T, t_0)\lambda_0 + \Psi_{\lambda x}(T, t_0)\mathbf{x}_0 \ ,
\end{aligned}
$$

and thus

$$
\lambda_0 = \left(\Psi_{\lambda\lambda} - \mathbf{Q}\Psi_{x\lambda}(T, t_0)\right)^{-1}\left(\mathbf{Q}\Psi_{xx}(T, t_0) - \Psi_{\lambda x}\right)\mathbf{x}_0 \ .
$$

Value function and relation $\lambda = \mathbf{P}\mathbf{x}$

Let the value function be

$$
V(\mathbf{x}_t, t; \mathbf{u}) = \frac{1}{2}\int_{\tau=t}^{T}\left\{\mathbf{x}_\tau^T\widetilde{\mathbf{Q}}_\tau\mathbf{x}_\tau + \mathbf{x}_\tau^T\mathbf{S}_\tau\mathbf{u}_\tau + \mathbf{u}_\tau^T\mathbf{S}_\tau^T\mathbf{x}_\tau + \mathbf{u}_\tau^T\widetilde{\mathbf{R}}_\tau\mathbf{u}_\tau\right\}d\tau \ + \frac{1}{2}\mathbf{x}_T^T\mathbf{Q}_T\mathbf{x}_T \ ,
$$

subject to the equations of motion as constraints $\dot{\mathbf{x}}_\tau = \mathbf{A}_\tau\mathbf{x}_\tau + \mathbf{B}_\tau\mathbf{u}_\tau$, and the initial condition $\mathbf{x}(t) = \mathbf{x}_t$. This constraint can be introduced either 1) expressing the state $\mathbf{x}_\tau$ as a function of the initial state and the input

$$
\mathbf{x}_\tau = \Phi(\tau, t)\mathbf{x}_t + \int_{\xi=t}^{\tau}\Phi(\tau, \xi)\mathbf{B}_\xi\mathbf{u}_\xi\, d\xi \ ,
$$

or 2) with the methods of Lagrange multipliers. A **co-state** λ_t - corresponding to the Lagrange multiplier - can be evaluated as

$$
\lambda_t = \nabla_{\mathbf{x}_t}V \ .
$$

Method 1.

$$
\begin{aligned}
V(\mathbf{x}_t, t; \mathbf{u}) = \frac{1}{2}\int_{\tau=t}^{T} &\left\{ \left(\Phi(\tau,t)\mathbf{x}_t + \int_{\xi=t}^{\tau} \Phi(\tau,\xi)\mathbf{B}_\xi \mathbf{u}_\xi \, d\xi \right)^T \widetilde{\mathbf{Q}}_\tau \left(\Phi(\tau,t)\mathbf{x}_t + \int_{\xi=t}^{\tau} \Phi(\tau,\xi)\mathbf{B}_\xi \mathbf{u}_\xi \, d\xi \right) \right\} + \\
&+ \frac{1}{2}\int_{\tau=t}^{T} \left\{ \left(\Phi(\tau,t)\mathbf{x}_t + \int_{\xi=t}^{\tau} \Phi(\tau,\xi)\mathbf{B}_\xi \mathbf{u}_\xi \, d\xi \right)^T \mathbf{S}_\tau \mathbf{u}_\tau \right\} d\tau + \\
&+ \frac{1}{2}\int_{\tau=t}^{T} \left\{ \mathbf{u}_\tau^T \mathbf{S}_\tau^T \left(\Phi(\tau,t)\mathbf{x}_t + \int_{\xi=t}^{\tau} \Phi(\tau,\xi)\mathbf{B}_\xi \mathbf{u}_\xi \, d\xi \right) \right\} d\tau + \\
&+ \frac{1}{2}\int_{\tau=t}^{T} \mathbf{u}_\tau^T \widetilde{\mathbf{R}}_\tau \mathbf{u}_\tau \, d\tau + \\
&+ \frac{1}{2}\left(\Phi(T,t)\mathbf{x}_t + \int_{\xi=t}^{T} \Phi(T,\xi)\mathbf{B}_\xi \mathbf{u}_\xi \, d\xi \right)^T \mathbf{Q}_T \left(\Phi(T,t)\mathbf{x}_t + \int_{\xi=t}^{T} \Phi(T,\xi)\mathbf{B}_\xi \mathbf{u}_\xi \, d\xi \right) =
\end{aligned}
$$

and thus

$$
\begin{aligned}
\lambda_t = \nabla_{\mathbf{x}_t} V &= \\
&= \int_{\tau=t}^{T} \left(\Phi^T(\tau,t)\widetilde{\mathbf{Q}}_\tau \mathbf{x}_\tau + \Phi^T(\tau,t)\mathbf{S}_\tau \mathbf{u}_\tau \right) d\tau + \Phi^T(T,t)\mathbf{Q}_T \mathbf{x}_T = \\
&= \left\{ \int_{\tau=t}^{T} \Phi(\tau,t)^T \widetilde{\mathbf{Q}}_\tau \Phi(\tau,t) \, d\tau + \Phi^T(T,t)\mathbf{Q}_T \Phi(T,t) \right\} \mathbf{x}_t + \int_{\tau=t}^{T} \Phi^T(\tau,t)\mathbf{S}_\tau \mathbf{u}_\tau \, d\tau \, .
\end{aligned}
$$

From optimization, $\mathbf{u}_\tau = -\widetilde{\mathbf{R}}_\tau^{-1} \left(\mathbf{B}_\tau^T \lambda_\tau + \mathbf{S}_\tau^T \mathbf{x}_\tau \right)$,

Method 2.

Optimal control

HJB optimality equation

$$
0 = \partial_t V^*(\mathbf{x}_t, t) + \min_{\mathbf{u}} \left\{ \partial_{\mathbf{x}} V^*(\mathbf{x}_t, t) \, \mathbf{f}(\mathbf{x}_t, \mathbf{u}_t) + C(\mathbf{x}_t, \mathbf{u}_t) \right\} \, .
$$

$$
\begin{aligned}
\mathbf{0} = \delta_{\mathbf{u}} V &= \\
&= \int_{\tau=t}^{T} \left\{ \int_{\xi=t}^{\tau} \delta\mathbf{u}_\xi^T \mathbf{B}_\xi^T \Phi^T(\tau,\xi) \, d\xi \, \left(\widetilde{\mathbf{Q}}_\tau \mathbf{x}_\tau + \mathbf{S}_\tau \mathbf{u}_\tau \right) + \delta\mathbf{u}_\tau^T \mathbf{S}_\tau^T \mathbf{x}_\tau + \delta\mathbf{u}_\tau^T \widetilde{\mathbf{R}} \mathbf{u}_\tau + \mathbf{B}_\tau^T \Phi^T(T,\tau)\mathbf{Q}_T \mathbf{x}_T \right\} d\tau = \\
&= \text{TODO...details about variable swap in the integrals of the first term} = \\
&= \int_{\tau=t}^{T} \delta\mathbf{u}_\tau^T \left\{ \mathbf{B}_\tau^T \underbrace{\left(\int_{\xi=\tau}^{T} \Phi^T(\xi,\tau) \left(\widetilde{\mathbf{Q}}_\xi \mathbf{x}_\xi + \mathbf{S}_\xi \mathbf{u}_\xi \right) d\xi + \Phi^T(T,\tau)\mathbf{Q}\mathbf{x}_T \right)}_{\lambda_\tau} + \mathbf{S}_\tau^T \mathbf{x}_\tau + \widetilde{\mathbf{R}} \mathbf{u}_\tau \right\} d\tau \, ,
\end{aligned}
$$

and thus, from the arbitrariety of $\delta\mathbf{u}_\tau$, and since $\widetilde{\mathbf{R}}$ is required to be invertible

$$
\mathbf{u}_\tau = -\widetilde{\mathbf{R}}_\tau^{-1} \left(\mathbf{B}_\tau^T \lambda_\tau + \mathbf{S}_\tau^T \mathbf{x}_\tau \right) \, .
$$

...

If (**todo** why?) $\lambda = \mathbf{P}\mathbf{x}$,

Using HJB equation.

$$V(\mathbf{x}_t, t; \mathbf{u}) = \frac{1}{2}\int_{\tau=t}^{T} \dots d\tau + \frac{1}{2}\mathbf{x}_T^T \mathbf{Q}_T \mathbf{x}_T + \dots$$

Proportional control. This should not be an assumption, but a result of the problem *!!!*

If $\mathbf{u} = -\mathbf{G}\mathbf{x}$, the solution of the closed-loop system

$$\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{B}\mathbf{G})\,\mathbf{x}$$

reads $\mathbf{x}(\tau) = \Psi_c(\tau, t)\mathbf{x}(t)$. The value function becomes

$$\begin{aligned} V(\mathbf{x}_t, t; \mathbf{G}) &= \frac{1}{2}\mathbf{x}_t^T \int_{\tau=t}^{T} \Phi_c^T(\tau, t)\left(\widetilde{\mathbf{Q}} - \mathbf{G}^T\mathbf{S}^T - \mathbf{S}\mathbf{G} + \mathbf{G}^T\widetilde{\mathbf{R}}\mathbf{G}\right)\Phi_c(\tau, t)\,d\tau\,\mathbf{x}_t + \frac{1}{2}\mathbf{x}_t^T\Phi_c^T(T, t)\mathbf{Q}_T\Phi_c(T, t)\mathbf{x}_t = \\ &= \frac{1}{2}\mathbf{x}_t^T \left\{\int_{\tau=t}^{T} \Phi_c^T(\tau, t)\left(\widetilde{\mathbf{Q}} - \mathbf{G}^T\mathbf{S}^T - \mathbf{S}\mathbf{G} + \mathbf{G}^T\widetilde{\mathbf{R}}\mathbf{G}\right)\Phi_c(\tau, t)\,d\tau + \Phi_c^T(T, t)\mathbf{Q}_T\Phi_c(T, t)\right\}\mathbf{x}_t = \\ &= \frac{1}{2}\mathbf{x}_t^T\mathbf{P}(t)\mathbf{x}_t\ . \end{aligned}$$

Thus, the costate reads $\lambda = \nabla_{\mathbf{x}_t} V = \mathbf{P}\mathbf{x}$.

Matrix $\mathbf{P}$ satisfies a *Lyapunov equation*,

$$\dot{\mathbf{P}} = \mathbf{A}^T\mathbf{P} + \mathbf{P}\mathbf{A} + \left(\widetilde{\mathbf{Q}} - \mathbf{G}^T\mathbf{S}^T - \mathbf{S}\mathbf{G} + \mathbf{G}^T\widetilde{\mathbf{R}}\mathbf{G}\right)\ ,$$

as it can be easily found by direct computation of the time derivative of $\mathbf{P}$.

Decoupled weights

Let $\mathbf{S} = \mathbf{0}$, then $\mathbf{u} = -\widetilde{\mathbf{R}}^{-1}\mathbf{B}^T\lambda$, and thus the control is a linear combination of the co-state.

The transformation $\mathbf{v} := \mathbf{u} + \widetilde{\mathbf{R}}^{-1}\mathbf{S}^T\mathbf{x}$, make a coupled objective function for the original system a decoupeld objective function for a modified system.

todo

- Are Ψ matrices invertible?
- Properties of the Hamilton matrix…
- What's a conjugate point?
- …

Coupled state-input weights

$$J = \frac{1}{2}\int_{\tau=0}^{T} \begin{bmatrix}\mathbf{x}\\ \mathbf{u}\end{bmatrix}^T \begin{bmatrix}\mathbf{Q} & \mathbf{S}\\ \mathbf{S}^T & \mathbf{R}\end{bmatrix}\begin{bmatrix}\mathbf{x}\\ \mathbf{u}\end{bmatrix} d\tau + \frac{1}{2}\mathbf{x}_T^T\mathbf{Q}_T\mathbf{x}_T$$

Decoupled state-input weights

The problem can be decoupled with a transformation

$$\begin{bmatrix}\mathbf{x}\\ \mathbf{u}\end{bmatrix} = \begin{bmatrix}\mathbf{I} & \mathbf{0}\\ \mathbf{T} & \mathbf{I}\end{bmatrix}\begin{bmatrix}\tilde{\mathbf{x}}\\ \tilde{\mathbf{u}}\end{bmatrix}$$

so that

$$\begin{aligned}\begin{bmatrix}\mathbf{x}\\ \mathbf{u}\end{bmatrix}^T \begin{bmatrix}\mathbf{Q} & \mathbf{S}\\ \mathbf{S}^T & \mathbf{R}\end{bmatrix}\begin{bmatrix}\mathbf{x}\\ \mathbf{u}\end{bmatrix} &= \begin{bmatrix}\tilde{\mathbf{x}}\\ \tilde{\mathbf{u}}\end{bmatrix}^T \begin{bmatrix}\mathbf{I} & \mathbf{T}^T\\ \mathbf{0} & \mathbf{I}\end{bmatrix}\begin{bmatrix}\mathbf{Q} & \mathbf{S}\\ \mathbf{S}^T & \mathbf{R}\end{bmatrix}\begin{bmatrix}\mathbf{I} & \mathbf{0}\\ \mathbf{T} & \mathbf{I}\end{bmatrix}\begin{bmatrix}\tilde{\mathbf{x}}\\ \tilde{\mathbf{u}}\end{bmatrix} = \\ &= \begin{bmatrix}\tilde{\mathbf{x}}\\ \tilde{\mathbf{u}}\end{bmatrix}^T \begin{bmatrix}\mathbf{Q}+\mathbf{T}^T\mathbf{S}^T & \mathbf{S}+\mathbf{T}^T\mathbf{R}\\ \mathbf{S}^T & \mathbf{R}\end{bmatrix}\begin{bmatrix}\mathbf{I} & \mathbf{0}\\ \mathbf{T} & \mathbf{I}\end{bmatrix}\begin{bmatrix}\tilde{\mathbf{x}}\\ \tilde{\mathbf{u}}\end{bmatrix} = \\ &= \begin{bmatrix}\tilde{\mathbf{x}}\\ \tilde{\mathbf{u}}\end{bmatrix}^T \begin{bmatrix}\mathbf{Q}+\mathbf{T}^T\mathbf{S}^T+\mathbf{S}\mathbf{T}+\mathbf{T}^T\mathbf{R}\mathbf{T} & \mathbf{S}+\mathbf{T}^T\mathbf{R}\\ \mathbf{S}^T+\mathbf{R}\mathbf{T} & \mathbf{R}\end{bmatrix}\begin{bmatrix}\tilde{\mathbf{x}}\\ \tilde{\mathbf{u}}\end{bmatrix},\end{aligned}$$

Extra-diagonal terms are identically zero if $\mathbf{T} = -\mathbf{R}^{-1}\mathbf{S}^T$. The coordinate transformation becomes

$$\begin{aligned}\mathbf{x} &= \tilde{\mathbf{x}}\\ \mathbf{u} &= -\mathbf{R}^{-1}\mathbf{S}^T\tilde{\mathbf{x}} + \tilde{\mathbf{u}}\end{aligned}$$

and the running cost reads

$$\begin{bmatrix}\tilde{\mathbf{x}}\\ \tilde{\mathbf{u}}\end{bmatrix}^T \begin{bmatrix}\mathbf{Q}-\mathbf{S}\mathbf{R}^{-1}\mathbf{S}^T & \mathbf{0}\\ \mathbf{0} & \mathbf{R}\end{bmatrix}\begin{bmatrix}\tilde{\mathbf{x}}\\ \tilde{\mathbf{u}}\end{bmatrix}.$$

The linear system

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}\\ \mathbf{y} &= \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}\end{aligned}$$

becomes

$$\begin{aligned}\dot{\tilde{\mathbf{x}}} &= \left(\mathbf{A}-\mathbf{B}\mathbf{R}^{-1}\mathbf{S}^T\right)\tilde{\mathbf{x}} + \mathbf{B}\tilde{\mathbf{u}}\\ \mathbf{y} &= \left(\mathbf{C}-\mathbf{D}\mathbf{R}^{-1}\mathbf{S}^T\right)\tilde{\mathbf{x}} + \mathbf{D}\tilde{\mathbf{u}}\end{aligned}$$

Decoupled system

$$J = \frac{1}{2}\int_{\tau=0}^{T}\begin{bmatrix}\mathbf{x}\\ \tilde{\mathbf{u}}\end{bmatrix}^T \begin{bmatrix}\hat{\mathbf{Q}} & \mathbf{0}\\ \mathbf{0} & \mathbf{R}\end{bmatrix}\begin{bmatrix}\mathbf{x}\\ \tilde{\mathbf{u}}\end{bmatrix} d\tau + \frac{1}{2}\mathbf{x}_T^T\mathbf{Q}_T\mathbf{x}_T - \int_{\tau=0}^{t}\lambda^T\left(\dot{\mathbf{x}} - \hat{\mathbf{A}}\mathbf{x} + \mathbf{B}\tilde{\mathbf{u}}\right)d\tau.$$

Optimization gives

$$\begin{aligned}\delta\lambda(t):&\quad \dot{\mathbf{x}} = \hat{\mathbf{A}}\mathbf{x} + \mathbf{B}\tilde{\mathbf{u}}\\ &\quad \mathbf{x}(0) = \mathbf{x}_0\\ \delta\mathbf{x}(t):&\quad \dot{\lambda} = -\hat{\mathbf{A}}^T\lambda - \hat{\mathbf{Q}}\mathbf{x}\\ \delta\mathbf{x}(T):&\quad \lambda(T) = \mathbf{Q}_T\mathbf{x}(T)\\ \delta\mathbf{u}(t):&\quad \mathbf{0} = \mathbf{R}\tilde{\mathbf{u}} + \mathbf{B}^T\lambda\,.\end{aligned} \tag{31.5}$$

The modified optimal control reads

$$\tilde{\mathbf{u}} = -\mathbf{R}^{-1}\mathbf{B}^T\lambda$$

and thus

$$\mathbf{u} = -\mathbf{R}^{-1}\left(\mathbf{S}^T\mathbf{x} + \mathbf{B}^T\lambda\right) \ .$$

$$\mathbf{x}(t) = \hat{\Phi}(t,t_0)\mathbf{x}_0 + \int_{\tau=t_0}^{t} \hat{\Phi}(t,\tau)\mathbf{B}_\tau\tilde{\mathbf{u}}(\tau)d\tau$$

$$\lambda_t = \partial_{\mathbf{x}_t} V = \left[\int_{\tau=t}^{T} \hat{\Phi}^T(\tau,t)\hat{\mathbf{Q}}_\tau\hat{\Phi}(\tau,t)\,d\tau + \hat{\Phi}^T(T,t)\mathbf{Q}_T\hat{\Phi}(T,t)\right]\mathbf{x}_t = \mathbf{P}(t,T)\mathbf{x}_t \ .$$

$$\partial_t\hat{\Phi}(t,\tau) = \hat{\mathbf{A}}(t)\hat{\Phi}(t,\tau) \ .$$

$$\partial_T\mathbf{P}$$

31.2.2 Full-state feedback (OLD)

Optimal control can be recast as a **constrained optimization problem**, J, where an extreme - optimum - of an objective function must be found, subject to constraints that include the equations of motion. Some constraints may be included into an augmented objective function $\widetilde{J}$ with the methods of *Lagrange multipliers*.

Finite time vs. Infinite time horizon.

…

Generic ODE without exogenous inputs

$$\mathbf{M}\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x},\mathbf{u}) \ ,$$

with initial condition $\mathbf{x}(0) = \mathbf{x}_0$.

The objective function combines (weights) the error on a desired performance and the control input, in order to get the desired behavior with feasible control (that can be provided by actuators, without saturation, avoiding unnecessary high power input and too sharp behavior,…)

As an example, if the goal of the control $\mathbf{u}$ is to keep the system around $\mathbf{x} = \mathbf{0}$, the cost function to be minimized can be designed as

$$J = \int_{t=0}^{T} \frac{1}{2}\begin{bmatrix}\mathbf{x}^T & \mathbf{u}^T\end{bmatrix}\begin{bmatrix}\mathbf{Q} & \mathbf{S}\\ \mathbf{S}^T & \mathbf{R}\end{bmatrix}\begin{bmatrix}\mathbf{x}\\ \mathbf{u}\end{bmatrix} dt + \frac{1}{2}\mathbf{x}^T(T)\mathbf{Q}_T\mathbf{x}(T) \ ,$$

with $\mathbf{Q} \geq 0$, and $\mathbf{R} > 0$ and symmetric.

Constrained optimization

$$\widetilde{J} = \frac{1}{2}\int_{t=0}^{T}\begin{bmatrix}\mathbf{x}\\ \mathbf{u}\end{bmatrix}^T\begin{bmatrix}\mathbf{Q} & \mathbf{S}\\ \mathbf{S}^T & \mathbf{R}\end{bmatrix}\begin{bmatrix}\mathbf{x}\\ \mathbf{u}\end{bmatrix} dt + \frac{1}{2}\mathbf{x}^T(T)\mathbf{Q}_T\mathbf{x}(T) - \int_{t=0}^{T}\lambda^T\left(\mathbf{M}\dot{\mathbf{x}} - \mathbf{f}(\mathbf{x},\mathbf{u})\right)$$

$$
\begin{aligned}
0 = \delta\widetilde{J} = & \\
&= \int_{t=0}^{T} \delta\mathbf{x}^T \left(\mathbf{Q}\mathbf{x} + \mathbf{S}\mathbf{u} + \partial_{\mathbf{x}}\mathbf{f}^T\lambda + \mathbf{M}^T\dot{\lambda}\right) \\
&+ \int_{t=0}^{T} \delta\mathbf{u}^T \left(\mathbf{S}^T\mathbf{x} + \mathbf{R}\mathbf{u} + \partial_{\mathbf{u}}\mathbf{f}^T\lambda\right) \\
&+ \delta\mathbf{x}^T(T)\mathbf{Q}_T\mathbf{x}(T) \\
&+ \int_{t=0}^{T} \delta\lambda^T(\mathbf{M}\dot{\mathbf{x}} - \mathbf{f}(\mathbf{x},\mathbf{u})) \\
&- \delta\mathbf{x}\mathbf{M}^T\lambda\Big|_{t=0}^{T}
\end{aligned}
$$

So that

$$
\begin{aligned}
\delta\lambda(t): &\quad \mathbf{M}\dot{\mathbf{x}} = \mathbf{f} \\
&\quad \mathbf{x}(0) = \mathbf{x}_0 \\
\delta\mathbf{x}(t): &\quad \mathbf{M}^T\dot{\lambda} = -\partial_{\mathbf{x}}\mathbf{f}^T\lambda - \mathbf{S}\mathbf{u} - \mathbf{Q}\mathbf{x} \\
\delta\mathbf{x}(T): &\quad \mathbf{M}^T\lambda(T) = \mathbf{Q}_T\mathbf{x}(T) \\
\delta\mathbf{u}(t): &\quad \mathbf{u} = -\mathbf{R}^{-1}\left(\mathbf{S}^T\mathbf{x} + \partial_{\mathbf{u}}\mathbf{f}^T\lambda\right) \ .
\end{aligned}
\tag{31.6}
$$

Solution of the problem

Replacing $\mathbf{u}$

$$
\begin{bmatrix} \mathbf{M} & \\ & \mathbf{M}^T \end{bmatrix} \begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\lambda} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ -\partial_{\mathbf{x}}\mathbf{f}^T\lambda - \mathbf{S}\mathbf{u} - \mathbf{Q}\mathbf{x} \end{bmatrix} ,
$$

with $\mathbf{u} = -\mathbf{R}^{-1}\left(\mathbf{S}^T\mathbf{x} + \partial_{\mathbf{u}}\mathbf{f}^T\lambda\right)$.

Gradient descent

- Start from a control law $\mathbf{u}^{(0)}(t)$,
- the state equation is integrated forward in time to get $\mathbf{x}^{(0)}(t)$,
- Lagrange multiplier equation is integrated backward in time to get $\lambda^{(0)}(t)$,
- The control law is updated with an increment $\delta\mathbf{u}$ that's proportional to the gradient of the cost function (with "opposite direction" to get $\delta\widetilde{J} < 0$), i.e. $\mathbf{u}^{(1)}(t) = \mathbf{u}^{(0)}(t) + \delta\mathbf{u}^{(0)}(t)$ with $\delta\mathbf{u}^{(0)}(t) = -c\nabla_{\mathbf{u}(t)}J^{(0)}$, with a positive step $c > 0$ and $\nabla_{\mathbf{u}}J = \mathbf{S}^T\mathbf{x} + \mathbf{R}\mathbf{u} + \partial_{\mathbf{u}}\mathbf{f}^T\lambda$.
- Repeat previous steps with the updated control law, until convergence.

LTI

$$
\begin{gathered}
J = \int_{t=0}^{T} \frac{1}{2}\begin{bmatrix}\mathbf{x}^T & \mathbf{u}^T\end{bmatrix} \begin{bmatrix} \mathbf{Q} & \mathbf{S} \\ \mathbf{S}^T & \mathbf{R} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{u} \end{bmatrix} dt + \frac{1}{2}\mathbf{x}^T(T)\mathbf{Q}_T\mathbf{x}(T) \\
\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \end{cases}
\end{gathered}
\tag{31.7}
$$

Constrained optimization

As $\mathbf{f}(\mathbf{x},\mathbf{u}) = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$, here $\partial_{\mathbf{x}}\mathbf{f} = \mathbf{A}$ and $\partial_{\mathbf{u}}\mathbf{f} = \mathbf{B}$. For the linear system (31.7) $\mathbf{M} = \mathbf{I}$. Thus the equations (31.6) become

$$\begin{aligned}
\delta\lambda(t): &\quad \mathbf{M}\dot{\mathbf{x}} = \mathbf{f} \\
&\quad \mathbf{x}(0) = \mathbf{x}_0 \\
\delta\mathbf{x}(t): &\quad \mathbf{M}^T\dot{\lambda} = -\mathbf{A}^T\lambda - \mathbf{S}\mathbf{u} - \mathbf{Q}\mathbf{x} \\
\delta\mathbf{x}(T): &\quad \mathbf{M}^T\lambda(T) = \mathbf{Q}_T\mathbf{x}(T) \\
\delta\mathbf{u}(t): &\quad \mathbf{u} = -\mathbf{R}^{-1}\left(\mathbf{S}^T\mathbf{x} + \mathbf{B}^T\lambda\right) \ .
\end{aligned} \tag{31.8}$$

Infinite-horizon full-state feedback

No need for an observer. The system is assumed to be stable. The augmented cost function reads

$$\widetilde{\mathcal{J}}(\mathbf{x},\mathbf{u};\lambda) = \int_{t=0}^{+\infty} \left\{ \frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} + \mathbf{x}^T\mathbf{S}\mathbf{u} + \frac{1}{2}\mathbf{u}^T\mathbf{R}\mathbf{u} - \lambda^T(\dot{\mathbf{x}} - \mathbf{A}\mathbf{x} - \mathbf{B}\mathbf{u}) \right\} dt \ ,$$

with given initial conditions $\mathbf{x}(0) = \mathbf{x}_0$, so that $\delta\mathbf{x}_0 = \mathbf{0}$. Using *calculus of variations*, the variations of the cost function w.r.t. $\mathbf{x}$, $\mathbf{u}$, λ read

$$\begin{aligned}
\delta_{\mathbf{x}}: &\quad \mathbf{Q}\mathbf{x} + \mathbf{S}\mathbf{u} + \dot{\lambda} + \mathbf{A}^T\lambda = \mathbf{0} \\
\delta_{\mathbf{u}}: &\quad \mathbf{S}^T\mathbf{x} + \mathbf{R}\mathbf{u} + \mathbf{B}^T\lambda = \mathbf{0} \\
\delta_{\lambda}: &\quad \dot{\mathbf{x}} - \mathbf{A}\mathbf{x} - \mathbf{B}\mathbf{u} = \mathbf{0}
\end{aligned}$$

From the variation w.r.t. $\mathbf{u}$, since $\mathbf{R} > 0$ and thus innvertible,

$$\mathbf{u} = -\mathbf{R}^{-1}\left(\mathbf{S}^T\mathbf{x} + \mathbf{B}^T\lambda\right)$$

Now, assuming the relation $\lambda = \mathbf{P}\mathbf{x}$, it follows

$$\begin{aligned}
\dot{\lambda} &= -\mathbf{Q}\mathbf{x} - \mathbf{S}\mathbf{u} - \mathbf{A}^T\lambda = \\
&= \left\{-\mathbf{Q} + \mathbf{S}\mathbf{R}^{-1}\left(\mathbf{S}^T + \mathbf{B}^T\mathbf{P}\right) - \mathbf{A}^T\mathbf{P}\right\}\mathbf{x} \\
\dot{\lambda} &= \dot{\mathbf{P}}\mathbf{x} + \mathbf{P}\dot{\mathbf{x}} = \\
&= \dot{\mathbf{P}}\mathbf{x} + \mathbf{P}\left(\mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}\right) = \\
&= \left\{\dot{\mathbf{P}} + \mathbf{P}\left[\mathbf{A} - \mathbf{B}\mathbf{R}^{-1}\left(\mathbf{S}^T + \mathbf{B}^T\mathbf{P}\right)\right]\right\}\mathbf{x} \ ,
\end{aligned}$$

and comparing the two different expressions of $\dot{\lambda}$, if the equality holds for any $\mathbf{x}$, the **dynamical Riccati equation** for $\mathbf{P}$ is derived as

$$\dot{\mathbf{P}} + \mathbf{P}\widetilde{\mathbf{A}} + \widetilde{\mathbf{A}}^T\mathbf{P} - \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P} + \widetilde{\mathbf{Q}} = \mathbf{0} \ ,$$

where $\widetilde{\mathbf{A}} = \mathbf{A} - \mathbf{B}\mathbf{R}^{-1}\mathbf{S}^T$ and $\widetilde{\mathbf{Q}} = \mathbf{Q} - \mathbf{S}\mathbf{R}^{-1}\mathbf{S}^T$. Riccati equation is a non-linear dynamical matrix equation in $\mathbf{P}$. Algorithms for computing the solution of dynamical and algebraic equation exists, see *Example 31.2.1*.

Once $\mathbf{P}$ is evaluated, the control law reads

$$\mathbf{u} = -\mathbf{R}^{-1}\left(\mathbf{S}^T + \mathbf{B}^T\mathbf{P}\right)\mathbf{x} \ .$$

For infinite-horizone, the algebraic equation (ARE) for the steady state is solved after setting $\dot{\mathbf{P}} = \mathbf{0}$, the solution for a LTI system is a constant matrix $\mathbf{P}$, and thus the control law is a proportional feedback on the full-state of the system,

$$\mathbf{u} = -\mathbf{G}\mathbf{x} \ ,$$

with $\mathbf{G} = \mathbf{R}^{-1}\left(\mathbf{S}^T + \mathbf{B}^T\mathbf{P}\right)$.

Example 31.2.1 (Solution of Riccati equation)

…

Properties. todo

- $\mathbf{P}$ symmetric? definite positive? …
- …

31.2.3 Optimal observer for deterministic disturbances

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} + \mathbf{B}_d\mathbf{d} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} + \mathbf{D}_d\mathbf{d} + \mathbf{D}_r\mathbf{r} \end{cases}$$

$$\begin{cases} \dot{\mathbf{o}} = \mathbf{A}\mathbf{o} + \mathbf{B}\mathbf{u} + \mathbf{L}\left(\mathbf{y} - \mathbf{y}_o\right) \\ \mathbf{y}_o = \mathbf{C}\mathbf{o} + \mathbf{D}\mathbf{u} \end{cases}$$

The error is defined as $\mathbf{e} := \mathbf{o} - \mathbf{x}$, and it's governed by the dynamical equation

$$\dot{\mathbf{e}} = \left(\mathbf{A} - \mathbf{L}\mathbf{C}\right)\mathbf{e} + \left(\mathbf{B}_d - \mathbf{L}\mathbf{D}_d\right)\mathbf{d} - \mathbf{L}\mathbf{D}_r\mathbf{r}\ .$$

This equation shows that the dynamics is asymptotically stable if the matrix $\mathbf{A} - \mathbf{L}\mathbf{C}$ is asymptotically stable. The matrix $\mathbf{L}$ is involved in the dynamics from the exogenous input $\mathbf{d}$ and measurement error $\mathbf{r}$. The design of $\mathbf{L}$ is driven by some requiremenets:

- asymptotic stability of $\mathbf{A} - \mathbf{L}\mathbf{C}$,
- low sensitivity to exogenous input and measurement error.

31.2.4 Sub-optimal control for output feedback

31.2.5 Combination of controller and observer

Full-state feed-back or the observed state builds the controller on the observed state. As an example, an optimal full-state controller is designed as the state was available, to find matrix $\mathbf{G}$; a state observer is designed to get the observed state $\mathbf{o}(t)$, through the design of a matrix $\mathbf{L}$ as in *Kalman filter*; the feed-back on the observed state produces the control $\mathbf{u}(t) = -\mathbf{G}\mathbf{o}(t)$.

Two different cases are discussed here:

1. the whole state is used to design the control law, as it's the case of small-systems with few states/degrees of freedom. In this situation, a **separation principle** holds, i.e. the state estimator design and the control design can be performed independently, as the eigenvalues of the closed-loop system are the eigenvalues of the closed-loop matrix $\mathbf{A} - \mathbf{B}\mathbf{G}$ and the observer $\mathbf{A} - \mathbf{L}\mathbf{C}$
2. a reduced state is used to design the control law, as it's the case of large-dimensional systems or continuous systems (as deformable bodies, whose nomber of degrees of freedom is infinite). For these systems, a subset of states is used for control design, in order to reduce the computational burden. In this situation, separation principle doesn't hold and **spill-over** may occur between states.

Separation principle

System

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}_u\mathbf{u} + \mathbf{B}_d\mathbf{d} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}_u\mathbf{u} + \mathbf{D}_d\mathbf{d} + \mathbf{D}_r\mathbf{r} \end{cases}$$

Observer

$$\begin{cases} \dot{\mathbf{o}} = \mathbf{A}\mathbf{o} + \mathbf{B}_u\mathbf{u} + \mathbf{L}(\mathbf{y} - \mathbf{y}_o) \\ \mathbf{y}_o = \mathbf{C}\mathbf{o} + \mathbf{D}_u\mathbf{u} \ . \end{cases}$$

with the error dynamics, $\mathbf{e} := \mathbf{x} - \mathbf{o}$

$$\dot{\mathbf{e}} = (\mathbf{A} - \mathbf{L}\mathbf{C})\,\mathbf{e} + (\mathbf{B}_d - \mathbf{L}\mathbf{D}_d)\,\mathbf{d} - \mathbf{L}\mathbf{D}_r\mathbf{r}$$

and with the control law $\mathbf{u} = -\mathbf{G}\mathbf{o} = -\mathbf{G}\,(\mathbf{x} - \mathbf{e})$. The augmented system with state $\{\mathbf{x}, \mathbf{e}\}$ reads

$$\frac{d}{dt}\begin{bmatrix}\mathbf{x}\\ \mathbf{e}\end{bmatrix} = \begin{bmatrix}\mathbf{A} - \mathbf{B}_u\mathbf{G} & \mathbf{B}_u\mathbf{G} \\ \mathbf{0} & \mathbf{A} - \mathbf{L}\mathbf{C}\end{bmatrix}\begin{bmatrix}\mathbf{x}\\ \mathbf{e}\end{bmatrix} + \begin{bmatrix}\mathbf{B}_d & \mathbf{0} \\ \mathbf{B}_d - \mathbf{L}\mathbf{D}_d & -\mathbf{L}\mathbf{D}_r\end{bmatrix}\begin{bmatrix}\mathbf{d}\\ \mathbf{r}\end{bmatrix} \ .$$

The matrix of the system is a block triangular matrix, and thus its eigenvalues are the union of the eigenvalues of the diagonal blocks, i.e. the eigenvalues of the closed-loop matrix $\mathbf{A} - \mathbf{B}_u\mathbf{G}$, and the matrix of the observer $\mathbf{A} - \mathbf{L}\mathbf{C}$. Thus, for the modal stability of the system, the observer and the ccontroller can be designed independently.

Spillover

Design reduced system and full system

Let $\mathbf{x}_1$ the state of the reduced system used for the design, and $\mathbf{x}_2$ the "residual" state of the large-dimensional system used for verification, or as a detailed mathematical model of the system of interest. Let the dynamical system be

$$\begin{cases} \dot{\mathbf{x}}_1 = \mathbf{A}_1\mathbf{x}_1 + \mathbf{B}_{1u}\mathbf{u} \\ \dot{\mathbf{x}}_2 = \mathbf{A}_2\mathbf{x}_2 + \mathbf{B}_{2u}\mathbf{u} \\ \mathbf{y} = \mathbf{C}_1\mathbf{x}_1 + \mathbf{C}_2\mathbf{x}_2 + \mathbf{D}_u\mathbf{u} \end{cases}$$

and the optimal control designed on the reduced system, so that the input is $\mathbf{u} = -\mathbf{G}\mathbf{x}_1$. The equations governing the closed-loop dynamics of the detailed system reads

$$\frac{d}{dt}\begin{bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{bmatrix} = \begin{bmatrix}\mathbf{A}_1 - \mathbf{B}_{1u}\mathbf{G} & \cdot \\ -\mathbf{B}_{2u}\mathbf{G} & \mathbf{A}_2\end{bmatrix}\begin{bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{bmatrix}$$
$$\mathbf{y} = \begin{bmatrix}\mathbf{C}_1 - \mathbf{D}_u\mathbf{G} & \mathbf{C}_2\end{bmatrix}\begin{bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{bmatrix}$$

The matrix of the system is block triangular, so that the eigenvalues of the system are the union of the eigenvalues of the closed loop matrix designed on the reduced system $\mathbf{A}_1 - \mathbf{B}_{1u}\mathbf{G}$, and the matrix of the residual detailed system $\mathbf{A}_2$. In order to get a stable system, the reduced system must contain all the open-loop unstable dynamics, as the matrix $\mathbf{A}_2$ must be stable.

Design reduced system, observer and full system

Let $\mathbf{x}_1$ the state of the reduced system used for the design, and $\mathbf{x}_2$ the "residual" state of the large-dimensional system used for verification, or as a detailed mathematical model of the system of interest. Let the dynamical system be

$$\begin{cases} \dot{\mathbf{x}}_1 = \mathbf{A}_1\mathbf{x}_1 + \mathbf{B}_{1u}\mathbf{u} \\ \dot{\mathbf{x}}_2 = \mathbf{A}_2\mathbf{x}_2 + \mathbf{B}_{2u}\mathbf{u} \\ \mathbf{y} = \mathbf{C}_1\mathbf{x}_1 + \mathbf{C}_2\mathbf{x}_2 + \mathbf{D}_u\mathbf{u} \end{cases}$$

and the observer designed on the low-dimensional system

$$\begin{cases} \dot{\mathbf{o}} = \mathbf{A}_1\mathbf{o} + \mathbf{B}_{1u}\mathbf{u} + \mathbf{L}\left(\mathbf{y} - \mathbf{y}_o\right) \\ \mathbf{y}_o = \mathbf{C}_1\mathbf{o} + \mathbf{D}_{1u}\mathbf{u} \end{cases}$$

so that

$$\dot{\mathbf{o}} = \left(\mathbf{A}_1 - \mathbf{L}\mathbf{C}_1\right)\mathbf{o} + \mathbf{L}\mathbf{C}_1\mathbf{x}_1 + \mathbf{L}\mathbf{C}_2\mathbf{x}_2 + \left(\mathbf{B}_{1u} - \mathbf{L}\mathbf{D}_u\right)\mathbf{u} \ ,$$

and the dynamical equation of the error $\mathbf{e} := \mathbf{x}_1 - \mathbf{o}$ is

$$\dot{\mathbf{e}} = \left(\mathbf{A}_1 - \mathbf{L}\mathbf{C}_1\right)\mathbf{e} - \mathbf{L}\mathbf{C}_2\mathbf{x}_2 \ .$$

The input is proportional to the observed state, $\mathbf{u} = -\mathbf{G}\mathbf{o}$. Thus, the augmented system reads

$$\frac{d}{dt}\begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \mathbf{e} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_1 - \mathbf{B}_{1u}\mathbf{G} & \cdot & \mathbf{B}_{1u}\mathbf{G} \\ -\mathbf{B}_{2u}\mathbf{G} & \mathbf{A}_2 & \mathbf{B}_{2u}\mathbf{G} \\ \cdot & -\mathbf{L}\mathbf{C}_2 & \mathbf{A}_1 - \mathbf{L}\mathbf{C}_1 \end{bmatrix}\begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \mathbf{e} \end{bmatrix} \ .$$

The matrix of the system is not block triangular, as the dynamics of the "residual" state of the large-dimensional system and the reconstruction error are coupled. The eigenvalue of the systems are non-trivial as they're not anymore the eigenvalue of the diagonal blocks.

31.2.6 Kalman filter

Kalman filter is an optimal state estimator, in terms of minimum-variance error reconstruction. It has the structure of a state observer. Let the linear system

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} + \mathbf{B}_d\mathbf{d} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}_u\mathbf{u} + \mathbf{D}_d\mathbf{d} + \mathbf{D}_r\mathbf{r} \ , \end{cases}$$

Kalman filter for time-continuous systems has the structure

$$\begin{cases} \dot{\mathbf{o}} = \mathbf{A}\mathbf{o} + \mathbf{B}\mathbf{u} + \mathbf{L}(\mathbf{y} - \mathbf{y}_o) \\ \mathbf{y}_o = \mathbf{C}\mathbf{o} + \mathbf{D}_u\mathbf{u} \end{cases}$$

Error dynamics

Let the error be defined as $\mathbf{e} := \mathbf{x} - \mathbf{o}$, its dynamical equation reads

$$\begin{aligned} \dot{\mathbf{e}} &= \left(\mathbf{A} - \mathbf{L}\mathbf{C}\right)\mathbf{e} + \left(\mathbf{B}_d - \mathbf{L}\mathbf{D}_d\right)\mathbf{d} - \mathbf{L}\mathbf{D}_r\mathbf{r} \\ \dot{\mathbf{e}} &= \mathbf{A}_e\mathbf{e} + \mathbf{B}_{ed}\mathbf{d} + \mathbf{B}_{er}\mathbf{r} \\ \dot{\mathbf{e}} &= \mathbf{A}_e\mathbf{e} + \xi \ , \end{aligned} \tag{31.9}$$

with the definition of the noise input in the dynamical equation of the error, $\xi = \left(\mathbf{B}_d - \mathbf{L}\mathbf{D}_d\right)\mathbf{d} - \mathbf{L}\mathbf{D}_r\mathbf{r}$.

Details

$$
\begin{aligned}
\dot{\mathbf{e}} &= \dot{\mathbf{x}} - \dot{\mathbf{o}} = \\
&= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} + \mathbf{B}_d\mathbf{d} - (\mathbf{A}\mathbf{o} + \mathbf{B}\mathbf{u} + \mathbf{L}(\mathbf{y} - \mathbf{y}_o)) = \\
&= (\mathbf{A} - \mathbf{L}\mathbf{C})\,\mathbf{e} + (\mathbf{B}_d - \mathbf{L}\mathbf{D}_d)\,\mathbf{d} - \mathbf{L}\mathbf{D}_r\mathbf{r}
\end{aligned}
$$

Let the exogenous input **d** and the measurement input **r** be random signals, thus the input ξ of the error dynamical equation (31.9) is a random signal with correlation

$$
\begin{aligned}
\mathbf{R}_{\xi\xi}(t,\tau) := \mathbb{E}\left[\xi(t)\xi^*(\tau)\right] = \mathbf{B}_{ed}\mathbf{R}_{\mathbf{dd}}(t,\tau)\mathbf{B}_{ed}^* + \mathbf{B}_{ed}\mathbf{R}_{\mathbf{dr}}(t,\tau)\mathbf{B}_{er}^* + \\
+ \mathbf{B}_{er}\mathbf{R}_{\mathbf{rd}}(t,\tau)\mathbf{B}_{ed}^* + \mathbf{B}_{er}\mathbf{R}_{\mathbf{rr}}(t,\tau)\mathbf{B}_{er}^*
\end{aligned}
$$

Using the general results for linear systems for **white noise** as an input ξ, the covariance matrix $\mathbf{P}_{\mathbf{e}}(t) := \mathbf{R}_{\mathbf{ee}}(t,t)$ of the error - governed by eqaution (31.9) - satisfies the Lyapunov equation (23.3)

$$
\dot{\mathbf{P}}_{\mathbf{e}} = \mathbf{A}_e\mathbf{P}_{\mathbf{e}} + \mathbf{P}_{\mathbf{e}}\mathbf{A}_e^* + \sigma_\xi^2 \ ,
$$

being

$$
\sigma_{xi}^2 = ...
$$

Optimality conditions

…

$$
\mathbf{L} =
$$

- duality with the optimal control…
- …

31.2.7 Hamilton Bellman Jacobi equation

Let a dynamical system

$$
\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x},\mathbf{u}) \ ,
$$

with initial conditions $\mathbf{x}(0) = \mathbf{x}_0$. Let a **value function** $V(\mathbf{x}_t, t)$ defined as

$$
V(\mathbf{x}_t, t) = \int_{\tau=t}^{T} C(\mathbf{x}(\tau), \mathbf{u}(\tau))d\tau + D(\mathbf{x}(T)) \ ,
$$

quantifying a cost for $\tau \in [t, T]$, subject to the dynamical equations. Value function has two arguments, the "initial time" t and the state of the system at that time $\mathbf{x}_t$. The dynamical equations are a constraint that can be introduced into the value function with the method of Lagrange multipliers

$$
\widetilde{V}(\mathbf{x}_t, t, \lambda) = \int_{\tau=t}^{T} C(\mathbf{x}(\tau), \mathbf{u}(\tau))d\tau + D(\mathbf{x}(T)) - \int_{\tau=t}^{T} \lambda^T(\tau)\left(\dot{\mathbf{x}}(\tau) - \mathbf{f}(\mathbf{x}(\tau), \mathbf{u}(\tau))\right) d\tau \ .
$$

Method 1. Non-augmented value function

Method 2. Augmented value function

Part XI

Partial Differential Equations

CHAPTER

THIRTYTWO

INTRODUCTION TO PARTIAL DIFFERENTIAL EQUATIONS

Partial differential equations usually comes from balance equations in **continuum mechanics**. Integral equations are the most general form of these equations, and an equivalent differential problem only exists if the fields involved in the equations are regular enough, for their derivatives to exist - and to apply theorems requiring some regularity of the functions.

Classical numerical methods:

- **FVM**: directly solves the **integral problem**, solving integral balance equations for cells in which the domain is divided
- **FDM**: given the problem in **differential form**, FDM directly approximates space derivatives of the **strong formulation** of the problem
- **FEM**: given the problem in **differential form**, FEM projects the **weak formulation** of the problem on a finite-dimensional space
- **BEM**: *integro-differential equation*, *singularities*,…
- **Spectral methods**,…
- **SEM**,…

32.1 Examples

In Physics:

- Advection equation

$$\partial_t u + \vec{a} \cdot \nabla u = f$$

- Diffusion equation

$$\partial_t u - \nu \nabla^2 u = f$$

- Hyperbolic equation/system of equations

$$\partial_t \mathbf{u} + \nabla \cdot \mathbf{F}(\mathbf{u}) = \mathbf{f}$$

- Wave equation

$$\frac{1}{c^2} \partial_{tt} u - \nabla^2 u = f$$

32.2 Balance equations in physics

- Small-strain continuum mechanics

$$\rho \partial_{tt} \vec{s} = \rho_0 \vec{g} + \nabla \cdot \boldsymbol{\sigma}(\boldsymbol{\varepsilon})$$

- Heat conduction
- Fluid dynamics
  - Navier-Stokes for compressible fluids (conservative or convective equations)

$$\{$$

  - Navier-Stokes for incompressible fluids (convective form,…)

$$\begin{cases} \rho \partial_t \vec{u} + \rho (\vec{u} \cdot \nabla) \vec{u} - \mu \nabla^2 \vec{u} + \nabla P = \rho \vec{g} \\ \nabla \cdot \vec{u} = 0 \end{cases}$$

todo

- Different forms of equations may be more or less convenient for different solution approaches
- Most of the physical laws comes from integral balance equation of the form

$$\frac{d}{dt} \int_{V_t} \rho \mathbf{u} = \int_{V_t} \rho \mathbf{r} + \oint_{\partial V_t} \hat{n} \cdot \mathbf{T}(\mathbf{u})$$

  whose local - differential - form (in case of differentiable functions) readily follows from the application of Reynolds' transport theorem and divergence theorem to transform time derivative and boundary terms

$$\begin{aligned} &\partial_t (\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \vec{u}) = \rho \mathbf{r} + \nabla \cdot \mathbf{T}(\mathbf{u}) \\ &\partial_t (\rho \mathbf{u}) + \nabla \cdot \mathbf{F}(\mathbf{u}) = \rho \mathbf{r} \,, \end{aligned}$$

  and the physical meaning of each term is evident and readily expalinable as flux or volume or surface sources.

- Further manipulation/simplification may cover the clear meaning of the terms of the differential equation. As an example, the conservative form of Navier-Stokes equations for incompressible fluids with constant and uniform density read

$$\begin{cases} \partial_t (\rho \vec{u}) + \nabla \cdot (\rho \vec{u} \otimes \vec{u}) = \rho \vec{g} + \nabla \cdot \mathbb{T} \\ \nabla \cdot \vec{u} = 0 \,, \end{cases}$$

  where the stress tensor for a Newtonian fluid reads

$$\begin{aligned} \mathbb{T} &= -p \mathbb{I} + 2\mu \mathbb{D} + \lambda (\nabla \cdot \vec{u}) \mathbb{I} \\ &= -p \mathbb{I} + \mu (\nabla \vec{u} + \nabla^T \vec{u}) + \lambda (\nabla \cdot \vec{u}) \mathbb{I} \end{aligned}$$

  Using the incompressibility constraint $\nabla \cdot \vec{u}$, and treating the density ρ as a constant and uniform parameter, the convective form of the Navier-Stokes equations reads

$$\begin{cases} \rho \partial_t \vec{u} + \rho (\vec{u} \cdot \nabla) \vec{u} = \rho \vec{g} - \nabla P + 2\mu \nabla \cdot \mathbb{D} \\ \nabla \cdot \vec{u} = 0 \,. \end{cases}$$

  The divergence of the viscous stress tensor becomes

$$2\mu \nabla \cdot \mathbb{D} = \mu \nabla \cdot (\nabla \vec{u} + \nabla^T \vec{u}) = \mu \left(\nabla^2 \vec{u} + \nabla \underbrace{(\nabla \cdot \vec{u})}_{=0} \right) = \mu \nabla^2 \vec{u} \,,$$

so that one of the most common form of incompressible Navier-Stokes equations follows

$$\begin{cases} \rho\partial_t\vec{u} + \rho(\vec{u}\cdot\nabla)\vec{u} - \mu\nabla^2\vec{u} + \nabla P = \rho\vec{g} \\ \nabla\cdot\vec{u} = 0\ . \end{cases}$$

It should be evident that in the latter form of Navier-Stokes equations no divergence explicitly appears, so that the right expression of surface source terms can't be found immediately. In momentum equation, surface source terms come from surface stress acting on the boundary of the domain, whose expression reads

$$\begin{aligned} \vec{t}_n &= \hat{n}\cdot\mathbb{T} = \\ &= \hat{n}\cdot(-P\mathbb{I} + 2\mu\mathbb{D}) = \\ &= \hat{n}\cdot\left(-P\mathbb{I} + \mu\left(\nabla\vec{u} + \nabla^T\vec{u}\right)\right) = \\ &= -P\hat{n} + \hat{n}\cdot\left(\mu\left(\nabla\vec{u} + \nabla^T\vec{u}\right)\right) = \end{aligned}$$

As an example, in weak formulation of incompressible Navier-Stokes problem the **natural boundary condition** arising in the method depends on the expression of the strong formulation of the NS problem. If one needs to prescribe stress boundary conditions, it could be an idea to start from NS equations w/o extra simplifications.

CHAPTER

THIRTYTHREE

ELLIPTIC EQUATIONS

33.1 Poisson equation

Given the volume density source $f(\vec{r})$ and the diffusivity $\nu(\vec{r})$, Poisson equation for the scalar field $\phi(\vec{r})$ reads

$$-\nabla \cdot (\nu \nabla \phi) = f \qquad \vec{r} \in V$$

with proper boundary conditions on ∂V. As an example, tipical boundary conditions are:

$\phi(\vec{r}) = g(\vec{r})$	$\vec{r} \in S_D$	esserntial - Dirichlet b.c.
$\nu \hat{n} \cdot \nabla \phi(\vec{r}) = h(\vec{r})$	$\vec{r} \in S_N$	natural - Neumann b.c.
$a\phi(\vec{r}) + \nu \hat{n} \cdot \nabla \phi(\vec{r}) = b(\vec{r})$	$\vec{r} \in S_R$	Robin b.c.

Numerical methods

1-dimensional Poisson equation:

- *Finite Element Methods*
- *Finite Volume Methods*
- *Finite Difference Methods*
- *Boundary Element Methods*
- *Spectral Methods* and *Spectral Element Methods*

33.1.1 Weak formulation

For $\forall w \in$ ... (functional space, recall some results about existence and uniqueness of the solution, Lax-Milgram theorem,…)

$$\begin{aligned} 0 &= \int_V w \, \{\nabla \cdot (\nu \nabla \phi) + f\} = \\ &= \oint_{\partial V} w \hat{n} \cdot (\nu \nabla \phi) + \int_V \{-\nu \nabla \vec{w} \cdot \nabla \phi + w f\} = \end{aligned}$$

Splitting boundary contribution as the sum from single contributions from different regions, and applying boundary conditions, setting $w = 0$ for $\vec{r} \in S_D$ (see the ways to prescribe essential boundary conditions),

$$0 = \int_{S_D} \underbrace{w}_{=0} \, \hat{n} \cdot (\nu \nabla \phi) + \int_{S_N} w \, \underbrace{\hat{n} \cdot (\nu \nabla \phi)}_{=h} + \int_{S_R} w \, \underbrace{\hat{n} \cdot (\nu \nabla \phi)}_{=b-a\phi} + \int_V \{-\nu \nabla \vec{w} \cdot \nabla \phi + w f\} \ .$$

and rearranging the equation separating terms containing unknowns from known contributions,

$$\int_V \nu \nabla w \cdot \nabla \phi + \int_{S_R} wa\phi = \int_V wf + \int_{S_N} wh + \int_{S_R} wb \qquad \forall w \in \dots \ ,$$

and $\phi = g$, for $\vec{r} \in S_D$.

Different ways to prescribe essential boundary conditions

Strong formulation.

Using Lagrance multiplier - weak formulation of essential boundary conditions. Adding a the essential boundary condition as a constraint with Lagrange multipliers in the weak formulation of the problem,

$$\dots + \int_{S_D} w_D(\phi - g) \ ,$$

…

33.1.2 Existence and uniqueness

Assuming two solutions $u_1(\mathbf{r})$, $u_2(\mathbf{r})$ exist for the Poisson problem

$$\begin{cases} -\nabla \cdot (\nu \nabla u) = f & \mathbf{r} \in V \\ u|_{S_D} = g \\ \nu \hat{n} \cdot \nabla u|_{S_N} = h \end{cases}$$

Their difference $\delta u(\mathbf{r}) := u_2(\mathbf{r}) - u_1(\mathbf{r})$ then satisfies the homogeneous problem

$$\begin{cases} -\nabla \cdot (\nu \nabla \delta u) = 0 & \mathbf{r} \in V \\ \delta u|_{S_D} = 0 \\ \nu \hat{n} \cdot \nabla \delta u|_{S_N} = 0 \end{cases}$$

The norm of the gradient of the difference of the solution reads

$$\begin{aligned} \int_V \nu |\nabla \delta u|^2 &= \int_V \nu \nabla \delta u \cdot \nabla \delta u = \\ &= \int_V \nabla \cdot (\delta u \, \nu \nabla \delta u) - \int_V \delta u \, \underbrace{\nabla \cdot (\nu \nabla \delta u)}_{=0} = \\ &= \oint_{\partial V} \delta u \, \nu \hat{n} \cdot \nabla \delta u = \\ &= \int_{S_D} \underbrace{\delta u}_{\delta u|_{S_D}=0} \nu \hat{n} \cdot \nabla \delta u + \int_{S_N} \delta u \, \underbrace{\nu \hat{n} \cdot \nabla \delta u}_{\nu \hat{n} \cdot \nabla \delta u|_{S_N}=0} = \\ &= 0 \ . \end{aligned}$$

If $\nu(\mathbf{r}) > 0$ for $\forall \mathbf{r} \in V$[1], it follows that

$$\nabla \delta u = \nabla (u_2 - u_1) = 0 \ ,$$

[1] As an example, this condition appears in physical systems with non-zero and positive diffusion coefficients in diffusion problems, like thermal conduction or specie diffusion via Fick's law.

and thus the two solutions differs at most by an additive constant,

$$u_2(\mathbf{r}) - u_1(\mathbf{r}) = c \ .$$

If the Dirichlet boundary has non-null dimension, it forces the value of the functions to coincide on that boundary, $u_2(\mathbf{r}) = u_1(\mathbf{r})$ on S_D, and thus sets the value of the additive constant c to be zero, $c = 0$, and

$$u_2(\mathbf{r}) = u_1(\mathbf{r}) \ ,$$

thus proving the uniqueness of the solution of the Poisson problem, with non-negative diffusion coefficient and non-zero dimension of Dirichlet boundary.

CHAPTER

THIRTYFOUR

PARABOLIC EQUATIONS

34.1 Heat equation

Heat equation for a scalar field $\phi(\vec{r},t)$ can be interpreted as the unsteady equation of a *Poisson equation*,

$$\partial_t \phi - \nabla \cdot (\nu \nabla \phi) = f \qquad (\vec{r},t) \in V \times [0,T] \ ,$$

with proper boundary and initial conditions, $\phi(\vec{r},0) = \phi_0(\vec{r})$. Common boundary conditions are the same as the one discussed for Poisson problem.

34.1.1 Weak formulation

For $\forall w \in \ldots$ (functional space, recall some results about existence and uniqueness of the solution, Lax-Milgram theorem,…)

$$\begin{aligned} 0 &= \int_V w \left\{ -\partial_t \phi + \nabla \cdot (\nu \nabla \phi) + f \right\} = \\ &= \oint_{\partial V} w \hat{n} \cdot (\nu \nabla \phi) + \int_V \left\{ -\partial_t \phi - \nu \nabla \vec{w} \cdot \nabla \phi + w f \right\} = \end{aligned}$$

Splitting boundary contribution as the sum from single contributions from different regions, and applying boundary conditions, setting $w = 0$ for $\vec{r} \in S_D$ (see the ways to prescribe essential boundary conditions),

$$0 = \int_{S_D} \underbrace{w}_{=0} \hat{n} \cdot (\nu \nabla \phi) + \int_{S_N} w \underbrace{\hat{n} \cdot (\nu \nabla \phi)}_{=h} + \int_{S_R} w \underbrace{\hat{n} \cdot (\nu \nabla \phi)}_{=k-\phi} + \int_V \left\{ -\partial_t \phi - \nu \nabla \vec{w} \cdot \nabla \phi + w f \right\} \ .$$

and rearranging the equation separating terms containing unknowns from known contributions,

$$\int_V w \partial_t \phi + \int_V \nu \nabla w \cdot \nabla \phi + \int_{S_R} w \phi = \int_V w f + \int_{S_N} w h + \int_{S_R} w k \qquad \forall w \in \ldots \ ,$$

and $\phi = g$, for $\vec{r} \in S_D$.

CHAPTER

THIRTYFIVE

HYPERBOLIC EQUATIONS

- Integral equations
- Jump conditions
- Differential equations
- Method of characteristics: domains of influence
- "Inviscid" limit, multiple solutions, entropy condition and unique physical solution

Some useful script on GDrive `basics-books/colabs/math/pde/fvm`. So far 1-dimensional problem only: linear equation, linear system (linearized p-sys), non linear system (p-sys).

Problems

1-dimensional domain

- *Scalar linear*
- *Vector linear*
- *Scalar non-linear*
- Vector non-linear

n-dimensional domain

- Scalar linear
- Vector linear
- Scalar non-linear
- Vector non-linear

todo

- hyperbolic problems as the **ideal limit of dissipative** (parabolic?) problems. In non-linear problems, multiple (infinite?) solutions of the hyperbolic problem exist, but there's only one (? Prove it! Or just provide the idea) solution that is the limit of the solution of the dissipative problem, in the *ideal* limit: when do jumps occur? When do rerefaction fans occur?
  - show multiple solutions
  - show limit of the dissipative solution

  on some simple problems, maybe starting from a scalar problem in 1-dimesnional domain, and then generalize to "more difficult" problems
- domains of **dependence** and **influence**. Show them in both unsteady and steady problems?

- Example with compressible flows (or the p-sys) comparing unsteady and steady problems, and discussing the nature of the equations in the subsonic and supersonic regimes

35.1 Hyperbolic problems and conservation laws

todo

- In fluid dynamics: no viscosity, no heat conduction
- In this limit, the equation may produce non-physical equations, i.e. solutions that are not the limit of the solution of the full problem in the limit of $\mathbf{F}^d(\nabla\mathbf{u}) \to \mathbf{0}$
- How many equations in how many variables?
- Linear diffusion flux $\mathbf{F}^d = \ldots$
- Entropy condition, to select the physical solution from all the possible solutions

35.1.1 Integral equations

Many laws can be written as integral balance equations for a material volume V_t,

$$\frac{d}{dt}\int_{V_t} \mathbf{u} + \oint_{\partial V_t} \hat{\mathbf{n}} \cdot \mathbf{F}(\mathbf{u}, \nabla\mathbf{u}) = \int_{V_t} \mathbf{s}$$
$$\frac{d}{dt}\int_{V_t} u_i + \oint_{\partial V_t} n_j\, F_{ji} = \int_{V_t} s_i$$

or for generic volume v_t, using *Reynolds' transport theorem*,

$$\frac{d}{dt}\int_{v_t} \mathbf{u} + \oint_{\partial v_t} \mathbf{u}\,(\mathbf{v} - \mathbf{v}_b) \cdot \hat{\mathbf{n}} + \oint_{\partial v_t} \hat{\mathbf{n}} \cdot \mathbf{F}(\mathbf{u}, \nabla\mathbf{u}) = \int_{v_t} \mathbf{s}\ ,$$

being $\mathbf{v}$ the velocity of the continuous medium and $\mathbf{v}_b$ the velocity of the boundary ∂v_t.

The flux can be often written as a sum of a *conservative part* and a *diffusion part* as

$$\mathbf{F}(\mathbf{u}, \nabla\mathbf{u}) = \mathbf{F}(\mathbf{u}) + \mathbf{F}^d(\nabla\mathbf{u})\ ,$$

with slight abuse of notation in writing the conservative flux with the same letter as the full flux.

Under some conditions, the diffusion part goes to zero.

35.1.2 Jump relations

Collapsing the volume v_t on a surface, volume terms vanish faster than the surface terms. Jump conditions follow from the arbitrariety of this surface

$$\begin{aligned}\mathbf{0} &= \hat{\mathbf{n}}_1 \cdot [(\mathbf{v}_1 - \mathbf{v}_b)\mathbf{u} + \mathbf{F}_1] + \hat{\mathbf{n}}_2 \cdot [(\mathbf{v}_2 - \mathbf{v}_b)\mathbf{u} + \mathbf{F}_2] = \\ &= \hat{\mathbf{n}} \cdot [(\mathbf{v} - \mathbf{v}_b)\mathbf{u} + \mathbf{F}] = \\ &= [v_n^{rel}\mathbf{u} + \mathbf{F}_n]\ .\end{aligned}$$

35.1.3 Differential equations

If the functions are differentiable, with divergence theorem and exploiting the arbitrariety of the volume v_t

$$\partial_t u_i + \partial_j F_{ji} = s_i$$
$$\partial_t \mathbf{u} + \nabla \cdot \mathbf{F} = \mathbf{s} \ .$$

35.2 Hyperbolic problems - dimensions

Hyperbolic problems often come from a small-amplitude linearization, or as the non-diffusion (or inviscid) limit of a more general problem.

As a result of these simplification, these problems may experience **shocks** (i.e. discontinuity in the solution, where the differential equations stop to hold, and integral equations and jump conditions are required). **todo** *classification of discontinuities on the massflow across the surface*

The very nature of these problem also suggest methods for the solution or the analysis of these equations, like **characteristic method**.

35.2.1 Scalar linear

1-dimensional

$$\partial_t u(x,t) + a\partial_x u(x,t) = f(x,t)$$

Caracteristic method. $U(t) = u(X(t),t)$, with the caracteristic curves $X(t)$ defined as those curves where the PDE becomes a ODE. Evaluating the time derivative of the function $u(X(t),t)$, the hyperbolic equation can be recast as

$$\frac{dU}{dt} + \left[a(X(t),t) - \frac{dX}{dt}\right]\partial_x u = f(X(t),t) \ .$$

The equation of characteristic lines is

$$\frac{dX}{dt} = a(X(t),t) \ ,$$

and the PDE on characteristic line becomes the ODE

$$\frac{dU}{dt}(X(t),t) = f(X(t),t) \ .$$

35.2.2 Scalar non-linear

35.2.3 System linear

1-dimensional

$$\mathbf{u}(x,t)$$

$$\partial_t \mathbf{u} + \mathbf{A}\partial_x \mathbf{u} = \mathbf{f}$$

Method of characteristics

Characteristics. $\mathbf{U}(t) = \mathbf{u}(X(t), t)$

$$\frac{d\mathbf{U}}{dt} - \frac{dX}{dt}\partial_x \mathbf{u} + \mathbf{A}\partial_x \mathbf{u} = \mathbf{f}$$

In order to get the equations of characteristic lines where PDE turns into ODEs, the eigenproblem

$$\mathbf{A}\partial_x \mathbf{u} = \frac{dX}{dt}\partial_x \mathbf{u} \ ,$$

holds. This problem has non trivial solution if $\frac{dX}{dt}$ and $\partial_x \mathbf{u}$ are pairs of eigenvalues and (right) eigenvectors of the array $\mathbf{A}$.

Diagonalization.

$$\mathbf{A} = \mathbf{R}\mathbf{\Lambda}\mathbf{L}$$

$$\mathbf{L}\left[\partial_t \mathbf{u} + \mathbf{R}\mathbf{\Lambda}\mathbf{L}\partial_x \mathbf{u}\right] = \mathbf{L}\mathbf{f}$$

Since $\mathbf{L} = \mathbf{R}^{-1}$, and defining the **characteristic variables** by $d\mathbf{q} = \mathbf{L}d\mathbf{u}$ - in linear problems matrix $\mathbf{A}$ is constant, and so its spectral decomoposition, and thus $\mathbf{q} = \mathbf{L}\mathbf{u}$ - , it's possible to recast the original problem in diagonal form

$$\partial_t \mathbf{q} + \mathbf{\Lambda}\partial_x \mathbf{q} = \mathbf{L}\mathbf{f}$$

$$\partial_t q_i + \Lambda_i \partial_x q_i = \sum_k L_{ik}\, f_k =: F_i \ .$$

Thus, on the ith family of characteristic lines, $\dfrac{dX}{dt} = \lambda_i$, $Q_i(t) = q_i(x(t), t)$ evolves as

$$\frac{dQ_i}{dt} = F_i \ .$$

If $F_i = [\mathbf{L}\mathbf{f}]_i = 0$, the characteristic variable q_i is constant along the characteristic lines. Once the characteristic variables are determined, the conservative variables are evalauted as $\mathbf{u}(x,t) = \mathbf{R}\mathbf{q}(x,t)$.

Domain of influence and domain of dependence

Riemann problem

A Riemann problem is defined as the evolution of the initial state

$$\mathbf{u}(x, t_0) = \begin{cases} \mathbf{u}_a \ , & x < x_0 \\ \mathbf{u}_b \ , & x > x_0 \end{cases}$$

This problem is quite useful in quite a wide range of numerical methods for hyperbolic problems - Godunov schemes in Finite Volume Methods -, to evaluate the **boundary state** to be used numerical flux.

For linear problems, the matrix $\mathbf{A}$ is constant ad so it is its spectral decomposition, $\mathbf{A} = \mathbf{R}\mathbf{\Lambda}\mathbf{L}$, and the solution of a Riemann problem of an homogeneous linear hyperbolic system can be easily determined analytically with the method of characteristics,

Let's change the origin of space and time, so that the initial state is in $t = 0$, and the jump in the initial condiiton in $x = 0$. Each charactersitic variable $q_k(x,t)$ is constant on its family of characteristic lines, $x = X_k(t) = x_{0,k} + \lambda_k t$.

$$q_k(x,t) = q_k(x_{0,k} + \lambda_k t, t) = q_k(x_{0,k}, 0) = q_k(x - \lambda_k t, 0) = L_{ki} u_j(x - \lambda_k t, 0) \ .$$

Thus, the solution in conservative variables $\mathbf{u}(x,t)$ in x at time t reads

$$\begin{aligned}\mathbf{u}(x,t) &= \mathbf{R}\mathbf{q}(x,t)\\ u_i(x,t) &= R_{ik}q_k(x,t) = R_{ik}q_k(x-\lambda_k t,0) = R_{ik}L_{kj}u_j(x-\lambda_k t,0)\end{aligned}$$

In a Riemann problem for a N-dimensional linear system the solution shows $N+1$ homogeneous regions (at most, in general the same number as the number of the non-coincident eigenvalues $+1$), delimited by the characteristic lines with origin in the discontinuity. Sorting the eigenvalues in increasing order

$$\lambda_1 > \lambda_2 > \cdots > \lambda_N \ ,$$

and defining the homogeneous regions

$$\begin{aligned}S_0 &: \frac{x}{t} \in (-\infty, \lambda_1)\\ S_1 &: \frac{x}{t} \in (\lambda_1, \lambda_2)\\ &\cdots\\ S_i &: \frac{x}{t} \in (\lambda_i, \lambda_{i+1})\\ &\cdots\\ S_{N-1} &: \frac{x}{t} \in (\lambda_{N-1}, \lambda_N)\\ S_N &: \frac{x}{t} \in (\lambda_N, +\infty)\end{aligned}$$

the solution is in the S_i region is

$$u_i(x,t) = \sum_{\lambda_k > \frac{x}{t}} R_{ik}q_{a,k} + \sum_{\lambda_k < \frac{x}{t}} R_{ik}q_{b,k}$$

Example 35.2.1 (Linear(ized) P-system)

The linear(ized) P-system around a uniform reference state $\overline{\rho}$, $\overline{u}$ in convective form reads

$$\partial_t \begin{bmatrix}\rho\\ u\end{bmatrix} + \begin{bmatrix}\overline{u} & \overline{\rho}\\ \frac{a^2}{\overline{\rho}} & \overline{u}\end{bmatrix} \partial_x \begin{bmatrix}\rho\\ u\end{bmatrix} = \mathbf{0} \ .$$

Spectral decomposition.

$$0 = |-\lambda\mathbf{I} + \mathbf{A}| = (\overline{u}-\lambda)^2 - a^2$$

$$\lambda_{12} = \overline{u} \mp a \qquad , \qquad \mathbf{r}_{12} = \begin{bmatrix}\overline{\rho}\\ \mp a\end{bmatrix}$$

$$\begin{aligned}\mathbf{R} &= \begin{bmatrix}\overline{\rho} & \overline{\rho}\\ -a & a\end{bmatrix}\\ \mathbf{L} = \mathbf{R}^{-1} &= \frac{1}{2\overline{\rho}a}\begin{bmatrix}a & -\overline{\rho}\\ a & \overline{\rho}\end{bmatrix}\end{aligned}$$

Reference state.

$$|u| \ : \ \begin{cases} = 0 & \text{at rest}\\ < a & \text{subsonic flow}\\ > a & \text{supersonic flow to the left/right}\end{cases}$$

Subsonic: the two families of characteristic lines have opposite direction; supersonic: the two families of characteristic lines have the same direction.

Example 35.2.2 (Linearized shallow water equations)

Example 35.2.3 (Linearized Euler equations (acoustics))

Example 35.2.4 (Wave equation)

A wave equation arises in many different fields of science. As an example, 1-dimensional wave equation descrives the axial dynamics of a truss

$$m\partial_{tt}u - EA\partial_{xx}u = f \ ,$$

that can be recast in the general expression of wave equation

$$\partial_{tt}u - c^2\partial_{xx}u = F$$

The 2nd order differential operator appearing in 1-dimensional wave equation can be factored as the "product" of 2 1st order differentail operators,

$$\left(\partial_{tt} - c^2\partial_{xx}\right)u = \left(\partial_t - c\partial_x\right)\left(\partial_t + c\partial_x\right)u$$

and thus a wave equation can be written as

$$\begin{cases} \partial_t u + c\partial_x u - w = 0 \\ \partial_t w - c\partial_x w = F \end{cases}$$

In the regime of small displacement, the velocity field is the partial time derivative of the dispalcement field, $v = \partial_t u$, and the axial force reads $N = EA\partial_x u$. Exploiting Schwartz's theorem about mixed partial derivatives to write $\partial_t N = EA\partial_x v$, it's possible to write the wave function as the following system of hyperbolic equations in the physical unknowns v, N

$$\begin{cases} \partial_t N - EA\partial_x v = 0 \\ \partial_t v - \frac{1}{m}\partial_x N = f \end{cases}$$

P-system and wave equation - reference state at rest, $\overline{u} = 0$.

$$\begin{cases} \partial_t \rho + \overline{\rho}\partial_x u = 0 \\ \partial_t u + \frac{a^2}{\overline{\rho}}\partial_x \rho = 0 \end{cases}$$

Taking time partial derivative of the first and space partial derivative of the second equation times $\overline{\rho}$, and evaluating their difference, a wave equation for rho appears

$$\partial_{tt}\rho - a^2\partial_{xx}\rho = 0 \ .$$

Analogously, taking space derivative of the first and time derivative of the second, a wave equation for the velocity field appears

$$\partial_{tt}u - a^2\partial_{xx}u = 0 \ .$$

35.2.4 System non-linear

1-dimensional space

$$\mathbf{u}(x,t)$$
$$\begin{aligned} \partial_t \mathbf{u} + \partial_x \mathbf{F}(\mathbf{u}) &= \mathbf{f} && \text{(conservative form)} \\ \partial_t \mathbf{u} + \partial_{\mathbf{u}} \mathbf{F}(\mathbf{u}) \partial_x \mathbf{u} &= \mathbf{f} && \text{(convective form)} \end{aligned}$$

35.2.5 n-dimensional space

$$\mathbf{u}(\vec{r},t)$$
$$\begin{aligned} \partial_t \mathbf{u} + \nabla \cdot \mathbf{F}(\mathbf{u}) &= \mathbf{f} && \text{(conservative form)} \\ \partial_t \mathbf{u} + \nabla \mathbf{u} \cdot \partial_{\mathbf{u}} \mathbf{F}(\mathbf{u}) &= \mathbf{f} && \text{(convective form)} \end{aligned}$$

Different descritpions of integral problem,

$$\begin{aligned} \frac{d}{dt}\int_V \mathbf{u} + \oint_{\partial V} \hat{n} \cdot \mathbf{F}(\mathbf{u}) &= \int_V \mathbf{f} && \text{(Eulerian)} \\ \frac{d}{dt}\int_{V_t} \mathbf{u} - \oint_{\partial V_t} \mathbf{u}\vec{u} \cdot \hat{n} + \oint_{\partial V_t} \hat{n} \cdot \mathbf{F}(\mathbf{u}) &= \int_{V_t} \mathbf{f} && \text{(Lagrangian)} \\ \frac{d}{dt}\int_{v_t} \mathbf{u} - \oint_{\partial v_t} \mathbf{u}\vec{u}_b \cdot \hat{n} + \oint_{\partial v_t} \hat{n} \cdot \mathbf{F}(\mathbf{u}) &= \int_{v_t} \mathbf{f} && \text{(arbitrary)} \end{aligned}$$

todo *in coordinates*

$$\begin{aligned} f_i &= \partial_t u_i + \partial_{x_k} F_{ki}(u_l) = \\ &= \partial_t u_i + \partial_{x_k} u_m \partial_{u_m} F_{ki}(u_l) = \end{aligned}$$

Example 35.2.5 (P-system in 1-dimensional domain)

$$\begin{cases} \partial_t \rho + u \partial_x \rho + \rho \partial_x u = 0 \\ \rho \partial_t u + \rho u \partial_x u + \partial_x P = 0 \end{cases}$$

with $\partial_x P = a^2 \partial_x \rho$,

Convective form

$$\partial_t \begin{bmatrix} \rho \\ u \end{bmatrix} + \begin{bmatrix} u & \rho \\ \frac{a^2}{\rho} & u \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ u \end{bmatrix} = \underline{0}\,.$$

Conservative form

$$\partial_t \begin{bmatrix} \rho \\ \rho u \end{bmatrix} + \partial_x \begin{bmatrix} \rho u \\ \rho u^2 + \rho a^2 \end{bmatrix} = \underline{0}\,.$$

Spectral decomposition of $\mathbf{A}(\mathbf{u})$ gives

$$0 = \left| \begin{bmatrix} u - s & \rho \\ \frac{a^2}{\rho} & u - s \end{bmatrix} \right| = (u-s)^2 - a^2$$

$$
\begin{aligned}
s_{1,2} &= u \mp a \\
\mathbf{R} &= \begin{bmatrix} \rho & \rho \\ a & -a \end{bmatrix} \\
\mathbf{L} &= \frac{1}{2\rho a} \begin{bmatrix} a & \rho \\ a & -\rho \end{bmatrix}
\end{aligned}
$$

Example 35.2.6 (Euler equations in 1-dimensional domain)

Conservative form

$$
\partial_t \begin{bmatrix} \rho \\ \rho u \\ \rho e^t \end{bmatrix} + \partial_x \begin{bmatrix} \rho u \\ \rho u^2 + P \\ \rho u h^t \end{bmatrix} = \underline{0} \,,
$$

with $h^t = e^t + \frac{P}{\rho}$ and $e^t = e + \frac{u^2}{2}$, and the pressure field can be written as a function of the other thermodynamic variables. As an example, using conservative variables $(\rho, m, E^t) = (\rho, \rho u, \rho e^t) = \left(\rho, \rho u, \rho \left(e + \frac{u^2}{2}\right)\right)$

$$
P(\rho, e) = P\left(\rho, \frac{E^t}{\rho} - \frac{m^2}{2\rho^2}\right) = \Pi\left(\rho, m, E^t\right)
$$

so that

$$
\begin{aligned}
\partial_\rho \Pi &= \partial_\rho P\big|_e + \partial_e P\big|_\rho \left(-\frac{E^t}{\rho^2} + \frac{m^2}{\rho^3}\right) = \\
&= \partial_\rho P\big|_e + \partial_e P\big|_\rho \left(-\frac{e^t}{\rho} + \frac{u^2}{\rho}\right) \\
&= c^2 - \frac{P}{\rho^2} \partial_e P\big|_\rho + \partial_e P\big|_\rho \left(-\frac{e^t}{\rho} + \frac{u^2}{\rho}\right) \\
&= c^2 + \partial_e P\big|_\rho \left(-\frac{h^t}{\rho} + \frac{u^2}{\rho}\right) \\
\partial_m \Pi &= \partial_e P\big|_\rho \left(-\frac{m}{\rho^2}\right) \\
\partial_{E^t} \Pi &= \partial_e P\big|_\rho \left(\frac{1}{\rho}\right)
\end{aligned}
$$

The speed of sound reads

$$
\begin{aligned}
c^2 &= \partial_\rho P\big|_s = \\
&= \partial_\rho P\big|_e + \partial_e P\big|_\rho \partial_\rho e\big|_s = \\
&= \partial_\rho P\big|_e + \frac{P}{\rho^2} \partial_e P\big|_\rho \,,
\end{aligned}
$$

Conservative form in conservative variables.

$$
\partial_t \begin{bmatrix} \rho \\ m \\ E^t \end{bmatrix} + \partial_x \begin{bmatrix} m \\ \frac{m^2}{\rho} + \Pi(\rho, m, E^t) \\ \frac{m}{\rho}\left(E^t + \Pi(\rho, m, E^t)\right) \end{bmatrix} = \underline{0} \,,
$$

Convective form in conservative variables.

$$
\partial_t \begin{bmatrix} \rho \\ m \\ E^t \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 \\ -\frac{m^2}{\rho^2} + \partial_\rho \Pi & \frac{2m}{\rho} + \partial_m \Pi & \partial_{E^t} \Pi \\ -\frac{m}{\rho^2}(E^t + \Pi) + \frac{m}{\rho} \partial_\rho \Pi & \frac{1}{\rho}(E^t + \Pi) + \frac{m}{\rho} \partial_m \Pi & \frac{m}{\rho}\left(1 + \partial_{E^t} \Pi\right) \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ m \\ E^t \end{bmatrix} = \underline{0} \,,
$$

Spectral decomposition of $\mathbf{A}(\mathbf{u})$

$$
\begin{aligned}
0 = & \left|\begin{bmatrix} -s & 1 & 0 \\ -u^2+\partial_\rho\Pi & 2u+\partial_m\Pi - s & \partial_{E^t}\Pi \\ -u\left(e^t+\frac{P}{\rho}\right)+u\partial_\rho\Pi & e^t+\frac{P}{\rho}+u\partial_m\Pi & u\left(1+\partial_{E^t}\Pi\right)-s \end{bmatrix}\right| = \\
& = -s\left[\left(2u+\partial_m\Pi - s\right)\left(u\left(1+\partial_{E^t}\Pi\right)-s\right) - \partial_{E^t}\Pi\left(h^t+u\partial_m\Pi\right)\right] + \\
& \quad - uh^t\partial_{E^t}\Pi + u\partial_\rho\Pi\partial_{E^t}\Pi + \\
& \quad + \left(u^2-\partial_\rho\Pi\right)\left(u(1+\partial_{E^t}\Pi)-s\right) = \\
& = -s^3 + \\
& \quad + s^2\left(2u+\partial_m\Pi+u+u\partial_{E^t}\Pi\right) + \\
& \quad + s\left(-2u^2-2u^2\partial_{E^t}\Pi - u\partial_m\Pi - u\partial_m\Pi\,\partial_{E^t}\Pi + \partial_{E^t}\Pi\,h^t + u\partial_{E^t}\Pi\,\partial_m\Pi - u^2 + \partial_\rho\Pi\right) + \\
& \quad + \left(-uh^t\partial_{E^t}\Pi + u\partial_\rho\Pi\,\partial_{E^t}\Pi + u^3 + u^3\partial_{E^t}\Pi - u\partial_\rho\Pi - u\partial_\rho\Pi\,\partial_{E^t}\Pi\right) + \\
& = -s^3 + \\
& \quad + s^2\left(3u+\partial_m\Pi+u\partial_{E^t}\Pi\right) + \\
& \quad + s\left(-3u^2-2u^2\partial_{E^t}\Pi - u\partial_m\Pi + \partial_{E^t}\Pi\,h^t + \partial_\rho\Pi\right) + \\
& \quad + \left(u^3 - uh^t\partial_{E^t}\Pi + u^3\partial_{E^t}\Pi - u\partial_\rho\Pi\right) = \\
& = -(s-u)^3 + (s-u)c^2 = \\
& = (s-u)\left[-(s-u)^2+c^2\right]
\end{aligned}
$$

being

$$
\begin{aligned}
\partial_m\Pi + u\partial_{E^t}\Pi &= \left(-\frac{u}{\rho}+\frac{u}{\rho}\right)\partial_\rho P\big|_e = 0 \\
-2u^2\partial_{E^t}\Pi - u\partial_m\Pi + \partial_{E^t}\Pi\,h^t + \partial_\rho\Pi &= -\frac{u^2}{\rho}\partial_\rho P\big|_e + \frac{1}{\rho}\partial_e P\big|_\rho h^t + \partial_\rho P\big|_e + \partial_e P\big|_\rho\left(-\frac{e^t}{\rho}+\frac{u^2}{\rho}\right) = \\
&= \partial_e P\big|_\rho\frac{P}{\rho^2} + \partial_\rho P\big|_e = \\
&= c^2 \\
-u\partial_\rho\Pi + u^3\partial_{E^t}\Pi - uh^t\partial_{E^t}\Pi &= u\left(-\partial_\rho P\big|_e - \partial_e P\big|_\rho\left(-\frac{e^t}{\rho}+\frac{u^2}{\rho}\right) + \frac{u^2}{\rho}\partial_e P\big|_\rho - \frac{h^t}{\rho}\partial_e P\big|_\rho\right) \\
&= u\left(-\partial_\rho P\big|_e - \frac{P}{\rho^2}\partial_e P\big|_\rho\right) = \\
&= -uc^2\ .
\end{aligned}
$$

Thus,

$$
s_{1,3} = u \mp c \quad , \quad s_2 = u
$$

$$
\mathbf{r}_{1,3} = \begin{bmatrix} 1 \\ u\mp c \\ \dots \end{bmatrix}\hat{\rho} \quad , \quad \mathbf{r}_2 = \begin{bmatrix} \dots \\ 0 \\ \dots \end{bmatrix}\hat{\rho}
$$

being

$$
\begin{aligned}
\hat{E}^t\partial_{E^t}\Pi &= \left[u^2 - \partial_\rho\Pi + (-u\pm c - \partial_m\Pi)(u\mp c)\right]\hat{\rho} = \\
&= \left[u^2 - \partial_\rho\Pi - u^2 + \pm 2uc - c^2 - \partial_m\Pi(u\mp c)\right]\hat{\rho} = \\
&= \left[-c^2 - \partial_e P\big|_\rho\left(-\frac{h^t}{\rho}+\frac{u^2}{\rho}\right) \pm 2uc - c^2 + \frac{u}{\rho}\partial_e P\big|_\rho(u\mp c)\right]\hat{\rho} = \\
&= \left[-2c^2 + \partial_e P\big|_\rho\frac{h^t}{\rho} \pm 2uc \mp \frac{uc}{\rho}\partial_e P\big|_\rho\right]\hat{\rho} =
\end{aligned}
$$

$$\mathbf{R} = \ldots$$

$$\mathbf{L} = \ldots$$

Example 35.2.7 (Shallow water equation in 1-dimensional domain)

Let $b(x)\ldots$, $h(x)$ the height of the free surface, $\eta(x) = h(x) - b(x)$ the depth.

Derivative of integrals with non-constant extremes

$$\partial_x \int_{z=0}^{\eta(x,t)} \rho u \, dz = \int_{z=0}^{\eta(x,t)} \partial_x(\rho u) \, dz + \rho u(x, \eta(x,t), t) \partial_x \eta(x,t) \; .$$

Continuity equation reads

$$\partial_t \rho + \partial_x(\rho u) + \partial_z(\rho w) = 0 \; ,$$

for fluids with constant and uniform density

$$\begin{aligned} 0 &= \int_{z=0}^{\eta(x,t)} \left(\partial_t \rho + \partial_x(\rho u) + \partial_z(\rho w)\right) \, dz = \\ &= \partial_x \int_{z=0}^{\eta(x,t)} (\rho u) \, dz - \rho u(x, \eta, t) \partial_x \eta + \rho w(x, \eta(x,t), t) = \\ &= \partial_x \int_{z=0}^{\eta(x,t)} (\rho u) \, dz + \rho \partial_t \eta = \qquad\qquad \simeq \partial_x \left(\rho \eta u\right) + \partial_t \left(\rho \eta\right) \; . \end{aligned}$$

having linked the velocity to the material derivative of the position, whose vertical component reads

$$w(x, \eta(x,t), t) = \frac{D\eta}{Dt} = \partial_t \eta(x,t) + u(x, \eta(x,t), t) \partial_x \eta \; .$$

Assuming hydrostatic pressure distribution, $p = P_a + \rho g z$ at depth z under the level of local free surface,

Momentum equation reads

$$0 = \partial_t(\rho u) + \partial_x(\rho u^2) + \partial_z(\rho u w) + \partial_x P \; .$$

and integration in z-direction **todo** Explicitly treat the z term

$$\begin{aligned} 0 &= \partial_t(\rho \eta u) + \partial_x(\rho u^2 \eta) + \partial_x \int_{z=0}^{\eta(x)} (P_a + \rho g z) \, dz = \\ &= \partial_t(\rho \eta u) + \partial_x \left(\rho u^2 \eta + \frac{1}{2} \rho g \eta^2\right) \; . \end{aligned}$$

Conservative form of the equations.

$$\begin{cases} \partial_t(\eta) + \partial_x m = 0 \\ \partial_t m + \partial_x \left(\frac{m^2}{\eta} + \frac{g\eta^2}{2}\right) = 0 \end{cases}$$

Convective form of the equations.

$$\partial_t \begin{bmatrix} \eta \\ m \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ -\frac{m^2}{\eta^2} + g\eta & 2\frac{m}{\eta} \end{bmatrix} \partial_x \begin{bmatrix} \eta \\ m \end{bmatrix} = \underline{0}$$

Spectrum of matrix $\mathbf{A}(\mathbf{u})$.

$$0 = |\mathbf{A}(\mathbf{u}) - s^2\mathbf{I}| = -s\,(2u - s) + u^2 - g\eta = (s-u)^2 - g\eta\ .$$

Example 35.2.8 (P-system in n-dimensional domain)

- Conservative variables: $(\rho, \vec{m})$
- Physical variables: e.g. $(\rho, \vec{u})$, $(P, \vec{u})$,…

$$\begin{cases} \partial_t \rho + \nabla \cdot \vec{m} = 0 \\ \partial_t \vec{m} + \nabla \cdot \left[\frac{\vec{m} \otimes \vec{m}}{\rho} + \rho a^2 \mathbb{I} \right] = 0 \end{cases}$$

Example 35.2.9 (Euler system in n-dimensional domain)

- Conservative variables: $(\rho, \vec{m}, E^t)$
- Physical variables: e.g. $(\rho, \vec{u}, e)$,…

$$\begin{cases} \partial_t \rho + \nabla \cdot \vec{m} = 0 \\ \partial_t \vec{m} + \nabla \cdot \left[\frac{\vec{m} \otimes \vec{m}}{\rho} + \Pi\, \mathbb{I} \right] = \vec{0} \\ \partial_t E^t + \nabla \cdot \left[\frac{\vec{m}(E^t + \Pi)}{\rho} \right] = 0 \end{cases}$$

where Π represents the pressure field as a function of the conservative varaibles,

$$\Pi\left(\rho, \vec{m}, E^t\right) = P\left(\rho, e\right) = P\left(\rho, \frac{E^t}{\rho} - \frac{|\vec{m}|^2}{\rho^3}\right)\ ,$$

and P the pressure field expressed by the **equation of state of the fluid** as a function of density and internal energy per unit mass as the pair of independent variables determining the thermodynamic state.

Example 35.2.10 (Shallow water equations in 2-dimensional domain)

$$\begin{cases} \partial_t(\rho\eta) + \nabla \cdot (\rho\eta\vec{u}) = 0 \\ \partial_t(\rho\eta\vec{u}) + \nabla \cdot \left(\rho\eta\vec{u}\vec{u} + \frac{1}{2}\rho g \eta^2 \mathbb{I}\right) = 0 \end{cases}$$

35.3 Scalar linear equation

Differential equation.

$$\begin{aligned} r &= \partial_t u + a\partial_x u && \text{convective form} \\ &= \partial_t u + \partial_x(au) && \text{conservative form} \end{aligned}$$

Integral equation. On a control volume V at rest

$$0 = \frac{d}{dt}\int_V u + \oint_{\partial V} n_x a u\ .$$

Integral equation in an arbitrary domain v_t can be derived using *Reynolds' transport theorem*.

Method of characteristics. Let $U(t) = u(X(t), t)$. From derivative of composite functions,

$$d_t U(t) = \partial_t u|_{X(t),t} + \dot{X}(t)\partial_x u|_{X(t),t} \ ,$$

and thus the PDE can be recast as a system of two ODEs

$$\begin{cases} \dot{X}(t) = a \\ \dot{U}(t) = r \end{cases}$$

…

35.4 Linear vector equation

Differential equation.

$$\begin{aligned} \mathbf{r} &= \partial_t \mathbf{u} + \mathbf{A}\partial_x \mathbf{u} && \text{convective form} \\ &= \partial_t \mathbf{u} + \partial_x (\mathbf{A}\mathbf{u}) && \text{conservative form} \end{aligned}$$

Integral eqaution. On a control volume V at rest

$$\int_V \mathbf{r} = \frac{d}{dt}\int_V \mathbf{u} + \oint_{\partial V} n_x \mathbf{A}\mathbf{u} \ .$$

Integral equation in an arbitrary domain v_t can be derived using *Reynolds' transport theorem*.

Method of characteristics. Let $\mathbf{A}$ be diagonalizable,

$$\begin{aligned} \mathbf{AR} &= \mathbf{RS} \\ \mathbf{A} &= \mathbf{RSL} \end{aligned}$$

Using spectral decomposition and multiplying the PDE by $\mathbf{L}$, the PDE is transformed in a set of scalar PDE equations,

$$\begin{aligned} \mathbf{Lr} &= \mathbf{L}\left(\partial_t \mathbf{u} + \mathbf{RSL}\partial_x \mathbf{u}\right) = \\ &= \mathbf{L}\partial_t \mathbf{u} + \mathbf{SL}\partial_x \mathbf{u} = \\ &= \partial_t \mathbf{v} + \mathbf{S}\partial_x \mathbf{v} \ , \end{aligned}$$

or

$$\partial_t v_i + s_i \partial_x v_i = L_{ik} r_k \ ,$$

having introduced the characteristic variables $\mathbf{v}(x,t) := \mathbf{Lu}(x,t)$. If the source is not function of the unknown function $\mathbf{u}(x,t)$, this is a system of decoupled PDEs. The solution of the vector PDE follows from the solution of n scalar PDE linear equations. Using methods of characteristics, the solution of the ith scalar PDE equation can be recast as the solution of pairs of ODEs, after writing $V_i(t) = v_i(X_i(t), t)$, being $X_i(t)$ the equation of a curve belonging to the ith familiy of characteristics,

$$\begin{cases} \dot{X}_i = s_i \\ \dot{V}_i = L_{ik} r_k \ . \end{cases}$$

As the spectral decomposition of a constant matrix is constant, the characteristic lines are straight lines $X_i(t) = s_i t + c$. In general, the solution of the dynamical equations of the characteristic variables V_i is a full system of n ODEs.

Particular case: no source. As a particular case, if the source term is identically zero, $\mathbf{r}(x,t) = \mathbf{0}$, characteristic variable $V_i(t)$ is constant along each characteristic line $X_i(t; \dots)$ of the ith family of characteristics.

…

$$u_k(x,t) = R_{ki} v_i(x,t) = R_{ri} v_i(x_{0,i} + s_i(t - t_{0,i}), t) \ .$$

35.5 Scalar non-linear equation

todo list

- When does a jump occur? I.e. when is the physical solution discontinuous? Have non physical jumps already been discussed?
- Boundary conditions for hyperbolic problems
- …

35.5.1 Examples

Burgers's equation

Differential equation and integral balance

Burgers' differential equation reads

$$\begin{aligned} 0 &= \partial_t u + u \partial_x u = && \text{(convective form)} \\ &= \partial_t u + \partial_x \frac{u^2}{2} = && \text{(conservative form)} \\ &= \partial_t u + \partial_x F(u) \ , \end{aligned}$$

being the flux $F(u) = \frac{u^2}{2}$, and it can be derived as the local counterpart of the integral balance equation,

$$0 = \frac{d}{dt} \int_V u + \oint_{\partial V} n_x \frac{u^2}{2} = \frac{d}{dt} \int_V u + F(u(x))\Big|_a^b \ ,$$

for a control volume at rest $V = [a, b]$. For a control volume with arbitrary motion $v_t = [a_t, b_t]$ the integral balance equation reads

$$\begin{aligned} 0 &= \frac{d}{dt} \int_{v_t} u - \oint_{\partial v_t} \hat{\mathbf{n}} \cdot \mathbf{v}_b \, u + \oint_{\partial v_t} n_x F(u) = \\ &= \frac{d}{dt} \int_{v_t} u - \oint_{\partial v_t} n_x v_{b,x} \, u + \oint_{\partial v_t} n_x F(u) = \\ &= \frac{d}{dt} \int_{v_t} u + \big[F(u(x)) - v_{b,x} u(x)\big]\Big|_a^b \ , \end{aligned}$$

having used Reynolds' transport theorem[1], and the unit normal vector in the only space direction $n_x(x = a) = -1$ on the left extreme and $n_x(x = b) = 1$ on the right extreme of the domain $V = [a, b]$.

[1] For a generic volume v_t, $d_t \int_{v_t} f = \int_{v_t} \partial_t f + \oint_{\partial v_t} \hat{\mathbf{n}} \cdot \mathbf{v}_b \, f$. Thus, for a control volume at rest V, $d_t \int_V f = \int_V \partial_t f$. Comparing the two expressions, it immediately follows that $d_t \int_{v_t} f = d_t \int_{V \equiv v_t} f + \oint_{\partial v_t} \hat{\mathbf{n}} \cdot \mathbf{v}_b \, f$.

Jump condition

Jump condition immediately follows from the integral form of the balance equations for the arbitrary volume v_t.

$$[F] = v_{b,x}[u] \ .$$

For Burgers' equation, $F(u) = \frac{u^2}{2}$, and thus the jump relation becomes

$$\frac{u_2^2}{2} - \frac{u_1^2}{2} = v_{b,x}\left(u_2 - u_1\right) \ ,$$

and thus, except for the trivial condition $u_1 = u_2$ where no jump occurs,

$$v_{b,x} = \frac{1}{2}\left(u_2 + u_1\right) \ ,$$

the velocity of the point where the discontinuity appears - the shock - is the average of the velocity u on its sides.

todo *When does a jump occur? I.e. when is the physical solution discontinuous? Have non physical jumps already been discussed?*

Characteristics

Let $U(t) = u(X(t), t)$. Burgers' equation can be recast as[2]

$$0 = d_t U - \dot{X}\partial_x u + u\partial_x u \ .$$

A characteristic line is defined as a curve for which the coefficient of the spatial partial differential equation is zero, i.e. $\dot{X}(t) = U(X(t), t)$. On these lines, the PDE becomes a ODE, so that it's somehow equivalent **todo** *be more precies* to the system of ODEs

$$\begin{cases} \dot{X}(t) = U(X(t), t) \\ \dot{U}(t) = 0 \ . \end{cases}$$

These equations need initial and boundary conditions.

todo *Boundary conditions for hyperbolic problems*

Let x_0, t_0 a point in space and time where the solution is known, $u_0 = u(x_0, t_0)$. On the characteristic line originating from x_0, t_0, the velocity is constant in time as $\dot{U}(t) = 0$, and thus $U(t)_{X(t;x_0,t_0)} = u_0$. The characteristic line satisfying the first equation and the initial conditions is the line with equation

$$X(t; t_0, x_0, u_0) = u_0(t - t_0) + x_0 \ .$$

Riemann problem

Riemann problem is defined as the PDE with a discontinuity in a piecewise constant initial conditons. For an unbounded domain $V = (-\infty, +\infty)$[3]

$$u(x, 0) = \begin{cases} u_1 & , \quad x < 0 \\ u_2 & , \quad x > 0 \end{cases}$$

Two cases must be discussed here: 1) $u_1 < u_2$, 2) $u_1 > u_2$.

[2] Using derivation of composite functions, $d_t U(t) = d_t u(X(t), t) = \partial_t u + d_t X \partial_x u = \partial_t u|_{(x,t)=(X(t),t)} + \dot{X}(t)\partial_x u|_{(x,t)=(X(t),t)}$.

[3] In an unbounded domain, there's no need to treat boundary conditions…**todo** *add link to boundary conditions for hyperbolic problems.*

Expansion fan, $u_1 < u_2$. A self-similar solution exists between lines $r_1 : x = u_1 t$ and $r_2 : x = u_2 t$. Defining $\xi := \frac{x}{t}$, the solution becomes $u(x,t) = U(\xi(x,t))$, so that

$$\partial_t U = \partial_t \xi U'(\xi) = -\frac{x}{t^2} U'(\xi)$$
$$\partial_x U = \partial_x \xi U'(\xi) = \frac{1}{t} U'(\xi) \ ,$$

so that Burgers' equation in the rarefaction fan becomes

$$0 = \partial_t u + u \partial_x u = \frac{1}{t} \left[-\xi + U(\xi) \right] U'(\xi) \ .$$

Looking for a continuous solution blending the two states, the condition $U'(\xi) = 0$ is not feasible. Then,

$$U(\xi) = \xi \ ,$$

or

$$u(x,t) = \frac{x}{t} \quad , \quad \frac{x}{t} \in [u_1, u_2] \ .$$

The solution of the Riemann problem reads

$$u(x,t) = \begin{cases} u_1 & , \quad \frac{x}{t} \leq u_1 \\ \frac{x}{t} & , \quad \frac{x}{t} \in [u_1, u_2] \\ u_2 & , \quad \frac{x}{t} \geq u_2 \end{cases}$$

Shock, $u_1 > u_2$. Characteristic method fails at a shock originating from the origin. Characteristic lines originating from $x < 0$ have a slope $\dot{X}(t) = u_1$ larger than the slope of the characteristic lines originating from $x > 0$: characteristic lines collapses into the shock, that has velocity

$$\dot{s} = \frac{u_1 + u_2}{2} \ .$$

The solution of the Riemann problem reads

$$u(x,t) = \begin{cases} u_1 & , \quad \frac{x}{t} < \dot{s} \\ u_2 & , \quad \frac{x}{t} > \dot{s} \end{cases}$$

Boundary conditions

In order to have a well-defined hyperbolic problem, boundary conditions are **required** on the regions of the boundary **where the information enters the domain**. For a scalar problem, the local direction of the information is determined by the velocity of the local characteristic lines. For Burgers' equation, the direction of the local intormation is just the sign of the velocity of the characteristics, that is equal to the local velocity. Thus, boundary conditions are strictly needed on the left boundary $x = a$ if $u(a,t) > 0$ and on the right boundary $x = b$ if $u(b,t) < 0$.

35.6 Method of characteristics

This page is an old version of the material. It contains some useful and (incredibly?) interesting things, but it will be used to write new pages first and eventually deleted. Check all the **cross-references**

35.6.1 Scalar multi-dimensional problem

$$\mathbf{a}^T(u(\mathbf{x}),\mathbf{x})\,\nabla u(\mathbf{x}) = c(u(\mathbf{x}),\mathbf{x}) \quad , \quad \mathbf{x}\in D$$
$$a_k(u(\mathbf{x}),\mathbf{x})\frac{\partial}{\partial x_k}u(\mathbf{x}) = c(u(\mathbf{x}),\mathbf{x})$$

The method of characteristics aims at transforming the PDE over the domain D, in a set of ODEs over lines - the characteristic lines. Both the coordinates of these lines and the unknown function on these lines are written in parametric form as

$$\mathbf{x} = \mathbf{X}(s)$$
$$u(\mathbf{X}(s)) = U(s)\ .$$

Now, taking the ordinary derivative of $U'(s)$,

$$U'(s) = \frac{dX_k}{ds}(s)\frac{\partial u}{\partial x_k}\left(\mathbf{X}(s)\right)\ .$$

If the characteristic lines are defined by the differential equation $X_k'(s) = a_k(U(s),X(s))$, the set of ODEs become

$$U'(s) = c\left(U(s),\mathbf{X}(s)\right)$$
$$\mathbf{X}'(s) = \mathbf{a}\left(U(s),\mathbf{X}(s)\right)\ .$$

35.6.2 Vector multi-dimensional problem

todo *Characteristic method fails…characteristic - eikonal - equation. Uncomment*

Let

$$\mathbf{u}(t,\mathbf{r}) : \mathbb{R}\times\mathbb{R}^d \to \mathbb{R}^n$$
$$\mathbf{F}(\mathbf{u}) : \mathbb{R}^n \to \mathbb{R}^d\times\mathbb{R}^n$$
$$\mathbf{s}(t,\mathbf{r}) : \mathbb{R}\times\mathbb{R}^d \to \mathbb{R}^n$$

the conservative form of a hyperbolic problem reads

$$\partial_t\mathbf{u} + \nabla\cdot\mathbf{F}(\mathbf{u}) = \mathbf{s}$$
$$\partial_t u_i + \partial_j F_{ji} = s_i\ .$$

Expanding the divergence of the flux, the convective (quasi-linear? It makes little to no sense to me…) form follows

$$s_i = \partial_t u_i + \sum_{j=1}^{d}\partial_j F_{ji} =$$
$$= \partial_t u_i + \sum_{j=1}^{d}\sum_{k=1}^{n}\partial_j u_k \partial_{u_k} F_{ji} =$$
$$= \partial_t u_i + \sum_{j=1}^{d}\sum_{k=1}^{n} A_{ik}^j(\mathbf{u})\,\partial_j u_k\ ,$$

for every component $i = 1 : n$, or

$$\mathbf{s} = \partial_t \mathbf{u} + \sum_{j=1}^{d} \mathbf{A}^j \partial_j \mathbf{u} \ .$$

This problem can be made a little more general with by defining $\mathbf{x} = (t, \mathbf{r}) = (x_0, \mathbf{r})$, as

$$\sum_{j=0}^{d} \mathbf{A}^j \, \partial_j \mathbf{u} = \mathbf{s} \ .$$

The former expression of the hyperbolic problem immediately follows if $\mathbf{A}_0 = \mathbf{I}_n$, $A^0_{ik} = \delta_{ik}$, and

$$\widetilde{\nabla} \cdot \tilde{\mathbf{F}} = \partial_0 \mathbf{F}_0 + \sum_{j=1}^{d} \partial_j \mathbf{F}_j = \widetilde{\nabla} \cdot [\ \mathbf{u} \mid \mathbf{F}] \ .$$

as, for $i = 1 : n$,

$$\delta_{ik} = \partial_{u_k} F_{0i} \quad \rightarrow \quad F_{0i} = u_k \delta_{ik} = u_i \ .$$

Taylor expansion of a solution

Starting from the solution on a manifold determined by the equation $S : f(\mathbf{x}) = 0$, $\mathbf{x}_0 \in S$, if the solution is differentiable, the solution in a point $\mathbf{x} = \mathbf{x}_0 + \Delta\mathbf{x}$ reads

$$\begin{aligned} \mathbf{u}(\mathbf{x}) &\sim \mathbf{u}(\mathbf{x}_0) + \Delta\mathbf{x} \cdot \nabla\mathbf{u}(\mathbf{x}_0) \\ u_i(\mathbf{x}) &\sim u_i(\mathbf{x}_0) + \Delta_j \frac{\partial u_i}{\partial x_j} \ . \end{aligned}$$

Now, with a change of coordinates from $\mathbf{x}$ to $\xi = (n, \mathbf{t})$, i.e. to local normal and tangential direction,

$$\nabla\mathbf{u} = \hat{\mathbf{x}}_i \frac{\partial}{\partial x_i} \mathbf{u} = \hat{\mathbf{x}}_i \frac{\partial \mathbf{u}}{\partial x_i} = \hat{\mathbf{x}}_i \frac{\partial \mathbf{u}}{\partial \xi_k} \frac{\partial \xi_k}{\partial x_i} = \hat{\xi}_k \frac{\partial \mathbf{u}}{\partial \xi_k}$$

Inserting in the hyperbolic equation

$$\begin{aligned} s_i &= \sum_{j=0}^{d} A^j_{ik} \frac{\partial u_k}{\partial x_j} = \\ &= \sum_{j=0}^{d} A^j_{ik} \sum_{\ell=0}^{d} \frac{\partial u_k}{\partial \xi_\ell} \frac{\partial \xi_\ell}{\partial x_j} = \\ &= \sum_{\ell=0}^{d} \sum_{j=0}^{d} A^j_{ik} \frac{\partial \xi_\ell}{\partial x_j} \frac{\partial u_k}{\partial \xi_\ell} = \\ &= \sum_{\ell=0}^{d} \sum_{j=0}^{d} A^j_{ik} \xi^\ell_j \frac{\partial u_k}{\partial \xi_\ell} \ , \end{aligned}$$

and separating the normal $\ell = 0$ from the tangential $\ell = 1 : d$ coordinates,

$$\begin{aligned} s_i &= \sum_{j=0}^{d} A^j_{ik} \xi^0_j \frac{\partial u_k}{\partial \xi_0} + \sum_{\ell=1}^{d} \sum_{j=0}^{d} A^j_{ik} \xi^t_j \frac{\partial u_k}{\partial \xi_t} = \\ &= \sum_{j=0}^{d} A^j_{ik} n_j \frac{\partial u_k}{\partial n} + \sum_{\ell=1}^{d} \sum_{j=0}^{d} A^j_{ik} t^\ell_j \frac{\partial u_k}{\partial t_\ell} = \\ &= \sum_{j=0}^{d} n_j \mathbf{A}^j \partial_n \mathbf{u} + \sum_{\ell=1}^{d} \sum_{j=0}^{d} t^\ell_j \mathbf{A}^j \partial_{t_\ell} \mathbf{u} \ . \end{aligned}$$

On a smooth surface where the solution is known, all the tangential derivatives of the solution are known as well. In order to evaluate the normal derivative $\partial_n \mathbf{u}$ from the PDE, the (formal) inversion of the matrix

$$\mathbf{A}_{\hat{\mathbf{n}}} := \sum_{j=0}^{d} n_j \, \mathbf{A}^j \ ,$$

must be invertible, to formally get

$$\partial_n \mathbf{u} = \mathbf{A}_{\hat{\mathbf{n}}}^{-1} \left(\mathbf{s} - \sum_{\ell=1}^{d} \mathbf{A}_{\hat{\mathbf{t}}_\ell} \partial_{t_\ell} \mathbf{u} \right) \ .$$

This expression of the normal derivative of the solution - whenever it exists - can be used to find the approximation of the solution in normal direction w.r.t. the surface S, i.e. with $\Delta \mathbf{x} = \Delta \ell \hat{\mathbf{n}}$ as

$$\begin{aligned} \mathbf{u}(\mathbf{x}) &\sim \mathbf{u}(\mathbf{x}_0) + \Delta \ell \, \hat{\mathbf{n}} \cdot \nabla \mathbf{u}(\mathbf{x}_0) = \\ &= \mathbf{u}(\mathbf{x}_0) + \Delta \ell \, \partial_n \mathbf{u}(\mathbf{x}_0) \ . \end{aligned}$$

If the matrix $\mathbf{A}_{\hat{\mathbf{n}}}$ is diagonalizable, it's invertible if it has no zero eigenvalue. Let $\hat{\mathbf{n}}_i$ be a unit vector so that the matrix $\mathbf{A}_{\hat{\mathbf{n}}}$ has an eigenvalue equal to zero $s_0 = 0$, with right and left eigenvectors $\mathbf{r}_0$, $\mathbf{l}_0$. Recalling the expression of the PDE in normal and tangential components

$$\mathbf{s} = \mathbf{A}_{\hat{\mathbf{n}}} \partial_n \mathbf{u} + \sum_{\ell=1}^{d} \mathbf{A}_{\hat{\mathbf{d}}_\ell} \partial_{t_\ell} \mathbf{u} \ ,$$

left-multiplying by $\mathbf{l}$ gives

$$\mathbf{l}_0^T \mathbf{s} = \mathbf{l}_0^T \sum_{\ell=1}^{d} \mathbf{A}_{\hat{\mathbf{d}}_\ell} \partial_{t_\ell} \mathbf{u} = \qquad \dots \qquad = \mathbf{l}_0^T \sum_{k=0}^{d} \mathbf{A}^k \partial_k \mathbf{u} \ .$$

todo

- *compatibility conditions*
- *eikonal equation*
- *explicitly treating steady vs. unsteady problems*

Examples

Isothermal compressible flow - 2d

- Conservative variables: $(\rho, \vec{m})$
- Physical variables: e.g. $(\rho, \vec{u})$, $(P, \vec{u})$,…

In isothermal compressible flow model, pressure is a function of density only and

$$dp = a^2 d\rho \ ,$$

with constant a, speed of sound.

Conservative form reads

$$\begin{cases} \partial_t \rho + \nabla \cdot \mathbf{m} = 0 \\ \partial_t \mathbf{m} + \nabla \cdot \left[\frac{\mathbf{m} \otimes \mathbf{m}}{\rho} + \rho a^2 \mathbf{I} \right] = 0 \end{cases}$$

Convective form in a 2-dimensional domain reads

$$\partial_t \begin{bmatrix} \rho \\ m_x \\ m_y \end{bmatrix} + \begin{bmatrix} \cdot & 1 & \cdot \\ a^2 - \frac{m_x m_x}{\rho^2} & \frac{2m_x}{\rho} & \cdot \\ -\frac{m_x m_y}{\rho^2} & \frac{m_y}{\rho} & \frac{m_x}{\rho} \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ m_x \\ m_y \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & 1 \\ -\frac{m_x m_y}{\rho^2} & \frac{m_y}{\rho} & \frac{m_x}{\rho} \\ a^2 - \frac{m_y m_y}{\rho^2} & \cdot & \frac{2m_y}{\rho} \end{bmatrix} \partial_y \begin{bmatrix} \rho \\ m_x \\ m_y \end{bmatrix} = \mathbf{0}$$

Some algebra

$$
\begin{aligned}
0 &= \rho_{/t} + m_{x/x} + m_{y/y} \\
0 &= m_{i/t} + \partial_j \left(\frac{m_j m_i}{\rho} + \rho a^2 \delta_{ij} \right) \ .
\end{aligned}
$$

Thus,

$$
\begin{aligned}
\mathbf{A}_{\hat{\mathbf{n}}} &= \begin{bmatrix} \cdot & n_x & n_y \\ a^2 n_x - u_n u_x & u_x n_x + u_n & u_x n_y \\ a^2 n_y - u_n u_y & u_y n_x & u_n + u_y n_y \end{bmatrix} = \\
&= \begin{bmatrix} 0 & \mathbf{n}^T \\ a^2 \mathbf{n} - (\mathbf{n}^T \mathbf{u}) \mathbf{u} & \mathbf{n}^T \mathbf{u} \mathbf{I} + \mathbf{u} \mathbf{n}^T \end{bmatrix} = \\
&= \begin{bmatrix} 0 & \hat{\mathbf{n}}^T \\ a^2 \mathbf{n} - (\mathbf{n} \cdot \mathbf{u}) \mathbf{u} & (\mathbf{n} \cdot \mathbf{u}) \mathbb{I} + \mathbf{u} \otimes \mathbf{n} \end{bmatrix} \ .
\end{aligned}
$$

Eigenvalues. Details

The eigenvalue decomposition of the matrix $\mathbf{A}_{\hat{\mathbf{n}}}$ follows from

$$
\begin{aligned}
0 = |\mathbf{A}_{\hat{\mathbf{n}}} - s\mathbf{I}| &= \left| \begin{bmatrix} -s & \mathbf{n}^T \\ a^2 \mathbf{n} - u_n \mathbf{u} & (u_n - s)\mathbf{I} + \mathbf{u} \mathbf{n}^T \end{bmatrix} \right| = \\
&= -s(u_n + u_x n_x - s)(u_n + u_y n_y - s) + n_x u_x n_y (a^2 n_y - u_n u_y) + n_y u_y n_x (a^2 n_x - u_n u_x) + \\
&- (a^2 n_y - u_n u_y)(u_x n_x + u_n - s) n_y - (a^2 n_x - u_n u_x)(u_y n_y + u_n - s) n_x - s u_x u_y n_x n_y = \\
&= s^3(-1) + \\
&\quad + s^2(u_n + u_y n_y + u_n + u_x n_x) + \\
&\quad + s(-(u_n + u_x n_x)(u_n + u_y n_y) + n_y (a^2 n_y - u_n u_y) + n_x (a^2 n_x - u_n u_x) - u_x n_x u_y n_y) + \\
&\quad + 1 \cdot (n_x u_x n_y (a^2 n_y - u_n u_y) + n_y u_y n_x (a^2 n_x - u_n u_x) - n_y (a^2 n_y - u_n u_y)(u_x n_x + u_n) - n_x (a^2 n_x - u_n u_x)(u_y n_y + u_n)) = \\
&= -s^3 + 3 u_n s^2 + s(-u_n^2 - u_n (u_x n_x + u_y n_y) - u_x n_x u_y n_y - u_n u_y n_y - u_n u_x n_x + a^2 - u_x n_x u_y n_y) + \\
&\quad + (a^2 \left(n_y^2 u_x n_x + n_x^2 n_y u_y - n_y^2 u_x n_x - n_y^2 u_n - n_x^2 u_y n_y - u_n n_x^2 \right) - u_n u_x n_x u_y n_y - u_n u_x n_x u_y n_y + u_n u_x n_x u_y n_y + u_n^2 u_y n_y + \\
&= -s^3 + 3 u_n s^2 + s(a^2 - 3 u_n^2) - u_n (a^2 - u_n^2) = \\
&= -(s - u_n)^3 + (s - u_n) a^2 = \\
&= -(s - u_n)((s - u_n)^2 - a^2) \ .
\end{aligned}
$$

Thus the eigenvalues of the matrix $\mathbf{A}_{\hat{\mathbf{n}}}$ are

$$
s_{1,3} = u_n \mp a \quad , \quad s_2 = u_n \ ,
$$

and the right and left eigenvectors follow

$$
\begin{aligned}
\mathbf{R} &= \\
\mathbf{L} &=
\end{aligned}
$$

Right and left eigenvectors. Details

For $s_{1,3} = u_n \mp a$,

$$\mathbf{0} = \begin{bmatrix} -u_n \pm a & \mathbf{n}^T \\ a^2\mathbf{n} - u_n\mathbf{u} & \pm a\mathbf{I} + \mathbf{u}\mathbf{n}^T \end{bmatrix} \begin{bmatrix} \hat{\rho} \\ \hat{\mathbf{m}} \end{bmatrix}$$

From the first equation it follows $\hat{m}_n = (u_n \mp a)\hat{\rho}$, and thus the momentum equation gives

$$\begin{aligned} \mathbf{0} &= \hat{\rho}(a^2\mathbf{n} - u_n\mathbf{u}) \pm a\hat{\mathbf{m}} + \mathbf{u}\hat{m}_n = \\ &= \hat{\rho}(a^2\mathbf{n} - u_n\mathbf{u}) \pm a\hat{\mathbf{m}} + \mathbf{u}(u_n \mp a)\hat{\rho} = \\ &= \hat{\rho}(a^2\mathbf{n} \mp a\mathbf{u}) \pm a\hat{\mathbf{m}} \ , \end{aligned}$$

and thus

$$\hat{\mathbf{m}}_{1,3} = \hat{\rho}_{1,3}(\mathbf{u} \mp a\mathbf{n}) \ .$$

For $s_2 = u_n$,

$$\mathbf{0} = \begin{bmatrix} -u_n & \mathbf{n}^T \\ a^2\mathbf{n} - u_n\mathbf{u} & \mathbf{u}\mathbf{n}^T \end{bmatrix} \begin{bmatrix} \hat{\rho} \\ \hat{\mathbf{m}} \end{bmatrix}$$

From the first equation it follows $\hat{m}_n = u_n\hat{\rho}$, and thus the momentum equation gives

$$\begin{aligned} \mathbf{0} &= \hat{\rho}(a^2\mathbf{n} - u_n\mathbf{u}) + \mathbf{u}\,\hat{m}_n = \\ &= \hat{\rho}(a^2\mathbf{n} - u_n\mathbf{u}) + \mathbf{u}\,u_n\,\hat{\rho} = \\ &= \hat{\rho}a^2\mathbf{n} \ , \end{aligned}$$

and thus $\hat{\rho}_2 = 0$. The components of the momentum readily follows from mass equation that is equivalent to the condition

$$\mathbf{n}^T\hat{\mathbf{m}}_2 = 0 \ ,$$

i.e. as an example $\hat{\mathbf{m}}_2 = \begin{bmatrix} n_y \\ -n_x \end{bmatrix}$.

Assuming positive density, it's possible to write all the eigenvectors with the proper physical dimensions, and collect them in the matrix of right eigenvectors,

$$\mathbf{R} = \begin{bmatrix} \rho & 0 & \rho \\ \rho(u - an_x) & \rho a n_y & \rho(u + an_x) \\ \rho(v - an_y) & -\rho a n_x & \rho(v + an_y) \end{bmatrix} \ .$$

The determinant reads

$$\frac{1}{\rho^3 a^2}|\mathbf{R}| = n_y(v + n_y) - (u - n_x)n_x - n_y(v - n_y) + n_x(u + n_x) = 2 \ .$$

The inverse matrix reads

$$\begin{aligned} \mathbf{L} &= \frac{1}{|\mathbf{R}|}\,\rho^2 \begin{bmatrix} a(a + u_n) & -an_x & -an_y \\ 2a(n_x v - n_y u) & 2an_y & -2an_x \\ a(a - u_n) & an_x & an_y \end{bmatrix} = \\ &= \frac{1}{2\rho a^2} \begin{bmatrix} a(a + u_n) & -an_x & -an_y \\ 2a(n_x v - n_y u) & 2an_y & -2an_x \\ a(a - u_n) & an_x & an_y \end{bmatrix} \ . \end{aligned}$$

as

$$
\begin{aligned}
L_{11} &\propto an_y(v+an_y)+an_x(u+an_x) = au_n + a^2 \\
-L_{21} &\propto (u-an_x)(v+an_y)-(v-an_y)(u+an_x) = 2aun_y - 2an_xv \\
L_{31} &\propto -an_x(u-an_x)-an_y(v-an_y) = -au_n + a^2 \\
-L_{12} &\propto an_x \\
L_{22} &\propto (v+an_y)-(v-an_y) = 2an_y \\
-L_{32} &\propto -an_x \\
L_{13} &\propto -an_y \\
-L_{23} &\propto (u+an_x)-(u-an_x) = 2an_x \\
L_{33} &\propto an_y
\end{aligned}
$$

Some check

$$
\begin{aligned}
\mathbf{LR} &= \frac{1}{2\rho a^2}\begin{bmatrix} a(a+u_n) & -a\mathbf{n}^T \\ 2a(n_xv-n_yu) & 2a\mathbf{t}^T \\ a(a-u_n) & a\mathbf{n}^T \end{bmatrix}\begin{bmatrix} 1 & 0 & 1 \\ \mathbf{u}-a\mathbf{n} & a\mathbf{t} & \mathbf{u}+a\mathbf{n} \end{bmatrix}\rho = \\
&= \frac{1}{2a^2}\begin{bmatrix} a^2+au_n-au_n+a^2 & -a\mathbf{n}^T\mathbf{t} & a^2+au_n-au_n-a^2 \\ 2a(n_xv-n_yu)+2a(n_yu-n_xv) & 2a^2 & 2a(n_xv-n_yu)+2a(n_yu-n_xv) \\ a^2-au_n+au_n-a^2 & a\mathbf{n}^T\mathbf{t} & a^2-au_n+au_n+a^2 \end{bmatrix} = \mathbf{I}_3 \ .
\end{aligned}
$$

Normal vectors of the characteristic surfaces. For **locally supersonic flows**, $|\mathbf{u}| > a$, there are three directions, i.e. three unit vectors $\hat{\mathbf{n}}_i$ that make the eigenvalues equal to zero,

$$
\hat{\mathbf{n}}_{1,3}\cdot\mathbf{u} = \pm a \quad , \quad \hat{\mathbf{n}}_2\cdot\mathbf{u} = 0 \ ,
$$

or equivalently

$$
\hat{\mathbf{n}}_{1,3}\cdot\left(\mathbf{u}\mp a\hat{\mathbf{n}}_{1,3}\right) = 0 \quad , \quad \hat{\mathbf{n}}_2\cdot\mathbf{u} = 0 \ .
$$

Let $\hat{\mathbf{u}}$ the unit vector along the local velocity field, it follows

$$
u\hat{\mathbf{u}}\cdot\hat{\mathbf{n}}_{1,3} = \pm a \ ,
$$

i.e.

$$
\cos\theta_{1,3} = \pm\frac{a}{u} = \pm\frac{1}{M} \ .
$$

For **locally subsonic flows**, $|\mathbf{u}| < a$, the eigenvalues $s_{1,3}$ are always negative and positive respectively, while the eigenvalue s_2 becomes equal to zero for unit vectors that are orthogonal w.r.t. the local velocity.

Right and left eigenvectors

Right eigenvector problem reads

$$
\mathbf{AR} = \mathbf{RS} \ ,
$$

s.t. $\mathbf{Ar}_i = s_i\mathbf{r}_i$, with $\mathbf{r}_i$ the ith column of matrix $\mathbf{R}$.

Left eigenvector problem reads

$$
\mathbf{LA} = \mathbf{SL} \ ,
$$

s.t. $\mathbf{l}_i^T\mathbf{A} = s_i\mathbf{l}_i^T\mathbf{A}$, with $\mathbf{l}_i^T$ the ith row of matrix $\mathbf{L}$.

Compatibility equations. For **locally supersonic flows**, for $s_{1,3} = u_n \mp a$, $u_n = \pm a$ to get $s_{1,3} = 0$, and thus the left eigenvectors for those choices of unit vector become

$$\begin{aligned}\mathbf{l}_{0;1,3}^T &= \frac{1}{2\rho a^2}\left[\, a(a \pm u_n) \mid \mp a\mathbf{n}_{1,3}^T\right] = \\ &= \frac{1}{2\rho a}\left[\, 2a \mid \mp \mathbf{n}_{1,3}^T\right] \,.\end{aligned}$$

The quasi linear form of the equations becomes

todo *Check algebra, and UNCOMMENT!*

$$0 = \frac{1}{2\rho a}\left[\, 2a \mid \mp \mathbf{n}_{1,3}^T \,\right]\left(\partial_t \begin{bmatrix}\rho\\ m_x\\ m_y\end{bmatrix} + \begin{bmatrix}\cdot & 1 & \cdot\\ a^2 - \frac{m_x m_x}{\rho^2} & \frac{2m_x}{\rho} & \cdot\\ -\frac{m_x m_y}{\rho^2} & \frac{m_y}{\rho} & \frac{m_x}{\rho}\end{bmatrix}\partial_x \begin{bmatrix}\rho\\ m_x\\ m_y\end{bmatrix} + \begin{bmatrix}\cdot & \cdot & 1\\ -\frac{m_x m_y}{\rho^2} & \frac{m_y}{\rho} & \frac{m_x}{\rho}\\ a^2 - \frac{m_y m_y}{\rho^2} & \cdot & \frac{2m_y}{\rho}\end{bmatrix}\partial_y \begin{bmatrix}\rho\\ m_x\\ m_y\end{bmatrix} = \mathbf{0}\right) =$$

$$\begin{aligned}0 = \ldots\, &\partial_t \cdots + \\ &+ \left[\mp a^2 n_x \pm u_n u_x \mid 2a \mp u_x n_x \mp u_n \mid \mp n_y u_x \,\right]\partial_x \begin{bmatrix}\rho\\ \mathbf{m}\end{bmatrix} + \\ &+ \left[\mp a^2 n_y \pm u_n u_y \mid \mp n_x u_y \mid 2a \mp u_y n_y \mp u_n \,\right]\partial_y \begin{bmatrix}\rho\\ \mathbf{m}\end{bmatrix} = \\ = &\left[\left(\mp a^2 n_x \pm u_n u_x\right)\partial_x + \left(\mp a^2 n_y \pm u_n u_y\right)\partial_y\right]\rho + \\ &+ \left[\left(2a \mp u_x n_x \mp u_n\right)\partial_x + \left(\mp n_x u_y\right)\partial_y\right] m_x + \\ &+ \left[\left(\mp n_y u_x\right)\partial_x + \left(2a \mp u_y n_y \mp u_n\right)\partial_y\right] m_y =\end{aligned}$$

Comparing directional derivative of ρ and m_x

By direct comparison of the coefficients of the differential operator applied to physical quantities, it follows that these differential operators are proportional to the same directional derivative. Here, as an example, the ccoefficients of the differential operators acting on ρ and m_x, $X^\rho\partial_x + Y^\rho\partial_y$ and $X^{m_x}\partial_x + Y^{m_x}\partial_y$, are compared with the goal of proving that

$$\frac{X^\rho}{Y^\rho} = \frac{X^{m_x}}{Y^{m_x}} \,,$$

or, avoiding singularities, $X^\rho Y^{m_x} = Y^\rho X^{m_x}$. With the explicit expressions of the coefficients,

$$\begin{aligned}&(\mp a^2 n_x \pm u_n u_x)(\mp n_x u_y) = ? = (\mp a^2 n_y \pm u_n u_y)(2a \mp u_x n_x \mp u_n) \\ 0\ ? &= a^2 n_x^2 u_y - u_n u_x n_x u_y \pm 2a^3 n_y - a^2 u_x n_x n_y - a^2 u_n n_y \mp 2a u_n u_y + u_n u_x u_y n_x + u_n^2 u_y \\ 0\ ? &= a^2 n_x^2 u_y \pm 2a^3 n_y - a^2 u_x n_x n_y - a^2 u_n n_y \mp 2a u_n u_y + u_n^2 u_y \\ 0\ ? &= a^2 u_y - a^2 n_y^2 u_y \pm 2a^3 n_y - a^2 u_x n_x n_y - a^2 u_n n_y \mp 2a u_n u_y + u_n^2 u_y\end{aligned}$$

and if $u_n = \pm a$,

$$\begin{aligned}0\ ? &= a^2 u_y - a^2 n_y^2 u_y \pm 2a^3 n_y - a^2 u_x n_x n_y - a^2 u_n n_y \mp 2a u_n u_y + u_n^2 u_y \\ 0\ ? &= a^2 u_y - a^2 n_y^2 u_y \pm 2a^3 n_y - a^2 u_x n_x n_y \mp a^3 n_y - 2a^2 u_y + a^2 u_y \\ 0\ ? &= -a^2 n_y(n_y u_y + n_x u_x) \pm 2a^3 n_y \mp a^3 n_y \\ 0\ ? &= \mp a^3 n_y \pm 2a^3 n_y \mp a^3 n_y = 0 \,.\end{aligned}$$

Ratio of the coefficients of the differential operators

With $a = \pm u_n$, the comparison of coefficients in differential operator acting on ρ and m_x

$$
\begin{aligned}
\frac{X^\rho}{X^{m_x}} &= \frac{\mp a^2 n_x \pm u_n u_x}{2a \mp u n_x \mp u_n} = \\
&= \frac{\mp a^2 n_x + a u}{2a \mp u n_x - a} = \\
&= a \frac{\mp a n_x + u}{a \mp u n_x} = \\
&= a \frac{-u_n n_x + u}{\pm u_n \mp u n_x} = \\
&= \pm a \frac{-(u n_x + v n_y) n_x + u}{u n_x + v n_y - u n_x} = \\
&= \pm a \frac{u n_y^2 - v n_y n_x}{+v n_y} = \\
&= \pm a \frac{u n_y - v n_x}{v} \, .
\end{aligned}
$$

$$
\begin{aligned}
\frac{Y^\rho}{Y^{m_x}} &= \frac{\mp a^2 n_y \pm u_n v}{\mp n_x v} = \\
&= \frac{\mp a \cdot a n_y \pm u_n v}{\mp n_x v} = \\
&= \frac{-a u_n n_y + a v}{\mp n_x v} = \\
&= \pm a \frac{u_n n_y - v}{n_x v} = \\
&= \pm a \frac{(u n_x + v n_y) n_y - v}{n_x v} = \\
&= \pm a \frac{u n_x n_y - v n_x^2}{n_x v} = \\
&= \pm a \frac{u n_y - v n_x}{v} \, .
\end{aligned}
$$

The comparison of coefficients in differential operator acting on ρ and m_y

$$
\begin{aligned}
\frac{X^\rho}{X^{m_y}} &= \frac{\mp a^2 n_x + a u}{\mp n_y u} = \\
&= \frac{\mp a \cdot a n_x \pm u_n u}{\mp n_y u} = \\
&= \frac{-a u_n n_x + a u}{\mp n_y u} = \\
&= \pm a \frac{u_n n_x - u}{n_y u} = \\
&= \pm a \frac{(u n_x + v n_y) n_x - u}{n_y u} = \\
&= \pm a \frac{-u n_y + v n_x}{u} = \\
&= \mp a \frac{u n_y - v n_x}{u} \, .
\end{aligned}
$$

$$\frac{Y^\rho}{Y^{m_y}} = \quad =$$

Thus it follows that on the characteristic lines,

$$0 = \pm a(un_y - vn_x)d\rho + vdm_x - udm_y \ ,$$

or, recalling that $m_x = \rho u$, $m_y = \rho v$

$$\begin{aligned} 0 &= \pm a(un_y - vn_x)d\rho + vdm_x - udm_y = \\ &= \pm a(un_y - vn_x)d\rho + \rho vdu + uvd\rho - \rho udv - uvd\rho = \\ &= \pm a(un_y - vn_x)d\rho + \rho vdu - \rho udv = \end{aligned}$$

or evaluating the ratio of these coefficients

$$\frac{\rho v}{\pm a(un_y - vn_x)}$$

todo *Find the general relation.*

Prandtl-Meyer relation

$$0 = \pm a(un_y - vn_x)d\rho + \rho vdu - \rho udv \ .$$

Defining θ as the flow angle w.r.t. the x-axis, s.t. and V the absolute value of the local velocity, it follows

$$\begin{aligned} u &= V\cos\theta \\ v &= V\sin\theta \ . \end{aligned}$$

From the relation $\mathbf{u} \cdot \hat{\mathbf{n}} = \pm a$ on Mach lines, being $\hat{\mathbf{n}}$ the unit vector orthogonal to the Mach lines.,

$$u_n = \pm a \quad , \quad V\cos\mu = \pm a \ ,$$

with μ the angle between the Mach line and the flow. The components of the unit normal vector read

$$\pm a = n_x u + n_y v = V\left(n_x\cos\theta + n_y \sin theta\right)$$

$$\begin{aligned} \pm\sin\mu &= n_x\cos\theta + n_y\sin\theta = \\ &= \pm\sin(\mu - \theta + \theta) = \\ &= \pm\left[\sin(\mu-\theta)\cos\theta + \cos(\mu-\theta)\sin\theta\right] = \end{aligned}$$

so that …

todo *Treat properly $\mp$ and $\pm$ for the 2 families of Mach lines*

$$\begin{aligned} n_x &= \sin(\theta \mp \mu) \\ n_y &= -\cos(\theta \mp \mu) \ . \end{aligned}$$

Now

$$\begin{aligned} un_y - vn_x &= V\left(-\cos\theta\cos(\theta\mp\mu) - \sin\theta\sin(\theta\mp\mu)\right) = \\ &= -V\cos(\theta - \theta \pm \mu) = -V\cos\mu \ . \end{aligned}$$

The differentials of the Cartesian components of the velocity in polar coordinates become

$$\begin{aligned} du &= d(V\cos\theta) = dV\cos\theta - V\sin\theta d\theta \\ dv &= d(V\sin\theta) = dV\sin\theta + V\cos\theta d\theta \end{aligned}$$

So that, putting everything together,

$$
\begin{aligned}
0 &= \pm a(un_y - vn_x)d\rho + \rho v du - \rho u dv = \\
&= \mp aV\cos\mu d\rho + \rho V\left(\sin\theta\cos\theta - \cos\theta\sin\theta\right)dV + \rho V^2\left(-\sin\theta\sin\theta - \cos\theta\cos\theta\right)d\theta = \\
&= \mp aV\cos\mu d\rho - \rho V^2 d\theta \ ,
\end{aligned}
$$

i.e.

$$
0 = \pm a\cos\mu\, d\rho + \rho V\, d\theta \ .
$$

Now, as $\sin\mu = \frac{a}{V} = \frac{1}{M}$, $\cos\mu = \sqrt{1-\sin^2\mu} = \sqrt{1-\frac{1}{M^2}}$, Prandtl-Meyer relation can be recast as

$$
\begin{aligned}
0 &= \pm\frac{1}{M}\sqrt{1-\frac{1}{M^2}}d\rho + \rho d\theta \\
0 &= \pm\sqrt{M^2-1}d\rho + M^2\rho d\theta \ .
\end{aligned}
$$

For an isothermal and irrotational flow, momentum equation reads

$$
\begin{aligned}
\mathbf{0} &= \mathbf{u}\cdot\nabla\mathbf{u} + a^2\frac{\nabla\rho}{\rho} = \\
&= \nabla\frac{|\mathbf{u}|^2}{2} + \omega\times\mathbf{u} + a^2\frac{\nabla\rho}{\rho} = \\
&= \nabla\frac{|\mathbf{u}|^2}{2} + a^2\frac{\nabla\rho}{\rho} \ ,
\end{aligned}
$$

and thus, as $\frac{d\rho}{\rho} = -\frac{1}{a^2}d\frac{V^2}{2} = -\frac{1}{a^2}VdV$, Prandtl-Meyer relation becomes

$$
\begin{aligned}
0 &= \mp\sqrt{M^2-1}\frac{1}{a^2}VdV + M^2\, d\theta \\
0 &= \mp\sqrt{M^2-1}\,\underbrace{\frac{V^2}{a^2}}_{=M^2}\,\frac{VdV}{V^2} + M^2\, d\theta \\
0 &= \mp\sqrt{M^2-1}\frac{dV}{V} + d\theta \\
0 &= \mp\sqrt{M^2-1}\frac{dM}{M} + d\theta \ ,
\end{aligned}
$$

with the last equation containing only non-dimensional quantities.

…

For $s_2 = u_n$, $u_n = 0$ to get $s_2 = 0$, and thus the corresponding left eigenvector reads

$$
\mathbf{l}_{0;2}^T = \frac{1}{\rho a}\left[-\mathbf{t}^T\mathbf{u} \mid \mathbf{t}^T\right] \ ,
$$

with $\mathbf{t} = \begin{bmatrix} n_y \\ -n_x \end{bmatrix}$, so that $\mathbf{t}^T\mathbf{n} = 0$. The quasi linear form of the equations becomes

$$0 = \frac{1}{\rho a}\left[\, -\mathbf{t}^T\mathbf{u} \mid \mathbf{t}^T \,\right]\left(\partial_t \begin{bmatrix} \rho \\ m_x \\ m_y \end{bmatrix} + \begin{bmatrix} \cdot & 1 & \cdot \\ a^2 - \frac{m_x m_x}{\rho^2} & \frac{2m_x}{\rho} & \cdot \\ -\frac{m_x m_y}{\rho^2} & \frac{m_y}{\rho} & \frac{m_x}{\rho} \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ m_x \\ m_y \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & 1 \\ -\frac{m_x m_y}{\rho^2} & \frac{m_y}{\rho} & \frac{m_x}{\rho} \\ a^2 - \frac{m_y m_y}{\rho^2} & \cdot & \frac{2m_y}{\rho} \end{bmatrix} \partial_y \begin{bmatrix} \rho \\ m_x \\ m_y \end{bmatrix} = \mathbf{0}\right) =$$

$$\begin{aligned}
0 &= \left[\, -\mathbf{t}^T\mathbf{u} \mid \mathbf{t}^T \,\right]\partial_t \begin{bmatrix} \rho \\ \mathbf{m} \end{bmatrix} + \\
&\quad + \left[\, t_x a^2 - \mathbf{t}^T\mathbf{u} u_x \mid -\mathbf{t}^T\mathbf{u} + t_x u_x + \mathbf{t}^T\mathbf{u} \mid t_y u_x \,\right]\partial_x \begin{bmatrix} \rho \\ \mathbf{m} \end{bmatrix} + \\
&\quad + \left[\, t_y a^2 - \mathbf{t}^T\mathbf{u} u_y \mid t_x u_y \mid -\mathbf{t}^T\mathbf{u} + t_y u_y + \mathbf{t}^T\mathbf{u} \,\right]\partial_y \begin{bmatrix} \rho \\ \mathbf{m} \end{bmatrix} = \\
0 &= \left[\, -\mathbf{t}^T\mathbf{u} \mid \mathbf{t}^T \,\right]\partial_t \begin{bmatrix} \rho \\ \mathbf{m} \end{bmatrix} + \\
&\quad + a^2\left(t_x\partial_x + t_y\partial_y\right)\rho - u_t\left(u_x\partial_x + u_y\partial_y\right)\rho + t_x\left(u_x\partial_x + u_y\partial_y\right)m_x + t_y\left(u_x\partial_x + u_y\partial_y\right)m_y = \\
&=
\end{aligned}$$

or choosing $\hat{\mathbf{t}} = \hat{\mathbf{u}}$, s.t. $\mathbf{u} = u\hat{\mathbf{t}}$,

$$\begin{aligned}
0 &= \ldots\,\partial_t \cdots + a^2\partial_u\rho - u^2\partial_u\rho + ut_x\partial_u m_x + ut_y\partial_u m_y = \\
&= \ldots\,\partial_t \cdots + a^2\partial_u\rho - u^2\partial_u\rho + \rho\,\underbrace{ut_x}_{u_x}\,\partial_u u_x + \rho\,\underbrace{ut_y}_{u_y}\,\partial_u u_y + \underbrace{uu_x t_x\partial_u\rho + uu_y t_y\partial_u\rho}_{=u^2\partial_u\rho} = \\
&= \ldots\,\partial_t \cdots + a^2\,\partial_u\rho + \rho\partial_u\frac{|\mathbf{u}|^2}{2}\,.
\end{aligned}$$

In **steady conditions**, dividing by $\rho \neq 0$

$$0 = a^2\,\frac{\partial_u\rho}{\rho} + \partial_u\frac{u^2}{2} = \partial_u\left(a^2\ln\rho + \frac{u^2}{2}\right)\,.$$

This is the isothermal version of the Bernoulli's theorem along a streamline for compressible flows, in absence of volume force. This can be derived from momentum equation in steady conditions, after scalar multiplication by the velocity field

$$\begin{aligned}
0 &= \mathbf{u}\cdot\left[\rho\mathbf{u}\cdot\nabla\mathbf{u} + \nabla p\right] = \\
&= \mathbf{u}\cdot\left[\rho\nabla\frac{|\mathbf{u}|^2}{2} + \omega\times\mathbf{u} + \nabla p\right] = \\
&= \rho\mathbf{u}\cdot\left[\nabla\frac{|\mathbf{u}|^2}{2} + a^2\frac{\nabla\rho}{\rho}\right] = \\
&= \rho\mathbf{u}\cdot\nabla\left(\frac{|\mathbf{u}|^2}{2} + a^2\ln\rho\right)\,.
\end{aligned}$$

Euler equations for homoentropic inviscid compressible flows - 2d

Homoentropic flows have constant and uniform entropy, $s(\mathbf{r}, t) = \overline{s}$. This condition immediately follows from the entropy equation under the assumptions of:

1. negligible viscous effects
2. negligible heat conductivity
3. uniform entropy at the boundaries

so that, there's no source in the governing equation of the entropy of material particles (because of 1. and 2., $D_t s = 0$) and the value of the entropy of material particles entering the domain is constant and uniform (because of 3., $s = \overline{s}$).

todo *Check if anything changes from the very beginning, e.g. transforming $\nabla p = a^2 \nabla \rho$ and collecting in the flux term $\nabla \cdot (\rho a^2 \mathbf{I})$.*

Under these conditions, all the computation done for the isothermal flow model are valid except for the ones assuming constant speed of sound a. In general the speed of sound is a function of the thermodynamic state. As an example, using ρ, s as the pair of thermodynamic independent variables, the speed of sound reads

$$a^2(\rho, s) = \left(\frac{\partial p}{\partial \rho}\right)_s .$$

Here a is a function of the density only, as the constant value of entropy can be treated as a constant parameter.

Euler equations for inviscid compressible flows - 2d

1. Using conservative variables.

Direct computation with conservative variables

Conservative form reads

$$\begin{cases} \partial_t \rho + \nabla \cdot \mathbf{m} = 0 \\ \partial_t \mathbf{m} + \nabla \cdot \left[\frac{\mathbf{m} \otimes \mathbf{m}}{\rho} + p\mathbf{I}\right] = 0 \\ \partial_t E^t + \nabla \cdot \left(\frac{E^t + p}{\rho}\mathbf{m}\right) = 0 \,. \end{cases}$$

Convective form in a 2-dimensional domain reads

$$\partial_t \begin{bmatrix} \rho \\ m_x \\ m_y \\ E^t \end{bmatrix} + \begin{bmatrix} \cdot & 1 & \cdot & \cdot \\ \partial_\rho p - \frac{m_x m_x}{\rho^2} & \frac{2m_x}{\rho} + \partial_{m_x} P & \partial_{m_y} P & \partial_{E^t} p \\ -\frac{m_x m_y}{\rho^2} & \frac{m_y}{\rho} & \frac{m_x}{\rho} & \cdot \\ \frac{-h^t + \partial_\rho p}{\rho} m_x + \frac{\partial_\rho P}{\rho} & h^t + \frac{\partial_{m_x} P}{\rho} m_x & \frac{\partial_{m_y} P}{\rho} m_x & \frac{1 + \partial_{E^t} P}{\rho} m_x \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ m_x \\ m_y \\ E^t \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & 1 \\ -\frac{m_x m_y}{\rho^2} & \frac{m_y}{\rho} & \frac{m_x}{\rho} \\ \partial_\rho p - \frac{m_y m_y}{\rho^2} & \partial_{m_x} P & \frac{2m_y}{\rho} + \partial_{m_y} P \\ \frac{-h^t + \partial_\rho p}{\rho} m_y & \frac{\partial_{m_x} P}{\rho} m_y & h^t + \frac{\partial_{m_y} P}{\rho} m_y \end{bmatrix}$$

Thus,

$$\begin{aligned} \mathbf{A}_n &= \begin{bmatrix} 0 & \mathbf{n}^T & 0 \\ \partial_\rho p \mathbf{n} - u_n \mathbf{u} & u_n \mathbf{I} + \mathbf{u}\mathbf{n}^T + \mathbf{n}\partial_{\mathbf{m}} P & \partial_{E^t} p \mathbf{n} \\ (-h^t + \partial_\rho p) u_n & h^t \mathbf{n}^T + u_n \partial_{\mathbf{m}} P & (1 + \partial_{E^t} P)\, u_n \end{bmatrix} = \\ &= \begin{bmatrix} 0 & n_x & n_y & 0 \\ \partial_\rho p\, n_x - u_n u_x & u_n + n_x u + n_x \partial_{m_x} P & n_y u + n_x \partial_{m_y} P & \partial_{E^t} p\, n_x \\ \partial_\rho p\, n_y - u_n u_y & n_x v + n_y \partial_{m_x} P & u_n + n_y v + n_y \partial_{m_y} P & \partial_{E^t} p\, n_y \\ (-h^t + \partial_\rho P) u_n & h^t n_x + u_n \partial_{m_x} P & h^t n_y + u_n \partial_{m_y} P & (1 + \partial_{E^t} P)\, u_n \end{bmatrix} . \end{aligned}$$

with

$$\begin{aligned} \partial_{m_i} P &= \partial_e P|_\rho \left(-\frac{m_i}{\rho^2}\right) \\ \partial_{\mathbf{m}} P &= -\frac{\mathbf{m}}{\rho^2}\, \partial_e P|_\rho = -\frac{\mathbf{u}}{\rho} \partial_e P|_\rho \end{aligned}$$

and thus

$$
\begin{aligned}
\mathbf{A}_n &= \begin{bmatrix} 0 & n_x & n_y & 0 \\ \partial_\rho P\, n_x - u_n u_x & u_n + n_x u(1-\partial_{E^t} P) & n_y u - n_x v \partial_{E^t} P & \partial_{E^t} P\, n_x \\ \partial_\rho P\, n_y - u_n u_y & n_x v - n_y u \partial_{E^t} P & u_n + n_y v(1-\partial_{E^t} P) & \partial_{E^t} P\, n_y \\ (-h^t + \partial_\rho P) u_n & h^t n_x + u_n \partial_{m_x} P & h^t n_y + u_n \partial_{m_y} P & (1+\partial_{E^t} P)\, u_n \end{bmatrix} \\
&= \begin{bmatrix} 0 & \mathbf{n}^T & 0 \\ \partial_\rho P \mathbf{n} - u_n \mathbf{u} & u_n \mathbf{I} + \mathbf{u}\mathbf{n}^T - \mathbf{n}\mathbf{u}^T \partial_{E^t} P & \partial_{E^t} P \mathbf{n} \\ (-h^t + \partial_\rho P) u_n & h^t \mathbf{n}^T - u_n \mathbf{u}^T \partial_{E^t} P & (1+\partial_{E^t} P)\, u_n \end{bmatrix} .
\end{aligned}
$$

Derivative of the pressure function $P(\rho, \mathbf{m}, E^t)$

As $E^t = \rho\left(e + \frac{|\mathbf{u}|^2}{2}\right) = \rho e^t + \frac{|\mathbf{m}|^2}{2\rho}$, $e^t = \frac{E^t}{\rho} - \frac{|\mathbf{m}|^2}{2\rho^2}$

$$P(\rho, \mathbf{m}, E^t)$$

$$P(\rho, e) = P\left(\rho, e\left(\rho, \mathbf{m}, E^t\right)\right) \ ,$$

$$
\begin{aligned}
\partial_\rho P\big|_{\mathbf{m},E^t} &= \partial_\rho P\big|_e + \partial_e P\big|_\rho\, \partial_\rho e\big|_{\mathbf{m},E^t} &&= \partial_\rho P\big|_e + \partial_e P\big|_\rho \left(-\frac{E^t}{\rho^2} + \frac{|\mathbf{m}|^2}{\rho^3}\right) \\
\partial_{m_i} P\big|_{\rho,E^t} &= \partial_e P\big|_\rho\, \partial_{m_i} e\big|_{\rho,E^t} &&= \partial_e P\big|_\rho \left(-\frac{m_i}{\rho^2}\right) \\
\partial_{E^t} P\big|_{\rho,\mathbf{m}} &= \partial_e P\big|_\rho\, \partial_{E^t} e\big|_{\rho,\mathbf{m}} &&= \partial_e P\big|_\rho \left(\frac{1}{\rho}\right)
\end{aligned}
$$

and by direct comparison,

$$
\begin{aligned}
\partial_\rho P\big|_{\mathbf{m},E^t} &= \partial_\rho P\big|_e + \partial_{E^t} P\big|_{\rho,\mathbf{m}} \left(-e^t + |\mathbf{u}|^2\right) \\
\partial_{\mathbf{m}} P\big|_{\rho,E^t} &= -\mathbf{u}\, \partial_{E^t} P\big|_{\rho,\mathbf{m}}
\end{aligned}
$$

Speed of sound

…link to an updated version of Thermodynamics:potentials:Maxwell's relations

$$a^2(\rho, s) = \frac{\partial p}{\partial \rho}\bigg|_s$$

Independent variables $f(\rho, e)$

$$a^2(\rho, e) = \ ?$$

$$
\begin{aligned}
a^2(\rho, e(\rho, s)) &= \frac{\partial p}{\partial \rho}\bigg|_s = \\
&= \frac{\partial p}{\partial \rho}\bigg|_e + \frac{\partial p}{\partial e}\bigg|_\rho \frac{\partial e}{\partial \rho}\bigg|_s = \\
&= \frac{\partial p}{\partial \rho}\bigg|_e + \frac{\partial p}{\partial e}\bigg|_\rho \frac{p}{\rho^2} \ .
\end{aligned}
$$

Thermodynamic potentials, partial derivatives and Maxwell's relations

$$de = T ds + \frac{p}{\rho^2} d\rho \quad , \quad T = \frac{\partial p}{\partial s}\bigg|_\rho \quad , \quad \frac{p}{\rho^2} = \frac{\partial e}{\partial \rho}\bigg|_s$$

Independent variables $f(\rho, \mathbf{m}, E^t)$ With

$$E^t = \rho\left(\frac{|\mathbf{u}|^2}{2} + e\right) = \frac{|\mathbf{m}|^2}{2\rho} + \rho e\,.$$

$$a^2(\rho, \mathbf{m}, E^t(\rho, \mathbf{m}, s))$$

$$\begin{aligned}
a^2(\rho, \mathbf{m}, E^t) &= \left.\frac{\partial p}{\partial \rho}\right|_s (\rho, \mathbf{m}, E^t(\rho, \mathbf{m}, e(\rho, s))) = \\
&= \left.\frac{\partial p}{\partial \rho}\right|_{\mathbf{m}, E^t} + \left.\frac{\partial p}{\partial E^t}\right|_{\rho, \mathbf{m}} \left[\left.\frac{\partial E^t}{\partial \rho}\right|_{\mathbf{m}, e} + \left.\frac{\partial E^t}{\partial e}\right|_{\rho, \mathbf{m}} \left.\frac{\partial e}{\partial \rho}\right|_s\right] = \\
&= \left.\frac{\partial p}{\partial \rho}\right|_{\mathbf{m}, E^t} + \left.\frac{\partial p}{\partial E^t}\right|_{\rho, \mathbf{m}} \left[\left(-\frac{|\mathbf{m}|^2}{2\rho^2} + e\right) + \rho \left.\frac{\partial e}{\partial \rho}\right|_s\right] =
\end{aligned}$$

Eigenvalues

$$0 = |\mathbf{A}_n - s\mathbf{I}| = \begin{bmatrix} -s & n_x & n_y & 0 \\ \partial_\rho p\, n_x - u_n u_x & -s + u_n + n_x u & n_y u & \partial_{E^t} p\, n_x \\ \partial_\rho p\, n_y - u_n u_y & n_x v & -s + u_n + n_y v & \partial_{E^t} p\, n_y \\ (-h^t + \partial_\rho P) u_n & h^t n_x + u_n \partial_{m_x} P & h^t n_y + u_n \partial_{m_y} P & -s + (1 + \partial_{E^t} P)\, u_n \end{bmatrix}.$$

Using physical variables $(\rho, \mathbf{u}, s)$ and variable transformation

Convection matrix

Let $\mathbf{u} = (\rho, \mathbf{m}, E^t)$ the vector of conservative variables, and $\mathbf{v} = (\rho, s\mathbf{u}, e)$ the vector of physical variables. As proved in the following section, the relation between the convection matrices reads

$$\mathbf{A}_\mathbf{n}^\mathbf{u} = \mathbf{U}_\mathbf{v} \mathbf{A}_\mathbf{n}^\mathbf{v} \mathbf{V}_\mathbf{u}\,,$$

with

$$\mathbf{A}_\mathbf{n}^\mathbf{v} = \begin{bmatrix} u_n & \rho \mathbf{n}^T & \cdot \\ \frac{1}{\rho}\partial_\rho p|_e \mathbf{n} & u_n \mathbf{I} & \frac{1}{\rho}\partial_e p|_\rho \mathbf{n} \\ \cdot & \frac{p}{\rho}\mathbf{n}^T & u_n \end{bmatrix},$$

and

$$\mathbf{U}_\mathbf{v} = \begin{bmatrix} 1 & \cdot & \cdot & \cdot \\ u & \rho & \cdot & \cdot \\ v & \cdot & \rho & \cdot \\ e + \frac{|\mathbf{u}|^2}{2} & \rho u & \rho v & \rho \end{bmatrix} = \begin{bmatrix} 1 & \cdot & \cdot \\ \mathbf{u} & \mathbf{I}_2 & \cdot \\ e^t & \rho \mathbf{u}^T & \rho \end{bmatrix}$$

$$\mathbf{V}_\mathbf{u} = \mathbf{U}_\mathbf{v}^{-1} = \begin{bmatrix} 1 & \cdot & \cdot & \cdot \\ -\frac{u}{\rho} & \frac{1}{\rho} & \cdot & \cdot \\ -\frac{v}{\rho} & \cdot & \frac{1}{\rho} & \cdot \\ \frac{1}{\rho}\left(\frac{|\mathbf{u}|^2}{2} - e\right) & -\frac{u}{\rho} & -\frac{v}{\rho} & \frac{1}{\rho} \end{bmatrix} = \begin{bmatrix} 1 & \cdot & \cdot \\ -\frac{\mathbf{u}}{\rho} & \frac{1}{\rho}\mathbf{I}_2 & \cdot \\ \frac{1}{\rho}\left(|\mathbf{u}|^2 - e^t\right) & -\frac{\mathbf{u}^T}{\rho} & \frac{1}{\rho} \end{bmatrix}$$

the gradient and its inverse (easy to evaluate for the triangular structure of the $\mathbf{U}_\mathbf{v}$ matrix) of the change of variables

$$\mathbf{u} = \begin{bmatrix} \rho \\ m_x \\ m_y \\ E^t \end{bmatrix} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho\left(e + \frac{|\mathbf{u}|^2}{2}\right) \end{bmatrix}$$

Right and left eigenvectors

With the right and left eigenvectors using physical variables $(\rho, \mathbf{u}, e)$,

$$\mathbf{R}^{\mathbf{v}} = \begin{bmatrix} \rho & \rho & 0 & \rho \\ -an_x & 0 & an_y & an_x \\ -an_y & 0 & -an_x & an_y \\ \frac{p}{\rho} & -\rho\frac{\partial_\rho p|_e}{\partial_e p|_\rho} & 0 & \frac{p}{\rho} \end{bmatrix} = \begin{bmatrix} \rho & \rho & 0 & \rho \\ -a\mathbf{n} & \mathbf{0} & a\mathbf{t} & a\mathbf{n} \\ \frac{p}{\rho} & -\rho\frac{\partial_\rho p|_e}{\partial_e p|_\rho} & 0 & \frac{p}{\rho} \end{bmatrix}$$

$$\mathbf{L}^{\mathbf{v}} = \begin{bmatrix} \frac{P_\rho}{2\rho a^2} & -\frac{1}{2a}\mathbf{n}^T & \frac{P_e}{2\rho a^2} \\ \frac{P}{\rho^2}\frac{P_e}{\rho a^2} & \mathbf{0}^T & -\frac{P_e}{\rho a^2} \\ 0 & \frac{\mathbf{t}^T}{a} & 0 \\ \frac{P_\rho}{2\rho a^2} & +\frac{1}{2a}\mathbf{n}^T & \frac{P_e}{2\rho a^2} \end{bmatrix}$$

Right eigenvectors w.r.t. conservative variables are

$$\mathbf{R}^{\mathbf{u}} = \mathbf{U}_{\mathbf{v}}\mathbf{R}^{\mathbf{v}} = \begin{bmatrix} \rho & \rho & 0 & \rho \\ \rho u - \rho a n_x & \rho u & \rho a n_y & \rho u + \rho a n_x \\ \rho u - \rho a n_y & \rho v & -\rho a n_x & \rho u + \rho a n_y \\ \rho e^t - \rho u_n a + p & \rho e^t - \rho^2\frac{P_\rho}{P_e} & \rho a u_t & \rho e^t + \rho u_n a + p \end{bmatrix} =$$

$$= \begin{bmatrix} \rho & \rho & 0 & \rho \\ \rho u - \rho a n_x & \rho u & \rho a n_y & \rho u + \rho a n_x \\ \rho u - \rho a n_y & \rho v & -\rho a n_x & \rho u + \rho a n_y \\ \rho h^t - \rho u_n a & \rho e^t - \rho^2\frac{P_\rho}{P_e} & \rho a u_t & \rho h^t + \rho u_n a \end{bmatrix},$$

with $h^t = e^t + \frac{|\mathbf{u}|^2}{2}$, and

$$\rho e^t - \rho^2\frac{P_\rho}{P_e} = \rho e^t + P - P - \rho^2\frac{P_\rho}{P_e} = \rho h^t - \frac{\rho^2}{P_e}\left(\frac{P}{\rho^2}P_e + P_\rho\right) = \rho h^t - \frac{\rho^2}{P_e}a^2 \ .$$

Or using the derivatives of $\Pi(\rho, \mathbf{m}, E^t)$,

$$\begin{aligned} E^t - \rho^2\frac{P_\rho}{P_e} &= E^t - \rho^2\frac{\partial_\rho\Pi - \partial_{E^t}\Pi(-e^t + |\mathbf{u}|^2)}{\rho\partial_{E^t}\Pi} = \\ &= E^t - \rho\frac{\partial_\rho\Pi}{\partial_{E^t}\Pi} - E^t + \rho|\mathbf{u}|^2 = \\ &= \rho|\mathbf{u}|^2 - \rho\frac{\partial_\rho\Pi}{\partial_{E^t}\Pi} \ . \end{aligned}$$

Left eigenvectors w.r.t. conservative variables are

$$\mathbf{L}^{\mathbf{u}} = \mathbf{L}^{\mathbf{v}}\mathbf{V}_{\mathbf{u}} = \begin{bmatrix} \frac{P_\rho}{2\rho a^2} + \frac{u_n}{2\rho a} + \frac{1}{2\rho a^2}\frac{P_e}{\rho}(|\mathbf{u}|^2 - e^t) & -\frac{1}{2\rho a}n_x - \frac{u}{2\rho a^2}\frac{P_e}{\rho} & -\frac{1}{2\rho a}n_y - \frac{v}{2\rho a^2}\frac{P_e}{\rho} & \frac{P_e}{2\rho^2 a^2} \\ \frac{P}{\rho^2}\frac{P_e}{\rho a^2} - \frac{P_e}{\rho a^2}\frac{|\mathbf{u}|^2 - e^t}{\rho} & \frac{P_e}{\rho a^2}\frac{u}{\rho} & \frac{P_e}{\rho a^2}\frac{v}{\rho} & -\frac{P_e}{\rho^2 a^2} \\ -\frac{u_t}{\rho a} & \frac{t_x}{\rho a} & \frac{t_y}{\rho a} & 0 \\ \frac{P_\rho}{2\rho a^2} - \frac{u_n}{2\rho a} + \frac{1}{2\rho a^2}\frac{P_e}{\rho}(|\mathbf{u}|^2 - e^t) & \frac{1}{2\rho a}n_x - \frac{u}{2\rho a^2}\frac{P_e}{\rho} & \frac{1}{2\rho a}n_y - \frac{v}{2\rho a^2}\frac{P_e}{\rho} & \frac{P_e}{2\rho^2 a^2} \end{bmatrix} =$$

$$= \frac{1}{2\rho a^2}\begin{bmatrix} \Pi_\rho + a u_n & -a n_x + \Pi_{m_x} & -a n_y + \Pi_{m_y} & \Pi_{E^t} \\ 2a^2 - 2\Pi_\rho & -2\Pi_{m_x} & 2 - \Pi_{m_y} & -2\Pi_{E^t} \\ -2a u_t & 2a t_x & 2a t_y & 0 \\ \Pi_\rho - a u_n & a n_x + \Pi_{m_x} & a n_y + \Pi_{m_y} & \Pi_{E^t} \end{bmatrix}$$

being

$$\Pi_\rho = P_\rho + \frac{P_e}{\rho}\left(-e^t + |\mathbf{u}|^2\right)$$
$$\Pi_\mathbf{m} = -\frac{\mathbf{u}}{\rho}P_e$$
$$\Pi_{E^t} = \frac{1}{\rho}P_e$$

$$2\frac{P}{\rho^2}P_e - 2P_e\frac{|\mathbf{u}|^2 - e^t}{\rho} = 2\frac{P}{\rho^2}P_e + 2P_\rho - 2\Pi_\rho = \\ = 2a^2 - 2\Pi_\rho$$

and the speed of sound

$$a^2(\rho, s) = \partial_\rho P|_s$$

with $P(\rho, e(\rho, s))$

$$a^2 = \partial_\rho P|_e + \partial_\rho e|_s\, \partial_e P|_\rho = \\ = \partial_\rho P|_e + \frac{P}{\rho^2}\partial_e\, P|_\rho\ .$$

2. Using physical variables $(\rho, \mathbf{u}, s)$**.** So that speed of sound $a^2(\rho, s) = \left(\partial_\rho p\right)_s$ naturally appears.

Change of coordinates and convective form

Starting from a set of variables $\mathbf{u}$ and the convective form of the equations

$$\mathbf{s} = \partial_t \mathbf{u} + \sum_k \mathbf{A}^k \partial_k \mathbf{u}\ ,$$

using another set of variables $\mathbf{v}$, the equations become

$$\mathbf{s} = \partial_t \mathbf{u} + \sum_k \mathbf{A}^k \partial_k \mathbf{u}$$
$$\mathbf{s} = \mathbf{U}_\mathbf{v}(\mathbf{v})\partial_t \mathbf{v} + \sum_k \mathbf{A}^k \mathbf{U}_\mathbf{v}(\mathbf{v})\partial_k \mathbf{v}\ ,$$

being $d\mathbf{u} = \mathbf{U}_\mathbf{v} d\mathbf{v}$. If the transformation is non-singular, and $\mathbf{U}_\mathbf{v}^{-1} = \mathbf{V}_\mathbf{u}(\mathbf{v})$, then the differential equations can be recast as

$$\partial_t \mathbf{v} + \sum_k \mathbf{V}_\mathbf{u}\mathbf{A}^k\mathbf{U}_\mathbf{v}\partial_k \mathbf{v} = \mathbf{V}_\mathbf{u}\mathbf{s}$$

With a coordinate transformation, the normal and tangential matrices become

$$\mathbf{A}_\mathbf{n}^\mathbf{v} = \sum_k n_k \mathbf{V}_\mathbf{u}\mathbf{A}^k\mathbf{U}_\mathbf{v} = \\ = \mathbf{V}_\mathbf{u} \underbrace{\sum_k n_k \mathbf{A}^k}_{\mathbf{A}_\mathbf{n}^\mathbf{u}} \mathbf{U}_\mathbf{v} = \mathbf{V}_\mathbf{u}\mathbf{A}_\mathbf{n}^\mathbf{u}\mathbf{U}_\mathbf{v}\ ,$$

$$\mathbf{A}_{\mathbf{t}_i}^\mathbf{v} = \cdots = \mathbf{V}_\mathbf{u}\mathbf{A}_{\mathbf{t}_i}^\mathbf{u}\mathbf{U}_\mathbf{v}\ .$$

If the change of variables is non-singular, the Jacobian matrices $\mathbf{V_u}$, $\mathbf{U_v}$ are non singular and reciprocally inverse. Thus, the **determinant** of the matrix $\mathbf{A_n}$ is independent from the set of chosen variables, as

$$\begin{aligned}\det\left(\mathbf{A_n^v}\right) &= \det\left(\mathbf{V_u}\right)\det\left(\mathbf{A_n^u}\right)\det\left(\mathbf{U_v}\right) = \\ &= \det\left(\mathbf{V_u}\right)\det\left(\mathbf{A_n^u}\right)\det\left(\mathbf{V_u}\right)^{-1} = \\ &= \det\left(\mathbf{A_n^u}\right)\ .\end{aligned}$$

Thus, the **eigenvalues** are independent from the set of variables as well

$$\begin{aligned}0 &= \det\left(s\mathbf{I} - \mathbf{A_n^v}\right) = \\ &= \det\left(\mathbf{V_u}\right)\det\left(s\mathbf{I} - \mathbf{A_n^u}\right)\det\left(\mathbf{U_v}\right) = \\ &= \det\left(s\mathbf{I} - \mathbf{A_n^u}\right)\ .\end{aligned}$$

The relation between **right eigenvectors** immediately follows

$$\begin{aligned}\mathbf{A_n^v}\mathbf{R^v} &= \mathbf{R^v}\mathbf{S} \\ \mathbf{V_u}\mathbf{A_n^u}\mathbf{U_v}\mathbf{R^v} &= \mathbf{R^v}\mathbf{S} \\ \mathbf{A_n^u}\underbrace{\mathbf{U_v}\mathbf{R^v}}_{\mathbf{R^u}} &= \underbrace{\mathbf{U_v}\mathbf{R^v}}_{\mathbf{R^u}}\mathbf{S} \\ \mathbf{A_n^u}\mathbf{R^u} &= \mathbf{R^u}\mathbf{S}\ ,\end{aligned}$$

with $\mathbf{R^u} = \mathbf{U_v}\mathbf{R^v}$, or for any individual eigenvector $\mathbf{r_i^u} = \mathbf{U_v}\mathbf{r_i^v}$.

The relation between **left eigenvectors** analogously follows

$$\begin{aligned}\mathbf{L^v}\mathbf{A_n^v} &= \mathbf{S}\mathbf{L^v} \\ \mathbf{L^v}\mathbf{V_u}\mathbf{A_n^u}\mathbf{U_v} &= \mathbf{S}\mathbf{L^v} \\ \underbrace{\mathbf{L^v}\mathbf{V_u}}_{\mathbf{L^u}}\mathbf{A_n^u} &= \mathbf{S}\underbrace{\mathbf{L^v}\mathbf{V_u}}_{\mathbf{L^u}} \\ \mathbf{L^u}\mathbf{A_n^u} &= \mathbf{S}\mathbf{L^u}\ ,\end{aligned}$$

with $\mathbf{L^u} = \mathbf{L^v}\mathbf{U_v}$, or for any individual eigenvector $\left(\mathbf{l_i^u}\right)^T = \left(\mathbf{l_i^v}\right)^T\mathbf{U_v}$, or $\mathbf{l_i^u} = \mathbf{U_v}^T\mathbf{l_i^v}$.

Convective form in a 2-dimensional domain reads

$$\begin{aligned}D_t\rho + \rho\nabla\cdot\mathbf{u} &= 0 \\ D_t\mathbf{u} + \frac{1}{\rho}\nabla p &= \mathbf{0} \\ D_t e + \frac{p}{\rho}\nabla\cdot\mathbf{u} &= 0\ .\end{aligned}$$

$$\begin{aligned}0 &= \partial_t\rho + u\partial_x\rho + \rho\partial_x u + v\partial_y\rho + \rho\partial_y v \\ 0 &= \partial_t u + u\partial_x u + v\partial_y u + \frac{1}{\rho}\partial_x p = \\ &= \partial_t u + u\partial_x u + v\partial_y u + \frac{1}{\rho}\partial_\rho p|_e\partial_x\rho + \frac{1}{\rho}\partial_e p|_\rho\partial_x e \\ 0 &= \partial_t v + u\partial_x v + v\partial_y v + \frac{1}{\rho}\partial_y p = \\ &= \partial_t v + u\partial_x v + v\partial_y v + \frac{1}{\rho}\partial_\rho p|_e\partial_y\rho + \frac{1}{\rho}\partial_e p|_\rho\partial_y e \\ 0 &= \partial_t e + u\partial_x e + v\partial_y e + \frac{p}{\rho}\left(\partial_x u + \partial_y v\right)\ .\end{aligned}$$

Convective form in a 2-dimensional domain reads

$$\partial_t \begin{bmatrix} \rho \\ u \\ v \\ e \end{bmatrix} + \begin{bmatrix} u & \rho & \cdot & \cdot \\ \frac{1}{\rho}\partial_\rho p & u & \cdot & \frac{1}{\rho}\partial_e p \\ \cdot & \cdot & u & \cdot \\ \cdot & \frac{p}{\rho} & \cdot & u \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ u \\ v \\ e \end{bmatrix} + \begin{bmatrix} v & \cdot & \rho & \cdot \\ \cdot & v & \cdot & \cdot \\ \frac{1}{\rho}\partial_\rho p & \cdot & v & \frac{1}{\rho}\partial_e p \\ \cdot & \cdot & \frac{p}{\rho} & v \end{bmatrix} \partial_y \begin{bmatrix} \rho \\ u \\ v \\ e \end{bmatrix} = \mathbf{0}$$

Thus,

$$\begin{aligned} \mathbf{A}_n = \begin{bmatrix} u_n & \rho\mathbf{n}^T & \cdot \\ \frac{1}{\rho}\partial_\rho p\,\mathbf{n} & u_n\mathbf{I} & \frac{1}{\rho}\partial_e p\,\mathbf{n} \\ \cdot & \frac{p}{\rho}\mathbf{n}^T & u_n \end{bmatrix} = \\ = \ldots \end{aligned}$$

Eigenvalues of the $\mathbf{A_n}$ matrix using variables $(\rho, \mathbf{u}, e)$.

Let $\mathbf{M} = \mathbf{A}_n - \lambda\mathbf{I}$. We expand $\det(\mathbf{M})$ along the **first row**:

$$\det(\mathbf{M}) = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13} - a_{14}M_{14}$$

Identification of Minors Substituting the elements from the first row $[\chi, \rho n_x, \rho n_y, 0]$:

$$\det(\mathbf{M}) = \chi \det \underbrace{\begin{bmatrix} \chi & 0 & \beta n_x \\ 0 & \chi & \beta n_y \\ \eta n_x & \eta n_y & \chi \end{bmatrix}}_{M_{11}} - \rho n_x \det \underbrace{\begin{bmatrix} \alpha n_x & 0 & \beta n_x \\ \alpha n_y & \chi & \beta n_y \\ 0 & \eta n_y & \chi \end{bmatrix}}_{M_{12}} + \rho n_y \det \underbrace{\begin{bmatrix} \alpha n_x & \chi & \beta n_x \\ \alpha n_y & 0 & \beta n_y \\ 0 & \eta n_x & \chi \end{bmatrix}}_{M_{13}} - 0$$

Evaluating Minor M_{11} Expanding M_{11} along its first row:

$$\det(M_{11}) = \chi(\chi^2 - \beta\eta n_y^2) - 0 + \beta n_x(0 - \chi\eta n_x) = \chi^3 - \chi\beta\eta n_y^2 - \chi\beta\eta n_x^2$$

Since $n_x^2 + n_y^2 = 1$:

$$\det(M_{11}) = \chi(\chi^2 - \beta\eta)$$

Evaluating Minor M_{12} Expanding M_{12} along its first row:

$$\det(M_{12}) = \alpha n_x(\chi^2 - \beta\eta n_y^2) - 0 + \beta n_x(\alpha n_x \eta n_y - 0) = \alpha\chi^2 n_x$$

Note. Cross terms often cancel in symmetric projections)}

Note. In the final summation, the ρn_x multiplier simplifies this to $\alpha\rho n_x^2\chi^2$.

Evaluating Minor M_{13} Expanding M_{13} along its first row:

$$\det(M_{13}) = \alpha n_x(0 - \beta\eta n_x) - \chi(\alpha n_y \chi - 0) + \beta n_x(\alpha n_y \eta n_x - 0) = -\alpha\chi^2 n_y$$

Total Characteristic Polynomial Combining the terms:

$$\begin{aligned} 0 &= \chi[\chi(\chi^2 - \beta\eta)] - \rho n_x[\alpha n_x \chi^2] + \rho n_y[-\alpha n_y \chi^2] = \\ &= \chi^2(\chi^2 - \beta\eta) - \alpha\rho\chi^2(n_x^2 + n_y^2) = \\ &= \chi^2(\chi^2 - \beta\eta - \alpha\rho) \end{aligned}$$

Final Substitution Recall $\alpha\rho = \partial_\rho p$ and $\beta\eta = \frac{p}{\rho^2}\partial_e p$. The speed of sound is $a^2 = \alpha\rho + \beta\eta$:

$$\chi^2(\chi^2 - a^2) = 0$$

Thus the solutions $\chi = u_n - s$ are:

- $\chi_{1,2} = \pm a \implies s_{1,2} = u_n \mp a$
- $\chi_{3,4} = 0 \implies s_{3,4} = u_n$.

Right eigenvectors

The right eigenvectors $\mathbf{R}$ satisfy the equation $(\mathbf{A}_n - \lambda\mathbf{I})\mathbf{R} = 0$. Using the substituted variable $\chi = u_n - \lambda$, the system for any eigenvector $\mathbf{R} = [R_\rho, R_u, R_v, R_e]^T$ is:

1. $\chi R_\rho + \rho n_x R_u + \rho n_y R_v = 0$
2. $\alpha n_x R_\rho + \chi R_u + \beta n_x R_e = 0$
3. $\alpha n_y R_\rho + \chi R_v + \beta n_y R_e = 0$
4. $\eta n_x R_u + \eta n_y R_v + \chi R_e = 0$

Convective Eigenvectors ($\lambda_{1,2} = u_n \implies \chi = 0$) When $\chi = 0$, the system collapses to:

- From (1) and (4): $n_x R_u + n_y R_v = 0$ (Velocity change is purely tangential).
- From (2) and (3): $\alpha R_\rho + \beta R_e = 0 \implies R_\rho = -\frac{\beta}{\alpha} R_e$.

We choose two linearly independent vectors representing the non-acoustic transport:

Entropy Wave ($\mathbf{R}_1$): Only density changes; pressure (and thus velocity/energy) remains constant.

$$\mathbf{R}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -\alpha/\beta \end{bmatrix}$$

Vorticity/Shear Wave ($\mathbf{R}_2$): Only the tangential velocity changes; no change in density or pressure.

$$\mathbf{R}_2 = \begin{bmatrix} 0 \\ n_y \\ -n_x \\ 0 \end{bmatrix}$$

Acoustic Eigenvectors ($\lambda_{3,4} = u_n \pm a \implies \chi = \mp a$) Substituting $\chi = \mp a$ into the system:

1. $\mp a R_\rho + \rho(n_x R_u + n_y R_v) = 0 \implies n_x R_u + n_y R_v = \pm\frac{a}{\rho} R_\rho$
2. From (2) and (3): The velocity changes R_u, R_v must be proportional to n_x, n_y (the wave is longitudinal). Let $R_u = k n_x$ and $R_v = k n_y$.
3. Then $k = \pm\frac{a}{\rho} R_\rho$.

For the energy component from (4): $\eta k(n_x^2 + n_y^2) \mp a R_e = 0 \implies R_e = \pm\frac{\eta k}{a} = \frac{\eta}{\rho} R_\rho$.

Setting $R_\rho = \rho$ as a normalization:

Right-Running Acoustic Wave (C^+):

$$\mathbf{R}_3 = \begin{bmatrix} \rho \\ a n_x \\ a n_y \\ p/\rho \end{bmatrix}$$

Left-Running Acoustic Wave (C^-):

$$\mathbf{R}_4 = \begin{bmatrix} \rho \\ -a n_x \\ -a n_y \\ p/\rho \end{bmatrix}$$

Left eigenvectors

The left eigenvectors $\mathbf{L}$ are found by solving $\mathbf{L}(\mathbf{A}_n - \lambda\mathbf{I}) = 0$. Using $\chi = u_n - \lambda$, $\alpha = \frac{1}{\rho}\partial_\rho p$, and $\beta = \frac{1}{\rho}\partial_e p$:

Convective Eigenvectors ($\chi = 0$)

- $\mathbf{L}_1$ **(Entropy):** Projects onto the state where pressure is constant.

$$\mathbf{L}_1 = [1, 0, 0, -\alpha/\beta]$$

- $\mathbf{L}_2$ **(Vorticity):** Projects onto the tangential velocity change.

$$\mathbf{L}_2 = [0, n_y, -n_x, 0]$$

Acoustic Eigenvectors ($\chi = \mp a$) For the acoustic waves, the weights for u and v align with the normal $\mathbf{n}$.

- $\mathbf{L}_3$ **($u_n + a$):**

$$\mathbf{L}_3 = \left[\frac{\alpha}{2a^2}, \frac{n_x}{2a}, \frac{n_y}{2a}, \frac{\beta}{2a^2}\right]$$

- $\mathbf{L}_4$ **($u_n - a$):**

$$\mathbf{L}_4 = \left[\frac{\alpha}{2a^2}, -\frac{n_x}{2a}, -\frac{n_y}{2a}, \frac{\beta}{2a^2}\right]$$

Compatibility Equations Multiplying the primitive variable differential vector $d\mathbf{V} = [d\rho, du, dv, de]^T$ by these eigenvectors yields:

$$\frac{\alpha}{a^2}d\rho + \frac{\beta}{a^2}de \pm \frac{1}{a}(n_x du + n_y dv) = 0$$

Using $a^2 d\rho_{total} = \alpha\rho d\rho + \beta\rho de$, this simplifies to the physical form:

$$\frac{dp}{\rho a} \pm du_n = 0$$

Shallow water - 2d

35.7 Hyperbolic problems in multi-dimensional domains

Examples

- *draft dummy page with the scheme used for all the systems*
- *P-system*
- *Euler equations*
- *Shallow water*

35.7.1 General hyperbolic problem

Integral equations

For a volume V at rest

$$\frac{d}{dt}\int_V \mathbf{u} + \oint_{\partial V} \hat{n}\cdot\mathbf{F} = \int_V \mathbf{s}$$

Integral equations for a control volume V at rest

Integral equation for a control volume at rest V

$$\frac{d}{dt}\int_V \mathbf{u} + \oint_{\partial V} \mathbf{u}\vec{u}\cdot\hat{n} = \int_{\partial V} \hat{n}\cdot\mathbf{f} + \int_V \mathbf{s}$$
$$\frac{d}{dt}\int_V \mathbf{u} + \oint_{\partial V} \hat{n}\cdot\mathbf{F} = \int_V \mathbf{s}$$

being $\mathbf{u}$ the vector of the conservative variables, $\mathbf{u} \in \mathbb{R}^n$, and $\vec{u}$ the velocity field in the d-dimensional domain, and $\mathbf{F} := -\mathbf{f} + \vec{u}\mathbf{u}$.

Integral equations for an arbitary domain v_t

Integral equation for an arbitrary domain v_t, with *Reynolds' transport theorem*

$$\frac{d}{dt}\int_{v_t} \mathbf{u} + \oint_{\partial v_t} \mathbf{u}(\vec{u}-\vec{v}_b)\cdot\hat{n} = \int_{\partial v_t} \hat{n}\cdot\mathbf{f} + \int_{v_t} \mathbf{s}$$
$$\frac{d}{dt}\int_{v_t} \mathbf{u} - \oint_{\partial v_t} \mathbf{u}\vec{v}_b\cdot\hat{n} + \int_{\partial v_t} \hat{n}\cdot\mathbf{F} = \int_{v_t} \mathbf{s}$$

Jump conditions

$$\hat{n}\cdot\vec{v}_b\,[\,\mathbf{u}\,] = \hat{n}\cdot[\,\mathbf{F}\,] \tag{35.1}$$

From integral equations on v_t to jump relations

Collapsing v_t on a surface only the boundary terms on the two sides of the surface persist and the balance equations becomes

$$\begin{aligned}
\mathbf{0} &= \oint_{\partial v_t} (-\hat{n}\cdot\vec{v}_b\mathbf{u} + \hat{n}\cdot\mathbf{F}) = \\
&= \int_{S_-} (-\hat{n}_-\cdot\vec{v}_b\mathbf{u}_- + \hat{n}_-\cdot\mathbf{F}_-) + \int_{S_+} (-\hat{n}_+\cdot\vec{v}_b\mathbf{u}_+ + \hat{n}_+\cdot\mathbf{F}_+) = \\
&= \int_S (-\hat{n}_-\cdot\vec{v}_b(\mathbf{u}_- - \mathbf{u}_+) + \hat{n}_-\cdot(\mathbf{F}_- - \mathbf{F}_+))
\end{aligned}$$

being $\hat{n}_- = -\hat{n}_+$. From the arbitrariness of the surface S, jump relations follow

$$\hat{n}\cdot\vec{v}_b\,[\,\mathbf{u}\,] = [\,\mathbf{F}\,]\ ,$$

being $[\,\cdot\,] = (\cdot)_+ - (\cdot)_-$ the jump across the surface.

Differential equations

In regions where the fields are smooth, differential equations follow from integral equations using *theorems of differential calculus*.

Conservative form

Conservative form of the differential equations reads

$$\partial_t \mathbf{u} + \nabla \cdot \mathbf{F}(\mathbf{u}) = \mathbf{s} \ . \tag{35.2}$$

From integral equations on V to conservative form of differential equations

For a volume V at rest

$$\frac{d}{dt}\int_V \mathbf{u} + \oint_{\partial V} \hat{n} \cdot \mathbf{F} = \int_V \mathbf{s}$$
$$\int_V \partial_t \mathbf{u} + \oint_{\partial V} \hat{n} \cdot \mathbf{F} = \int_V \mathbf{s}$$

or using components

$$\int_V \partial_t u_i + \oint_{\partial V} n_k F_{ki} = \int_V s_i \ ,$$

and using [divergence theorem] under the assumption of sufficient regularity of the flux

$$\begin{aligned} 0 &= \int_V \partial_t u_i + \oint_{\partial V} n_k F_{ki} - \int_V s_i = \\ &= \int_V \partial_t u_i + \int_V \partial_k F_{ki} - \int_V s_i = \\ &= \int_V \left\{ \partial_t u_i + \partial_k F_{ki} - s_i \right\} \ , \end{aligned}$$

and from the aribitrariness of the domain V,

$$\partial_t u_i + \partial_k F_{ki} = s_i \ ,$$

or using abstract vector notation

$$\partial_t \mathbf{u} + \nabla \cdot \mathbf{F} = \mathbf{s} \ .$$

$$\partial_t \mathbf{u} + \sum_{k=1}^{d} \mathbf{A}^{(k)}(\mathbf{u})\, \partial_k \mathbf{u} = \mathbf{s} \ ,$$

with the convection matrices

$$\left\{ \mathbf{A}^{(k)} \right\}_{i\ell} = \partial_{u_\ell} F_{ki} \ .$$

From conservative form to convective form of differential equations

Starting from the component expression {eq}`` of the convective differential equations

$$
\begin{aligned}
s_i &= \partial_t u_i + \partial_k F_{ki}(u_\ell) = \\
&= \partial_t u_i + \partial_{u_\ell} F_{ki}(u_\ell)\partial_k u_\ell = \\
&= \partial_t u_i + A^{(k)}_{i\ell}(u_\ell)\partial_k u_\ell = \\
&= \partial_t u_i + \sum_{k=1}^{d} A^{(k)}_{i\ell}(u_\ell)\partial_k u_\ell \ ,
\end{aligned}
$$

having defined $A^{(k)}_{i\ell} = \partial_{u_\ell} F_{ki}$, or using vector notation

$$
\partial_t \mathbf{u} + \sum_{k=1}^{d} \mathbf{A}^{(k)}(\mathbf{u})\partial_k \mathbf{u} = \mathbf{s} \ .
$$

Method of characteristics

Spectral decomposition

Summary of method of characteristics in multi-dimensional domains.

Let

$$
\begin{aligned}
\mathbf{u}(t,\vec{r}) &: \mathbb{R}\times\mathbb{R}^d \to \mathbb{R}^n \\
\mathbf{F}(\mathbf{u}) &: \mathbb{R}^n \to \mathbb{R}^d\times\mathbb{R}^n \\
\mathbf{s}(t,\vec{r}) &: \mathbb{R}\times\mathbb{R}^d \to \mathbb{R}^n
\end{aligned}
$$

the conservative form of a hyperbolic problem reads

$$
\begin{aligned}
\partial_t \mathbf{u} + \nabla\cdot\mathbf{F}(\mathbf{u}) &= \mathbf{s} \\
\partial_t u_i + \partial_j F_{ji} &= s_i \ .
\end{aligned}
$$

Expanding the divergence of the flux, the convective (quasi-linear? It makes little to no sense to me…) form follows

$$
\begin{aligned}
s_i &= \partial_t u_i + \sum_{j=1}^{d} \partial_j F_{ji} = \\
&= \partial_t u_i + \sum_{j=1}^{d}\sum_{k=1}^{n} \partial_j u_k \partial_{u_k} F_{ji} = \\
&= \partial_t u_i + \sum_{j=1}^{d}\sum_{k=1}^{n} A^j_{ik}(\mathbf{u})\,\partial_j u_k \ ,
\end{aligned}
$$

for every component $i = 1:n$, or

$$
\mathbf{s} = \partial_t \mathbf{u} + \sum_{j=1}^{d} \mathbf{A}^j \partial_j \mathbf{u} \ .
$$

This problem can be made a little more general with by defining $\mathbf{x} = (t,\mathbf{r}) = (x_0,\vec{r})$, as

$$
\sum_{j=0}^{d} \mathbf{A}^j\,\partial_j \mathbf{u} = \mathbf{s} \ .
$$

The former expression of the hyperbolic problem immediately follows if $\mathbf{A}_0 = \mathbf{I}_n$, $A^0_{ik} = \delta_{ik}$, and

$$\widetilde{\nabla} \cdot \tilde{\mathbf{F}} = \partial_0 \mathbf{F}_0 + \sum_{j=1}^{d} \partial_j \mathbf{F}_j = \widetilde{\nabla} \cdot [\, \mathbf{u} \mid \mathbf{F}] \ .$$

as, for $i = 1 : n$,

$$\delta_{ik} = \partial_{u_k} F_{0i} \quad \rightarrow \quad F_{0i} = u_k \delta_{ik} = u_i \ .$$

Taylor expansion of a solution. Starting from the solution on a manifold determined by the equation $S : f(\mathbf{x}) = 0$, $\mathbf{x}_0 \in S$, if the solution is differentiable, the solution in a point $\mathbf{x} = \mathbf{x}_0 + \Delta\mathbf{x}$ reads

$$\begin{aligned} \mathbf{u}(\mathbf{x}) &\sim \mathbf{u}(\mathbf{x}_0) + \Delta\mathbf{x} \cdot \nabla\mathbf{u}(\mathbf{x}_0) \\ u_i(\mathbf{x}) &\sim u_i(\mathbf{x}_0) + \Delta x_j \frac{\partial u_i}{\partial x_j}(\mathbf{x}_0) \ . \end{aligned}$$

Now, with a change of coordinates from $\mathbf{x}$ to $\xi = (n, \mathbf{t})$, i.e. to local normal and tangential directions,

$$\nabla\mathbf{u} = \hat{\mathbf{x}}_i \frac{\partial}{\partial x_i}\mathbf{u} = \hat{\mathbf{x}}_i \frac{\partial \mathbf{u}}{\partial x_i} = \hat{\mathbf{x}}_i \frac{\partial \mathbf{u}}{\partial \xi_k}\frac{\partial \xi_k}{\partial x_i} = \hat{\xi}_k \frac{\partial \mathbf{u}}{\partial \xi_k}$$

Inserting in the hyperbolic equation

$$\begin{aligned} s_i &= \sum_{j=0}^{d} A^j_{ik} \frac{\partial u_k}{\partial x_j} = \\ &= \sum_{j=0}^{d} A^j_{ik} \sum_{\ell=0}^{d} \frac{\partial u_k}{\partial \xi_\ell}\frac{\partial \xi_\ell}{\partial x_j} = \\ &= \sum_{\ell=0}^{d}\sum_{j=0}^{d} A^j_{ik} \frac{\partial \xi_\ell}{\partial x_j}\frac{\partial u_k}{\partial \xi_\ell} = \\ &= \sum_{\ell=0}^{d}\sum_{j=0}^{d} A^j_{ik} \xi^\ell_j \frac{\partial u_k}{\partial \xi_\ell} \ , \end{aligned}$$

and separating the normal $\ell = 0$ from the tangential $\ell = 1 : d$ coordinates,

$$\begin{aligned} s_i &= \sum_{j=0}^{d} A^j_{ik} \xi^0_j \frac{\partial u_k}{\partial \xi_0} + \sum_{\ell=1}^{d}\sum_{j=0}^{d} A^j_{ik} \xi^t_j \frac{\partial u_k}{\partial \xi_t} = \\ &= \sum_{j=0}^{d} A^j_{ik} n_j \frac{\partial u_k}{\partial n} + \sum_{\ell=1}^{d}\sum_{j=0}^{d} A^j_{ik} t^\ell_j \frac{\partial u_k}{\partial t_\ell} = \\ &= \sum_{j=0}^{d} n_j \mathbf{A}^j \partial_n \mathbf{u} + \sum_{\ell=1}^{d}\sum_{j=0}^{d} t^\ell_j \mathbf{A}^j \partial_{t_\ell}\mathbf{u} \ . \end{aligned}$$

On a smooth surface where the solution is known, all the tangential derivatives of the solution are known as well. In order to evaluate the normal derivative $\partial_n \mathbf{u}$ from the PDE, the (formal) inversion of the matrix

$$\mathbf{A}_{\hat{\mathbf{n}}} := \sum_{j=0}^{d} n_j \, \mathbf{A}^j \ ,$$

must be feasible, and thus $\mathbf{A}_{\hat{\mathbf{n}}}$ must be invertible, to formally get

$$\partial_n \mathbf{u} = \mathbf{A}^{-1}_{\hat{\mathbf{n}}} \left(\mathbf{s} - \sum_{\ell=1}^{d} \mathbf{A}_{\hat{\mathbf{t}}_\ell} \partial_{t_\ell} \mathbf{u} \right) \ .$$

This expression of the normal derivative of the solution - whenever it exists - can be used to find the approximation of the solution in normal direction w.r.t. the surface S, i.e. with $\Delta\mathbf{x} = \Delta\ell\hat{\mathbf{n}}$ as

$$\begin{aligned}\mathbf{u}(\mathbf{x}) \sim \mathbf{u}(\mathbf{x}_0) + \Delta\ell\,\hat{\mathbf{n}}\cdot\nabla\mathbf{u}(\mathbf{x}_0) = \\ = \mathbf{u}(\mathbf{x}_0) + \Delta\ell\,\partial_n\mathbf{u}(\mathbf{x}_0)\ .\end{aligned}$$

If the matrix $\mathbf{A}_{\hat{\mathbf{n}}}$ is diagonalizable, it's invertible if it has no zero eigenvalue. Let $\hat{\mathbf{n}}_i$ be a unit vector so that the matrix $\mathbf{A}_{\hat{\mathbf{n}}}$ has an eigenvalue equal to zero $s_0 = 0$, with right and left eigenvectors $\mathbf{r}_0$, $\mathbf{l}_0$. Recalling the expression of the PDE in normal and tangential components

$$\mathbf{s} = \mathbf{A}_{\hat{\mathbf{n}}}\partial_n\mathbf{u} + \sum_{\ell=1}^{d}\mathbf{A}_{\hat{\mathbf{d}}_\ell}\partial_{t_\ell}\mathbf{u}\ ,$$

left-multiplying by **l** gives

$$\begin{aligned}\mathbf{l}_0^T\mathbf{s} = \mathbf{l}_0^T\left(\mathbf{A}_{\hat{\mathbf{n}}}\partial_{\hat{n}}\mathbf{u} + \sum_{\ell=1}^{d}\mathbf{A}_{\hat{\mathbf{d}}_\ell}\partial_{t_\ell}\mathbf{u}\right) = \\ = \mathbf{l}_0^T\sum_{\ell=1}^{d}\mathbf{A}_{\hat{\mathbf{d}}_\ell}\partial_{t_\ell}\mathbf{u} = \\ = \mathbf{l}_0^T\sum_{k=0}^{d}\mathbf{A}^k\partial_k\mathbf{u}\ ,\end{aligned}$$

i.e. the **compatibility condition** on the surface with unit normal $\hat{\mathbf{n}}$ that makes $s_0 = 0$,

$$\mathbf{l}_0^T\mathbf{s} = \mathbf{l}_0^T\sum_{k=0}^{d}\mathbf{A}^k\partial_k\mathbf{u} = \mathbf{l}_0^T\widetilde{\nabla}\cdot\tilde{\mathbf{F}}\ .$$

Surfaces with unit normal vectors $\hat{\mathbf{n}}$ making eigenvalues of the matrix $\mathbf{A}_{\mathbf{n}}$ equal to zero are defined as **characteristic surfaces**. On each characteristic surface, the corresponding compatibility condition holds.

Change of variables

Let $\mathbf{u}$ the set of conservative variables[1] and let $\mathbf{v}$ another set of variables. Let the transfomration of variables $\mathbf{v}(\mathbf{u})$ be invertible and let $\mathbf{u}(\mathbf{v})$ be the inverse transformation. Let $\mathbf{U}_{\mathbf{v}}$ the gradient of the transformation $\mathbf{u}(\mathbf{v})$ w.r.t. variables $\mathbf{v}$ and $\mathbf{V}_{\mathbf{u}}$ the gradient of the inverse transformation $\mathbf{v}(\mathbf{u})$, so that the differentials of the variables read

$$d\mathbf{u} = \mathbf{U}_{\mathbf{v}}d\mathbf{v} \quad , \quad d\mathbf{v} = \mathbf{V}_{\mathbf{u}}d\mathbf{u}\ .$$

These gradients are mutually inverse, $\mathbf{U}_{\mathbf{v}} = \mathbf{V}_{\mathbf{u}}^{-1}$. The convective form of the equations using $\mathbf{u}$ or $\mathbf{v}$ as primary variables are respectively

$$\begin{aligned}\partial_t\mathbf{u} + \sum_{k=1}^{d}\mathbf{A}^{\mathbf{u},k}\partial_k\mathbf{u} = \mathbf{s}^{\mathbf{u}} \\ \partial_t\mathbf{v} + \sum_{k=1}^{d}\mathbf{A}^{\mathbf{v},k}\partial_k\mathbf{v} = \mathbf{s}^{\mathbf{v}}\end{aligned}$$

with

$$\begin{aligned}\mathbf{A}^{\mathbf{v},k} = \mathbf{V}_{\mathbf{u}}\mathbf{A}^{\mathbf{u},k}\mathbf{U}_{\mathbf{v}} \\ \mathbf{s}^{\mathbf{v}} = \mathbf{V}_{\mathbf{u}}\mathbf{s}^{\mathbf{u}}\ .\end{aligned}$$

The equations using $\mathbf{v}$ immediately follows after introducing the relation between the differentials and pre-multiplication of the equations by $\mathbf{V}_{\mathbf{u}}$,

[1] Once the problem is written in convective form, this set of variable could be an arbitrary set of variables and not only the conservative variables.

Change of variables and spectral decomposition of the matrix $\mathbf{A}_{\hat{n}}$.

Starting from a set of variables $\mathbf{u}$ and the convective form of the equations

$$\mathbf{s} = \partial_t \mathbf{u} + \sum_k \mathbf{A}^k \partial_k \mathbf{u} \ ,$$

using another set of variables $\mathbf{v}$, the equations become

$$\begin{aligned}
\mathbf{s} &= \partial_t \mathbf{u} + \sum_k \mathbf{A}^k \partial_k \mathbf{u} \\
\mathbf{s} &= \mathbf{U}_\mathbf{v}(\mathbf{v}) \partial_t \mathbf{v} + \sum_k \mathbf{A}^k \mathbf{U}_\mathbf{v}(\mathbf{v}) \partial_k \mathbf{v} \ ,
\end{aligned}$$

being $d\mathbf{u} = \mathbf{U}_\mathbf{v} d\mathbf{v}$. If the transformation is non-singular, and $\mathbf{U}_\mathbf{v}^{-1} = \mathbf{V}_\mathbf{u}(\mathbf{v})$, then the differential equations can be recast as

$$\partial_t \mathbf{v} + \sum_k \mathbf{V}_\mathbf{u} \mathbf{A}^k \mathbf{U}_\mathbf{v} \partial_k \mathbf{v} = \mathbf{V}_\mathbf{u} \mathbf{s}$$

With a coordinate transformation, the normal and tangential matrices become

$$\begin{aligned}
\mathbf{A}_\mathbf{n}^\mathbf{v} &= \sum_k n_k \mathbf{V}_\mathbf{u} \mathbf{A}^k \mathbf{U}_\mathbf{v} = \\
&= \mathbf{V}_\mathbf{u} \underbrace{\sum_k n_k \mathbf{A}^k}_{\mathbf{A}_\mathbf{n}^\mathbf{u}} \mathbf{U}_\mathbf{v} = \mathbf{V}_\mathbf{u} \mathbf{A}_\mathbf{n}^\mathbf{u} \mathbf{U}_\mathbf{v} \ , \\
\mathbf{A}_{\mathbf{t}_i}^\mathbf{v} &= \cdots = \mathbf{V}_\mathbf{u} \mathbf{A}_{\mathbf{t}_i}^\mathbf{u} \mathbf{U}_\mathbf{v} \ .
\end{aligned}$$

If the change of variables is non-singular, the Jacobian matrices $\mathbf{V}_\mathbf{u}$, $\mathbf{U}_\mathbf{v}$ are non singular and reciprocally inverse. Thus, the **determinant** of the matrix $\mathbf{A}_\mathbf{n}$ is independent from the set of chosen variables, as

$$\begin{aligned}
\det\left(\mathbf{A}_\mathbf{n}^\mathbf{v}\right) &= \det\left(\mathbf{V}_\mathbf{u}\right) \det\left(\mathbf{A}_\mathbf{n}^\mathbf{u}\right) \det\left(\mathbf{U}_\mathbf{v}\right) = \\
&= \det\left(\mathbf{V}_\mathbf{u}\right) \det\left(\mathbf{A}_\mathbf{n}^\mathbf{u}\right) \det\left(\mathbf{V}_\mathbf{u}\right)^{-1} = \\
&= \det\left(\mathbf{A}_\mathbf{n}^\mathbf{u}\right) \ .
\end{aligned}$$

Thus, the **eigenvalues** are independent from the set of variables as well

$$\begin{aligned}
0 &= \det\left(s\mathbf{I} - \mathbf{A}_\mathbf{n}^\mathbf{v}\right) = \\
&= \det\left(\mathbf{V}_\mathbf{u}\right) \det\left(s\mathbf{I} - \mathbf{A}_\mathbf{n}^\mathbf{u}\right) \det\left(\mathbf{U}_\mathbf{v}\right) = \\
&= \det\left(s\mathbf{I} - \mathbf{A}_\mathbf{n}^\mathbf{u}\right) \ .
\end{aligned}$$

The relation between **right eigenvectors** immediately follows

$$\begin{aligned}
\mathbf{A}_\mathbf{n}^\mathbf{v} \mathbf{R}^\mathbf{v} &= \mathbf{R}^\mathbf{v} \mathbf{S} \\
\mathbf{V}_\mathbf{u} \mathbf{A}_\mathbf{n}^\mathbf{u} \mathbf{U}_\mathbf{v} \mathbf{R}^\mathbf{v} &= \mathbf{R}^\mathbf{v} \mathbf{S} \\
\mathbf{A}_\mathbf{n}^\mathbf{u} \underbrace{\mathbf{U}_\mathbf{v} \mathbf{R}^\mathbf{v}}_{\mathbf{R}^\mathbf{u}} &= \underbrace{\mathbf{U}_\mathbf{v} \mathbf{R}^\mathbf{v}}_{\mathbf{R}^\mathbf{u}} \mathbf{S} \\
\mathbf{A}_\mathbf{n}^\mathbf{u} \mathbf{R}^\mathbf{u} &= \mathbf{R}^\mathbf{u} \mathbf{S} \ ,
\end{aligned}$$

with $\mathbf{R}^\mathbf{u} = \mathbf{U}_\mathbf{v} \mathbf{R}^\mathbf{v}$, or for any individual eigenvector $\mathbf{r}_i^\mathbf{u} = \mathbf{U}_\mathbf{v} \mathbf{r}_i^\mathbf{v}$.

The relation between **left eigenvectors** analogously follows

$$\begin{aligned}
\mathbf{L}^\mathbf{v} \mathbf{A}_\mathbf{n}^\mathbf{v} &= \mathbf{S} \mathbf{L}^\mathbf{v} \\
\mathbf{L}^\mathbf{v} \mathbf{V}_\mathbf{u} \mathbf{A}_\mathbf{n}^\mathbf{u} \mathbf{U}_\mathbf{v} &= \mathbf{S} \mathbf{L}^\mathbf{v} \\
\underbrace{\mathbf{L}^\mathbf{v} \mathbf{V}_\mathbf{u}}_{\mathbf{L}^\mathbf{u}} \mathbf{A}_\mathbf{n}^\mathbf{u} &= \mathbf{S} \underbrace{\mathbf{L}^\mathbf{v} \mathbf{V}_\mathbf{u}}_{\mathbf{L}^\mathbf{u}} \\
\mathbf{L}^\mathbf{u} \mathbf{A}_\mathbf{n}^\mathbf{u} &= \mathbf{S} \mathbf{L}^\mathbf{u} \ ,
\end{aligned}$$

with $\mathbf{L}^\mathbf{u} = \mathbf{L}^\mathbf{v} \mathbf{U}_\mathbf{v}$, or for any individual eigenvector $\left(\mathbf{l}_i^\mathbf{u}\right)^T = \left(\mathbf{l}_i^\mathbf{v}\right)^T \mathbf{U}_\mathbf{v}$, or $\mathbf{l}_i^\mathbf{u} = \mathbf{U}_\mathbf{v}^T \mathbf{l}_i^\mathbf{v}$.

- Eigenvalues, right and left eigenvectors
- Characteristic lines/surfaces, and compatibility conditions

Riemann problem

- Useful in numerical schemes in finite volume methods, using Godunov flux

Linearization - Roe intermediate state

- Local linearization of the problem, to reduce the computational cost of solving non-linear Riemann problems at all the interfaces in a grid in FVM

Boundary conditions

- characteristic-based
- wall
- …

35.7.2 P-system

35.7.3 Euler equations

Thermodynamics and constitutive equations

todo *Reference to Physics: Thermodynamics bbook*

Thermodynamics

Principles of thermodynamics and differential of the internal energy

The differential of the internal energy as a function of s, ρ as independent thermodynamic variables, $e(\rho, s)$ reads

$$de = T\,ds + \frac{p}{\rho^2} d\rho \,, \tag{35.3}$$

from the principles of thermodynamics, and the definition of the entropy s in classical thermodynamics. Following Gibbs' mathematical foundation of classical thermodynamics, the partial derivatives of the internal energy can be related to temperature, pressure and density

$$T = \left(\frac{\partial e}{\partial s}\right)_\rho \quad , \quad \frac{p}{\rho^2} = \left(\frac{\partial e}{\partial \rho}\right)_s \,. \tag{35.4}$$

Speed of sound, $a(\rho, s)$.

As it should be clear from the study of Euler equations as an hyperbolic system the speed of sound a can be defined as

$$a^2(\rho, s) = \left(\frac{\partial p}{\partial \rho}\right)_s . \tag{35.5}$$

As an example, the definition of the speed of sound appears in two of the eigenvalues of the system, as shown in (35.16) using the convective form (35.15) of the equations using $(\rho, \vec{u}, s)$ as primary variables.

Speed of sound, and change of independent thermodynamic variables

All the relations in this box mainly exploit **rule of derivatives of composite functions** and basic thermodynamic relations.

As an example, if the pressure is defined as a function of (ρ, e), and the relation $e(\rho, s)$ is known, with pressure can be written as $p(\rho, e) = p(\rho, e(\rho, s))$. With the derivation of the composite function, an alternative expression of the speed of sound follows

$$\begin{aligned} a^2 &= \left(\frac{\partial p}{\partial \rho}\right)_s = \\ &= \left(\frac{\partial p}{\partial \rho}\right)_e + \left(\frac{\partial e}{\partial \rho}\right)_s \left(\frac{\partial p}{\partial e}\right)_\rho = \\ &= \left(\frac{\partial p}{\partial \rho}\right)_e + \frac{p}{\rho^2}\left(\frac{\partial p}{\partial e}\right)_\rho . \end{aligned} \tag{35.6}$$

Conservative variables of Euler equations are $(\rho, \vec{m}, E^t) = \left(\rho, \rho\vec{u}, \rho\left(e + \frac{|\vec{u}^2}{2}\right)\right)$. Pressure as a function of conservative variable reads

$$p = \Pi\left(\rho, \vec{m}, E^t\right) = \Pi\left(\rho, \rho\vec{u}, \rho e(\rho, s) + \rho\frac{|\vec{u}|^2}{2}\right)$$

Constitutive equations for an ideal Newtonian fluid

- Stress tensor $\mathbb{T} = -p\mathbb{I}$
- Stress vector $\vec{t}_{\hat{n}} = \hat{n} \cdot \mathbb{T} = -p\hat{n}$
- Heat conduction flux $\vec{q} = \vec{0}$

Integral equations

Conservative variables and flux are respectively

$$\mathbf{u} = \begin{bmatrix} \rho \\ \vec{m} \\ E^t \end{bmatrix} \tag{35.7}$$

$$\mathbf{F} = \begin{bmatrix} \vec{m} \\ \frac{\vec{m}\otimes\vec{m}}{\rho} + p\mathbb{I} \\ H^t\frac{\vec{m}}{\rho} \end{bmatrix} \tag{35.8}$$

Integral equations for a control volume V at rest

Integral equation for a control volume at rest V immediately follows from the three principles of classical mechanics: conservation of mass, 2-nd principle of Newtonian dynamics, 1-st principle of thermodynamics

$$\begin{aligned}
\frac{d}{dt}\int_V \rho + \oint_{\partial V} \rho\vec{u}\cdot\hat{n} &= 0 \\
\frac{d}{dt}\int_V \rho\vec{u} + \oint_{\partial V} \rho\vec{u}\vec{u}\cdot\hat{n} &= -\oint_{\partial V} p\hat{n} \\
\frac{d}{dt}\int_V \rho e^t + \oint_{\partial V} \rho e^t\vec{u}\cdot\hat{n} &= -\oint_{\partial V} p\vec{u}\cdot\hat{n}
\end{aligned}$$

with the need for an equation of state with the expression of pressure p as a function of the dynamical variables $p = \Pi(\rho, \vec{u}, e^t)$. Using conservative variables $(\rho, \vec{m}, E^t) = (\rho, \rho\vec{u}, \rho e^t)$

$$\begin{aligned}
\frac{d}{dt}\int_V \rho + \oint_{\partial V} \vec{m}\cdot\hat{n} &= 0 \\
\frac{d}{dt}\int_V \vec{m} + \oint_{\partial V} \left(\frac{\vec{m}\otimes\vec{m}}{\rho} + p\mathbb{I}\right)\cdot\hat{n} &= 0 \\
\frac{d}{dt}\int_V E^t + \oint_{\partial V} H^t\frac{\vec{m}}{\rho}\cdot\hat{n} &= 0\,.
\end{aligned}$$

having defined the total enthalpy per unit volume

$$H^t = \rho h^t = \rho\left(e^t + \frac{p}{\rho}\right) = \rho\left(e + \frac{p}{\rho} + \frac{|\vec{u}|^2}{2}\right) = \rho\left(h + \frac{|\vec{v}|^2}{2}\right)\,.$$

Integral equations for an arbitrary domain v_t

Integral equation for an arbitrary domain v_t, with *Reynolds' transport theorem*

$$\begin{aligned}
\frac{d}{dt}\int_{v_t} \rho + \oint_{\partial v_t} \rho(\vec{u}-\vec{v}_b)\cdot\hat{n} &= \\
\frac{d}{dt}\int_{v_t} \rho\vec{u} + \oint_{\partial v_t} \rho\vec{u}(\vec{u}-\vec{v}_b)\cdot\hat{n} &= \\
\frac{d}{dt}\int_{v_t} \rho e^t + \oint_{\partial v_t} \rho e^t(\vec{u}-\vec{v}_b)\cdot\hat{n} &=
\end{aligned}$$

Jump conditions

Jump conditions (35.1) for Euler equations read

$$\vec{v}_b\cdot\hat{n}\left[\begin{pmatrix}\rho\\ \vec{m}\\ E^t\end{pmatrix}\right] = \hat{n}\cdot\left[\begin{pmatrix}\vec{m}\\ \frac{\vec{m}\vec{m}}{\rho} + p\\ H^t\frac{\vec{m}}{\rho}\end{pmatrix}\right] \tag{35.9}$$

Differential equations

In regions where the fields are smooth, from integral to differential equations using *theorems of differential calculus*

Conservative form

Conservative form of the differential equations immediately follows introducing the expressions (35.7) of the conservative variables and (35.8) of the flux in the general expression of the conservative form of hyperbolic problems (35.2)

$$\partial_t \begin{bmatrix} \rho \\ \vec{m} \\ E^t \end{bmatrix} + \nabla \cdot \begin{bmatrix} \vec{m} \\ \frac{\vec{m}\otimes\vec{m}}{\rho} + p\mathbb{I} \\ H^t \frac{\vec{m}}{\rho} \end{bmatrix} = \mathbf{0} \,. \tag{35.10}$$

Convective forms

Convective form with conservative variables

Using Cartesian coordinates for a 2-dimensional domain,

$$\begin{cases} \partial_t \rho + \partial_x m_x + \partial_y m_y = 0 \\ \partial_t m_x + \partial_x \left(\frac{m_x^2}{\rho} + \Pi \right) + \partial_y \left(\frac{m_x m_y}{\rho} \right) = 0 \\ \partial_t m_y + \partial_x \left(\frac{m_x m_y}{\rho} \right) + \partial_y \left(\frac{m_y^2}{\rho} + \Pi \right) = 0 \\ \partial_t E^t + \partial_x \left((E^t + \Pi) \frac{m_x}{\rho} \right) + \partial_y \left((E^t + \Pi) \frac{m_y}{\rho} \right) = 0 \end{cases}$$

$$\begin{cases} \partial_t \rho + \partial_x m_x + \partial_y m_y = 0 \\ \partial_t m_x - \frac{m_x^2}{\rho^2} \partial_x \rho + 2 \frac{m_x}{\rho} \partial_x m_x + \partial_x \Pi - \frac{m_x m_y}{\rho^2} \partial_y \rho + \frac{m_y}{\rho} \partial_y m_x + \frac{m_x}{\rho} \partial_y m_y = 0 \\ \partial_t m_x - \frac{m_x m_y}{\rho^2} \partial_x \rho + \frac{m_y}{\rho} \partial_x m_x + \frac{m_x}{\rho} \partial_x m_y - \frac{m_y^2}{\rho^2} \partial_y \rho + 2 \frac{m_y}{\rho} \partial_y m_y + \partial_y \Pi = 0 \\ \partial_t E^t - u h^t \partial_x \rho + h^t \partial_x m_x + u \partial_x E^t + u \partial_x \Pi - v h^t \partial_y \rho + h^t \partial_y m_y + v \partial_y E^t + v \partial_y \Pi = 0 \end{cases}$$

and using the matrix formalism

$$\begin{aligned} \partial_t \begin{bmatrix} \rho \\ m_x \\ m_y \\ E^t \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 & 0 \\ -u^2 + \Pi_\rho & 2u + \Pi_u & \Pi_v & \Pi_{E^t} \\ -uv & v & u & \cdot \\ (-h^t + \Pi_\rho) u & h^t + u \Pi_{m_x} & u \Pi_{m_y} & u(1 + \Pi_{E^t}) \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ m_x \\ m_y \\ E^t \end{bmatrix} + \\ + \begin{bmatrix} 0 & 0 & 1 & 0 \\ -uv & v & u & \cdot \\ -v^2 + \Pi_\rho & \Pi_u & 2v + \Pi_v & \Pi_{E^t} \\ (-h^t + \Pi_\rho) v & v \Pi_{m_x} & h^t + v \Pi_{m_y} & v(1 + \Pi_{E^t}) \end{bmatrix} \partial_y \begin{bmatrix} \rho \\ m_x \\ m_y \\ E^t \end{bmatrix} = \mathbf{0} \end{aligned} \tag{35.11}$$

Convective form with primary variables $(\rho, \vec{u}, e)$.

$$\begin{cases} \partial_t \rho + \rho \nabla \cdot \vec{u} + \vec{u} \cdot \nabla \rho = 0 \\ \partial_t \vec{u} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{0} \\ \partial_t e + \vec{u} \cdot \nabla e + \frac{1}{\rho} p \nabla \cdot \vec{u} = 0 \,. \end{cases} \tag{35.12}$$

or using Cartesian coordinates and matrix formalism

$$\partial_t \begin{bmatrix} \rho \\ u \\ v \\ e \end{bmatrix} + \begin{bmatrix} u & \rho & \cdot & \cdot \\ \frac{p_\rho|_e}{\rho} & u & \cdot & \frac{p_e|_\rho}{\rho} \\ \cdot & \cdot & u & \cdot \\ \cdot & \frac{p}{\rho} & \cdot & u \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ u \\ v \\ e \end{bmatrix} + \begin{bmatrix} v & \cdot & \rho & \cdot \\ \cdot & v & \cdot & \cdot \\ \frac{p_\rho|_e}{\rho} & \cdot & v & \frac{p_e|_\rho}{\rho} \\ \cdot & \cdot & \frac{p}{\rho} & v \end{bmatrix} \partial_y \begin{bmatrix} \rho \\ u \\ v \\ e \end{bmatrix} = \mathbf{0} \tag{35.13}$$

Convective form with primary variables $(\rho, \vec{u}, s)$.

Using the thermodynamic relation $de = Tds + \frac{p}{\rho^2} d\rho$ and combining mass and internal energy equations, the differential equation $D_t s = 0$ follows for the entropy, having introduced the material derivative $D_{t_} = \partial_{t_} + \vec{u} \cdot \nabla_$. The convective form of the equations becomes

$$\begin{cases} \partial_t \rho + \rho \nabla \cdot \vec{u} + \vec{u} \cdot \nabla \rho = 0 \\ \partial_t \vec{u} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{0} \\ \partial_t s + \vec{u} \cdot \nabla s = 0 \, . \end{cases} \tag{35.14}$$

or using Cartesian coordinates and matrix formalism in a 2-dimensional domain

$$\partial_t \begin{bmatrix} \rho \\ u \\ v \\ s \end{bmatrix} + \begin{bmatrix} u & \rho & \cdot & \cdot \\ \frac{p_\rho|_s}{\rho} & u & \cdot & \frac{p_s|_\rho}{\rho} \\ \cdot & \cdot & u & \cdot \\ \cdot & \cdot & \cdot & u \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ u \\ v \\ s \end{bmatrix} + \begin{bmatrix} v & \cdot & \rho & \cdot \\ \cdot & v & \cdot & \cdot \\ \frac{p_\rho|_s}{\rho} & \cdot & v & \frac{p_s|_\rho}{\rho} \\ \cdot & \cdot & \cdot & v \end{bmatrix} \partial_y \begin{bmatrix} \rho \\ u \\ v \\ s \end{bmatrix} = \mathbf{0} \tag{35.15}$$

Using this expression of the equations, it's pretty easy to find the eigenvalues of the system, evaluating the eigenvalues of the matrix

$$\mathbf{A}^{\mathbf{v}}_{\hat{n}} = n_x \mathbf{A}^{\mathbf{v}}_x + n_y \mathbf{A}^{\mathbf{v}}_y = \begin{bmatrix} u_n & \rho \mathbf{n}^T & \cdot \\ \frac{p_\rho|_s}{\rho} \mathbf{n} & u_n \mathbf{I}_2 & \frac{p_s|_\rho}{\rho} \mathbf{n} \\ \cdot & \cdot & u_n \end{bmatrix}$$

as the zeros of the following equation,

$$0 = |\mathbf{A}^{\mathbf{v}}_{\hat{n}} - s\mathbf{I}_4| = (u_n - s)^2 \left((u_n - s)^2 - \left. \frac{\partial p}{\partial \rho} \right|_s \right) = (u_n - s)^2 \left((u_n - s)^2 - a^2 \right) \, ,$$

having used the definition of the spped of sound (35.5). Thus the **eigenvalues of the system** are

$$\begin{aligned} s_{1,4} &= u_n \mp a \\ s_{2,3} &= u_n \, . \end{aligned} \tag{35.16}$$

Spectral decomposition

It may be convenient to evaluate the spectrum using physical variables $\mathbf{v} = (\rho, \vec{u}, e)$ and then use the rules for the spectral decomposition after a *change of variables* to get the spectrum in terms of the conservative variables.

Transformation of variables. The tranformation between physical variables $\mathbf{v} = (\rho, \vec{u}, e)$ and the conservative variables $\mathbf{u} = (\rho, \vec{m}, E^t)$ reads

$$\mathbf{v}(\mathbf{u}) = \begin{bmatrix} \rho \\ \vec{u} \\ e \end{bmatrix} = \begin{bmatrix} \rho \\ \frac{\vec{m}}{\rho} \\ \frac{E^t}{\rho} - \frac{|\vec{m}|^2}{2\rho^2} \end{bmatrix} \quad , \quad \mathbf{u}(\mathbf{v}) = \begin{bmatrix} \rho \\ \vec{m} \\ E^t \end{bmatrix} = \begin{bmatrix} \rho \\ \rho \vec{u} \\ \rho \left(e + \frac{|\vec{u}|^2}{2} \right) \end{bmatrix} \, .$$

Gradients of the transformations.

$$\mathbf{U}_{\mathbf{v}} = \begin{bmatrix} 1 & \cdot & \cdot & \cdot \\ u & \rho & \cdot & \cdot \\ v & \cdot & \rho & \cdot \\ e + \frac{|\mathbf{u}|^2}{2} & \rho u & \rho v & \rho \end{bmatrix} = \begin{bmatrix} 1 & \cdot & \cdot \\ \mathbf{u} & \mathbf{I}_2 & \cdot \\ e^t & \rho \mathbf{u}^T & \rho \end{bmatrix}$$

$$\mathbf{V}_{\mathbf{u}} = \mathbf{U}_{\mathbf{v}}^{-1} = \begin{bmatrix} 1 & \cdot & \cdot & \cdot \\ -\frac{u}{\rho} & \frac{1}{\rho} & \cdot & \cdot \\ -\frac{v}{\rho} & \cdot & \frac{1}{\rho} & \cdot \\ \frac{1}{\rho}\left(\frac{|\mathbf{u}|^2}{2} - e\right) & -\frac{u}{\rho} & -\frac{v}{\rho} & \frac{1}{\rho} \end{bmatrix} = \begin{bmatrix} 1 & \cdot & \cdot \\ -\frac{\mathbf{u}}{\rho} & \frac{1}{\rho}\mathbf{I}_2 & \cdot \\ \frac{1}{\rho}\left(|\mathbf{u}|^2 - e^t\right) & -\frac{\mathbf{u}^T}{\rho} & \frac{1}{\rho} \end{bmatrix}$$

Spectral decomposition of $\mathbf{A}^{\mathbf{v}}_{\hat{n}}$, using $\mathbf{v} = (\rho, \vec{u}, e)$. The normal convective matrix using physical variables $\mathbf{v}$ reads

$$\mathbf{A}^{\mathbf{v}}_{\hat{n}} = \begin{bmatrix} u_n & \rho \mathbf{n}^T & \cdot \\ \frac{1}{\rho}\partial_\rho p|_e \mathbf{n} & u_n \mathbf{I} & \frac{1}{\rho}\partial_e p|_\rho \mathbf{n} \\ \cdot & \frac{p}{\rho}\mathbf{n}^T & u_n \end{bmatrix} ,$$

Eigenvalues

In a 2-dimensional domain, the system of equations is 4-dimensional. As shown with the convective form (35.15) using $(\rho, \vec{u}, s)$ as primary variables, the eigenvalues of the system are

$$\{u_n - a, u_n, u_n, u_n + a\} .$$

In a 3-dimensional domain, the system of equations is 5-dimensional, and its eigenvalues are

$$\{u_n - a, u_n, u_n, u_n, u_n + a\} .$$

Right eigenvectors

In a 2-dimensional domain

$$\mathbf{R}^{\mathbf{v}} = \begin{bmatrix} \rho & \rho & 0 & \rho \\ -an_x & 0 & an_y & an_x \\ -an_y & 0 & -an_x & an_y \\ \frac{p}{\rho} & -\rho\frac{\partial_\rho p|_e}{\partial_e p|_\rho} & 0 & \frac{p}{\rho} \end{bmatrix} = \begin{bmatrix} \rho & \rho & 0 & \rho \\ -a\mathbf{n} & \mathbf{0} & a\mathbf{t} & a\mathbf{n} \\ \frac{p}{\rho} & -\rho\frac{\partial_\rho p|_e}{\partial_e p|_\rho} & 0 & \frac{p}{\rho} \end{bmatrix}$$

Details for d-dimensional domain

Given the eigenvalues of the system in a d-dimensional ($d = 2, 3$), the right eigenvectors of the matrix $\mathbf{A}^{\mathbf{v}}_{\hat{n}}$,

$$\mathbf{A}^{\mathbf{v}}_{\hat{n}} = n_i \mathbf{A}^{\mathbf{v}}_i = \begin{bmatrix} u_n & \rho \mathbf{n}^T & \cdot \\ \frac{p_\rho|_s}{\rho}\mathbf{n} & u_n \mathbf{I}_d & \frac{p_s|_\rho}{\rho}\mathbf{n} \\ \cdot & \frac{p}{\rho}\mathbf{n}^T & u_n \end{bmatrix}$$

are evaluated. For $s_{1,d+2} = u_n \mp a$, the singular system

$$\mathbf{0} = \begin{bmatrix} \pm a & \rho \mathbf{n}^T & \cdot \\ \frac{p_\rho|_e}{\rho}\mathbf{n} & \pm a \mathbf{I}_3 & \frac{p_e|_\rho}{\rho}\mathbf{n} \\ \cdot & \frac{p}{\rho}\mathbf{n}^T & \pm a \end{bmatrix} \begin{bmatrix} \tilde{\rho} \\ \tilde{\mathbf{u}} \\ \tilde{e} \end{bmatrix} ,$$

has the non-trivial solution - it should be easy to prove it recalling the expression (35.6) of the speed of sound

$$\begin{bmatrix} \tilde{\rho}_{1,d+2} \\ \tilde{\tilde{u}}_{1,d+2} \\ \tilde{e}_{1,d+2} \end{bmatrix} = \begin{bmatrix} \rho \\ \mp a\mathbf{n} \\ \frac{p}{\rho} \end{bmatrix} .$$

For $s_{2:d+1} = u_n$, the solution of singular system

$$\mathbf{0} = \begin{bmatrix} \cdot & \rho\mathbf{n}^T & \cdot \\ \frac{p_\rho|_e}{\rho}\mathbf{n} & \mathbf{0}_d & \frac{p_e|_\rho}{\rho}\mathbf{n} \\ \cdot & \frac{p}{\rho}\mathbf{n}^T & \cdot \end{bmatrix} \begin{bmatrix} \tilde{\rho} \\ \tilde{\mathbf{u}} \\ \tilde{e} \end{bmatrix} ,$$

is a linear combination of the basis of the kernel of the matrix. It shouldn't be hard to prove that the kernel is spanned by the right eigenvectors

$$\mathbf{r}_2 = \begin{bmatrix} \rho \\ \mathbf{0} \\ -\rho\frac{p_\rho|_e}{p_e|_\rho} \end{bmatrix} ,$$

$$\mathbf{r}_{2+i} = \begin{bmatrix} \cdot \\ a\mathbf{t}_i \\ \cdot \end{bmatrix} \quad , \quad i = 1:d ,$$

with $\mathbf{t}_i$ $d-1$ independent "tangent vectors", orthogonal w.r.t. the unit "normal vector" $\mathbf{n}$.

Left eigenvectors

In a 2-dimensional domain

$$\mathbf{L}^{\mathbf{v}} = \begin{bmatrix} \frac{P_\rho}{2\rho a^2} & -\frac{1}{2a}\mathbf{n}^T & \frac{P_e}{2\rho a^2} \\ \frac{P}{\rho^2}\frac{P_e}{\rho a^2} & \mathbf{0}^T & -\frac{P_e}{\rho a^2} \\ 0 & \frac{\mathbf{t}^T}{a} & 0 \\ \frac{P_\rho}{2\rho a^2} & +\frac{1}{2a}\mathbf{n}^T & \frac{P_e}{2\rho a^2} \end{bmatrix}$$

Spectral decomposition of $\mathbf{A}^{\mathbf{u}}_{\hat{n}}$, using $\mathbf{u} = (\rho, \vec{m}, E^t)$.

Right eigenvectors

In a 2-dimensional domain

$$\mathbf{R}^{\mathbf{u}} = \begin{bmatrix} \rho & \rho & 0 & \rho \\ \vec{m} - \rho a\hat{n} & \vec{m} & \rho a\hat{t} & \vec{m} + \rho a\hat{n} \\ H^t - m_n a & \frac{|\vec{m}|^2}{\rho} - \rho\frac{\Pi_\rho}{\Pi_{E^t}} & am_t & H^t + m_n a \end{bmatrix}$$

Left eigenvectors

In a 2-dimensional domain

$$\mathbf{L}^{\mathbf{u}} = \frac{1}{2\rho a^2} \begin{bmatrix} \Pi_\rho + au_n & -an_x + \Pi_{m_x} & -an_y + \Pi_{m_y} & \Pi_{E^t} \\ 2a^2 - 2\Pi_\rho & -2\Pi_{m_x} & 2 - \Pi_{m_y} & -2\Pi_{E^t} \\ -2au_t & 2at_x & 2at_y & 0 \\ \Pi_\rho - au_n & an_x + \Pi_{m_x} & an_y + \Pi_{m_y} & \Pi_{E^t} \end{bmatrix}$$

Some algebra

Right eigenvectors w.r.t. conservative variables are

$$\mathbf{R}^{\mathbf{u}} = \mathbf{U}_{\mathbf{v}}\mathbf{R}^{\mathbf{v}} = \begin{bmatrix} \rho & \rho & 0 & \rho \\ \rho u - \rho a n_x & \rho u & \rho a n_y & \rho u + \rho a n_x \\ \rho u - \rho a n_y & \rho v & -\rho a n_x & \rho u + \rho a n_y \\ \rho e^t - \rho u_n a + p & \rho e^t - \rho^2 \frac{P_\rho}{P_e} & \rho a u_t & \rho e^t + \rho u_n a + p \end{bmatrix} =$$

$$= \begin{bmatrix} \rho & \rho & 0 & \rho \\ \rho u - \rho a n_x & \rho u & \rho a n_y & \rho u + \rho a n_x \\ \rho u - \rho a n_y & \rho v & -\rho a n_x & \rho u + \rho a n_y \\ \rho h^t - \rho u_n a & \rho e^t - \rho^2 \frac{P_\rho}{P_e} & \rho a u_t & \rho h^t + \rho u_n a \end{bmatrix},$$

with $h^t = e^t + \frac{|\mathbf{u}|^2}{2}$, and

$$\rho e^t - \rho^2 \frac{P_\rho}{P_e} = \rho e^t + P - P - \rho^2 \frac{P_\rho}{P_e} = \rho h^t - \frac{\rho^2}{P_e}\left(\frac{P}{\rho^2}P_e + P_\rho\right) = \rho h^t - \frac{\rho^2}{P_e}a^2 \ .$$

Or using the derivatives of $\Pi(\rho, \mathbf{m}, E^t)$,

$$\begin{aligned} E^t - \rho^2 \frac{P_\rho}{P_e} &= E^t - \rho^2 \frac{\partial_\rho \Pi - \partial_{E^t}\Pi(-e^t + |\mathbf{u}|^2)}{\rho \partial_{E^t}\Pi} = \\ &= E^t - \rho \frac{\partial_\rho \Pi}{\partial_{E^t}\Pi} - E^t + \rho|\mathbf{u}|^2 = \\ &= \rho|\mathbf{u}|^2 - \rho \frac{\partial_\rho \Pi}{\partial_{E^t}\Pi} \ . \end{aligned}$$

Left eigenvectors w.r.t. conservative variables are

$$\mathbf{L}^{\mathbf{u}} = \mathbf{L}^{\mathbf{v}}\mathbf{V}_{\mathbf{u}} = \begin{bmatrix} \frac{P_\rho}{2\rho a^2} + \frac{u_n}{2\rho a} + \frac{1}{2\rho a^2}\frac{P_e}{\rho}(|\mathbf{u}|^2 - e^t) & -\frac{1}{2\rho a}n_x - \frac{u}{2\rho a^2}\frac{P_e}{\rho} & -\frac{1}{2\rho a}n_y - \frac{v}{2\rho a^2}\frac{P_e}{\rho} & \frac{P_e}{2\rho^2 a^2} \\ \frac{P}{\rho^2}\frac{P_e}{\rho a^2} - \frac{P_e}{\rho a^2}\frac{|\mathbf{u}|^2 - e^t}{\rho} & \frac{P_e}{\rho a^2}\frac{u}{\rho} & \frac{P_e}{\rho a^2}\frac{v}{\rho} & -\frac{P_e}{\rho^2 a^2} \\ -\frac{u_t}{\rho a} & \frac{t_x}{\rho a} & \frac{t_y}{\rho a} & 0 \\ \frac{P_\rho}{2\rho a^2} - \frac{u_n}{2\rho a} + \frac{1}{2\rho a^2}\frac{P_e}{\rho}(|\mathbf{u}|^2 - e^t) & \frac{1}{2\rho a}n_x - \frac{u}{2\rho a^2}\frac{P_e}{\rho} & \frac{1}{2\rho a}n_y - \frac{v}{2\rho a^2}\frac{P_e}{\rho} & \frac{P_e}{2\rho^2 a^2} \end{bmatrix} =$$

$$= \frac{1}{2\rho a^2}\begin{bmatrix} \Pi_\rho + a u_n & -a n_x + \Pi_{m_x} & -a n_y + \Pi_{m_y} & \Pi_{E^t} \\ 2a^2 - 2\Pi_\rho & -2\Pi_{m_x} & 2 - \Pi_{m_y} & -2\Pi_{E^t} \\ -2a u_t & 2a t_x & 2a t_y & 0 \\ \Pi_\rho - a u_n & a n_x + \Pi_{m_x} & a n_y + \Pi_{m_y} & \Pi_{E^t} \end{bmatrix}$$

being

$$\begin{aligned} \Pi_\rho &= P_\rho + \frac{P_e}{\rho}\left(-e^t + |\mathbf{u}|^2\right) \\ \Pi_{\mathbf{m}} &= -\frac{\mathbf{u}}{\rho}P_e \\ \Pi_{E^t} &= \frac{1}{\rho}P_e \end{aligned}$$

$$\begin{aligned} 2\frac{P}{\rho^2}P_e - 2P_e\frac{|\mathbf{u}|^2 - e^t}{\rho} &= 2\frac{P}{\rho^2}P_e + 2P_\rho - 2\Pi_\rho = \\ &= 2a^2 - 2\Pi_\rho \end{aligned}$$

and the speed of sound

$$a^2(\rho, s) = \partial_\rho P|_s$$

with $P(\rho, e(\rho, s))$

$$a^2 = \partial_\rho P|_e + \partial_\rho e|_s \, \partial_e P|_\rho =$$
$$= \partial_\rho P|_e + \frac{P}{\rho^2} \partial_e \, P|_\rho \, .$$

- Eigenvalues, right and left eigenvectors
- Characteristic lines/surfaces, and compatibility conditions

Riemann problem

- Useful in numerical schemes in finite volume methods, using Godunov flux

Linearization - Roe intermediate state

- Local linearization of the problem, to reduce the computational cost of solving non-linear Riemann problems at all the interfaces in a grid in FVM

Boundary conditions

- characteristic-based
- wall
- …

35.7.4 Shallow water

Integral equations

- Intgral equation for a control volume at rest V
- Intgral equation for an arbitrary domain v_t, with Reynolds' transport theorem

Jump conditions

- From the integral equation for an arbitrary domain v_t, collapsed on a surface

Differential equations

- In regions where the fields are smooth, from integral to differential equations with "divergence theorem"

Spectral decomposition

- Eigenvalues, right and left eigenvectors
- Characteristic lines/surfaces, and compatibility conditions

Riemann problem

- Useful in numerical schemes in finite volume methods, using Godunov flux

Linearization - Roe intermediate state

- Local linearization of the problem, to reduce the computational cost of solving non-linear Riemann problems at all the interfaces in a grid in FVM

Boundary conditions

- characteristic-based
- wall
- …

35.8 Riemann problems

Riemann problems in compressible fluid mechanics

35.9 Physical solution in hyperbolic problems

Topics

This section deals with:

- existence of infinite number of solution of hyperbolic problems, and existence of one physical solution of the problem as the inviscid limit of the solution of the diffusive problem
- definition of an entropy function, $\eta(u)$, and its flux, $q(\eta)$
- properties of the entropy functions, and fluxes: **convex**, and non-convex fluxes
- discussion of the general case, and examples (Euler equations, P-sys, shallow water)

35.9.1 Scalar non-linear equation in 1-dimensional domain

The hyperbolic equation for $u(x,t)$, $\partial_t u + \partial_x F(u) = 0$, is often the inviscid limit of the 2-nd order equation

$$\partial_t u + \partial_x F(u) - \partial_x \left(\varepsilon \partial_x u\right) = 0 \; .$$

Term $\varepsilon \partial_x u$ represents a diffusive flux, and the inviscid limit is defined by the condition $\varepsilon \to 0$.

Definition of an entropy function $\eta(u)$, and its flux $q(u)$. Let $\eta(u)$ be a function of the field u. Evaluating its time derivative using the viscous equation,

$$\begin{aligned}\partial_t \eta(u) &= \eta'(u)\partial_t u = \\ &= \eta'(u)\left[-\partial_x F(u) - \partial_x(\varepsilon \partial_x u)\right] = \\ &= -\eta'(u)F'(u)\partial_x u - \eta'(u)(\varepsilon u_{/x})_{/x} \; .\end{aligned}$$

If a function $q(u)$ satisfies the condition

$$q'(u) = \eta'(u)F'(u) \; ,$$

function $\eta(u)$ can be defined as an **entropy function** and $q(u)$ as the corresponding **entropy flux**.

Differential inequality. Let these functions exist. Then

$$\partial_t \eta + \partial_x q = -\eta' \left(\varepsilon u_{/x}\right)_{/x} \; ,$$

is a differential balance equation for the entropy. The last term can be manipulated to find

$$\begin{aligned}0 &= \partial_t \eta + \partial_x q - \eta' \left(\varepsilon u_{/x}\right)_{/x} = \\ &= \partial_t \eta + \partial_x q - \left(\varepsilon \eta'(u) u_{/x}\right)_{/x} + \eta''(u)\varepsilon \left(u_{/x}\right)^2 \; ,\end{aligned}$$

For $\varepsilon \geq 0$, for **convex entropy function** $\eta''(u)$, a differential inequality follows

$$\partial_t \eta + \partial_x q + \left(\varepsilon \eta'(u) u_{/x}\right)_{/x} \leq 0 \; .$$

Integral inequality and jump conditions. Differential inequality is the local counterpart of the integral equation for a volume V at rest

$$\frac{d}{dt}\int_V \eta + \oint_{\partial V} n_x \, q + \oint_{\partial V} \varepsilon q'(u) u_{/x} n_x \leq 0 \; ,$$

or for a generic volume v_t,

$$\frac{d}{dt}\int_{v_t} \eta - \oint_{\partial v_t} n_x v_b \eta + \oint_{\partial v_t} n_x \, q + \oint_{\partial v_t} \varepsilon q'(u) u_{/x} n_x \leq 0 \; ,$$

being v_b the velocity of the points of the boundary of the domain ∂v_t. Jump condition in the inviscid limit $\varepsilon \to 0$ follows from the collapse of the volume on a surface ($\frac{|v_t|}{|\partial v_t|} \to 0$), and from the arbitrariness of this surface

$$v_b[\eta] \geq [q] \; .$$

Weak shocks. Expanding in Taylor series jump condition for weak shocks provides an alternative version of the entropy condition, as a relation between the eigenvalues of the system and the shock speed. The shock speed reads $\dot{s} = \frac{[f]}{[u]} =$

$\frac{f_R - f_L}{u_R - u_L}$. Taylor expansion of jump condition relies on Taylor expansion of the jumps there in and of the speed of the shock

$$
\begin{aligned}
f_R - f_L &= f'(L)\Delta u + f''(L)\frac{\Delta u^2}{2} + f'''(L)\frac{\Delta u^3}{6} + o(\Delta u^3) \\
\dot{s} &= f'(L) + f''(L)\frac{\Delta u}{2} + f'''(L)\frac{\Delta u^2}{6} + o(\Delta u^2) \\
\eta_R - \eta_L &= \eta'(L)\Delta u + \eta''(L)\frac{\Delta u^2}{2} + \eta'''(L)\frac{\Delta u^3}{6} + o(\Delta u^3) \\
q_R - q_L &= q'(L)\Delta u + q''(L)\frac{\Delta u^2}{2} + q'''(L)\frac{\Delta u^3}{6} + o(\Delta u^3)
\end{aligned}
$$

and comparing terms of the same order

$$
0 \leq \Delta u(\eta'_L f'_L - q'_L) + \Delta u^2 \left(\frac{1}{2} f'_L \eta''_L + \frac{1}{2} f''_L \eta'_L - \frac{1}{2} q''_L\right) + \Delta u^3 \left(\frac{1}{6}\eta'''_L f'_L + \frac{1}{4}\eta''_L f''_L + \frac{1}{6}\eta'_L f'''_L - \frac{1}{6} q'''_L\right) + o(\Delta u^3) \; .
$$

Exploiting the relation $q' = \eta' f'$, and evaluating higher-order derivatives

$$
\begin{aligned}
q'' &= \eta'' f' + \eta' f'' \\
q''' &= \eta''' f' + 2\eta'' f'' + \eta' f'''
\end{aligned}
$$

it follows that the first two terms of the jump condition inequality are identically zero, so that it can be written as

$$
0 \leq -\frac{1}{12}\Delta u^3 \eta''_L f''_L \; .
$$

For a system with **convex entropy function** $\eta''(u) \geq 0$, and **convex flux** $f''(u) \geq 0$, it follows that $\Delta u \leq 0$, i.e. $u_R < u_L$. Using the Taylor expansion (either centered in the left or right state) of the speed of the shock,

$$
\begin{aligned}
\dot{s} &= f'_L + f''_L \frac{\Delta u}{2} \leq f'_L \\
\dot{s} &= f'_R - f''_R \frac{\Delta u}{2} \geq f'_R
\end{aligned}
$$

the relation between the eigenvalues of the systems in the left $\lambda_L = f'(u_L)$ and right state $\lambda_R = f'(u_R)$, and the speed of the shock immediately follows

$$
\lambda_L \geq \dot{s} \geq \lambda_R \; .
$$

Burgers' equation

35.9.2 Non-linear system of equations in 1-dimensional domain

$$
\partial_t \mathbf{u} + \partial_x \mathbf{F}(\mathbf{u}) = \mathbf{0} \; .
$$

If a shock originates from the ith characteristic family, the condition $\lambda_{i,L} \geq \dot{s}_i \geq \lambda_{i,R}$ holds.

Details

Let $\mathbf{A}(\mathbf{u}) = \nabla_{\mathbf{u}} \mathbf{F}(\mathbf{u})$ the convection matrix, $\mathbf{A} = \mathbf{R}\Lambda\mathbf{L}$ its spectral decomposition, and $\mathbf{v}$ the characteristic variables s.t. $d\mathbf{v} = \mathbf{L} d\mathbf{u}$, or $d\mathbf{u} = \mathbf{R} d\mathbf{v}$.

The convective form of the equation reads

$$
\partial_t \mathbf{u} + \mathbf{A}\partial_x \mathbf{u} = \mathbf{0} \; ,
$$

or in diagonal form using characteristic variables,

$$\partial_t \mathbf{v} + \Lambda \partial_x \mathbf{v} = \mathbf{0} \ .$$

A shock originating from the ith family of characteristics comes from the scalar equation

$$\partial_t v_i + \lambda_i \partial_x v_i = 0 \ .$$

The speed of the shock satisfies $\dot{s}_i [\mathbf{u}]_i = [\mathbf{F}(\mathbf{u})]_i$, with $[\mathbf{u}]_i = \mathbf{u}_i^+ - \mathbf{u}_i^-$, and

$$\begin{aligned} \mathbf{u}_i^- &= \mathbf{u}(\mathbf{v}_{\notin i}, v_i^-) \\ \mathbf{u}_i^+ &= \mathbf{u}(\mathbf{v}_{\notin i}, v_i^+) \end{aligned}$$

Taylor expansion of the relation gives (no sum on i, as Δv_i is the only non-zero component of $\Delta \mathbf{v}$ across a shock originating from the ith family of characteristics)

$$\begin{aligned} \dot{s}_i \Delta_i u_k = \Delta_i F_k = \\ &= \partial_{u_\ell} F_k \Delta_i u_\ell + \frac{1}{2} \partial_{u_\ell u_m} F_k \Delta_i u_\ell \Delta_i u_m = \\ &= A_{k\ell} R_{\ell i} \Delta v_i + \frac{1}{2} \partial_{u_\ell u_m} F_k \Delta_i u_\ell \Delta_i u_m \end{aligned}$$

Exploiting spectral decomposition of the convection matrix

$$\left(\dot{s}_i R_{ki} - R_{km} \Lambda_{mn} \underbrace{L_{n\ell} R_{\ell i}}_{\delta_{ni}} \right) \Delta v_i = \frac{1}{2} \partial_{u_\ell u_m} F_k \Delta_i u_\ell \Delta_i u_m$$

$$\begin{aligned} (\dot{s}_i - \lambda_i) \, R_{ki} \Delta v_i &= \frac{1}{2} \partial_{u_\ell u_m} F_k \Delta_i u_\ell \Delta_i u_m \\ (\dot{s}_i - \lambda_i) \, \Delta_i u_k &= \frac{1}{2} \partial_{u_\ell u_m} F_k \Delta_i u_\ell \Delta_i u_m \ . \end{aligned}$$

If the flux is convex (here the flux is a vector quantity. Is this the requirement for each of its components F_k?), i.e. the Hessian is (semi)definite positive, the right-hand side is always non-negative.

The following relations have been used above

$$du_l = R_{lm} dv_m \ .$$

$$\Delta_i u_l = \partial_{v_m} u_l \underbrace{\Delta_i v_m}_{=\delta_{im} \Delta v_i} = \partial_{v_i} u_l \Delta v_i = R_{li} \Delta v_i \ .$$

Entropy function for systems

Let $\eta(\mathbf{u})$ the entropy function of the system $\partial_t \mathbf{u} + \partial_x \mathbf{F}(\mathbf{u}) = \mathbf{0}$. It follows

$$\begin{aligned} \partial_t \eta &= \partial_{u_i} \eta \, \partial_t u_i \\ &= -\partial_{u_i} \eta \, \partial_x F_i = \\ &= -\partial_{u_i} \eta \, \partial_{u_j} F_i \, \partial_x u_j = \\ &= -\partial_{u_j} q \, \partial_x u_j = \\ &= -\partial_x q \ , \end{aligned}$$

i.e. the compatibility equation between the entropy function $\eta(\mathbf{u})$ and the entropy flux $q(\mathbf{u})$ reads

$$\partial_{u_j} q = \partial_{u_j} F_i \, \partial_{u_i} \eta = A_{ij} \partial_{u_i} \eta \ ,$$

or using vector formalism

$$\nabla_{\mathbf{u}} q = \nabla_{\mathbf{u}} \mathbf{F} \cdot \nabla_{\mathbf{u}} \eta = \mathbf{A} \cdot \nabla_{\mathbf{u}} \eta \ .$$

Entropy function in diffusive equations

Let $\partial_t \mathbf{u} + \partial_x \mathbf{F}(\mathbf{u}) = \partial_x(\mathbf{D}\partial_x \mathbf{u})$ the diffusive equation. The entropy equation becomes

$$\begin{aligned}
\partial_t \eta &= \partial_{u_i} \eta \, \partial_t u_i \\
&= -\partial_{u_i} \eta \, \partial_x F_i + \partial_{u_i} \eta \partial_x \left(D_{ik} \partial_x u_k \right) = \\
&= -\partial_{u_i} \eta \, \partial_{u_j} F_i \, \partial_x u_j + \partial_x \left(\partial_{u_i} \eta D_{ik} \partial_x u_k \right) - \partial_x \partial_{u_i} \eta D_{ik} \partial_x u_k = \\
&= -\partial_{u_j} q \, \partial_x u_j + \partial_x \left(\partial_{u_i} \eta D_{ik} \partial_x u_k \right) - \partial_x u_l \partial_{u_l u_i} \eta D_{ik} \partial_x u_k \, .
\end{aligned}$$

By definition, if the **diffusion term** is **compatible** with the entropy function, the matrix (is this a matrix, a tensor or a matrix of tensors?) $\nabla_{\mathbf{uu}} \cdot \mathbf{D}$ is semi-definite positive. If the diffusion term is compatible with the entropy function, in the inviscid limit $D_{ik} \to 0$, the entropy inequality becomes

$$\partial_t \eta - \partial_x q \leq 0 \, .$$

Entropy variables

Now, let the **entropy variables** be $\mathbf{w} := \nabla_{\mathbf{u}} \eta$, $w_i = \partial_{u_i} \eta$. The last term becomes

$$\begin{aligned}
\frac{\partial u_l}{\partial x} \frac{\partial u_k}{\partial x} \frac{\partial}{\partial u_l} \frac{\partial}{\partial u_i} \eta D_{ik} &= \frac{\partial u_l}{\partial x} \frac{\partial u_k}{\partial x} \frac{\partial}{\partial u_l} w_i D_{ik} = \\
&= \frac{\partial w_i}{\partial x} D_{ik} \frac{\partial u_k}{\partial x} = \\
&= \partial_x \mathbf{w} \cdot \mathbf{D} \cdot \left(\nabla_{\mathbf{uu}} \eta \right)^{-1} \cdot \partial_x \mathbf{w} \, ,
\end{aligned}$$

being

$$\partial_x \mathbf{u} = \partial_x \mathbf{w} \cdot \nabla_{\mathbf{w}} \mathbf{u} = \partial_x \mathbf{w} \cdot \left(\nabla_{\mathbf{u}} \mathbf{w} \right)^{-1} = \partial_x \mathbf{w} \cdot \left(\nabla_{\mathbf{uu}} \eta \right)^{-1}$$

or, by symmetry of the Hessian, $\partial_x \mathbf{u} = \left(\nabla_{\mathbf{uu}} \eta \right)^{-1} \cdot \partial_x \mathbf{w}$.

Compatibility between the diffusion and the entropy function needs $\mathbf{D} \cdot \left(\nabla_{\mathbf{uu}} \eta \right)^{-1}$ to be semi-definite positive.

$$\partial_x u_k = \partial_{w_j} u_k \partial_x w_j = \left[\partial_{u_k} w_j \right]^{-1} \partial_x w_j \, ,$$

$$\delta_{ij} = \partial_{u_i} w_k \partial_{w_k} u_j \, .$$

P-system

A common choice of *entropy function* in a P-system is the mechanical energy (the "reversible" part of the energy).

$$\eta = \rho \left(\frac{1}{2} u^2 + a^2 \ln \frac{\rho}{\rho_0} \right) \, .$$

This quantity decreases across shocks. Corresponding flux reads

$$q = \rho \left(\frac{1}{2} u^2 + a^2 \ln \frac{\rho}{\rho_0} + a^2 \right) u \, .$$

Details

Taking the time derivative of the entropy function,

$$
\begin{aligned}
\eta_{/t} &= \rho_{/t}\frac{u^2}{2} + \rho u u_{/t} + a^2\rho_{/t}\ln\frac{\rho}{\rho_0} + a^2\rho_{/t} = \\
&= \left(\frac{u^2}{2} + a^2\ln\frac{\rho}{\rho_0} + a^2\right)\rho_{/t} + u\rho u_{/t} = \\
&= -\left(\frac{u^2}{2} + a^2\ln\frac{\rho}{\rho_0} + a^2\right)(\rho u)_{/x} - u\left(\rho u u_{/x} + a^2\rho_{/x}\right) = \\
&= -\left(\rho\frac{u^2}{2}u\right)_{/x}\underbrace{-a^2\ln\frac{\rho}{\rho_0}\rho u_{/x} - a^2\ln\frac{\rho}{\rho_0}u\rho_{/x} - a^2\rho_{/x}u}_{=-\left(a^2\rho\ln\frac{\rho}{\rho_0}u\right)_{/x}}\underbrace{-a^2\rho u_{/x} - a^2\rho_{/x}u}_{=-(a^2\rho u)_{/x}} = \\
&= -\left[\left(\rho\frac{u^2}{2} + a^2\rho\ln\frac{\rho}{\rho_0} + \rho a^2\right)u\right]_{/x}.
\end{aligned}
$$

Shallow water

Euler equations

35.10 Convexity in hyperbolic problems

Convexity of the entropy function and the flux are assumption used discussing *physical solutions in hyperbolic problems*.

Convexity of the entropy function, $\eta''(u) \geq 0$, and the flux F, F"(u)$$, is a requirement for the entropy inequality.

Next section discusses problems with non-convex fluxes, and the **Olenik-criterion**.

35.10.1 Examples

Burgers' equation

- Flux: $F(u) = \frac{u^2}{2}$
- Entropy: $\eta(u) = \frac{u^2}{2}$
- Entropy flux: $q(u) = \frac{u^3}{3}$

so that the compatibility condition $q'(u) = F'(u)\eta'(u)$ is satisfied.

The flux is convex, $F''(u) = 1 > 0$; the entropy function is convex as well, $\eta''(u) = 1 > 0$.

P-system

$$\partial_t \begin{bmatrix} \rho \\ m \end{bmatrix} + \partial_x \begin{bmatrix} m \\ \frac{m^2}{\rho} + \rho a^2 \end{bmatrix} = \partial_x \begin{bmatrix} 0 \\ \mu \partial_x \left(\frac{m}{\rho} \right) \end{bmatrix}$$

- Flux: $F(\mathbf{u}) = \begin{bmatrix} m \\ \frac{m^2}{\rho} + \rho a^2 \end{bmatrix}$
- Entropy: $\eta(\mathbf{u}) = \frac{m^2}{2\rho} + a^2 \rho \ln \frac{\rho}{\rho_0}$
- Diffusion flux: $\begin{bmatrix} 0 & 0 \\ -\mu \frac{m}{\rho^2} & \mu \frac{1}{\rho} \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ m \end{bmatrix}$

Flux. First component

$$\nabla_{\mathbf{u}} F_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\nabla_{\mathbf{uu}} F_1 = \mathbf{0}_2 \, .$$

Second component

$$\nabla_{\mathbf{u}} F_2 = \begin{bmatrix} a^2 - \frac{m^2}{\rho^2} \\ 2 \frac{m}{\rho} \end{bmatrix}$$

$$\nabla_{\mathbf{uu}} F_2 = \begin{bmatrix} 2 \frac{m^2}{\rho^3} & -\frac{2m}{\rho^2} \\ -\frac{2m}{\rho^2} & \frac{2}{\rho} \end{bmatrix} \, .$$

Entropy. Gradient

$$\nabla_{\mathbf{u}} \eta = \begin{bmatrix} -\frac{m^2}{2\rho^2} + a^2 \left(1 + \ln \frac{\rho}{\rho_0} \right) \\ \frac{m}{\rho} \end{bmatrix}$$

and Hessian

$$\nabla_{\mathbf{uu}} \eta = \begin{bmatrix} \frac{m^2}{\rho^3} + \frac{a^2}{\rho} & -\frac{m}{\rho^2} \\ -\frac{m}{\rho^2} & \frac{1}{\rho} \end{bmatrix}$$

Compatibility of diffusion and entropy.

$$\nabla_{\mathbf{uu}} \eta \cdot \mathbf{D} = \mu \begin{bmatrix} \frac{m^2}{\rho^4} & -\frac{m}{\rho^3} \\ -\frac{m}{\rho^3} & \frac{1}{\rho^2} \end{bmatrix}$$

This matrix is semi-definite positive with one zero eigenvalue (associated with exact conservation and no dissipation in mass equation), and the other eigenvalue

Euler equations for compressible flows

Navier-Stokes equations

$$\begin{aligned} &\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0 \\ &\rho D_t \mathbf{u} = \nabla \cdot \mathbb{T} \\ &\rho D_t e^t = \nabla \cdot (\mathbb{T} \cdot \mathbf{u}) - \nabla \cdot \mathbf{q} \, , \end{aligned}$$

with $\mathbb{T} = -p\mathbb{I} + 2\mu\mathbb{D} + \lambda(\nabla \cdot \mathbf{u})\mathbb{I}$ the constitutive equation for Newtonian fluids, and $\mathbf{q} = -k\nabla T$ if Fourier law holds.

The conservative form of the equations in 1-dimensional domain reads

$$\partial_t \begin{bmatrix} \rho \\ m \\ E^t \end{bmatrix} + \partial_x \begin{bmatrix} m \\ \frac{m^2}{\rho} + p \\ \frac{m}{\rho}\left(E^t + p\right) \end{bmatrix} = \partial_x \begin{bmatrix} 0 \\ (\mu+\lambda)\partial_x\left(\frac{m}{\rho}\right) \\ (\mu+\lambda)\frac{m}{\rho}\partial_x\left(\frac{m}{\rho}\right) - q \end{bmatrix}$$

The convective form using (ρ, u, e) as primary variables reads

$$\ldots$$

Entropy function is chosen as the thermodynamic entropy volume density $\eta = \rho s$. Entropy equation naturally follows from the thermodynamic relation $ds = \frac{1}{T}\left(de - \frac{p}{\rho^2}d\rho\right)$,

$$\rho\partial_t s + \rho\mathbf{u}\cdot\nabla s = \frac{2\mu|\mathbb{D}|^2 + \lambda(\nabla\cdot\mathbf{u})^2}{T} + \frac{k|\nabla T|^2}{T^2} - \nabla\cdot\left(\frac{\mathbf{q}}{T}\right)$$

or in conservative form (using mass equation)

$$\partial_t\left(\rho s\right) + \nabla\cdot\left(\rho s\mathbf{u}\right) = \frac{2\mu|\mathbb{D}|^2 + \lambda(\nabla\cdot\mathbf{u})^2}{T} + \frac{k|\nabla T|^2}{T^2} - \nabla\cdot\left(\frac{\mathbf{q}}{T}\right)$$

- Flux: $F(\mathbf{u}) = \begin{bmatrix} m \\ \frac{m^2}{\rho} + p \\ \frac{m}{\rho}(E^t + p) \end{bmatrix}$
- Entropy: $\eta(\mathbf{u}) = \rho s\left(\rho, \mathbf{m}, E^t\right)$
- Diffusion flux: $\begin{bmatrix} 0 & 0 & 0 \\ -(\mu+\lambda)\frac{m}{\rho^2} & (\mu+\lambda)\frac{1}{\rho} & 0 \\ \ldots & \ldots & \ldots \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ m \\ E^t \end{bmatrix}$

Flux. First component

$$\nabla_{\mathbf{u}} F_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\nabla_{\mathbf{uu}} F_1 = \mathbf{0}_3 \ .$$

Second component

$$\nabla_{\mathbf{u}} F_2 = \begin{bmatrix} -\frac{m^2}{\rho^2} + \Pi_{/\rho} \\ 2\frac{m}{\rho} + \Pi_{/m} \\ \Pi_{E^t} \end{bmatrix}$$

$$\nabla_{\mathbf{uu}} F_2 = \begin{bmatrix} \ldots \\ \ldots \\ \ldots \end{bmatrix} \ .$$

Third component

$$\nabla_{\mathbf{u}} F_2 = \begin{bmatrix} \ldots \\ \ldots \\ \ldots \end{bmatrix}$$

$$\nabla_{\mathbf{uu}} F_2 = \begin{bmatrix} \ldots \\ \ldots \\ \ldots \end{bmatrix} \ .$$

Entropy. Gradient

$$\nabla_{\mathbf{u}}\eta = \begin{bmatrix} s + \rho s_{/\rho} \\ \rho s_{/m} \\ \rho s_{/E^t} \end{bmatrix}$$

and Hessian

$$\nabla_{\mathbf{uu}}\eta = \begin{bmatrix} 2s_{/\rho} + \rho s_{/\rho\rho} & s_{/m} + \rho s_{\rho m} & s_{/E^t} + \rho s_{\rho E^t} \\ \dots & \rho s_{mm} & \rho s_{mE^t} \\ \dots & \dots & \rho s_{E^t E^t} \end{bmatrix}$$

Compatibility of diffusion and entropy.

$$\nabla_{\mathbf{uu}}\eta \cdot \mathbf{D} = \mu \left[\dots \right]$$

This matrix is semi-definite positive with one zero eigenvalue (associated with exact conservation and no dissipation in mass equation), and the other eigenvalue

35.11 Non-convex hyperbolic problems

In this section, non-convex hyperbolic problems are briefly discussed using a non-linear scalar equation as a toy problem to show their characteristics. The main topic is the solution of a Riemann problem. *Depending on the initial conditions*, the solution of the Rieamnn problem for a non-convex hyperbolic scalar equation may involve not only simple waves (like shock or rarefaction), but also **composite waves**, combining both a shock and a rarefaction.

First, a *numerical simulation* of a Riemann problem the viscous counterpart of the hyperbolic equation is performed and the solution is discussed. Then, the solution is discussed introducing the ***Olenik criterion*** to find the physical solution.

35.11.1 Toy problem

The non-linear scalar equation

$$\partial_t u + \partial_x F(u) = 0 \ ,$$

with flux $F(u) = u^3$ is a hyperbolic equation with non-convex flux, as

$$F''(u) = 6u \begin{cases} > 0 & \text{for } u > 0 \\ \leq 0 & \text{for } u \leq 0 \end{cases}$$

Numerical simulation

The numerical simulation of two Riemann problem of the viscous counterpart of the hyperbolic equation,

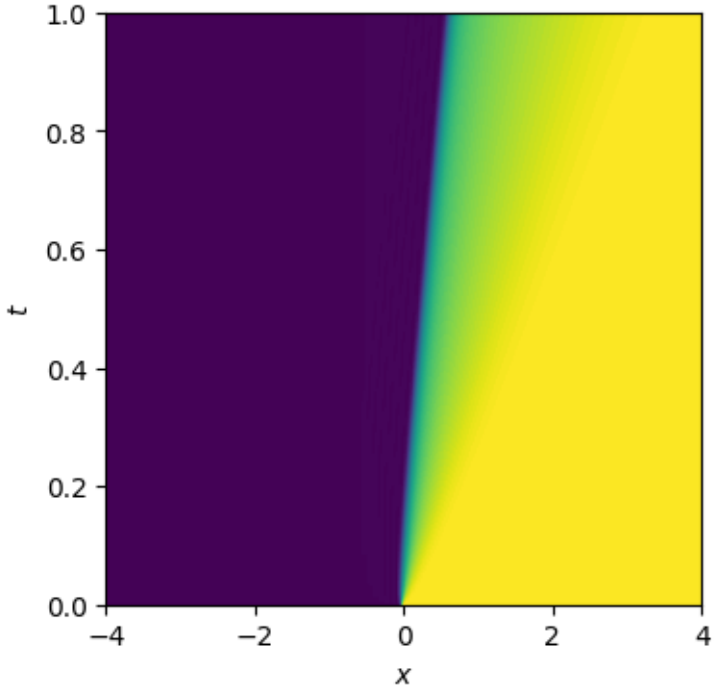
$$\partial_t u + \partial_x F(u) - \partial_x(\nu \partial_x u) = 0 \ ,$$

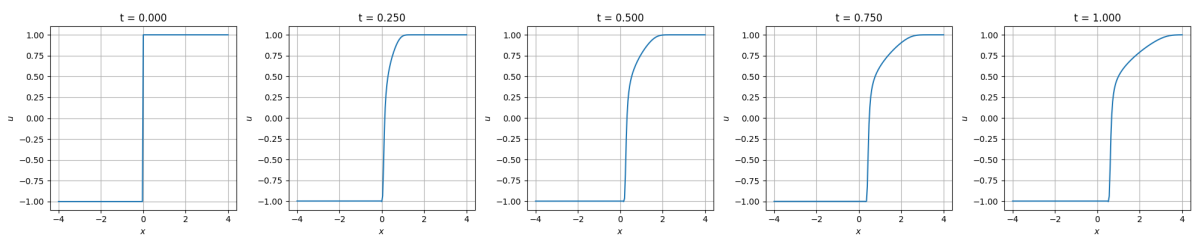
is performed here with a time-explicit finite-difference method for two different initial conditions: 1) $(u_L, u_R) = (-1, 1)$, 2) $(u_L, u_R) = (1, -1)$.

Here, the value of the diffusion coefficient is choosen as $\nu = .05$ as a compromise between thickness of the "shock" and the stability requirement of the numerical method.

Simulation with $(u_L, u_R) = (-1, 1)$

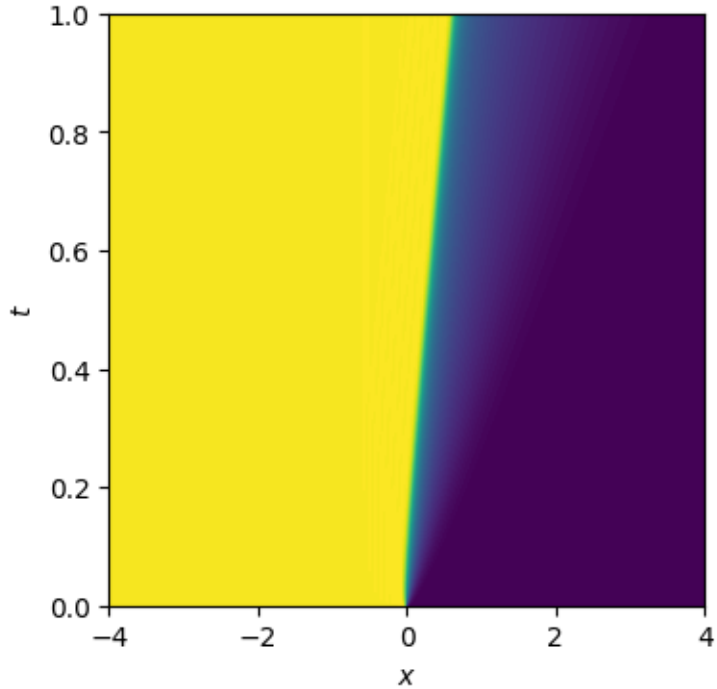
The solution is the combination of a left shock between state $u_L = -1$ and an intermediate state $u^* = \frac{1}{2}$, and a right fan between u^* and $u_R = 1$.

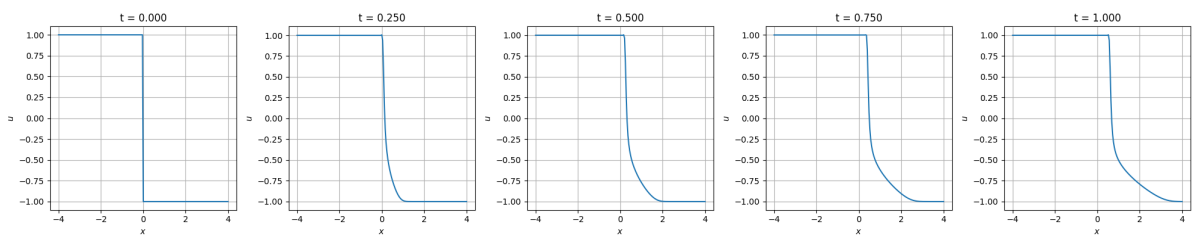




Simulation with $(u_L, u_R) = (1, -1)$

The solution is the combination of a left shock between state $u_L = 1$ and an intermediate state $u^* = \frac{1}{2}$, and a right fan between u^* and $u_R = -1$.





Olenik's criterion

In a Riemann problem with two values u_L, u_R, with $F''(u_L)F''(u_R) < 0$ - so that non-convexity manifests itself, and assuming that $F''(u)$ changes sign only once (see the comment below for more complex flux functions -, a composite wave occurs. This composite wave is composed of a rarefaction wave and a shock wave.

The intermediate state u^* is found with the condition that the speed of the shock equals the characteristic speed of the fan at the shock, i.e.

$$s(u_L, u^*) = F'(u^*) \ ,$$

or

$$\frac{F(u^*) - F(u_L)}{u^* - u_L} = F'(u^*) \ .$$

i.e. the slope $F'(u^*)$ of the tangent line at $F(u)$ in u^* equals the slope of the line connecting the state $(\cdot)^*$ with $(\cdot)_L$. This condition prevents characteristic lines from originating from the shock.

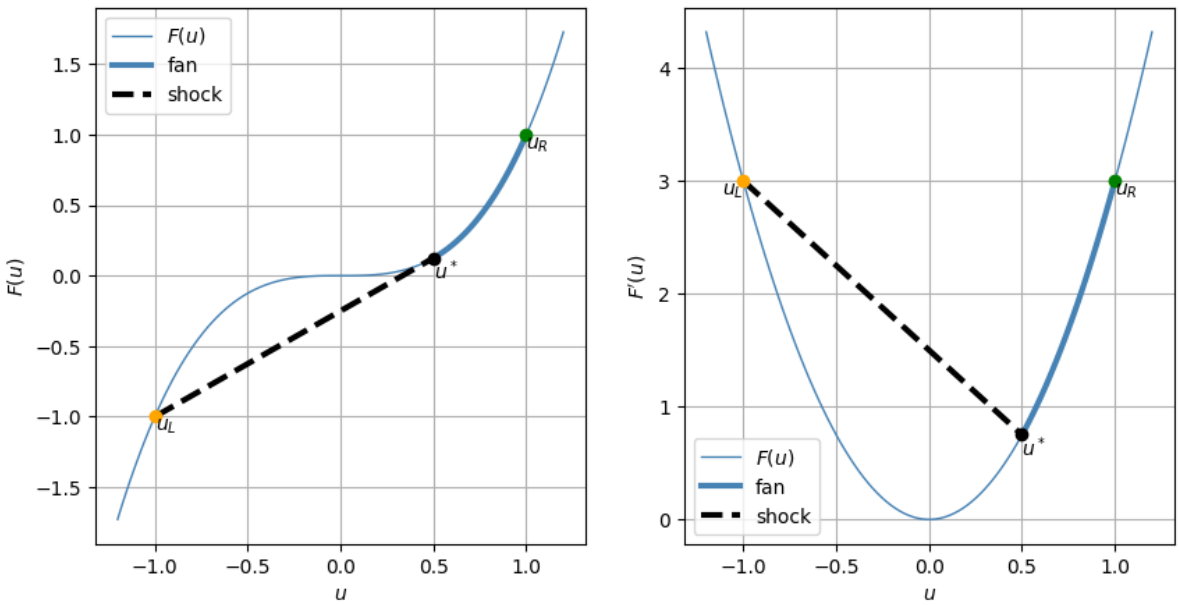
An alternative formulation of Olenik's criterion reads: for a shock to be stable, the shock speed s must be the slowest possible. For $u_L > u_R$,

$$\frac{f(u) - f(u_L)}{u - u_L} \geq s(u) \geq \frac{f(u) - f(u_R)}{u - u_R} \qquad s \in [u_L, u_R] \ .$$

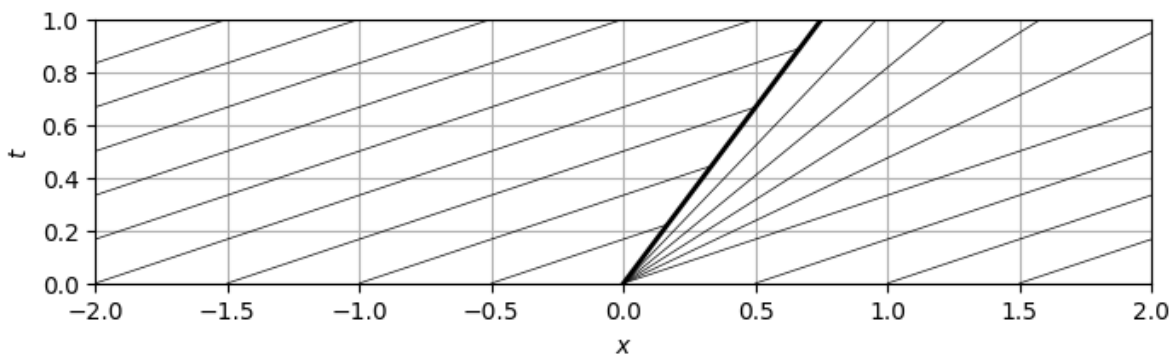
todo

- With more "complex" flux functions $F(u)$ (that have several ranges of u-values where $F''(u)$ changes sign), how many waves in the composite wave? Show how Olenik's criterion applies.

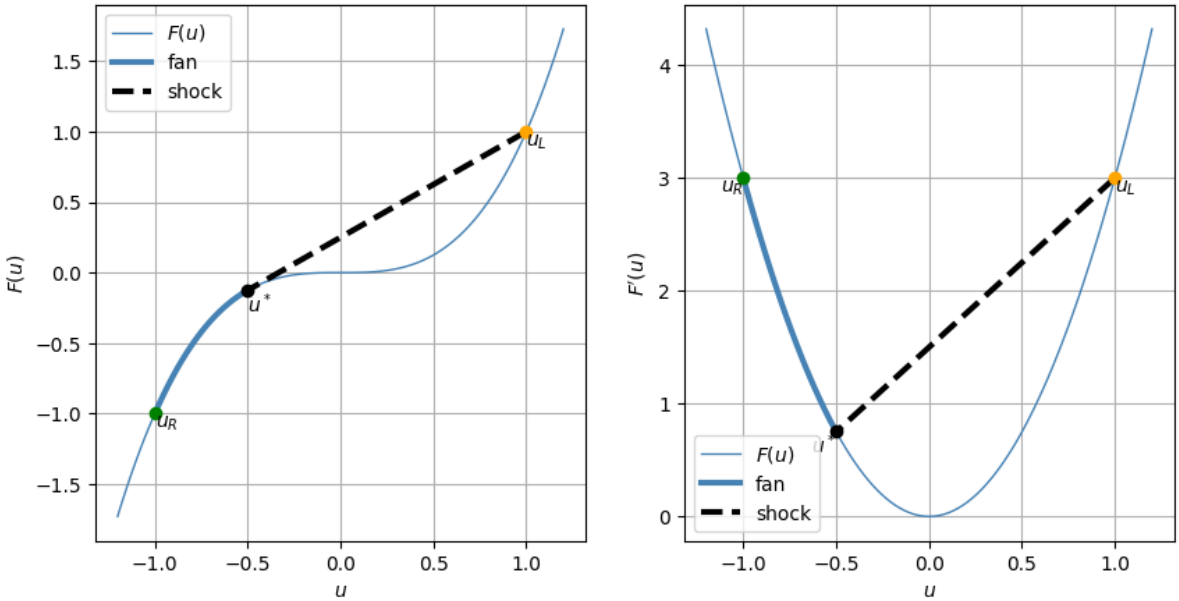
$(u_L, u_R) = (-1, 1)$



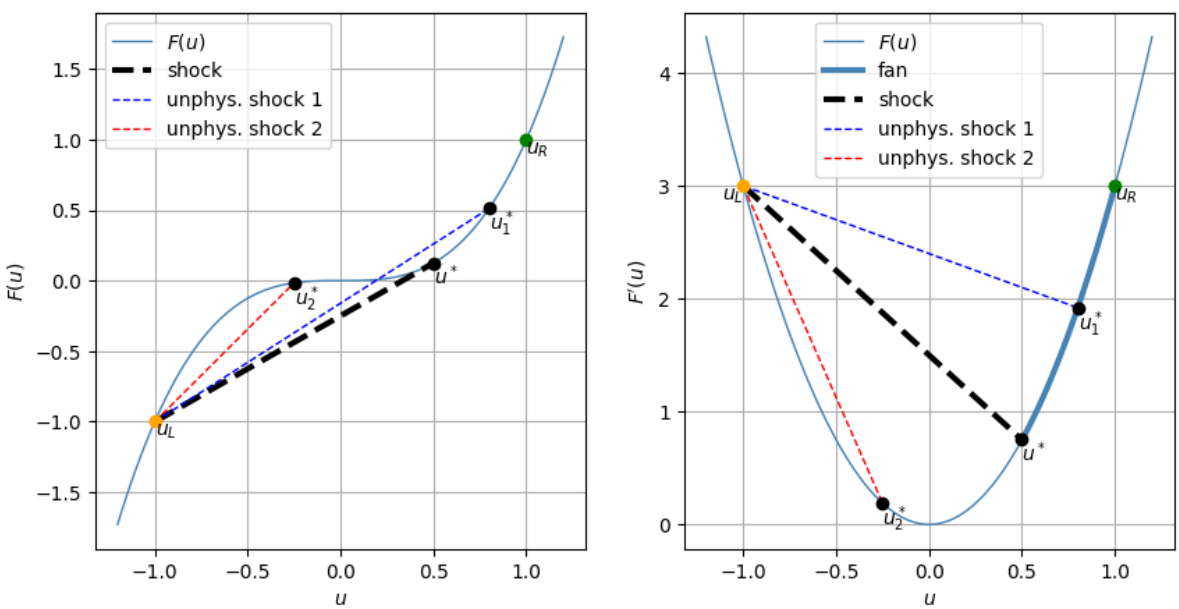
Characteristic lines in the $x - t$ plane

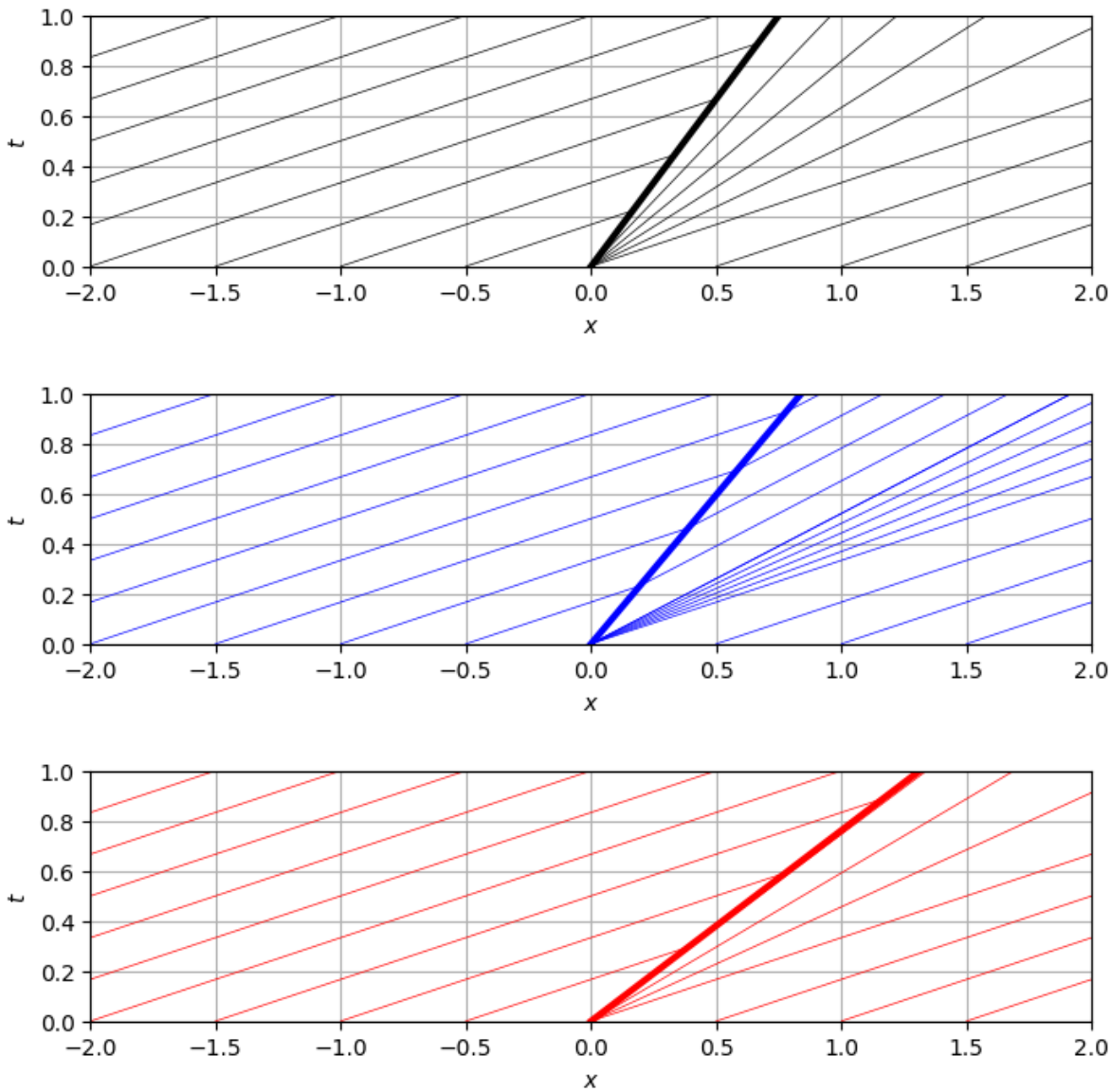


$(u_L, u_R) = (1, -1)$



Non-physical solutions for $(u_L, u_R) = (-1, 1)$





todo

- The blue unphysical solution looks unphysical for the characteristic lines orginating from the shock. But why is the red solution unphysical?
- Discussion about local stability
- Discussion with local production of entropy $\Sigma(u) = s(u)[\eta(u)] - [q(u)]$, and extreme points at the physical solution u^*? $\Sigma'(u^*) = 0$?

CHAPTER

THIRTYSIX

NAVIER-CAUCHY EQUATIONS

Navier-Cauchy equations are the differential balance equation of the momentum of an elastic isotropic medium in the regime of small strain and displacement,

$$\rho_0 \partial_{tt} \vec{s} = \rho_0 \vec{g} + \nabla \cdot \boldsymbol{\sigma} \ .$$

Stress tensor for an isotropic medium reads

$$\begin{aligned} \boldsymbol{\sigma} &= 2\mu \boldsymbol{\varepsilon} + \lambda \, \mathrm{tr}\left(\boldsymbol{\varepsilon}\right) \mathbb{I} = \\ &= \left(2\mu \boldsymbol{\varepsilon} - \frac{2}{3}\mu \, \mathrm{tr}(\boldsymbol{\varepsilon}) \mathbb{I}\right) + \left(\lambda + \frac{2}{3}\mu\right) \, \mathrm{tr}\left(\boldsymbol{\varepsilon}\right) \mathbb{I} \ , \end{aligned}$$

with the small strain tensor

$$\boldsymbol{\varepsilon} = \frac{1}{2} \left(\nabla \vec{s} + \nabla^T \vec{s}\right) \ .$$

Essential, natural and Robin boundary conditions read

$\vec{s} = \bar{\vec{s}}$	$\vec{r} \in S_D$	esserntial - Dirichlet b.c.
$\hat{n} \cdot \boldsymbol{\sigma} = \bar{\vec{t}}_n$	$\vec{r} \in S_N$	natural - Neumann b.c.
$a\vec{s} + \hat{n} \cdot \boldsymbol{\sigma} = \vec{b}$	$\vec{r} \in S_R$	Robin b.c.

36.1 Weak formulation

For $\forall \vec{w} \in \ldots$

$$\begin{aligned} 0 &= -\int_V \rho \vec{w} \cdot \partial_{tt} \vec{s} + \int_V \rho_0 \vec{w} \cdot \vec{g} + \int_V \vec{w} \cdot \nabla \cdot \boldsymbol{\sigma} = \\ &= -\int_V \rho \vec{w} \cdot \partial_{tt} \vec{s} + \int_V \rho_0 \vec{w} \cdot \vec{g} + \int_{\partial V} \hat{n} \cdot \boldsymbol{\sigma} \cdot \vec{w} - \int_V \nabla \vec{w} : \boldsymbol{\sigma} \end{aligned}$$

The volume integral containing the stress tensor can be written either as

$$\begin{aligned} \int_V \nabla \vec{w} : \boldsymbol{\sigma} &= \int_V w_{i/j} \left[\mu \left(s_{i/j} + s_{j/i}\right) + \lambda s_{k/k} \delta_{ij}\right] = \\ &= \int_V \mu w_{i/j} \left(s_{i/j} + s_{j/i}\right) + \int_V \lambda w_{j/j} s_{k/k} \end{aligned}$$

or

$$\begin{aligned} \int_V \frac{1}{2} \left(\nabla \vec{w} + \nabla^T \vec{w}\right) : \boldsymbol{\sigma} &= \int_V \frac{1}{2} \left(w_{i/j} + w_{j/i}\right) \left[\mu \left(s_{i/j} + s_{j/i}\right) + \lambda s_{k/k} \delta_{ij}\right] = \\ &= \int_V \frac{\mu}{2} \left(w_{i/j} + w_{j/i}\right) \left(s_{i/j} + s_{j/i}\right) + \int_V \lambda w_{j/j} s_{k/k} \end{aligned}$$

The weak formulation of the Navier-Cauchy equations reads

$$\int_V \rho_0 \vec{w} \cdot \partial_{tt} \vec{s} + \int_V 2\mu \frac{\nabla \vec{w} + \nabla^T \vec{w}}{2} : \frac{\nabla \vec{s} + \nabla^T \vec{s}}{2} + \int_V \lambda \nabla \cdot \vec{w} \, \nabla \cdot \vec{s} + \int_{S_R} \vec{w} \cdot a \vec{s} = \int_V \rho_0 \vec{w} \cdot \vec{g} + \int_{S_N} \vec{w} \cdot \bar{\vec{t}}_n + \int_{S_R} \vec{w} \cdot \vec{b} \, ,$$

for $\forall \vec{w} \in$ ..., and with $\vec{s} = \bar{\vec{s}}$ for $\vec{r} \in S_D$.

CHAPTER

THIRTYSEVEN

NAVIER-STOKES EQUATIONS

Incompressible Navier-Stokes equations read

$$\begin{cases} \rho\partial_t\vec{u} + \rho\left(\vec{u}\cdot\nabla\right)\vec{u} - \mu\nabla^2\vec{u} + \nabla P = \rho\vec{g} \\ \nabla\cdot\vec{u} = 0\ . \end{cases}$$

Mass balance equation is replaced by the incompressiblity kinematic constraint, $\nabla\cdot\vec{u} = 0$: this constraint is not dynamic, as time derivative of density does not appear in the equation. With the incompressibility constraint, mass equation tells us that material particles keep their density constant,

$$0 = \underbrace{\partial_t\rho + \vec{u}\cdot\nabla\rho}_{=\frac{D\rho}{Dt}} + \rho\underbrace{\nabla\cdot\vec{u}}_{=0} = \frac{D\rho}{Dt}\ ,$$

whose solution can be written using material coordinates $\vec{r}_0$ as $\rho(\vec{r}(\vec{r}_0,t),t) = \rho_0(\vec{r}_0,t)$.

37.1 Incompressibility constraint

Incompressibility constraint makes thermodynamic fade, while pressure field is replaced by/contains the contribution of a Lagrangian multiplier related to the incompressiblity constraint.

37.1.1 Wave-vector transformed space

Transforming the fields from physical space to the wave-vector space $\tilde{u}(\vec{\kappa},t) = \mathcal{F}\left\{\vec{u}(\vec{r},t)\right\}$, Navier-Stokes equations for incompressible fluids with uniform and constant density $\rho(\vec{r},t) = \rho$ becomes

$$\begin{cases} \rho\partial_t\tilde{u} + \mathcal{F}\left\{\left(\vec{u}\cdot\nabla\right)\vec{u}\right\} + \mu|\vec{\kappa}|^2\tilde{u} + i\vec{k}\widetilde{P} = \rho\tilde{g} \\ i\vec{\kappa}\cdot\tilde{u} = 0\ . \end{cases}$$

Taking the divergence of the momentum balance equation, i.e. taking the scalar product with $i\vec{\kappa}$ in the transformed space, and using the incompressibility constraint to set $i\vec{\kappa}\cdot\tilde{u} = 0$,

$$i\vec{\kappa}\cdot\mathcal{F}\left\{(\vec{u}\cdot\nabla)\vec{u}\right\} - |\vec{\kappa}|^2\widetilde{P} = i\vec{\kappa}\cdot\rho\tilde{g}\ ,$$

so that the transformed pressure field becomes

$$\widetilde{P} = \frac{i\vec{\kappa}}{|\vec{\kappa}|^2}\cdot\mathcal{F}\left\{\left(\vec{u}\cdot\nabla\right)\vec{u} - \rho\vec{g}\right\}\ ,$$

Replacing this expression in the transformed Navier-Stokes equations, the meaning of the pressure field as a Lagrange multiplier associated with incompressibility constraint becomes clear,

$$\rho\partial_t\tilde{u} + \mu|\vec{\kappa}|^2\tilde{u} = \left[1 - \frac{\vec{\kappa}\vec{\kappa}}{|\vec{\kappa}|^2}\right]\cdot\mathcal{F}\left\{-(\vec{u}\cdot\nabla)\vec{u} + \rho\tilde{g}\right\}$$

as the orthogonal projector $\left[1 - \frac{\vec{\kappa}\vec{\kappa}}{|\vec{\kappa}|^2}\right]$ onto the space of divergence-free functions acts on the non-linear and forcing terms.

37.2 Weak formulation of the problem

$$
\begin{aligned}
0 &= \int_V \vec{w}\cdot[\rho\partial_t\vec{u} + \rho(\vec{u}\cdot\nabla)\vec{u} - 2\mu\nabla\cdot\mathbb{D}(\vec{u}) + \nabla P - \rho\vec{g}] - \int_V v\nabla\cdot\vec{u} = \\
&= \int_V \vec{w}\cdot[\rho\partial_t\vec{u} + \rho(\vec{u}\cdot\nabla)\vec{u}] + \int_V 2\mu\nabla\vec{w}:\mathbb{D} - \int_V \nabla\cdot\vec{w}P - \int_V \vec{w}\cdot\rho\vec{g} - \int_V v\nabla\cdot\vec{u} - \int_{\partial V}\hat{n}\cdot(\mathbb{S} - P\mathbb{I})\cdot\vec{w}\,,
\end{aligned}
$$

37.2.1 Weak formulation and incompressibility constraint

$$
\vec{r}(\vec{r}_0, t) = \vec{r}(q(t), t)
$$

$$
\vec{u} = \frac{D\vec{r}}{Dt} = \dot{q}\frac{\partial\vec{r}}{\partial q} + \frac{\partial\vec{r}}{\partial t}
$$

In the weak formulation, using $\vec{w} = \frac{\partial\vec{r}}{\partial q} = \frac{\partial\vec{u}}{\partial\dot{q}}$

$$
0 = \int_V \vec{w}\cdot\rho\frac{D\vec{u}}{Dt} + \int_V 2\mu\nabla\vec{w}:\mathbb{D} - \int_V \nabla\cdot\vec{w}P - \int_V \rho\vec{w}\cdot\vec{g} - \int_V v\nabla\cdot\vec{u} - \int_{\partial V}\vec{t}_{\hat{n}}\cdot\vec{w}\,,
$$

$$
\begin{aligned}
\int_V \vec{w}\cdot\rho\frac{D\vec{u}}{Dt}\,dV &= \int_{V_0}\rho_0\frac{\partial\vec{u}}{\partial\dot{q}}\cdot\frac{D\vec{u}}{Dt} = \\
&= \int_{V_0}\rho_0\frac{D}{Dt}\left(\frac{\partial\vec{u}}{\partial\dot{q}}\cdot\vec{u}\right)dV_0 - \int_{V_0}\rho_0\frac{D}{Dt}\left(\frac{\partial\vec{r}}{\partial\dot{q}}\right)\cdot\vec{u}\,dV_0 = \\
&= \int_{V_0}\rho_0\frac{D}{Dt}\left(\frac{\partial}{\partial\dot{q}}\frac{|\vec{u}|^2}{2}\right)dV_0 - \int_{V_0}\rho_0\frac{\partial}{\partial q}\frac{|\vec{u}|^2}{2}\,dV_0 =
\end{aligned}
$$

…

37.3 Non-linear term

Different ways to treat the non-linear term:

- Semi-linear approximation of the non-linear term

  $$
  (\vec{u}(\vec{r}, t^n)\cdot\nabla)\vec{u}(\vec{r}, t^n) \sim (\vec{u}^*(\vec{r}, t^n)\cdot\nabla)\vec{u}(\vec{r}, t^n)\,,
  $$

  with $\vec{u}^*(\vec{r}, t^n)$ an approximation of $\vec{u}(\vec{r}, t^n)$ involving values of the velocity field at previous time-steps, as an example

  $$
  \vec{u}^*(\vec{r}, t^n) = \begin{cases} \vec{u}(\vec{r}, t^{n-1}) & 1^{st}\text{-order} \\ 2\vec{u}(\vec{r}, t^{n-1}) - \vec{u}(\vec{r}, t^{n-2}) & 2^{nd}\text{-order} \end{cases}
  $$

CHAPTER

THIRTYEIGHT

ARBITRARY LAGRANGIAN-EULERIAN DESCRIPTION

Reynold's transport theorem allows for the formulation of intergal equations, and grid-based methods like FVM, on moving grids and changing domains. Rules for derivatives of composite functions provide the relations between time derivatives in a Lagrangian, Eulerian, or arbitrary description,

$$\left.\frac{\partial f}{\partial t}\right|_{\vec{r}_0} = \left.\frac{\partial f}{\partial t}\right|_{\vec{r}} + \vec{u}\cdot\nabla f$$
$$\left.\frac{\partial f}{\partial t}\right|_{\vec{r}_b} = \left.\frac{\partial f}{\partial t}\right|_{\vec{r}} + \vec{u}_b\cdot\nabla f$$

Equations governing the motion of the grid are usually required as well. E.g.:

- known and prescribed motion of the grid;
- boundary conditions only without changing grids (for small displacements)
- pseudo-elastic deformation (usually good for small strain and displacement;
- for large displacements of/or models with complex geometry, sliding and/or overlapping grids could an option for grid-based methods.

38.1 Integral problem

Application of Reynolds theorem to the balance equation of the quantity **u** for a material volume V_t

$$\frac{d}{dt}\int_{V_t}\rho\mathbf{u} = \int_{V_t}\rho\mathbf{f} + \oint_{\partial V_t}\hat{n}\cdot\mathbf{T}\,.$$

provides the expression of the balance equation for a geometrical volume v_t in arbitrary motion,

$$\frac{d}{dt}\int_{v_t}\rho\mathbf{u} + \oint_{\partial v_t}\rho\mathbf{u}\left(\vec{u}-\vec{u}_b\right)\cdot\hat{n} = \int_{v_t}\rho\mathbf{f} + \oint_{\partial v_t}\hat{n}\cdot\mathbf{T}\,.$$

Here, the integral forulation of the problem will be applied to each element of the grid in arbitrary motion, for domains with variable geometry.

38.2 Differential problem

Rules for derivatives of composite functions allows to write the differential w.r.t. the variables associated with the points of a moving grid. A balance equation in convective form can be written as

$$\rho \frac{D\mathbf{u}}{Dt} = \rho \mathbf{f} + \nabla \cdot \mathbf{T}$$

$$\rho \left[\frac{\partial \mathbf{u}}{\partial t} + \vec{u} \cdot \nabla \mathbf{u} \right] =$$

$$\rho \left[\left. \frac{\partial \mathbf{u}}{\partial t} \right|_{\vec{r}_b} + (\vec{u} - \vec{u}_b) \cdot \nabla \mathbf{u} \right] =$$

Part XII

Numerical Methods for PDEs

CHAPTER

THIRTYNINE

INTRODUCTION TO NUMERICAL METHODS FOR PDES

Different numerical methods for PDEs rely on the discretization of different formulations of the continunuos problems. As an example,

- **FDM, Finite difference methods** rely on the approximation of derivatives of the **strong formulation** of the problem
- ***FEM, Finite element methods*** rely on a finite dimensional approximation of the **weak formulation** of the problem; usually the finite dimensional approximation can be interpreted as a projection of a infinite dimensional continuous problem onto a finite dimensional space, the space of the choosen finite elements
- ***FVM, Finite volume methods*** rely on an approximation of the **integral formulation** of the problem
- **BEM, Boundary elemement methods** rely on an approximation of a **boundary integral formulation** of the problem, when it's feasible and convenient
- *…spectral methods*, *spectral element methods*,…

Characteristics.

- grid: domain-grid-based, buondary-grid-based, grid-free methods
- range of interaction: short in physical space for FDV, FEM, FVM; long-range for boundary element methods, even though clustering techniques are available, like FMM; (usually) over the whole domain in space, short-range interaction in wave-number space for spectral methods;

Pros and cons. *More suited methods for each problems…; domain, order,…*

Let's take Poisson equations for a scalar function $u(\vec{r})$ to show all the possible approaches above. Here we start from the most general version of the equation, namely the integral form. As it's shown in Continuum Mechanics:Governing Equations, the most general form of balance equations is the integral form, while the differential form can be seamlessly derived only when the quantities involved are regular enough, to apply Stokes' theorem and for the derivatives appearing to exist.

The problem of interest can be interpreted as the problem of finding the temperature field $u(\vec{r})$ during steady heat conduction in the domain Ω, with distributed volume heat source $f(\vec{r})$. Fourier's law assume that condcution heat flux is proportional to the gradient of the temperature, $\vec{q} = -k\nabla u$. Temperature is prescribed, $u = g$ on the region of the boundary S_D, while the heat flux is prescribed $\hat{n} \cdot \vec{q} = h$ on the region of the boundary S_N.

The **integral formulation** of the problem for all the boundary reads

$$0 = -\oint_{\partial\Omega} \hat{n} \cdot \vec{q} + \int_{\Omega} f \ . \tag{39.1}$$

Example 39.1 (Finite volume methods)

Finite volume methods rely on a tassellation of the domain $\Omega = \cup_k \Omega_k$, to write and solve the integral balance equation (39.1) for all the elemantary domain Ω_k.

$$0 = -\oint_{\partial\Omega_k} \hat{n} \cdot \vec{q} + \int_{\Omega_k} f \ . \tag{39.2}$$

evaluating volume integrals with the internal variables of the cell Ω_k, and boundary flux with the variables of the cell Ω_k and the neighbouring cells $\Omega_i \in B_k$. Introducing the definition of numerical flux $F_{ik}(u_i, u_k)$ at the interface between a generic discrete version of the balance equation (39.2) for the kth element becomes

$$0 = -\sum_{\Omega_i \in B_k} F_{ik}(u_i, u_k) + S_k(u_k) \ .$$

Different numerical methods differs in the evaluation of fluxes and sources. FVM is **conservative** as it evaluate flux at interfaces and then distribute it to neighoring cells, see *Property 41.1*. A rough elementwise-uniform variable with jump at interfaces and diffusive flux,

$$F_{ik} = -A_{ik}\, k\, \frac{u_i - u_k}{d_{ik}} \ ,$$

with d_{ik} equal to the distance of the centers of elements i and k, leads to the balance equation for the internal volume Ω_k,

$$0 = \sum_i A_{ik}\, k\, \frac{u_i - u_k}{d_{ik}} + V_k f_k \ .$$

Volumes at the boundary are influenced by fluxes at the boundaries and boundary conditions in general.

For regular cubic mesh, $A_{ik} = \Delta x$, $d_{ik} = \Delta x$, $V_k = \Delta x$, and thus the equations for internal elements become

$$0 = k \sum_{i \in \Omega_k} \frac{u_i - u_k}{\Delta x^2} + f_k \ ,$$

where the summation is exactly equal to the center-difference stencil for the Laplacian, used in finite difference method *Example 39.2*.

If fields are regular enough to apply Stokes' theorem, it's possible to derive differential problem from the integral formulation,

$$0 = -\oint_{\partial V} \hat{n} \cdot \vec{q} + \int_V f = -\int_V \nabla \cdot \vec{q} + \int_V f = \int_V (-\nabla \cdot \vec{q} + f)$$

Exploiting the arbitrariety of the volume V, the **strong form** of the differential problem is the Poisson equation with suitable boundary conditions,

$$\begin{cases} -\nabla \cdot (k \nabla u) = f(\vec{r}) \ , & \vec{r} \in V \\ u = g(\vec{r}) \ , & \vec{r} \in S_D \\ \hat{n} \cdot \nabla u = h(\vec{r}) \ , & \vec{r} \in S_N \end{cases}$$

only holds in regions of the domain where the functions involved are continuous and differentiable, i.e. everything written in the problem is not meaningful, i.e. at least it exists.

Example 39.2 (Finite difference methods)

Finite different method approximates the strong form of the problem, building a stencil to evaluate an approximation of

the Laplacian. On a cubic regular grid ($\Delta x = \Delta y = \Delta z$), the Laplacian van be evaluated with a 7-point stencil

$$\begin{aligned}
\nabla^2 u &= \partial_{xx} u + \partial_{yy} u + \partial_{zz} u = \\
&= \frac{u_{i+1,j,k} - 2u_{i,j,k} + u_{i-1,j,k}}{\Delta x^2} + \frac{u_{i,j+1,k} - 2u_{i,j,k} + u_{i,j-1,k}}{\Delta y^2} + \frac{u_{i,j,k+1} - 2u_{i,j,k} + u_{i,j,k-1}}{\Delta z^2} = \\
&= \frac{1}{\Delta x^2}\left[-6u_{i,j,k} + u_{i+1,j,k} + u_{i-1,j,k} + u_{i,j+1,k} + u_{i,j-1,k} + u_{i,j,k+1} + u_{i,j,k-1}\right] = \\
&= \frac{1}{\Delta x^2} \sum_{i \in B_k} (u_i - u_k)
\end{aligned}$$

and it's equal to the same expression already provided for the rough finite volume method with regular grid, in *Example 39.1*

Starting from strong formulation of the differential problem, the **weak formulation** of the problem is derived:

- multiplying the strong problem by an arbitrary test function $w(\vec{r})$ compatible with the essential constraints
- integrating over the whole domain

$$\begin{aligned}
0 &= \int_V w(\vec{r}) \cdot [-\nabla \cdot (k\nabla u) - f(\vec{r})] = \\
&= \int_V \{k\nabla w(\vec{r}) \cdot \nabla u(\vec{r}) - w(\vec{r}) f(\vec{r})\} - \oint_{\partial V} w(\vec{r})\, \hat{n} \cdot \nabla u \qquad \forall w(\vec{r})\ .
\end{aligned}$$

Using a test function $w(\vec{r})$ equal to zero on the boundary where essential boundary conditions are prescribed $w|_{S_D} = 0$, and introducing the natural boundary conditions of the Neumann boundary S_N, $\hat{n} \cdot \nabla u|_{S_N} = h$,

$$\int_V k\nabla w \cdot \nabla u = \int_V wf + \int_{S_N} wh\ , \quad \forall w$$

compatible with essential boundary conditions.

Example 39.3 (Finite element methods)

Finite element methods build an N-dimensional systems:

- approximating the solution as a linear combination of N base functions, $u(\vec{r}) = \sum_j \phi_j(\vec{r}) u_j$
- testing the equation over N independent test functions $\psi_i(\vec{r})$

i.e. the ith equation becomes

$$\int_V k\nabla\psi_i(\vec{r}) \cdot \nabla\phi_j(\vec{r})\, u_j = \int_V \psi_i(\vec{r}) f + \int_{S_N} \psi_i(\vec{r}) h\ ,$$

or briefly

$$K_{ij} u_j = f_i \qquad , \qquad \mathbf{Ku} = \mathbf{f}\ .$$

A common choice uses the same functions both as test and base functions, $\psi_i(\vec{r}) = \phi_i(\vec{r})$.

The differential problem in strong form can be recast as a boundary element problem exploiting the properties of Green's functions $G(\vec{r}; \vec{r}_0)$. The expression of *Green's function $G(\vec{r}, \vec{r}_0)$* is known for some problems of interest like Poisson

equation, Helmholtz equation or wave equation. Exploiting the properties of Green's function and integration by parts, the integral buondary problem is derived as follow

$$
\begin{aligned}
E(\vec{r}_0)u(\vec{r}_0) &= \int_{\vec{r}\in V} u(\vec{r})\delta(\vec{r}-\vec{r}_0)\,dV = \\
&= -\int_{\vec{r}\in V} u(\vec{r})\nabla^2 G(\vec{r};\vec{r}_0)\,dV = \\
&= -\oint_{\vec{r}\in\partial V} u(\vec{r})\hat{n}(\vec{r})\cdot\nabla G(\vec{r};\vec{r}_0)\,dS + \oint_{\vec{r}\in\partial V} G(\vec{r};\vec{r}_0)\hat{n}(\vec{r})\cdot\nabla u(\vec{r})\,dS - \int_{\vec{r}\in V} G(\vec{r};\vec{r}_0)\nabla^2 u(\vec{r})\,dV = \\
&= -\oint_{\vec{r}\in\partial V} u(\vec{r})\hat{n}(\vec{r})\cdot\nabla G(\vec{r};\vec{r}_0)\,dS + \oint_{\vec{r}\in\partial V} G(\vec{r};\vec{r}_0)\hat{n}(\vec{r})\cdot\nabla u(\vec{r})\,dS + \int_{\vec{r}\in V} G(\vec{r};\vec{r}_0)f(\vec{r})\,dV\ .
\end{aligned}
$$

It must be paid attention that the integrals may be singular and may produce discontintuities in the fields across the surface ∂V (*so what's the value of a field in presence of a discontinuity?…virtual singularities, regularization close to the singularities…one way to correctly interpret them is via Cauchy principal value… evaluating some integrals in a check-point just inside or outside the domain may make the value of the function $E(\vec{r}_0)$ change, but the integral involving $\hat{n}\cdot\nabla G$ - related to the solid angle seen by the check-point of the surface - changes accordingly and keep the sum of these two terms constant=*.

The value of the unknown function or the flux $\hat{n}\cdot\nabla u$ are known on the Dirichlet and Neumann regions of the boundary respectively, and so the integro-differential problem becomes

$$
E(\vec{r}_0)u(\vec{r}_0) + \int_{S_N} u\hat{n}\cdot\nabla G\,dS - \int_{S_D} G\,\hat{n}\cdot\nabla u\,dS = \int_{S_D} g\hat{n}\cdot\nabla G\,dS - \int_{S_N} Gh\,dS + \int_{\vec{r}\in V} Gf\,dV
$$

The unknown function is approximated on the boundary of the domain ∂V as an N-dimensional approximation, as an example

$$
u(\vec{r}) = \sum_j \phi_j(\vec{r})u_j \qquad , \qquad \vec{r}\in\partial V\ ,
$$

and the integro-differential equation is evaluated in N different points $\vec{r}_{0,i}$ in order to get a N,N linear equation in the amplitudes u_j of the base functions,

$$
[\mathbf{E}+\mathbf{D}_N+\mathbf{S}_D]\,\mathbf{u} = \mathbf{D}_D\mathbf{g}+\mathbf{S}_N\mathbf{h}+\mathbf{S}_V\mathbf{f}\ .
$$

CHAPTER

FORTY

FINITE ELEMENT METHOD

Examples

- Structural mechanics of linear beam structures
- *Poisson equation, 1D*

40.1 1-dimensional Poisson equation

40.1.1 Strong form of the problem

$$-\left(\nu u'(x)\right)' = f(x) \quad , \qquad x \in [a,b]$$

Supplied with proper boundary conditions. As an example,

- Dirichlet boundary conditions: $u(\overline{x}) = \overline{u}$
- Neumann boundary conditions: $\nu u'(\overline{x}) = \overline{q}$
- Robin boundary conditions: $au(\overline{x}) + bu'(\overline{x}) = c$

40.1.2 Weak form of the problem

For $\forall w(x) \in \mathcal{H}^1([a,b])$[1] ,

$$\begin{aligned} 0 &= \int_{x=a}^{b} w(x) \left\{ -\left(\nu u'(x)\right)' - f(x) \right\} = \\ &= \int_{x=a}^{b} w'(x)\nu u'(x)\, dx - \int_{x=a}^{b} w(x) f(x)\, dx - \left[w(x)\nu u(x) \right]\Big|_{x=a}^{b} \end{aligned}$$

Boundary conditions…

[1] Which functional space? It would be nice to be as more precise as possible, without introducing too many "unnecessary" complications. Without going into details, 1) everything appears in the weak form should exists for the functions of that space: here, the functions must have L^2-integrable first-order derivative; 2) some results exist when test and unkwown functions belong to the same space: so, some "special" treatment of essential boundary conditions (on Dirichlet boundaries) is required.

Poisson equation with Dirichlet boundary conditions

By definition, test functions are identically zero on Dirichlet boundary conditions $w|_{S_D} = 0$.

Thus the weak form becomes

$$\int_{x=a}^{b} w'(x)\nu u'(x)\,dx = \int_{x=a}^{b} w(x)f(x)\,dx\ ,$$

along with the essential boundary conditions $u(a) = u_a$, $u(b) = u_b$.

Finite element method

Discretization of the domain. The domain $[a, b] =: \Omega$, is divided into elements Ω_i s.t.

$$\cup_i \Omega_i = \Omega$$
$$\Omega_i \cap \Omega_j = \emptyset \quad i \neq j$$

Exploiting additive property of integrals on the union of domains, integrals of the weak form become summation of integrals on the individual elements

$$\int_{\Omega} \cdots = \sum_i \int_{\Omega_i} \ldots$$

Choice of the base functions. *Lagrangian base functions* (1) $\phi_i(x_j) = \delta_{ij}$, 2) $\phi_i(x) = 0$, if $x \notin B(x_i)$; 3) $\sum_i \phi_i(x) = 1$, $\forall x \in \Omega$)

$$u(x) = \sum_i \phi_i(x) u_i\ .$$

Definition on a reference element $\xi \in [-1, 1]$ and then change of coordinates between "physical" and reference space to evaluate integrals.

Here, as the maximum order of the derivatives involved in the problem is 1, test functions must be piecewise-*linear* (or better, first degree polynomial) at least, in order to avoid numerically setting to zero some terms in the equation only due to poor approximation (here the term $\int_{x=a}^{b} w'(x)\nu u'(x)dx$ would be zero with piecewise-constant test functions).

First degree-polynomial Lagrangian base functions on the reference elements are

$$\widetilde{\varphi}_1(\xi) = \frac{1}{2}(1-\xi)$$
$$\widetilde{\varphi}_2(\xi) = \frac{1}{2}(1+\xi)$$

Evaluation of integrals. Polynomials functions are exactly evaluated using Gaussian integration, i.e. with a linear combination of the values of the function on a set of Gaussian nodes *x_g*

…

Analytical exact integration for first degree-polynomial. The coordinate transformation[2]

$$x = a_i \frac{1-\xi}{2} + b_i \frac{1+\xi}{2} = \frac{a_i + b_i}{2} + \xi \frac{\ell_i}{2}\ ,$$

transforms the reference element into physical elements: e.g. $x(\xi = -1) = a_i$, $x(\xi = 1) = b_i$. The Jacobian of this transformation is constant over the element,

$$\frac{dx}{d\xi} = \frac{\ell_i}{2}\ .$$

[2] Here, the transformation between physical and reference space is iso-parametric, as it uses the same base functions as those used for function approximations, $x(\xi) = a_i \widetilde{\phi}_1(\xi) + b_i \widetilde{\phi}_2(\xi)$.

On the element i,

$$
\begin{aligned}
K_{i,11} &= \int_{x=a_i}^{b_i} \frac{d\phi_1}{dx}(x)\frac{d\phi_1}{dx}(x)\,dx = \\
&= \int_{\xi=-1}^{\xi=1} \frac{d\xi}{dx}\frac{d\tilde{\phi}_1}{d\xi}\frac{d\xi}{dx}\frac{d\tilde{\phi}_1}{d\xi}\frac{dx}{d\xi}d\xi = \\
&= \int_{\xi=-1}^{\xi=1} \left(-\frac{1}{2}\right)\left(-\frac{1}{2}\right)d\xi\frac{2}{\ell_i} = \frac{1}{\ell_i}\,.
\end{aligned}
$$

Analogously,

$$
\begin{aligned}
K_{i,11} = K_{i,22} &= \frac{1}{\ell_i} \\
K_{i,12} = K_{i,21} &= -\frac{1}{\ell_i}
\end{aligned}
$$

$$
\begin{aligned}
M_{i,11} &= \int_{x=a_i}^{b_i} \phi_1(x)\phi_1(x)dx = \\
&= \int_{\xi=-1}^{1} \varphi_1(\xi)\varphi_1(\xi)\frac{dx}{d\xi}d\xi = \\
&= \int_{\xi=-1}^{1} \left[\frac{1}{2}(1-\xi)\right]^2 d\xi\frac{\ell_i}{2} = \\
&= \frac{\ell_i}{8}\left(2+\frac{2}{3}\right) = \frac{1}{3}\ell_i\,.
\end{aligned}
$$

and analogously

$$
\begin{aligned}
M_{i,11} = M_{i,22} &= \frac{\ell_i}{3} \\
M_{i,12} = M_{i,21} &= \frac{\ell_i}{6}
\end{aligned}
$$

```
import numpy as np
import scipy as sp

import matplotlib.pyplot as plt

#> Domain
a, b = 0, 1
l = b - a

#> Physical properties
nu = .01                        # Diffusion coefficient
f = 1.

#> Discretization
nel = 20             # n.of elements
nnodes = nel + 1
#> Dirichlet boundaries
i_dir = np.array([ 0, nel ])    # Global idx of nodes on Dirichlet boundaries
u_dir = np.array([ .0, .0 ])    # Value of the solution at nodes on Dir. boundaries

#> Volume forcing contribution
```

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```
f_nodal = np.ones(nel+1) * f    # Nodal values f(x_i) of volume forcing, here uniform␣
↪f(x) = f

#> Node coordinates array, rr, and element-node connectivity array, ee
rr = np.linspace(a,b, nel+1)
ee = np.array([[i, i+1] for i in np.arange(nel)])
#> Size of the elements
e_vol = np.array([ rr[ee[i,1]] - rr[ee[i,0]] for i in np.arange(nel) ])
```

```
#> Assembling stiffenss matrix, mass matrix, and forcing term
#> Stiffness matrix
ki = np.concatenate([[ee[i,0], ee[i,0], ee[i,1], ee[i,1]] for i in np.arange(nel)])
kj = np.concatenate([[ee[i,0], ee[i,1], ee[i,0], ee[i,1]] for i in np.arange(nel)])
ke = np.concatenate([nu * np.array([1., -1., -1., 1.,])/e_vol[i] for i in np.
↪arange(nel)])
K = sp.sparse.coo_array((ke, (ki, kj)),)

#> Mass matrix, here used for integration of the volume force
mi = np.concatenate([[ee[i,0], ee[i,0], ee[i,1], ee[i,1]] for i in np.arange(nel)])
mj = np.concatenate([[ee[i,0], ee[i,1], ee[i,0], ee[i,1]] for i in np.arange(nel)])
me = np.concatenate([np.array([2., 1., 1., 2.,])*e_vol[i]/6. for i in np.arange(nel)])
M = sp.sparse.coo_array((me, (mi, mj)),)

F = M @ f_nodal    # Volume force
```

Slicing the linear system

```
#> Apply essential boundary conditions, solve the linear system and retrieve the␣
↪solution
#> Method 1. Slicing matrices
# K u = f
# [ Kuu  Kud ] [ uu ] = [ fu ]
# [ Kdu  Kdd ] [ ud ]   [ fd ]
# with known: ud, fu; unknown: uu, fd found solving the problem
# Kuu * uu = fuu - Kud * ud  -> uu = ...
# fd = Kdu * uu + Kdd * ud
iD = i_dir.copy()
iU = np.array(list(set(np.arange(nnodes)) - set(iD)))
ud = u_dir.copy()

#> Slice stiffness matrix
Kuu = (K.tocsc()[:,iU]).tocsr()[iU,:]
Kud = (K.tocsc()[:,iD]).tocsr()[iU,:]
Kdu = (K.tocsc()[:,iU]).tocsr()[iD,:]
Kdd = (K.tocsc()[:,iD]).tocsr()[iD,:]

#> Slice forcing
fu = F[iU]

#> Solve the linear system
uu = sp.sparse.linalg.spsolve(Kuu, fu - Kud @ ud )
fd = Kdu @ uu + Kdd @ ud
```

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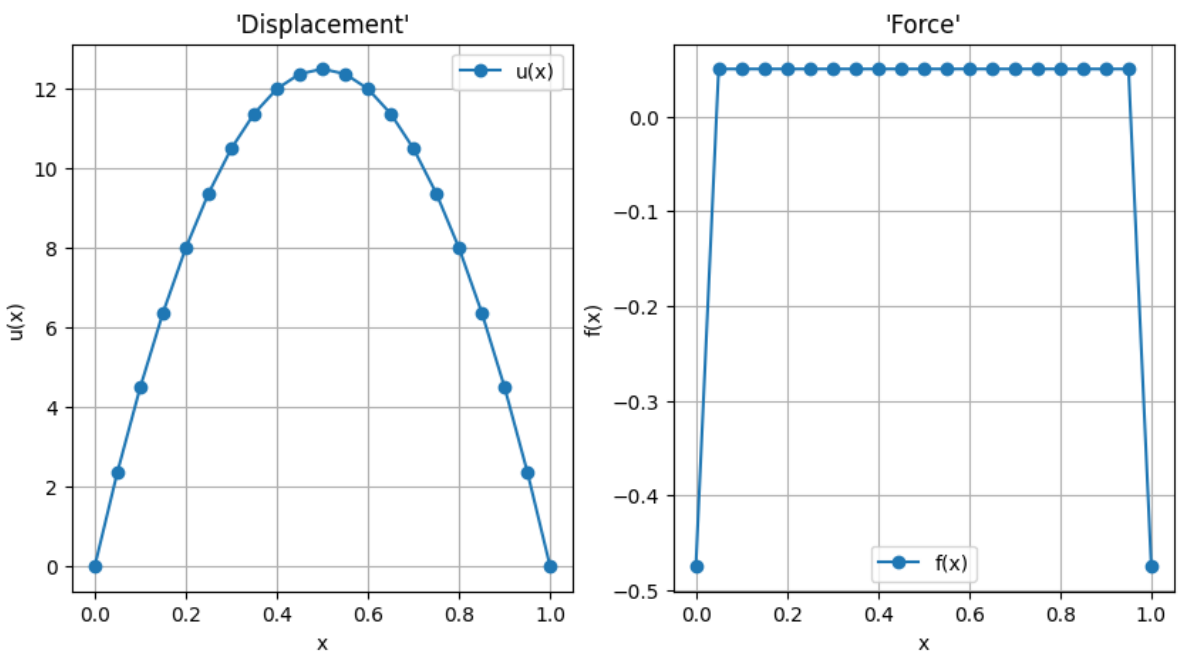
```
#> Re-assemble the solution
u = np.zeros(nnodes);  u[iU] = uu;  u[iD] = ud
f = np.zeros(nnodes);  f[iU] = fu;  f[iD] = fd
# print("uu: \n", uu)
# print("Fd: \n", fd)

#> Plot some results
fig, ax = plt.subplots(1,2, figsize=(10, 5))
ax[0].plot(rr, u, '-o', label='u(x)')
ax[0].legend()
ax[0].grid()
ax[0].set_title('\'Displacement\'')
ax[0].set_xlabel('x')
ax[0].set_ylabel('u(x)')
ax[1].plot(rr, f, '-o', label='f(x)')
ax[1].legend()
ax[1].grid()
ax[1].set_title('\'Force\'')
ax[1].set_xlabel('x')
ax[1].set_ylabel('f(x)')
```

```
Text(0, 0.5, 'f(x)')
```




```
#> Method 2. Augmented system
# ...
```

Augmenting the linear system

See discussion at the end of the *script implementing a basic finite volume method solution of 1-dimensional Poisson equation*.

Some comments and todos.

- **Convergence analysis.** It's possible to evaluate both convergence with the dimension of the elements and/or the degree of the polynomial base functions. The *exact solution* can be easily computed analytically via direct integration for simple distribution of load $f(x)$, and uniform ν
- **Result discussion.** "Force" contribution on Dirichlet boundary conditions involves both the node contribution to distributed load and the "constraint reaction". As an example, with a domain of size ℓ and uniform force distribution f, with $u(0) = u(\ell) = 0$, by *symmetry* and *global equilibrium* the force at boundaries are $-\frac{1}{2}q\ell = -\frac{1}{2}$. With the modelling choices done, the contribution of the uniform distributed load on an element on any of its nodes is $\frac{1}{2}q\ell_i = \frac{1}{2}q\frac{\ell}{n_{el}}$. Here with $n_{el} = 20$, this elementary contribution is $\frac{1}{40} = .025$: thus, the nodal contribution of internal nodes is twice (contribution from two neighboring elements) this value $.05$, and the force on each Dirichlet boundary is $-\frac{1}{2} + \frac{1}{40} = -0.475$.
- Try to set different values of Dirichlet boundary conditions, u_dir

CHAPTER

FORTYONE

FINITE VOLUME METHOD

Property 41.1

Evaluate flux on interfaces between cells, and distribute between neighboring cells.

41.1 1-dimensional Poisson equation

41.1.1 Integral form of the problem

The integral form of a Poisson problem reads, **for every** $V \in \Omega$,

$$-\oint_{\partial V} \hat{\mathbf{n}} \cdot (\nu \nabla u) = \int_V f \, .$$

supplied with proper boundary conditions. As an example,

- Dirichlet boundary conditions: $u(\overline{x}) = \overline{u}$
- Neumann boundary conditions: $\nu \hat{\mathbf{n}} \cdot \nabla u(\overline{x}) = \overline{q}$
- Robin boundary conditions: $au(\overline{x}) + bu'(\overline{x}) = c$

The multi-dimensional set may be useful also for 1-dimensional domains, to avoid confusions with signs

41.1.2 Domain, volume forcing and boundary conditions

```
import numpy as np
import scipy as sp

import matplotlib.pyplot as plt

#> Domain
a, b = 0, 1
l = b - a

#> Physical properties
nu = .01                        # Diffusion coefficient
f = 1.

#> Discretization
```

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```
nel = 10             # n.of elements
nnodes = nel + 1     # n.of nodes
nvol = nnodes        # n.finite volumes (node-centered FVM)

#> Dirichlet boundaries
i_dir = np.array([ 0, nvol-1])    # Global idx of nodes on Dirichlet boundaries
u_dir = np.array([ .0, .0 ])    # Value of the solution at nodes on Dir. boundaries

#> Volume forcing contribution
f_nodal = np.ones(nel+1) * f    # Nodal values f(x_i) of volume forcing, here uniform␣
↪f(x) = f

#> Node centered FVM
rr_n = np.linspace(a,b, nel+1)        # node coordinates (center of the node-centered␣
↪FVs)
rr_i = .5 * ( rr_n[:-1] + rr_n[1:] ) # coords of the inner cell interface
rr_b = np.array([ 2*rr_n[0]-rr_n[1], 2*rr_n[-1]-rr_n[-2] ])  # coords of the␣
↪boundaries

vols = np.array(
     [ np.abs(rr_i[0]-rr_n[0]) ] + \
       list( np.abs(rr_i[1:] - rr_i[:-1]) ) + \
     [ np.abs(rr_i[-1]-rr_n[-1]) ]
)

ni = len(rr_i)
nb = len(rr_b)

#> Interface-volume connectivity (ni, 2) (normal dir: first element -> second element)
iee = np.array([ [i, i+1] for i in np.arange(ni) ])
#> Boundary-volume connectivity (nb) (convention, normal dir: elem -> outside)
bee = np.array([0, nvol-1])
```

41.1.3 Discrete matheamtical problem

```
#> Evaluating boundary contributions
# Here assmbling a stiffness matrix. In many FVM applications, there's explicit time
# dependence and explicit time integration schemes are used, so there's no need to
# assemble any matrix while the algorithm just rely on accumulation of flux␣
↪contributions
# -> Add a link to a script implementing time-dependent "solver" for FVM, e.g.␣
↪hyperbolic problems

i_flux, j_flux, e_flux = [], [], []

#> Loop over internal interface
# accumulating and keep changing the dimension! May be very inefficient and slow
for i in np.arange(ni):
    flux_u = nu / np.abs( rr_n[ iee[i,1] ] - rr_n[ iee[i,0] ] )
    i_flux += [ iee[i,0], iee[i,0], iee[i,1], iee[i,1] ]
    j_flux += [ iee[i,0], iee[i,1], iee[i,0], iee[i,1] ]
    e_flux += [ -flux_u, flux_u, flux_u, -flux_u ]

e_flux = -np.array(e_flux)
```

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```
K = sp.sparse.coo_array((e_flux, (i_flux, j_flux)),)

#> Volume forcing
F = vols * f

#> Loop over boundary surfaces
# ...
# - Neumann: add a forcing term for finite volumes with sides on Neumann boundaries
# - Dirichlet: prescribe node value (e.g. 1) matrix slicing, 2) augmented system,...)
# - Robin: add a forcing term depending on the unknown value of the function ->␣
↪modify K matrix

# ...
```

41.1.4 Applying essential boundary conditions

The discrete counterpart of the continuous problem is a linear system that can be formally written as $\mathbf{A}\mathbf{u} = \mathbf{f}$. So far, the essential boundary conditions on Dirichlet boundary S_D have not been prescribed yet. Essential boundary conditions can be prescribed in (at least) two ways: 1) slicing the linear system; 2) augmenting the linear system.

Method 1. Slicing the linear system. The set of nodes $(\cdot)_d$ lying on Dirichlet boundaries can be distinguished from all the remaining nodes $(\cdot)_u$, and the linear system sliced accordingly

$$\begin{bmatrix} \mathbf{A}_{uu} & \mathbf{A}_{ud} \\ \mathbf{A}_{du} & \mathbf{A}_{dd} \end{bmatrix} \begin{bmatrix} \mathbf{u}_u \\ \mathbf{u}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_u \\ \mathbf{f}_d \end{bmatrix} .$$

On u-nodes, forcing $\mathbf{f}_u$ is known, while the value of the function $u(\mathbf{x})$ is unknown, and thus $\mathbf{u}_n$ is unknown; on the other hand, on d-nodes the value of the function $u(\mathbf{x})$ is known, and so $\mathbf{u}_d = \mathbf{u}_D$ is known, while the forcing $\mathbf{f}_d$ is not known (as it contains both a volume contribution and a "boundary contribution" required to prescribe essential boundary conditions, as it will be more clear later, hopefully).

After slicing, the problem is solved first finding $\mathbf{u}_u$ and then retrieving the value of $\mathbf{f}_d$,

$$\begin{aligned} \mathbf{A}_{uu}\mathbf{u} = \mathbf{f}_u - \mathbf{A}_{ud}\mathbf{u}_D \qquad &\rightarrow \qquad \mathbf{u}_u = \mathbf{A}_{uu}^{-1}\left(\mathbf{f}_u - \mathbf{A}_{ud}\mathbf{u}_D\right) \\ \mathbf{f}_d = \mathbf{A}_{du}\mathbf{u}_u + \mathbf{A}_{dd}\mathbf{u}_D \qquad &\rightarrow \qquad \mathbf{f}_d = \ldots \end{aligned}$$

Method 2. Augmenting the linear system. The linear system is augmented, a) explicityl adding equations $\mathbf{u}_d = \mathbf{u}_D$ for the essential boudary conditions, and splitting the forcing contribution on Dirichlet nodes as the sum of a known volume contribution and an unknown boundary contribution,

$$\mathbf{f} = \begin{bmatrix} \mathbf{f}_u^{vol} \\ \mathbf{f}_d^{vol} + \mathbf{f}_d^D \end{bmatrix} = \begin{bmatrix} \mathbf{f}_u^{vol} \\ \mathbf{f}_d^{vol} \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \mathbf{I} \end{bmatrix} \mathbf{f}_d^D .$$

Moving all the unkowns on the LHS, the augmented linear system reads

$$\begin{bmatrix} \mathbf{A}_{uu} & \mathbf{A}_{ud} & \mathbf{0} \\ \mathbf{A}_{du} & \mathbf{A}_{dd} & -\mathbf{I} \\ \mathbf{0} & -\mathbf{I} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{u}_u \\ \mathbf{u}_d \\ \mathbf{f}_d^D \end{bmatrix} = \begin{bmatrix} \mathbf{f}_u^{vol} \\ \mathbf{f}_d^{vol} \\ -\mathbf{u}_D \end{bmatrix} .$$

The (arbitrary) choice of the sign in the last block of equations (the ones prescribing the value of the unknown function on Dirichlet boundary) is usually chosen to preserve the symmetry of the original problem, if any.

todo *Discuss the reason why the system becomes determinate, after the application of the essential b.c.s*

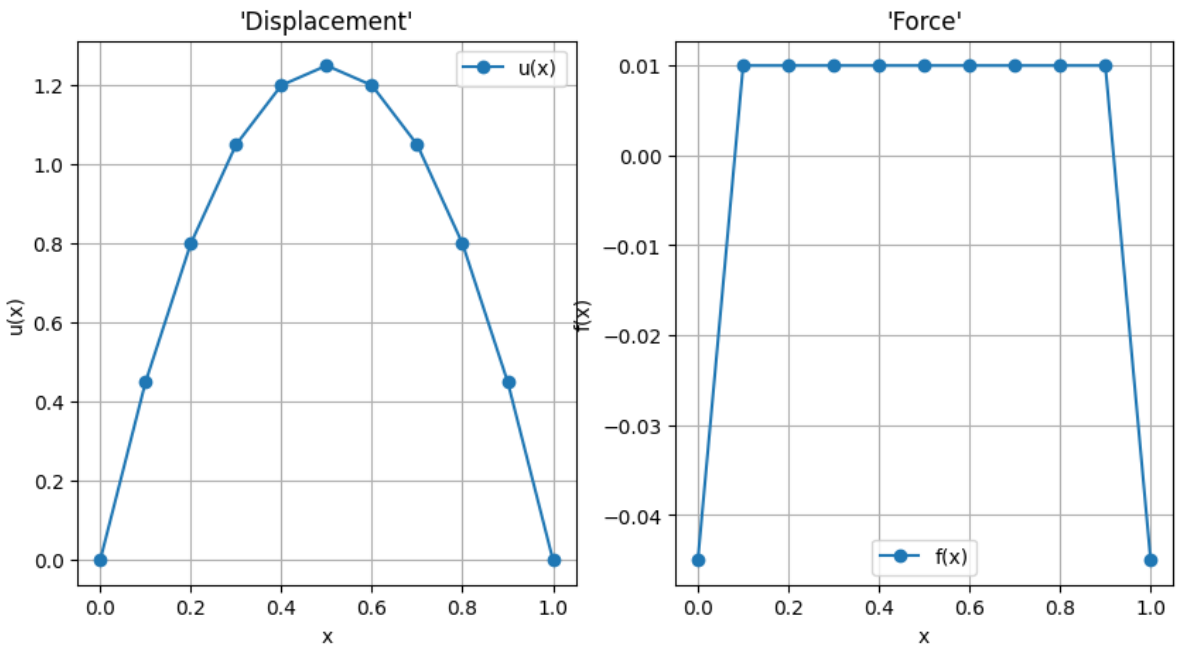
Slicing the linear system

Slice the linear system to prescribe the essential boundary conditions, solve the linear system, retrieve and plot the results.

```
#> Apply essential boundary conditions, solve the linear system and retrieve the␣
↪solution
#> Method 1. Slicing matrices
# K u = f
# [ Kuu  Kud ] [ uu ] = [ fu ]
# [ Kdu  Kdd ] [ ud ]   [ fd ]
# with known: ud, fu; unknown: uu, fd found solving the problem
# Kuu * uu = fuu - Kud * ud  -> uu = ...
# fd = Kdu * uu + Kdd * ud
iD = i_dir.copy()
iU = np.array(list(set(np.arange(nnodes)) - set(iD)))
ud = u_dir.copy()

print(iD)
print(iU)

#> Slice stiffness matrix
Kuu = (K.tocsc()[:,iU]).tocsr()[iU,:]
Kud = (K.tocsc()[:,iD]).tocsr()[iU,:]
Kdu = (K.tocsc()[:,iU]).tocsr()[iD,:]
Kdd = (K.tocsc()[:,iD]).tocsr()[iD,:]

#> Slice forcing
fu = F[iU]

#> Solve the linear system
uu = sp.sparse.linalg.spsolve(Kuu, fu - Kud @ ud )
fd = Kdu @ uu + Kdd @ ud

#> Re-assemble the solution
u = np.zeros(nnodes);  u[iU] = uu;  u[iD] = ud
f = np.zeros(nnodes);  f[iU] = fu;  f[iD] = fd
# print("uu: \n", uu)
# print("Fd: \n", fd)

#> Plot some results
fig, ax = plt.subplots(1,2, figsize=(10, 5))
ax[0].plot(rr_n, u, '-o', label='u(x)')
ax[0].legend()
ax[0].grid()
ax[0].set_title('\'Displacement\'')
ax[0].set_xlabel('x')
ax[0].set_ylabel('u(x)')
ax[1].plot(rr_n, f, '-o', label='f(x)')
ax[1].legend()
ax[1].grid()
ax[1].set_title('\'Force\'')
ax[1].set_xlabel('x')
ax[1].set_ylabel('f(x)')
```

```
[ 0 10]
[1 2 3 4 5 6 7 8 9]
```

```
Text(0, 0.5, 'f(x)')
```



Augmenting the linear system

Augmenting the linear system to prescribe the essential boundary conditions, solve the linear system, retrieve and plot the results.

todo

```
#> Method 2. Augmented system
# ...
```

Some comments and todos.

- **Convergence analysis.** It's possible to evaluate both convergence with the dimension of the elements and/or the degree of the polynomial base functions. The *exact solution* can be easily computed analytically via direct integration for simple distribution of load $f(x)$, and uniform ν
- **Result discussion.** "Force" contribution on Dirichlet boundary conditions involves both the node contribution to distributed load and the "constraint reaction". As an example, with a domain of size ℓ and uniform force distribution f, with $u(0) = u(\ell) = 0$, by *symmetry* and *global equilibrium* the force at boundaries are $-\frac{1}{2}q\ell = -\frac{1}{2}$. With the modelling choices done, the contribution of the uniform distributed load on an element on any of its nodes is $\frac{1}{2}q\ell_i = \frac{1}{2}q\frac{\ell}{n_{el}}$. Here with $n_{el} = 20$, this elementary contribution is $\frac{1}{40} = .025$: thus, the nodal contribution of internal nodes is twice (contribution from two neighboring elements) this value .05, and the force on each Dirichlet boundary is $-\frac{1}{2} + \frac{1}{40} = -0.475$.
- Try to set different values of Dirichlet boundary conditions, u_dir

41.2 FVM for hyperbolic problems

Approach

- Integral equations, often from balance or conservation laws
- First, problems in 1-dimensional domains are treated. Most of the features of a finite volume method are treated for 1-dimensional problems
  - flux: characteristics and common schemes
  - implementation: evaluation of flux at interfaces and distribution on cells
  - boundary conditions
  - …

  then everything is naturally generalized to multi-dimensional domain: each interface of a cell in a multi-dimensional domain is locally treated as a 1-dimensional domain.
- *Numerical flux*
  - features:
    * must be conservative…
    * must satisfy **entropy condition**, i.e. produce the physical solution (the limit of viscous problem for negligible viscosity) among the infinite possible solutions
  - schemes:
    * low-order
      · *Godunov*: the numerical flux is evaluated after solving a *Riemann problem* at each cell interface, taking the $x = 0$ state $\mathbf{u}(x = 0, t)$ to evaluate the flux $\mathbf{F}(\mathbf{u})$ at that cell boundary
      · *Roe* (~ linearized Godunov) + *entropy fix* required for the numerical method to select the physical solution
    * high-order schemes:
      · improve accuracy of differentiable solutions (no shock, no discontinuity) on regular grids
      · fail to represent the physical solution
    * high-resolution, (TVD schemes with flux limiters, MUSCL,…):
      · high-order methods where the solution is differentiable
      · low order method where the solution has discontinuities
      · high-resolution achieved as a linear combination of a low and high order flux

$$F(u_{i+1/2}) = f^{low}_{i+1/2} + \phi(r_{i-1,i}) \left(f^{high}_{i+1/2} - f^{low}_{i+1/2} \right) \ ,$$

        being $\phi(r_{i,i+1})$ the flux limiter, i.e. a function of the gradient r_i of the solution on the grid
- Theorems about numerics:
  - **CFL (Courant-Friedrichs-Lewy) condition** for explicit-time schemes (it could hold locally…),

$$\Delta t \leq \frac{\Delta x}{|\lambda|_{\max}}$$

  - *…results about fluxes…*

41.2.1 Numerical fluxes

Godunov flux

The numerical flux in Godunov scheme is the flux $\mathbf{F}$ evaluated with the state at the interface of two neighboring cells, resulting from the solution of the *Riemann problem* between the two cells.

For a piecewise constant approximation of the fields (like in standard finite volume methods), the evolution at each cell interface can be treated as a Riemann problem

In order to calculate the numerical flux at the interface i, separating cells A, B

1. The Riemann problem is solved at interface i separating the two cells with uniform fields $\mathbf{u}_A(t)$, $\mathbf{u}_B(t)$. Let $\mathbf{u}_i(x, t+\Delta t)$ be the solution, being x a local coordinate (orthogonal to the interface for multi-dimensional domains) with $x = 0$ at the interface, and $\Delta t > 0$. Riemann problem is a **non-linear problem**, so a non-linear solver is required. In order to avoid (expensive) non-linear methods, *Roe scheme* is introduced in the next section.
2. The state at interface $\mathbf{u}_i(x = 0, t + \Delta t)$ is retrieved from the solution of the Riemann problem
3. The numerical flux is evaluated with the state at the interface

$$\mathbf{F}_i(t) = \mathbf{F}(\mathbf{u}_i(0, t)) \ .$$

Roe flux

Instead of solving the original non-linear problem,

$$\begin{aligned} \mathbf{0} &= \partial_t \mathbf{u} + \partial_x \mathbf{F}(\mathbf{u}) = \\ &= \partial_t \mathbf{u} + \mathbf{A}(\mathbf{u}) \partial_x \mathbf{u} \ . \end{aligned}$$

a Roe solver aims at solving a linearized problem at each interface, namely

$$\mathbf{0} = \partial_t \mathbf{u} + \hat{\mathbf{A}}(\mathbf{u}_L, \mathbf{u}_R) \partial_x \mathbf{u} \ ,$$

being $\hat{\mathbf{A}}(\mathbf{u}_L, \mathbf{u}_R)$ an approximation of the convection matrix $\mathbf{A}(\mathbf{u})$ built with the states in the cells sharing the interface of interest: in time-explicit schemes, states in the cells $\mathbf{u}_L$, $\mathbf{u}_R$ are known, and so it is the linearized matrix $\hat{\mathbf{A}}(\mathbf{u}_L, \mathbf{u}_R)$.

Intermediate state, $\hat{\mathbf{u}}$. The linearized matrix can be written as the original matrix evaluated for an *intermediate state* $\hat{\mathbf{u}}(\mathbf{u}_L, \mathbf{u}_R)$,

$$\hat{\mathbf{A}}(\mathbf{u}_L, \mathbf{u}_R) = \mathbf{A}(\hat{\mathbf{u}}(\mathbf{u}_L, \mathbf{u}_R)) \ .$$

Roe flux. Roe flux looks similar to upwind scheme,

$$\begin{aligned} \mathbf{F}^{Roe} &= \frac{1}{2} \left(\mathbf{F}(\mathbf{u}_L) + \mathbf{F}(\mathbf{u}_R) - \left| \hat{\mathbf{A}}(\mathbf{u}_L, \mathbf{u}_R) \right| (\mathbf{u}_L - \mathbf{u}_R) \right) = \\ &= \frac{1}{2} \left(\mathbf{F}(\mathbf{u}_L) + \mathbf{F}(\mathbf{u}_R) - \hat{\mathbf{R}} \left| \hat{\Lambda} \right| \hat{\mathbf{L}} (\mathbf{u}_L - \mathbf{u}_R) \right) \ . \end{aligned}$$

Properties. Roe matrix needs to satisfy the following properties:

1. **diagonalization**: $\hat{\mathbf{A}}$ is diagonalizable
2. **consistency**: if $\mathbf{u}_L, \mathbf{u}_R \to \mathbf{u}$, then $\hat{\mathbf{u}} \to \mathbf{u}$
3. **conservation**: $\mathbf{A}(\hat{\mathbf{u}}) (\mathbf{u}_L - \mathbf{u}_R) = \mathbf{F}(\mathbf{u}_L) - \mathbf{F}(\mathbf{u}_R)$

Example 41.2.1 (P-sys. Intermediate state)

Intermediate state. For each value of $\hat{\rho}$,

$$\hat{u} = \frac{\sqrt{\rho_L} u_L + \sqrt{\rho_R} u_R}{\sqrt{\rho_L} + \sqrt{\rho_R}} \ ,$$

and thus

$$\hat{m} = \hat{\rho}\hat{u} \ .$$

Details

3. **Conservation**

$$\mathbf{A}(\hat{\mathbf{u}})(\mathbf{u}_L - \mathbf{u}_R) = \mathbf{F}_L - \mathbf{F}_R \ .$$

explicitly, using conservative variables

$$\begin{bmatrix} 0 & 1 \\ a^2 - \frac{\hat{m}^2}{\hat{\rho}^2} & 2\frac{\hat{m}}{\hat{\rho}} \end{bmatrix} \begin{bmatrix} \rho_L - \rho_R \\ m_L - m_R \end{bmatrix} = \begin{bmatrix} \rho_L - \rho_R \\ \frac{m_L^2}{\rho_L} + \rho_L a^2 - \frac{m_R^2}{\rho_R} - \rho_R a^2 \end{bmatrix} \ .$$

The first equation is identically satisfied. The second equation in $\hat{u}$ reads

$$\begin{aligned} 0 &= (a^2 - \hat{u}^2)(\rho_L - \rho_R) + 2\hat{u}\,(\rho_L u_L - \rho_R u_R) - \rho_L u_L^2 - \rho_L a^2 + \rho_R u_R^2 + \rho_R a^2 = \\ &= -\hat{u}^2(\rho_L - \rho_R) + 2\hat{u}\,(\rho_L u_L - \rho_R u_R) - \rho_L u_L^2 + \rho_R u_R^2 = \end{aligned}$$

and collecting terms

$$0 = \rho_L(\hat{u} - u_L)^2 - \rho_R(\hat{u} - u_R)^2$$

or, with $\rho > 0$,

$$\sqrt{\rho_L}(\hat{u} - u_L) = \mp\sqrt{\rho_R}(\hat{u} - u_R) \ ,$$

so that the solution that avoids singularities for $\rho_L = \rho_R$ (the one with the minus sign above) reads

$$\hat{u} = \frac{\sqrt{\rho_L} u_L + \sqrt{\rho_R} u_R}{\sqrt{\rho_L} + \sqrt{\rho_R}} \ ,$$

2. **Consistency.** From this choice of sign, consistency of the velocity follows immediately,

$$\lim_{\mathbf{u}_L, \mathbf{u}_R \to \mathbf{u}} \hat{\mathbf{u}}(\mathbf{u}_L, \mathbf{u}_R) = \mathbf{u} \ .$$

Arbitrary intermediate value of the density $\hat{\rho}$ needs to satisfy consistency, as well. Just as an example, if $\hat{\rho}$ is chosen to be the average of the densities, $\hat{\rho} = \frac{1}{2}(\rho_L + \rho_R)$, this is consistent choice.

Example 41.2.2 (Euler equations. Intermediate state)

Intermediate state. For each value of $\hat{\rho}$,

$$\begin{aligned} \hat{u} &= \frac{\sqrt{\rho_L} u_L + \sqrt{\rho_R} u_R}{\sqrt{\rho_L} + \sqrt{\rho_R}} \\ \hat{h}^t = \langle h^t \rangle_{\sqrt{\rho}} &= \frac{\sqrt{\rho_L} h_L^t + \sqrt{\rho_R} h_R^t}{\sqrt{\rho_L} + \sqrt{\rho_R}} \end{aligned}$$

while $\hat{\rho}$ can be found (if not trivial) solving an additional condition

$$\Delta P = \Delta\rho\, \partial_\rho \Pi(\hat{\rho}, \hat{\rho}\hat{u}, \hat{E}^t(\hat{\rho}, \hat{u}, \hat{h}^t)) + \Delta m\, \partial_m \Pi(\hat{\rho}, \hat{\rho}\hat{u}, \hat{E}^t(\hat{\rho}, \hat{u}, \hat{h}^t)) + \Delta E^t\, \partial_{E^t} \Pi(\hat{\rho}, \hat{\rho}\hat{u}, \hat{E}^t(\hat{\rho}, \hat{u}, \hat{h}^t)) \ .$$

If this condition is an identity - as it is for a perfect ideal gas - any value of $\hat{\rho}$ satisfying consistency is allowed.

Details

3. **Conservation**

$$\mathbf{A}(\hat{\mathbf{u}})(\mathbf{u}_L - \mathbf{u}_R) = \mathbf{F}_L - \mathbf{F}_R \ .$$

explicitly, using conservative variables

$$\begin{bmatrix} 0 & 1 & 0 \\ -\frac{m^2}{\rho^2} + \partial_\rho \Pi & \frac{2m}{\rho} + \partial_m \Pi & \partial_{E^t}\Pi \\ -\frac{m}{\rho^2}(E^t + \Pi) + \frac{m}{\rho}\partial_\rho \Pi & \frac{1}{\rho}(E^t + \Pi) + \frac{m}{\rho}\partial_m \Pi & \frac{m}{\rho}\left(1 + \partial_{E^t}\Pi\right) \end{bmatrix} \begin{bmatrix} \rho_L - \rho_R \\ m_L - m_R \\ E^t_L - E^t_R \end{bmatrix} = \left| \begin{bmatrix} \rho \\ \frac{m^2}{\rho} + \Pi(\rho, m, E^t) \\ \frac{m}{\rho}\left(E^t + \Pi(\rho, m, E^t)\right) \end{bmatrix} \right|_R^L$$

or

$$\begin{bmatrix} 0 & 1 & 0 \\ -\hat{u}^2 + \partial_\rho \Pi & 2\hat{u} + \partial_m \Pi & \partial_{E^t}\Pi \\ -\hat{u}\hat{h}^t + \hat{u}\partial_\rho \Pi & \hat{h}^t + \hat{u}\partial_m \Pi & \hat{u}\left(1 + \partial_{E^t}\Pi\right) \end{bmatrix} \begin{bmatrix} \rho_L - \rho_R \\ \rho_L u_L - \rho_R u_R \\ E^t_L - E^t_R \end{bmatrix} = \left| \begin{bmatrix} \rho \\ \rho u^2 + \Pi(\rho, m, E^t) \\ \rho u h^t \end{bmatrix} \right|_R^L$$

- The first equation is an identity
- The second equation in $\hat{u}$ gives

$$0 = -\left[\rho(\hat{u} - u)^2\right]\Big|_R^L + \left(\rho\partial_\rho \hat{\Pi} + \rho u \partial_m \hat{\Pi} + E^t \partial_{E^t}\hat{\Pi} - \Pi\right)\Big|_R^L \ ,$$

- The third equation in $\hat{h}^t$ gives

$$\begin{aligned} 0 = & \left[\rho(u - \hat{u})\hat{h}^t\right]\Big|_R^L + \left[\hat{u}\rho\partial_\rho \hat{\Pi} + \hat{u} m \partial_m \hat{\Pi} + \hat{u} E^t \partial_{E^t}\hat{\Pi} + \hat{u} \underbrace{E^t}_{=\rho h^t - p} - \rho u h^t\right]\Bigg|_L^R = \\ = & \left[\rho(u - \hat{u})\hat{h}^t + \rho h^t(\hat{u} - u)\right]\Big|_R^L + \hat{u}\left[\rho\partial_\rho \hat{\Pi} + m\partial_m \hat{\Pi} + E^t \partial_{E^t}\hat{\Pi} - \Pi\right]\Big|_L^R = \\ = & \left[\rho(u - \hat{u})(\hat{h}^t - h^t)\right]\Big|_R^L + \hat{u}\left[\rho\partial_\rho \hat{\Pi} + m\partial_m \hat{\Pi} + E^t \partial_{E^t}\hat{\Pi} - \Pi\right]\Big|_L^R \ . \end{aligned}$$

Adding the condition $\Delta P = \Delta\rho\partial_\rho \hat{\Pi} + m\partial_m \hat{\Pi} + E^t \partial_{E^t}\hat{\Pi}$, the second equation can be solved for $\hat{u}$ (choosing the solution providing consistency - see P-sys example for a brief discussion)

$$\hat{u} = \langle u \rangle_{\sqrt{\rho}} = \frac{\sqrt{\rho_L} u_L + \sqrt{\rho_R} u_R}{\sqrt{\rho_L} + \sqrt{\rho_R}}$$

and the third equation can be solved for $\hat{h}^t$ (using $\sqrt{\rho_L}(u_L - \hat{u}) = -\sqrt{\rho_R}(u_R - \hat{u})$ from the expression of $\hat{u}$ to get the equation $\sqrt{\rho}(\hat{h}^t - h^t_L) = -\sqrt{\rho_R}(\hat{h}^t - h^t_R)$),

$$\hat{h}^t = \langle h^t \rangle_{\sqrt{\rho}} = \frac{\sqrt{\rho_L} h^t_L + \sqrt{\rho_R} h^t_R}{\sqrt{\rho_L} + \sqrt{\rho_R}}$$

For a medium with generic equations of state, the condition about Δp introduced above can be eventually solved to find the value of $\hat{\rho}$. For a perfect ideal gas (PIG), this condition is an identity providing no additional information on the value of $\hat{\rho}$

2. **Consistency.** …

Entropy fix

Entropy fix is required to exploit numerical dissipation in choosing the physical solution, when some eigenvalue is close to zero. As an example, a threshold is set as a ratio of the local speed of sound a, e.g. $\delta = 0.1 \div 0.2$ and the Roe matrix is modified as

$$\hat{\mathbf{A}}^{s\,fix} = \hat{\mathbf{R}} \left| \hat{\Lambda}^{s\,fix} \right| \hat{\mathbf{L}} \,,$$

with

$$\left| \hat{\Lambda}_i^{s\,fix} \right| = \begin{cases} |\lambda_i| & |\lambda_i| \geq \delta a \\ \dfrac{|\lambda_i|^2}{2\delta a} + \dfrac{\delta a}{2} & |\lambda_i| < \delta a \end{cases}$$

todo *Add discussion about* **numerical dissipation** *and some numerical examples…*

41.3 Boundary conditions in hyperbolic problems

41.3.1 Wall

Domain at rest

Example 41.3.1 (Wall for Roe flux in 1-dimensional P-sys)

Roe intermediate state for a P-sys is

$$\begin{bmatrix} \hat{\rho} \\ \hat{m} \end{bmatrix} = \begin{bmatrix} \hat{\rho} \\ \hat{\rho}\hat{u} \end{bmatrix} \,,$$

with $\hat{\rho}$ satisfying consistency (as an example $\hat{\rho} = \sqrt{\rho_L \rho_R}$ or $\hat{\rho} = \frac{1}{2}\left(\rho_L + \rho_R\right)$), and

$$\hat{u} = \frac{\sqrt{\rho_L} u_L + \sqrt{\rho_R} u_R}{\sqrt{\rho_L} + \sqrt{\rho_R}} \,.$$

Wall boundary conditions for a P-sys implies no mass flux through the wall. The state of the ghost cell outside the domain is prescribed as

$$\mathbf{u}_g = \begin{bmatrix} \rho_g \\ \rho_g u_g \end{bmatrix} \,,$$

with $\rho_g = \rho_\partial$, $u_g = -u_\partial$. Roe intermediate state across the wall boundary interface has velocity

$$\hat{u}_\partial = \frac{\sqrt{\rho_g} u_g + \sqrt{\rho_\partial} u_\partial}{\sqrt{\rho_g} + \sqrt{\rho_\partial}} = 0 \,.$$

Upwind matrix in Roe flux. Using the spectral decomposition of the matrix

$$\mathbf{A} = \begin{bmatrix} u & \rho \\ \frac{a^2}{\rho} & u \end{bmatrix} \,,$$

i.e.

$$\mathbf{R} = \begin{bmatrix} \rho & \rho \\ a & -a \end{bmatrix} \quad , \quad \Lambda = \begin{bmatrix} u-a & 0 \\ 0 & u+a \end{bmatrix} \quad , \quad \mathbf{L} = \begin{bmatrix} \frac{1}{2\rho} & \frac{1}{2a} \\ \frac{1}{2\rho} & -\frac{1}{2a} \end{bmatrix}$$

the upwind matrix evaluated on the Roe intermediate state (no need for entropy fix; with entropy fix, the result doesn't change anyway, as there's no eigenvalue close to zero) reads

$$
\begin{aligned}
|\hat{\mathbf{A}}| = \hat{\mathbf{R}}\left|\hat{\Lambda}\right|\hat{\mathbf{L}} &= \begin{bmatrix} \rho & \rho \\ a & -a \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} \begin{bmatrix} \frac{1}{2\rho} & \frac{1}{2a} \\ \frac{1}{2\rho} & -\frac{1}{2a} \end{bmatrix} = \\
&= \begin{bmatrix} \rho a & \rho a \\ a^2 & -a^2 \end{bmatrix} \begin{bmatrix} \frac{1}{2\rho} & \frac{1}{2a} \\ \frac{1}{2\rho} & -\frac{1}{2a} \end{bmatrix} = \\
&= \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} .
\end{aligned}
$$

Roe flux. Putting everything together, Roe flux at wall reads

$$
\begin{aligned}
\mathbf{F}_\partial &= \frac{1}{2}\left[\mathbf{F}_g + \mathbf{F}_\partial - \left|\hat{\mathbf{A}}\right|\left(\mathbf{u}_g - \mathbf{u}_\partial\right)\right] = \\
&= \frac{1}{2}\left[\begin{bmatrix} -m_\partial \\ \rho_\partial\left(a^2 + u_\partial^2\right) \end{bmatrix} + \begin{bmatrix} m_\partial \\ \rho_g\left(a^2 + u_\partial^2\right) \end{bmatrix} - \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}\left(\begin{bmatrix} \rho_\partial \\ -m_\partial \end{bmatrix} - \begin{bmatrix} \rho_\partial \\ m_\partial \end{bmatrix}\right)\right] = \\
&= \begin{bmatrix} 0 \\ \rho_\partial (a + u_\partial)^2 \end{bmatrix} ,
\end{aligned}
$$

i.e. there's **no mass flux** across the wall, as the first component of the vector $\mathbf{F}_\partial$ is zero.

Example 41.3.2 (Wall for Roe flux in 1-dimensional Euler equations)

Moving domain

41.3.2 Characteristic-based boundary conditions

41.4 Examples of FVM for hyperbolic problems

1-dimensional P-sys

1-dimensional Euler equations for perfect ideal gas

Quasi 1-dimensional Euler equations for perfect ideal gas

1-dimensional Euler equations for perfect ideal gas on moving domains (ALE)

41.4.1 1-dimensional P-system

Useful classes for numerical methods

Basic libraries

Parent class

`HyperbolicSystem1d()` class with common functions (or just their templates) required to implement a finite volume method solver.

P-system class

Numerical simulation

System of equations

Initialized the system of equations to be solved

Domain

Boundary conditions

Simulation parameters

Time loop

Post-processing

41.4.2 1-dimensional Euler equations for Perfect Ideal Gas

Useful classes for numerical methods

Basic libraries

Parent class

`HyperbolicSystem1d()` class with common functions (or just their templates) required to implement a finite volume method solver.

Euler equations for perfect ideal gas class

Numerical simulation

System of equations

Initialized the system of equations to be solved

Domain

Boundary conditions

Initial conditions

Simulation parameters

Time loop

Post-processing

41.4.3 Quasi 1-dimensional Euler equations for Perfect Ideal Gas

Unsteady Euler equations for quasi-1 dimensional flows read

$$\partial_t \begin{bmatrix} \rho A \\ \rho u A \\ \rho e^t A \end{bmatrix} + \partial_z \begin{bmatrix} \rho u A \\ \frac{(\rho u A)^2}{\rho A} + pA \\ \frac{(\rho e^t A + pA)(\rho u A)}{\rho A} \end{bmatrix} = \begin{bmatrix} 0 \\ p\partial_z A \\ 0 \end{bmatrix}$$

Method 1. The same conservative variables as the 1-dimensional flow solver are used (ρ, m, E^t), and non-uniform

- appears as a source in momentum equation
- appears in discretization for cells A_i and as a factor multiplying fluxes $A_{i+1/2}F_{i+1/2}$

$$\mathbf{u}_i^{n+1} = \mathbf{u}_i^n - \frac{\Delta t}{\Delta A_i}\left(\mathbf{F}_{i+1/2}A_{i+1/2} - \mathbf{F}_{i-1/2}A_{i-1/2}\right) + \Delta t\, \mathbf{s}_i$$

Method 2. Another approach would be using $(\rho A, mA, E^t A)$ as conservative variables.

todo

Useful classes for numerical methods

Basic libraries

Parent class

`HyperbolicSystem1d()` class with common functions (or just their templates) required to implement a finite volume method solver.

Euler equations for perfect ideal gas class

Numerical simulation

System of equations

Initialized the system of equations to be solved

Domain

Boundary conditions

Initial conditions

Simulation parameters

Time loop

Post-processing

41.4.4 1-dimensional Euler equations for Perfect Ideal Gas on moving mesh (ALE)

Integral equations for general domain v_t. Starting from the balance equation for a control volume V at rest,

$$\frac{d}{dt}\int_V \mathbf{u} + \oint_{\partial V} \mathbf{F}(\mathbf{u}) \cdot \hat{\mathbf{n}} = \int_V \mathbf{s} \ ,$$

using Reynolds' transport theorem, the balance equation for an arbitrary volume v_t follows

$$\frac{d}{dt}\int_{v_t} \mathbf{u} - \oint_{\partial v_t} \mathbf{u}\mathbf{v}_b \cdot \hat{\mathbf{n}} + \oint_{\partial v_t} \mathbf{F}(\mathbf{u}) \cdot \hat{\mathbf{n}} = \int_{v_t} \mathbf{s} \ ,$$

or collecting all the boundary terms

$$\frac{d}{dt}\int_{v_t} \mathbf{u} + \oint_{\partial v_t} [\mathbf{F}(\mathbf{u}) - \mathbf{u}\mathbf{v}_b] \cdot \hat{\mathbf{n}} = \int_{v_t} \mathbf{s} \ .$$

Explicit Euler scheme

$$(\mathbf{u}_i^{n+1} V_i^{n+1}) = (\mathbf{u}_i^n V_i^n) - \Delta t \sum_{j \in B_i} \mathbf{F}_j^n \ ,$$

Roe flux …

Boundary conditions

Useful classes for numerical methods

Basic libraries

Parent class

`HyperbolicSystem1d()` class with common functions (or just their templates) required to implement a finite volume method solver.

Euler equations for perfect ideal gas class

Time dependent mesh

Time dependent mesh class, producing a mesh with uniform spacing, given the law of motion of the extreme points of the domain.

Numerical simulation

System of equations

Initialized the system of equations to be solved

Domain

VV of the moving mesh

Boundary conditions

Initial conditions

Simulation parameters

Time loop

Post-processing

CHAPTER

FORTYTWO

BOUNDARY ELEMENT METHOD

Part XIII

Boundary Methods for PDEs

CHAPTER

FORTYTHREE

GREEN'S FUNCTION METHOD

43.1 Poisson equation

General Poisson's problem

$$\begin{cases} -\nabla^2 \mathbf{u}(\mathbf{r},t) = \mathbf{f}(\mathbf{r},t) \\ + \text{b.c.} \end{cases}$$

with common boundary conditions

$$\begin{cases} \mathbf{u} = \mathbf{g} & \text{on } S_D \\ \hat{\mathbf{n}} \cdot \nabla \mathbf{u} = \mathbf{h} & \text{on } S_N \end{cases}$$

over Dirichlet and Neumann regions of the boundary.

Poisson's problem for Green's function, in infinite domain

$$-\nabla_{\mathbf{r}}^2 G(\mathbf{r};\mathbf{r}_0) = \delta(\mathbf{r}-\mathbf{r}_0)$$

Green's function method

$$\begin{aligned} E(\mathbf{r}_0,t)u_i(\mathbf{r}_0,t) &= \int_{\mathbf{r}\in\Omega} u_i(\mathbf{r},t)\delta(\mathbf{r}-\mathbf{r}_0) = \\ &= -\int_{\mathbf{r}\in\Omega} u_i(\mathbf{r},t)\nabla_{\mathbf{r}}^2 G(\mathbf{r}-\mathbf{r}_0) = \\ &= -\int_{\mathbf{r}\in\Omega} \nabla_{\mathbf{r}} \cdot (u_i \nabla_{\mathbf{r}} G - G\nabla_{\mathbf{r}} u_i) - \int_{\mathbf{r}\in\Omega} G\nabla^2 u_i = \\ &= -\oint_{\mathbf{r}\in\partial\Omega} \hat{\mathbf{n}} \cdot (u_i \nabla_{\mathbf{r}} G - G\nabla_{\mathbf{r}} u_i) + \int_{\mathbf{r}\in\Omega} G(\mathbf{r}-\mathbf{r}_0) f_i(\mathbf{r},t). \end{aligned}$$

An integro-differential boundary problem can be written using boundary conditions. As an example, using Dirichlet and Neumann boundary conditions, the integro-differential problem reads

$$\begin{aligned} &E(\mathbf{r}_0,t)\mathbf{u}(\mathbf{r}_0,t) + \int_{\mathbf{r}\in S_N} \mathbf{u}(\mathbf{r},t)\,\hat{\mathbf{n}} \cdot \nabla_{\mathbf{r}} G(\mathbf{r}-\mathbf{r}_0) - \int_{\mathbf{r}\in S_D} G(\mathbf{r}-\mathbf{r}_0)\,\hat{\mathbf{n}} \cdot \nabla_{\mathbf{r}} \mathbf{u}(\mathbf{r},t) = \\ &= -\int_{\mathbf{r}\in S_D} \mathbf{g}(\mathbf{r},t)\,\hat{\mathbf{n}} \cdot \nabla_{\mathbf{r}} G(\mathbf{r}-\mathbf{r}_0) + \int_{\mathbf{r}\in S_N} G(\mathbf{r}-\mathbf{r}_0)\,\mathbf{h}(\mathbf{r},t) + \int_{\mathbf{r}\in\Omega} G(\mathbf{r}-\mathbf{r}_0)\,\mathbf{f}(\mathbf{r},t). \end{aligned}$$

Green's function of the Poisson-Laplace equation reads

$$G(\mathbf{r};\mathbf{r}_0) = \frac{1}{4\pi}\frac{1}{|\mathbf{r}-\mathbf{r}_0|}\ .$$

Green's function of the Laplace equation

$$-\nabla^2 G = 0 \qquad \text{for } \mathbf{r} \neq \mathbf{r}_0$$

Solutions with spherical symmetry,

$$0 = \nabla^2 G = \frac{1}{r^2}\left(r^2 G'\right)' \quad \rightarrow \quad G'(r) = \frac{A}{r^2} \quad \rightarrow \quad G(r) = -\frac{A}{r} + B$$

Choosing $B = 0$ s.t. $G(r) \rightarrow 0$ as $r \rightarrow \infty$, and integrating over a sphere centered in $r = 0$ to get $A = -\frac{1}{4\pi}$,

$$1 = \int_V \delta(r) = -\int_V \nabla^2 G = -\oint_{\partial V} \hat{\mathbf{n}} \cdot \nabla G = -\oint_{\partial V} \hat{\mathbf{r}} \cdot \hat{\mathbf{r}} \frac{A}{r^2} = -4\pi\, A$$

43.2 Helmholtz equation

todo from Fourier to Laplace trasnform in the first lines of this section

A Helmholtz's equation can be thouoght as the time Fourier transform of a wave equation,

$$\begin{cases} \dfrac{1}{c^2}\partial_{tt}\mathbf{u}(\mathbf{r},t) - \nabla^2\mathbf{u}(\mathbf{r},t) = \mathbf{f}(\mathbf{r},t) \\ + \text{b.c.} \\ + \text{i.c.}\ , \end{cases}$$

Fourier transform in time of field $\mathbf{u}(\mathbf{r},t)$ reads

$$\tilde{\mathbf{u}}(\mathbf{r},\omega) = \mathcal{F}\{\mathbf{u}(\mathbf{r},t)\} = \int_{t=-\infty}^{+\infty} \mathbf{u}(\mathbf{r},t) e^{-i\omega t}\, d\omega$$

and, if $\mathbf{u}(\mathbf{r},t)$ is compact in time, Fourier transform of its time partial derivatives read

$$\begin{aligned} \mathcal{F}\{\dot{\mathbf{u}}(\mathbf{r},t)\} &= \int_{t=-\infty}^{+\infty} \dot{\mathbf{u}}(\mathbf{r},t) e^{-i\omega t}\, d\omega = \\ &= \mathbf{u}(\mathbf{r},t) e^{-i\omega t}\Big|_{t=-\infty}^{+\infty} + i\omega \int_{t=-\infty}^{+\infty} \mathbf{u}(\mathbf{r},t) e^{-i\omega t}\, d\omega = \\ &= i\omega \mathcal{F}\{\mathbf{u}(\mathbf{r},t)\} \end{aligned}$$

$$\mathcal{F}\{\partial_t^n \mathbf{u}(\mathbf{r},t)\} = (i\omega)^n \tilde{\mathbf{u}}\ .$$

The differential problem in the transformed domain thus reads

$$-\frac{\omega^2}{c^2}\tilde{\mathbf{u}} - \nabla^2\tilde{\mathbf{u}} = \tilde{\mathbf{f}}$$

Green's function of Helmholtz'e equation reads

$$G(\mathbf{r},s) = \alpha^+ \frac{e^{\frac{s|\mathbf{r}-\mathbf{r}_0|}{c}}}{|\mathbf{r}-\mathbf{r}_0|} + \alpha^- \frac{e^{-\frac{s|\mathbf{r}-\mathbf{r}_0|}{c}}}{|\mathbf{r}-\mathbf{r}_0|}$$

with $\alpha^+ + \alpha^- = \frac{1}{4\pi}$.

Being the Laplace transform,

$$\mathcal{L}\{f(t)\} = \int_{t=0^-}^{+\infty} f(t) e^{-st} dt\ ,$$

the Laplace transform of a causal function with time delay $\tau \geq 0$ reads

$$\mathcal{L}\{f(t-\tau)\} = \int_{t=0^-}^{+\infty} f(t-\tau)e^{-st}dt = \int_{z=-\tau}^{+\infty} f(z)e^{-s(z+\tau)}\,dz = e^{-s\tau}\int_{z=0}^{+\infty} f(z)e^{-sz}\,dz = e^{-s\tau}\,\mathcal{L}\{f(t)\}$$

having used causality $f(t) = 0$ for $t < 0$. Laplace transform of Dirac's delta $\delta(t)$ reads

$$\mathcal{L}\{\delta(t)\} = \int_{t=0^-}^{+\infty} \delta(t)\,dt = 1\ ,$$

so that $e^{-s\tau} = e^{-s\tau}\,1 = \mathcal{L}\{\delta(t-\tau)\}$.

Thus, Green's function for the wave equation reads

$$G(\mathbf{r},t;\mathbf{r}_0,t_0) = \alpha^+\frac{\delta\left(t-t_0+\frac{|\mathbf{r}-\mathbf{r}_0|}{c}\right)}{|\mathbf{r}-\mathbf{r}_0|} + \alpha^-\frac{\delta\left(t-t_0-\frac{|\mathbf{r}-\mathbf{r}_0|}{c}\right)}{|\mathbf{r}-\mathbf{r}_0|}$$

If $t \geq t_0$, and $G(\mathbf{r},t;\mathbf{r}_0,t_0)$ connects the past t_0 with the future t, the first term is not causal, and thus $\alpha^+ = 0$ and

$$G(\mathbf{r},t;\mathbf{r}_0,t_0) = \frac{1}{4\pi}\frac{\delta\left(t-t_0-\frac{|\mathbf{r}-\mathbf{r}_0|}{c}\right)}{|\mathbf{r}-\mathbf{r}_0|}\ .$$

Green's function of Helmholtz's equation

$$\frac{s^2}{c^2}G - \nabla^2 G = \delta(r)$$

$$G(r) = \frac{\alpha e^{kr} + \beta e^{-kr}}{r}$$

Proof:

- Gradient

$$\nabla G(r) = \hat{\mathbf{r}}\partial_r G = \hat{\mathbf{r}}\frac{\alpha(kr-1)e^{kr} + \beta(-kr-1)e^{-kr}}{r^2}$$

- Laplacian

$$\begin{aligned}\nabla^2 G(r) &= \frac{1}{r^2}\left(r^2 G'(r)\right)' = \\ &= \frac{1}{r^2}\left(\alpha(kr-1)e^{kr} + \beta(-kr-1)e^{-kr}\right)' = \\ &= \frac{1}{r^2}\left(\alpha k e^{kr} + \alpha k^2 r e^{kr} - \alpha k e^{kr} - \beta k e^{-kr} + \beta k^2 r e^{-kr} + \beta k e^{-kr}\right) = \\ &= \frac{1}{r}\left(\alpha e^{kr} + \beta e^{-kr}\right)k^2 = k^2 G(r)\ .\end{aligned}$$

  and thus $k^2 G(r) - \nabla^2 G = 0$, for $r \neq 0$;

- Unity

$$1 = \int_V \delta(r) = \int_V \left(k^2 G - \nabla^2 G\right) = \int_V k^2 G - \oint_{\partial V} \hat{\mathbf{n}}\cdot\nabla G$$

  the second term is the sum of two contributions of the form

$$\oint_{\partial V} \hat{\mathbf{n}}\cdot\nabla G^{\pm} = \oint_{\partial V} \frac{\alpha^{\pm}(\pm kr-1)e^{\pm kr}}{r^2} = 4\pi\alpha^{\pm}(\pm kr-1)e^{\pm kr}$$

the first term is the sum of two contributions of the form

$$
\begin{aligned}
k^2 \int_V G(r) = k^2 \int_V \frac{\alpha^{\pm} e^{\pm kr}}{r} = \\
&= k^2 \alpha^{\pm} \int_{R=0}^{r} \int_{\phi=0}^{\pi} \int_{\theta=0}^{2\pi} \frac{e^{\pm kR}}{R} R^2 \sin\phi \, dR \, d\phi \, d\theta = \\
&= k^2 \alpha^{\pm} \, 4\pi \int_{R=0}^{r} R \, e^{\pm kR} \, dR \, .
\end{aligned}
$$

the last integral can be evaluated with integration by parts

$$
\begin{aligned}
\int_{R=0}^{r} R \, e^{\pm kR} \, dR = \left[\frac{1}{\pm k} e^{\pm kR} R \right] \Big|_{R=0}^{r} \mp \frac{1}{k} \int_{R=0}^{r} e^{\pm kR} \, dR = \\
= \frac{1}{\pm k} e^{\pm kr} r - \frac{1}{k^2} e^{\pm kR} + \frac{1}{k^2} =
\end{aligned}
$$

Thus summing everything together,

$$
\begin{aligned}
1 = \alpha^{+} \left[4\pi k^2 \left(\frac{r}{k} e^{kr} - \frac{1}{k^2} e^{kr} + \frac{1}{k^2} \right) - 4\pi \left(kr - 1 \right) e^{kr} \right] + \alpha^{-} \left[\ldots \right] = \\
= 4\pi \left(\alpha^{+} + \alpha^{-} \right) \, .
\end{aligned}
$$

43.3 Wave equation

Wave equation general problem

$$
\begin{cases}
\frac{1}{c^2} \partial_{tt} \mathbf{u}(\mathbf{r}, t) - \nabla^2 \mathbf{u}(\mathbf{r}, t) = \mathbf{f}(\mathbf{r}, t) \\
+ \text{b.c.} \\
+ \text{i.c.}
\end{cases}
$$

Green's problem of the wave equation

$$
\frac{1}{c^2} \partial_{tt} G(\mathbf{r}, t; \mathbf{r}_0, t_0) - \nabla^2_{\mathbf{r}} G(\mathbf{r}, t; \mathbf{r}_0, t_0) = \delta(\mathbf{r} - \mathbf{r}_0) \delta(t - t_0)
$$

Integration by parts

$$
\begin{aligned}
E(\mathbf{r}_\alpha, t_\alpha) \mathbf{u}(\mathbf{r}_\alpha, t_\alpha) &= \int_{t \in T} \int_{\mathbf{r} \in V} \delta(t - t_\alpha) \delta(\mathbf{r} - \mathbf{r}_\alpha) \mathbf{u}(\mathbf{r}, t) = \\
&= \int_{t \in T} \int_{\mathbf{r} \in V} \left\{ \frac{1}{c^2} \partial_{tt} G - \nabla^2_{\mathbf{r}} G \right\} \mathbf{u} = \\
&= \int_{t \in T} \int_{\mathbf{r} \in V} \left\{ \frac{1}{c^2} \left[\partial_t \left(\mathbf{u} \partial_t G - G \partial_t \mathbf{u} \right) + G \partial_{tt} \mathbf{u} \right] - \nabla_{\mathbf{r}} \cdot \left(\nabla_{\mathbf{r}} G \, \mathbf{u} - G \nabla_{\mathbf{r}} \mathbf{u} \right) - G \, \nabla^2_{\mathbf{r}} \mathbf{u} \right\} = \\
&= \int_{\mathbf{r} \in V} \frac{1}{c^2} \left[\mathbf{u}(\mathbf{r}, t) \partial_t G(\mathbf{r}, t; \mathbf{r}_\alpha, t_\alpha) - G(\mathbf{r}, t; \mathbf{r}_\alpha, t_\alpha) \partial_t \mathbf{u}(\mathbf{r}, t) \right] \Big|_{t_0}^{t_1} + \\
&\quad + \int_{t \in T} \oint_{\mathbf{r} \in \partial V} \left\{ -\hat{\mathbf{n}}(\mathbf{r}, t) \cdot \nabla_{\mathbf{r}} G(\mathbf{r}, t; \mathbf{r}_\alpha, t_\alpha) \, \mathbf{u}(\mathbf{r}, t) + G(\mathbf{r}, t; \mathbf{r}_\alpha, t_\alpha) \, \hat{\mathbf{n}}(\mathbf{r}, t) \cdot \nabla_{\mathbf{r}} \mathbf{u}(\mathbf{r}, t) \right\} + \\
&\quad + \int_{t \in T} \int_{\mathbf{r} \in V} G(\mathbf{r}, t; \mathbf{r}_\alpha, t_\alpha) \underbrace{\left\{ \frac{1}{c^2} \partial_{tt} \mathbf{u}(\mathbf{r}, t) - \nabla^2_{\mathbf{r}} \mathbf{u}(\mathbf{r}, t) \right\}}_{= \mathbf{f}(\mathbf{r}, t)}
\end{aligned}
$$

$$
\int_{t \in T} \int_{\mathbf{r} \in V} \frac{1}{4\pi} \frac{\delta \left(t - t_\alpha + \frac{|\mathbf{r} - \mathbf{r}_\alpha|}{c} \right)}{|\mathbf{r} - \mathbf{r}_\alpha|} \mathbf{f}(\mathbf{r}, t) = \int_{\mathbf{r} \in V \cap B_{|\mathbf{r} - \mathbf{r}_\alpha| \leq c(t_\alpha - t)}} \frac{1}{4\pi |\mathbf{r} - \mathbf{r}_\alpha|} \mathbf{f} \left(\mathbf{r}, t_\alpha - \frac{|\mathbf{r} - \mathbf{r}_\alpha|}{c} \right)
$$

Part XIV

Optimization

CHAPTER

FORTYFOUR

OPTIMIZATION

- **Gradient-based methods**, and Hessian based, for continuos functions
- Free and constrained optimization
- …

44.1 Unconstrained optimization

Given $f(\mathbf{x})$, with $\mathbf{x} \in \mathbb{R}^n$, a **candidate local** extreme $\mathbf{x}^*$ is a point where the *gradient* $\partial_{\mathbf{x}} f$ of function f is zero

$$\partial_{\mathbf{x}} f(\mathbf{x}^*) = \mathbf{0} \; .$$

Among these candidates, local extreme may be:

- maximum: if the Hessian evalauted in $\mathbf{x}^*$ is definite negative,

$$\begin{cases} \partial_{\mathbf{x}} f(\mathbf{x}^*) = \mathbf{0} \\ \partial_{\mathbf{xx}} f(\mathbf{x}^*) > 0 \end{cases}$$

- minimum: if the Hessian evalauted in $\mathbf{x}^*$ is definite positive,

$$\begin{cases} \partial_{\mathbf{x}} f(\mathbf{x}^*) = \mathbf{0} \\ \partial_{\mathbf{xx}} f(\mathbf{x}^*) < 0 \end{cases}$$

Here the formalism $\partial_{\mathbf{xx}} f(\mathbf{x}^*) < 0$ indicates that the Hessian matrix is negative definite, see *Example 44.1.1*.

Example 44.1.1 (Positive/negative definite Hessian)

Local polynomial expansion of a differentiable function $\mathbf{x}$ around $\mathbf{x}^*$ reads

$$f(\mathbf{x}) = f(\mathbf{x}^*) + \Delta\mathbf{x}^T \partial_{\mathbf{x}} f(\mathbf{x}^*) + \frac{1}{2} \Delta\mathbf{x}^T \, \partial_{\mathbf{xx}} f(\mathbf{x}^*) \, \Delta\mathbf{x} + o(|\Delta\mathbf{x}|^2) \; ,$$

with $\Delta\mathbf{x} = \mathbf{x} - \mathbf{x}^*$.

Candidate extreme points are thos points $\mathbf{x}^*$ where the gradient vanishes $\partial_{\mathbf{x}} f \mathbf{x}^*) = 0$. If the Hessian $H(\mathbf{x}^*) := \partial_{\mathbf{xx}} f(\mathbf{x}^*)$ is positive definite, by definition,

$$\mathbf{v}^T \, H(\mathbf{x}^*) \mathbf{v} > 0 \; , \qquad \forall \mathbf{v} \neq \mathbf{0} \; ,$$

then

$$f(\mathbf{x}) = f(\mathbf{x}^*) + \frac{1}{2} \Delta\mathbf{x}^T H(\mathbf{x}^*) \, \Delta\mathbf{x} + o(|\Delta\mathbf{x}|^2) > f(\mathbf{x}^*) \; ,$$

as $|\Delta\mathbf{x}| \to 0$, and thus $\mathbf{x}^*$ is a point of minimum, as $f(\mathbf{x}) > f(\mathbf{x}^*)$ for all the neighoring points $\mathbf{x}$.

44.2 Constrained optimization

$$f(\mathbf{x}) \,, \qquad \mathbf{x} \in D \subseteq \mathbb{R}^n$$

with constraints, such as

$$\begin{aligned} \mathbf{g}(\mathbf{x}) &= \mathbf{0} \\ \mathbf{h}(\mathbf{x}) &\geq \mathbf{0} \end{aligned}$$

44.2.1 Method of Lagrange multipliers for equality constraints

$$\tilde{f}(\mathbf{x}; \lambda) = f(\mathbf{x}) - \lambda^T \mathbf{g}(\mathbf{x}) \,.$$

Part XV

Reinforcement Learning

CHAPTER

FORTYFIVE

INTRODUCTION TO REINFORCEMENT LEARNING

CHAPTER

FORTYSIX

MARKOV PROCESSES

Definition 46.1 (Stochastic process)

A stochastic process is as collection of random variables $X(t)$ defined on a common probability space $(\Omega, \mathcal{F}, P)$, indexed by some set T, with outcomes in a measurable space (S, Σ),

$$\{X(t) : t \in T\} \ .$$

Markov process is a memomry-less stochastic process (see def *Definition 46.1*). For discrete-time Markov processes, state at time n only depends on state at time $n-1$

$$\langle S, P, \mu \rangle$$

with

- S (finite) set of states
- **P** state transition probability matrix, $P_{ss'} = P(s'|s)$
- μ a set of initial probability, $\mu_i = P(X_0 = i), \forall i$

Properties of probability matrix.

A **stationary MP** is a process described by a constant state transition probability.

1-step transition.

Notation: probability vectors as row vectors

In treating stochastic processes, probability vectors are usually treated as row vectors. Probability distribution over states s' at time n is the sum of probabilities or reaching s' starting from s at time $n-1$

$$p_n(s') = \sum_s P(s'|s) p_{n-1}(s) = \sum_s p_{n-1}(s) P_{ss'} \ ,$$

or using matrix formalism

$$\mathbf{p}_n = \mathbf{p}_{n-1} \mathbf{P} \ .$$

Properties of probability matrix.

Since $P_{ss'} = P(s'|s)$ represents the probability of arriving in s' starting from s (i.e. the *conditional probability* of arriving in s', starting from s), the entries of a matrix representing a time-discrete process are

$$0 \leq P_{ss'} \leq 1 \ .$$

The sum over s' is 1 (sum over all the possibilities),

$$\sum_{s'} P(s'|s) = \sum_{s'} P_{ss'} = 1 \ , \quad \forall s \qquad \text{or} \qquad \mathbf{P1} = \mathbf{1} \ .$$

Thus, a probability matrix $\mathbf{P}$ has a eigenvalue $\lambda = 1$ with the uniform vector $\mathbf{1}$ as eigenvector.

Is this unique? *No, in general.*

Existence of a stationary distribution

Spectral radius of a probability matrix is $\rho(\mathbf{P}) = 1$.

Proof

By *Gershgorin circle theorem*,

$$|\lambda - P_{ii}| \leq \sum_{j \neq i} |P_{ij}| \ ,$$

for i s.t. the components of the (right) eigenvalue $\mathbf{v}$ satisfy $|x_i| \geq |x_j|, \forall j$. As $P_{ij} \geq 0$ and $\sum_j P_{ij} = P_{ii} + \sum_{j \neq i} P_{ij} = 1$, it follows

$$|\lambda - P_{ii}| \leq \sum_{j \neq i} P_{ij} = \sum_j P_{ij} - P_{ii} = 1 - P_{ii} \ ,$$

meaning that any eigenvalue λ is contained in a circle centered in P_{ii} with radius $1 - P_{ii}$. It's not hard to prove that such a circle is contained in a circle centered in 0 with radius 1[1]. Thus, for all the eigenvalues $|\lambda| \leq 1$ holds, and at least one eigenvalue with $|\lambda| = 1$, namely the $\lambda = 1$ eigenvalue shown above, exists. Thus, the spetral radius of a probability matrix is $\rho(\mathbf{P}) = 1$.

n-step transition. n-step transition in a stationary MP reads

$$\mathbf{p}_{m+n} = \mathbf{p}_{m+n-1}\mathbf{P} = \mathbf{p}_{m+n-2}\mathbf{PP} = \cdots = \mathbf{p}_m \mathbf{P}^n \ .$$

First passage. …; recurrent states

Classification of states: accessible, communicating (and dividing states into partitions, disjoint classes); positive recurrent, periodic, ergodic; absorbing or transient

Classification of MP: absorbing, ergodic, regular

Stationary distribution, a probability vector $\mathbf{p}$ so that probability doesn't change in the next timestep,

$$\mathbf{p} = \mathbf{pP} \ .$$

Fundamental matrix

Other definitions: mixing rate, spectral gap

46.1 Markov Reward Processes

$$\langle S, P, R, \gamma, \mu \rangle$$

with

[1] $1 - P_{ii} \geq |\lambda - P_{ii}| = |\lambda - 0 + 0 - P_i i| \geq ||\lambda| - P_i i|$; if $|\lambda| \geq P_{ii}$ it follows $1 \geq |\lambda|$; if $|\lambda| \leq P_{ii}$…**todo**

- R **reward function**; it could be a stochastic variable, depending on a state with probability $p(r|s)$ *(or on the transition $p(r|s,s')$?)*
- γ **discount factor**, $\gamma \in [0,1]$; it's used to represent the present value of future rewards ($\gamma \sim 0$ short-sighted, $\gamma \sim 1$ far-sighted)

Return

- Time horizion: finite, indefinite (until stopping criteria, or absorbing state), infinite
- Possible choices of cumulative reward:
  - total reward, $V = \sum_i r_i$
  - average reward, $V = \frac{1}{n}\sum_{i=1}^{n} r_i$
  - discounted reward $V = \sum_i \gamma^{i-1} r_i$

Return v_t is defined as the total discount reward from timestep t

$$v_t = \sum_{k=0}^{+\infty} \gamma^k r_{t+k+1} = r_{t+1} + \gamma r_{t+2} + ...$$

Value function Expected value of the return v_t from state $s_t = s$

$$V(s) := \mathbb{E}\left[v_t | s_t = s\right]$$

46.1.1 Bellman Equation for $V(s)$

$$\begin{aligned}
V(s) &= \mathbb{E}\left[v_t | s_t = s\right] = \\
&= \mathbb{E}\left[r_{t+1} + \gamma \sum_{k=0}^{+\infty} \gamma^k r_{t+2+k} \middle| s_t = s \right] = \\
&= \mathbb{E}\left[r_{t+1} + \gamma v_{t+1} | \, s_t = s \right] = \\
&= \mathbb{E}\left[r_{t+1} | s_t = s \right] + \gamma \mathbb{E}\left[v_{t+1} | s_t = s \right] = \\
&= R(s) + \gamma \sum_{s'} P(s'|s) \mathbb{E}\left[v_{t+1} | s_{t+1} = s' \right] = \\
&= R(s) + \gamma \sum_{s'} P(s'|s) \, V(s') \, .
\end{aligned}$$

or in matrix form

$$\mathbf{V} = \mathbf{R} + \gamma \mathbf{P} \mathbf{V} \, .$$

Solution of Bellman equation. Bellman equation for value function is linear. It can be solved directly, solving the linear problem

$$\left(\mathbf{I} - \gamma \mathbf{P}\right) \mathbf{V} = \mathbf{R} \, .$$

Solutions:

- direct solution of the linear system (feasible for small-dimensional MRP only?)
- iterative methods for large MRPs: DP (dynamic programming), MC (Monte-Carlo evaluation), TD (Temporal-Difference learning)

46.2 Markov Decision Processes

$$\langle S, A, P, R, \gamma, \mu \rangle$$

with

- A (finite) set of actions
- **P** state transition probability matrix, $P(s'|s,a)$
- R reward function, $R(s,a) = \mathbb{E}[r|s,a]$

46.2.1 Policies

A policy is a distribution over actions, given the initial state

$$\pi(a|s) = P(a|s)$$

Given a policy $\pi(a|s)$, a MDP becomes the MRP with

- transition probability

$$P^{\pi}(s'|s) = \sum_{a \in A} \pi(a|s) P(s'|s,a)$$

- reward (**todo** *Prove it!*)

$$R^{\pi}(s) = \sum_{a \in A} \pi(a|s) R(s,a)$$

Value functions.

- State value function, $V^{\pi}(s)$, expected return starting from s and following policy π

$$V^{\pi}(s) := \mathbb{E}_{\pi}[v_t | s_t = s]$$

- Action-value functon $Q^{\pi}(s,a)$, expected return starting from s, taking action a first and then following policy π

$$Q^{\pi}(s,a) := \mathbb{E}_{\pi}[v_t | s_t = s, a_t = a]$$

46.2.2 Bellman Equations

Bellman expectation equations. Bellman expectation equation for the state-value function $V^\pi(s)$ reads

$$
\begin{aligned}
V^\pi(s) &:= \mathbb{E}_\pi\left[v_t | s_t = s\right] = \\
&= \mathbb{E}_\pi\left[r_{t+1} + \gamma \sum_{k=0}^{+\infty} \gamma^k r_{t+2+k} \middle| s_t = s\right] = \\
&= \sum_{a\in A} \pi(a|s) \mathbb{E}_\pi\left[r_{t+1} + \gamma \sum_{k=0}^{+\infty} \gamma^k r_{t+2+k} \middle| s_t = s, a_t = a\right] = \\
&= \sum_{a\in A} \pi(a|s) \mathbb{E}\left[r_{t+1} | s_t = s, a_t = a\right] + \gamma \sum_{a\in A} \sum_{s'\in S} \pi(a|s) P(s'|s,a) \mathbb{E}_\pi\left[v_{t+1} | s_{t+1} = s'\right] = \\
&= \sum_{a\in A} \pi(a|s) R(s,a) + \gamma \sum_{s'\in S} P^\pi(s'|s) \mathbb{E}_\pi\left[v_{t+1} | s_{t+1} = s'\right] = \\
&= \sum_{a\in A} R(s,a) \left\{\pi(a|s) + \gamma \sum_{s'\in S} P(s'|s,a) V^\pi(s')\right\} = \\
&= R^\pi(s) + \gamma \sum_{s'\in S} P^\pi(s'|s) V^\pi(s') =
\end{aligned}
\tag{46.1}
$$

Bellman expectation equation for the action-value function $Q^\pi(s)$ reads

$$
\begin{aligned}
Q^\pi(s,a) &:= \mathbb{E}_\pi\left[v_t | s_t = s, a_t = a\right] = \\
&= \mathbb{E}\left[r_{t+1} | s_t = s, a_t = a\right] + \gamma \mathbb{E}_\pi\left[v_{t+1} | s_t = s, a_t = a\right] = \\
&= R(s,a) + \gamma \sum_{s'\in S} P(s'|s,a) \mathbb{E}_\pi\left[v_{t+1} | s_{t+1} = s'\right] = \\
&= R(s,a) + \gamma \sum_{s'\in S} P(s'|s,a) V^\pi(s') = \\
&= R(s,a) + \gamma \sum_{s'\in S} P(s'|s,a) \sum_{a'\in A} \pi(a'|s') \mathbb{E}_\pi[v_{t+1} | s_{t+1} = s', a_{t+1} = a'] = \\
&= R(s,a) + \gamma \sum_{s'\in S} P(s'|s,a) \sum_{a'\in A} \pi(a'|s') Q^\pi(s',a')
\end{aligned}
\tag{46.2}
$$

Bellman expectation operators todo

Optimal value functions, and optimal policies

Definition 46.2.1 (Optimal value functions)

Optimal state-value function $V^*(s)$ is defined as

$$V^*(s) = \max_\pi V^\pi(s) \ .$$

Optimal action-value function $Q^*(s,a)$ is defined as

$$Q^*(s,a) = \max_\pi Q^\pi(s,a) \ .$$

Usually, the goal of a problem involving MDP is the maximization of value functions, as the performance of the MDP.

Valus functions define a partial ordering of policies,

$$\pi \geq \pi' \text{ if } V^\pi(s) \geq V^{\pi'} \ \forall s \in S \ .$$

Theorem 46.2.1 (Optimal policies)

For a MDP,

- an optimal policy π^* exists, s.t. $\pi^* \geq \pi$, $\forall \pi$
- all optimal policies achieve the same value of optimal functions, namely

$$V^{\pi^*}(s) = V^*(s) \quad , \quad Q^{\pi^*}(s,a) = Q^*(s,a) \; .$$

- a deterministic optimal policy exists, and it's found optimizing action-value function over actions a,

$$\pi^*(a|s) = \begin{cases} 1 & , \quad \text{if } a = \text{argmax}_{a \in A} Q^*(s,a) \\ 0 & , \quad \text{otherwise} \end{cases}$$

Proof

todo

Property 46.2.1 (Property: optimal state-value and action-value functions)

$$V^*(s) = \max_a Q^*(s,a) \tag{46.3}$$

todo *Prove it!*

Bellman optimality equations. Using *Property 46.2.1* and the expression of action-value function in (46.2)

$$\begin{aligned}
V^*(s) &= \max_a Q^*(s,a) = \\
&= \max_a \max_\pi Q^\pi(s,a) = \\
&= \max_a \max_\pi \left\{ R(s,a) + \gamma \sum_{s' \in S} P(s'|s,a) V^\pi(s') \right\} = \\
&= \max_a \left\{ R(s,a) + \gamma \sum_{s' \in S} P(s'|s,a) V^*(s') \right\}
\end{aligned}$$

$$\begin{aligned}
Q^*(s,a) &= \max_\pi Q^\pi(s,a) = \\
&= \max_\pi \left\{ R(s,a) + \gamma \sum_{s' \in S} P(s'|s,a) V^\pi(s') \right\} = \\
&= R(s,a) + \max_\pi \left\{ \gamma \sum_{s' \in S} P(s'|s,a) V^\pi(s') \right\} = \\
&= R(s,a) + \gamma \sum_{s' \in S} P(s'|s,a) V^*(s') = \\
&= R(s,a) + \gamma \sum_{s' \in S} P(s'|s,a) \max_{a'} Q^*(s',a')
\end{aligned}$$

Bellman optimality operators

Properties of Bellman operators

Solving Bellman optimality equations

- non-linear
- no closed form solution
- iterative solutions:
  - Dynamic Programming (DP): Value Iteration (VI), Policy Iteration (PI)
  - Linear Programming (LP)
  - Reinforcement Learning (RL): Q-learning (off-policy), SARSA (on-policy),…

CHAPTER

FORTYSEVEN

METHODS OF SOLUTION OF MPD: DP AND LP

47.1 Brute force: policy search

47.2 Dynamic programming

Introduction to DP

- sequential or temporal approach to optimization
- solving complex problems breaking them down into sub-problems:
  - solve sub-problems
  - combine solutions of sub-problems
- for problems with 2 properties:
  - optimal substructure: optimal sol can be decomposed in sub-pbs
  - overlapping sub-problems: sub-pbs recur, sols can be cached and reused
- DP assumes full knowledge of the MDP
- used for planning (???)
- prediction: given MDP and policy, evaluate value function
- control: given MDP, evaluate optimal value function and policy

47.2.1 Policy iteration

Policy iteration alternates *policy evaluation* and *policy improvement* steps. When value function converges, iteration stops at **Bellman optimality condition**, as shown in the *algorithm* below.

Policy Evaluation

Bellman equation for a given policy π. Solution of Bellman equation (46.1),

$$V^\pi(s) = \sum_{a\in A} R(s,a) \left\{ \pi(a|s) + \gamma \sum_{s'\in S} P(s'|s,a) V^{pi}(s') \right\}$$

- direct solution. Complexity: …
- full policy-evaluation back-up

$$V_{k+1}(s) \leftarrow \sum_{a\in A} R(s,a) \left\{ \pi(a|s) + \gamma \sum_{s'\in S} P(s'|s,a) V_k^\pi(s') \right\}$$

  synchronous, asynchronous update; **todo** *convergence?*

Policy Improvement

Greedy update. Let π be a deterministic policy. Greedy update acts maximizing the action-value function for all state over all the actions,

$$\pi'(s) := \text{argmax}_{a\in A} Q^\pi(s,a) \ ,$$

and it "improves the value function" (which value function?) of any state,

$$Q^\pi(s,\pi'(s)) := \max_{a\in A} Q^\pi(s,a) \geq Q^\pi(s,\pi(s)) = V^\pi(s) \ . \tag{47.1}$$

In stochastic optimization, usually *ε-greedy improvement* is introduced to improve **exploration** and try to avoid converging to local optimum.

Theorem 47.2.1 (Policy improvement theorem)

Let π and π' a pair of deterministic policites s.t.

$$Q^\pi(s,\pi'(s)) \geq V^\pi(s) \ , \quad \forall s \in S$$

Then $\pi' \geq \pi$, i.e. $V^{\pi'}(s) \geq V^\pi(s)$, $\forall s \in S$.

Proof.

$$\begin{aligned}
V^\pi(s) &\leq Q^\pi(s,\pi'(s)) = \\
&=: \mathbb{E}_\pi \left[v_t | \, s_t = s, a_t = \pi'(s) \right] = \\
&= \mathbb{E}_\pi \left[r_{t+1} + \gamma v_{t+1} | \, s_t = s, a_t = \pi'(s) \right] = \\
(1) &= \mathbb{E}_{\pi'} \left[r_{t+1} | s_t = s \right] + \gamma \mathbb{E}_\pi \left[v_{t+1} | \, s_{t+1} = s', a_{t+1} = \pi(s) \right] = \\
&= \mathbb{E}_{\pi'} \left[r_{t+1} | s_t = s \right] + \gamma V^\pi(s) = \\
&\leq \mathbb{E}_{\pi'} \left[r_{t+1} | s_t = s \right] + \gamma Q^\pi(s,\pi'(s)) = \\
&= \mathbb{E}_{\pi'} \left[r_{t+1} | s_t = s \right] + \gamma \mathbb{E}_\pi \left[r_{t+2} + \gamma v_{t+2} | \, s_{t+1} = s, a_{t+1} = \pi'(s) \right] = \\
(2) &= \mathbb{E}_{\pi'} \left[r_{t+1} | s_t = s \right] + \gamma \mathbb{E}_{\pi'} \left[r_{t+2} | s_{t+1} = s \right] + \gamma^2 \mathbb{E}_\pi \left[v_{t+2} | \, s_{t+2} = s'', a_{t+2} = \pi(s) \right] = \\
&= \mathbb{E}_{\pi'} \left[r_{t+1} + \gamma r_{t+2} | s_t = s \right] + \gamma^2 V^\pi(s) = \\
&\leq \mathbb{E}_{\pi'} \left[r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \ldots | s_t = s \right] = \\
&\leq V^{\pi'}(s) \ .
\end{aligned}$$

with in (1) the first action a_t is drawn from $\pi'(s)$, s' results from the transition

$$P^{\pi}(s'|s) = P(s'|s,a)\pi(a|s) \ ,$$

and the following are drawn from π. The same notation is used in (2), and in all the following manipulations.

Policy Iteration algorithm

Policy iteration algorithm

$$\pi_0 \to V^{\pi_0} \to \pi_1 \to V^{\pi_1} \to ... \pi^* \to V^* \to \pi^*$$

Convergence is reached when $\pi_{n+1}(s) = \pi_n(s)$. Greedy update (47.1) gives

$$Q^{\pi_n}(s, \pi_{n+1}(s)) = Q^{\pi_n}(s, \pi_n(s)) = V^{\pi_n}(s) \ ,$$

namely, when the optimality condition (46.3) is reached.

47.2.2 Value Iteration

Unlike policy iteration, value iteration algorithm doesn't provide any explicit policy, but only optimal value of the value-functions.

An optimal policy for a MDP is built with an optimal action a^* first, followed by actions drawn from the optimal policy. Thus, it must satisfy

$$V^*(s) = \max_{a \in A} R(s,a) + V^*(s') \ ,$$

with $s_n = s$ and $s_{n+1} = s'$, resulting from transition $s'|s,a$. Subdividing the problem in sub-problems, **deterministic value iteration** proceeds from s' backwards to s,

$$V^*(s) \leftarrow \max_{a \in A} R(s,a) + V^*(s') \ .$$

or

$$V^*(s) \leftarrow \max_{a \in A} \left\{ R(s,a) + \gamma \sum_{s' \in S} P(s'|s,a) V(s') \right\} .$$

47.2.3 Dynamic programming: extensions

- synchronous, or in-place
- prioritized sweeping/update
- …

47.3 Linear programming

…

CHAPTER

FORTYEIGHT

METHODS OF SOLUTION OF MPD: RL

Introduction

- model-free / model-based
- on-policy / off-policy
- online / offline
- tabular / function approximation
- value-based / policy-based / actor-critic

Two main goals of methods in RL:

- *prediction or evaluation*: estimate the performance of a given policy
- *control*: find the best (or a good policy) to get the best performance

These two tasks usually rely on two processes: *evaluation* and *improvement*. Evaluation stage provides the perfomance of a given policy usually in terms of value functions. Improvement aims at finding a new policy with better performance.

Prediction involves only the evaluation step, while control usually involves an iteration over alternate evaluation and improvement processes.

48.1 Evaluation

48.1.1 Monte-Carlo RL

- model-free, learn from experience of complete episodes ((-) no bootstrapping)
- uses empirical mean return, instead of expected value
- first-visit MC (unbiased estimator) / every-visit MC (biased estimator)

First-visit MC

Initialize: π policy to be evaluated; V arbitrary state-value function, $R(s)$ empty list of returns

Loop until convergence or stopping criterion

- generate an episode using π
  - for each state s in the episode:
    - evalaute and append the return $r(s)$ following the first occurrence of s to the list of $R(s)$
    - update $V(s) = \text{average}(R(s))$

Every-visit MC

Initialize: π policy to be evaluated; V arbitrary state-value function, $R(s)$ empty list of returns

Loop until convergence or stopping criterion

- generate an episode using π
  - for each state s in the episode:
    - for each occurrence of state s in the episode:
      - evaluate and append the return $r(s)$ following the occurrence of s to the list of $R(s)$
      - update $V(s) = \text{average}(R(s))$

Using incremental mean update

$$\begin{aligned}
\mu_{N+1} &= \frac{x_1 + \cdots + x_N + x_{N+1}}{N+1} = \\
&= \frac{x_1 + \cdots + x_N}{N} \frac{N}{N+1} + \frac{x_{N+1}}{N+1} = \\
&= \frac{N}{N+1}\mu_N + \frac{1}{N+1}x_{N+1} = \\
&= \mu_N + \frac{1}{N+1}\left(x_{N+1} - \mu_N\right)
\end{aligned}$$

Incremental update of the average of the value function reads

$$V_k(s) \leftarrow V_{k-1}(s) + \frac{1}{k}\left(v_k - V_{k-1}(s)\right) \ ,$$

with v_k the return of state s_k.

A generalized update may follow, to introduce a free hyperparameter as a weight of older estimation,

$$\begin{aligned}
V_k(s) &\leftarrow V_{k-1}(s) + \alpha_k\left(v_k - V_{k-1}(s)\right) \\
&= (1-\alpha_k)V_{k-1}(s) + \alpha_k v_k
\end{aligned}$$

48.1.2 Temporal Difference (TD) Learning

- model-free, learn from experience of incomplete episodes (bootstrapping)
- …

48.1.3 TD prediction

- Starting from every visit MC

$$V(s_t) \leftarrow V(s_t) + \alpha(v_t - V(s_t))$$

- Update towards an **estimation of the return**
  - the simplest TD algorithm is $TD(0)$.
    - estimated return (called **TD target**) reads

$$v_t \sim r_{t+1} + \gamma V(s_{t+1})$$

  * **TD error** error δ_t is defined as

$$\delta_t := r_{t+1} + \gamma V(s_{t+1}) - V(s_t)$$

  * update step

$$\begin{aligned} V(s) &\leftarrow V(s) + \alpha_k \left(r_{t+1} + \gamma V(s_{t+1}) - V(s_{t+1})\right) = \\ &= V(s) + \alpha_k \, \delta_t \end{aligned}$$

- $TD(n)$ algorithm uses n-step prediction

$$v_t^{(n)} = r_{t+1} + \gamma r_{t+2} + ... \gamma r_{t+n} + \gamma^{n+1} V(s_{t+n})$$

- λ-return v_t^λ combines all n-step returns $v_t^{(n)}$ as

$$v_t^\lambda := (1-\lambda) \sum_{n=1}^{+\infty} \lambda^{n-1} v_t^{(n)}$$

  weights s.t. sum of weights $= 1$ and decay as $\lambda (< 1)$

- …eligibility traces, forward and backward TD,…

48.2 Model control

48.2.1 On- and Off-Policy Learning

- On-policy: learn policy π from experience sampled from π
- Off-policy: learn policy π from experience sampled from $\tilde{\pi}$

Greedy policy improvement

- over $V(s)$ requires model of MDP,

$$\pi'(s) = \text{argmax}_{a \in A} \left\{ R(s,a) + P(s'|s,a) V(s') \right\}$$

- **over $Q(s,a)$ is model-free**

$$\pi'(s) = \text{argmax}_{a \in A} Q(s,a) \; .$$

48.2.2 MC control

Note: Generalized policy iteration with MC evaluation

MC evaluation is model-free. To keep a model-free control task, a model-free policy improvement is required, like greedy policy improvement over $Q(s,a)$.

48.2.3 ϵ-greedy policy improvement

$$\pi(a|s) = \begin{cases} 1 - \varepsilon\left(1 - \dfrac{1}{N}\right) & , \quad \text{if } a = \text{argmax}_{a\in A} Q^{\pi}(s,a) \\ \dfrac{\varepsilon}{N} & , \quad \text{otherwise} \end{cases}$$

Theorem 48.2.1 (ε-greedy Policy Improvement)

For …, the ε-greedy policy π' w.r.t. Q^{π} is an improvement, $V^{\pi'}(s) \geq V^{\pi}(s)$.

Proof

$$\begin{aligned} Q^{\pi}(s,\pi'(s)) &= \sum_{a\in A} \pi'(a|s) Q^{\pi}(s,a) = \\ &= \sum_{a\in A, a\neq a^*} \frac{\varepsilon}{m} Q^{\pi}(s,a) + \left(1 - \varepsilon + \frac{\varepsilon}{m}\right) Q^{\pi}(s,a^*) = \\ &= \sum_{a\in A} \frac{\varepsilon}{m} Q^{\pi}(s,a) + (1-\varepsilon)\, Q^{\pi}(s,a^*) = \\ (1) &\geq \sum_{a\in A} \frac{\varepsilon}{m} Q^{\pi}(s,a) + (1-\varepsilon) \sum_{a\in A} \frac{\pi(a|s) - \frac{\varepsilon}{m}}{1-\varepsilon} Q^{\pi}(s,a) = \\ &= \sum_{a\in A} \pi(a|s) Q^{\pi}(s,a) = V^{\pi}(s) \end{aligned}$$

having used in (1) … And for the policy improvement theorem *Theorem 47.2.1*, $V^{\pi'}(s) \geq V(s)$. **todo** *Check it!*

GLIE (Greedy in the Limit of Infinite Exploration) …

GLIE MC Control

- k^{th} episode using \pi$$
- for each state s_t, and action a_t in the episode, update number of (s,a) pair occurence and action-value function

$$\begin{aligned} N(s_t,a_t) &\leftarrow N(s_t,a_t) + 1 \\ Q(s_t,a_t) &\leftarrow Q(s_t,a_t) + \frac{1}{N(s_t,a_t)}(v_t - Q(s_t,a_t)) \end{aligned}$$

- improve policy with ε-greedy method over action-value function Q,

$$\begin{aligned} \varepsilon &\leftarrow \frac{1}{k} \\ \pi &\leftarrow \varepsilon - \text{greedy}(Q) \end{aligned}$$

48.2.4 TD control

- (+): online, from incomplete episodes, lower variance
- use TD instead of MC for policy evaluation, **SARSA**:

$$Q(s,a) \leftarrow Q(s,a) + \alpha\left(r + \gamma Q(s',a') - Q(s,a)\right)$$

…

Variations. SARSA(λ),…

Note: SARSA is on-policy

48.2.5 Off-policy learning

Q-learning, is off-policy as a' is not taken from π

$$Q(s,a) \leftarrow Q(s,a) + \alpha\left(r + \gamma \max_{a'} Q(s',a') - Q(s,a)\right)$$

48.3 Actor-Critic

todo

48.4 Learning and planning

- Model-free RL: no model; **learn** policy and/or value function from real experience
- Model-based RL: learn a model from real experience; **plan** policy and/or value function from simulated experience
- Dyna: leanr a model from real experience; **learn and plan** a policy and/or value function from real and simulated experience

CHAPTER

FORTYNINE

LARGE OR CONTINUOUS MDPS

For small-dimensional problems, value functions can be efficiently represented by arrays or look-up tables,

$$\begin{aligned} V(s) \quad &\rightarrow \quad \mathbf{V} \in \mathbb{R}^{|S|} \\ Q(s,a) \quad &\rightarrow \quad \mathbf{Q} \in \mathbb{R}^{|S|\times|A|} \end{aligned}$$

For large problems with large number of discrete states $\sim 10^N$ ($N = 20$ for backgammon, $= 10^{170}$ for Go), or continuous state space, RL usually relies on **function approximation** of the value function,

$$\begin{aligned} V^\pi(s) &\sim V(s;\theta) \\ Q^\pi(s,a) &\sim Q(s,a;\theta) \end{aligned}$$

Parameters θ are updated during learning. Examples of function approximators, the most suitable depend on the problem itself (ANN, Fourier/wavelets basis, polynomial, piecewise-polynomial, coarse coding,…)

Optimizing/minimizing mean-squared error between $V(s,\theta)$ and $V^\pi(s)$

$$L(\theta) := \mathbb{E}_\pi \left[\left(V^\pi(s) - V(s;\theta)\right)^2 | s_t = s \right]$$

Gradient descent,

$$\begin{aligned} \Delta\theta = -\frac{1}{2}\alpha\nabla_\theta L(\theta) = \\ &= -\frac{1}{2}\alpha\nabla_\theta \, \mathbb{E}_\pi \left[\left(V^\pi(s) - V(s,\theta)\right)^2 | s_t = s \right] = \\ &= \alpha \, \mathbb{E}_\pi \big[\left(V^\pi(s) - V(s,\theta)\right) \nabla_\theta V(s,\theta) \mid s_t = s \big] \ , \end{aligned}$$

and the stochastic gradient decent update reads

$$\Delta\theta = \alpha \ \left(V^\pi(s) - V(s,\theta)\right) \ \nabla_\theta V(s,\theta) \ .$$

State representation with **features** $\phi(s)$,

$$\phi(s) = \begin{bmatrix} \phi_1(s) \\ \dots \\ \phi_n(s) \end{bmatrix}$$

State-value function as a linear combination of features

$$V(s;\theta) = \phi^T(s)\,\theta \ ,$$

and the gradient of the objective function directly follows from

$$\nabla_\theta V(s;\theta) = \phi(s) \ .$$

…

Action-value function as a linear combination of features

$$Q(s,a;\theta) = \phi^T(s,a)\,\theta \ ,$$

…

49.1 Value and Policy-Based RL

- value based
- policy based
- actor-critic

…

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